Beamforming

Theory, Algorithms, and Implementation

Joachim Nadal

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Preface

Beamforming is the art of turning an antenna array into a geometric instrument: a device able to listen selectively to some directions in space while rejecting others. At the intersection of wave physics, complex linear algebra, and statistical signal processing, it has become a central tool in radar systems, mobile communications, recent Wi-Fi, *massive* MIMO, microphone-array acoustics, and radio astronomy.

Aim of This Document

This book aims to provide a progressive understanding of the models, algorithms, and practical limits of modern beamforming. It is neither a collection of equations nor a catalogue of algorithms: each chapter systematically connects three inseparable levels,

$$\text{physics} \longrightarrow \text{mathematics} \longrightarrow \text{implementation.}$$

Physics explains why antennas receive different phases. Mathematics models this structure in terms of vectors, matrices, covariances, and subspaces. Implementation finally turns the models into executable code. Without the first level, formulas are only symbolic manipulations; without the second, intuitions remain vague; without the third, theory remains speculative.

The classical foundations of array processing are treated in detail in the books by Van Trees [1] and by Johnson and Dudgeon [2]. The present text builds on their central ideas with a more pedagogical and numerical orientation.

Who This Book Is For

The intended reader is an engineer, researcher, or student in signal processing, telecommunications, radar, SDR, or acoustics. Familiarity is assumed with the basics of linear algebra (vector spaces, eigenvalues, projections), classical signal processing (Fourier transforms, Shannon sampling), and probability (expectation, covariance, stationary processes). The remaining material is developed progressively.

Organization

The book is organized into seven parts that follow the logic of the subject.

Part I — Physical and Mathematical Foundations.
Wave propagation, delays, phases, the narrowband model, antenna-array geometry, spatial sampling, and the emergence of the *steering vector* as the spatial signature of a direction of arrival.

Part II — Deterministic Beamforming.
Beam pattern, Delay-and-Sum, and Null Steering. These methods, which do not rely on a statistical model of the environment, provide the geometric building blocks on which all more sophisticated approaches are based.

Part III — Statistical Beamforming.
Spatial Covariance, MVDR / Capon, LCMV and GSC Structure. The RF environment becomes a random object whose structure is learned in order to adapt the beamformer weights automatically.

Part IV — Iterative Adaptation.
Wiener Filter, LMS, NLMS and RLS. This part treats recursive weight updates in non-stationary environments.

Part V — Direction-of-Arrival Estimation.
Decomposition into signal and noise subspaces, MUSIC, ESPRIT, *Cramer-Rao bounds*, sparse methods, and Bayesian methods. This part goes beyond the resolution limit of conventional beamforming.

Part VI — Extensions.
Wideband beamforming, *MIMO* and *massive MIMO*, 2D/3D geometries, near-field propagation, and polarization. This part leaves the simplifying assumptions of Part I.

Part VII — Practical Implementation.
Calibration of a multi-antenna array, robustness to model errors, and an integrated anti-jamming chain. This is where theory meets hardware reality.

The three appendices collect reminders on complex linear algebra, perturbation analysis for eigenvalues and eigenvectors, and the Python code excerpts that appear throughout the book.

Code and Supplementary Material

The full Python code supporting the figures, simulations, and numerical examples is available at:

https://github.com/JoachimNadal/Adaptive-Beamforming-Simulation

The listings reproduced in the book are intentionally concise and pedagogical; the more complete executable version is in the repository.

Conventions

A single phase convention, stated in the notation chapter that follows this preface, is used throughout the book. A different convention is not wrong in itself, but it requires full consistency in the expressions for the *steering vector*, the *beam pattern*, the MUSIC pseudospectrum, and ESPRIT phases. This book adopts

$$a_m(\theta) = e^{-jmkd\sin\theta}$$

and keeps this convention throughout the developments that follow.

Contents

Conventions and Notation

Scalars, Vectors, and Matrices

Scalars are written in italic: x, t, f, θ, λ. Vectors are written in lowercase bold ($\boldsymbol{x}$, $\boldsymbol{w}$, $\boldsymbol{a}(\theta)$), and matrices in uppercase bold ($\boldsymbol{R}$, $\boldsymbol{A}$, $\boldsymbol{I}$). All vectors are column vectors. For an array of N antennas, a spatial *snapshot* is therefore

$$\boldsymbol{x}(n) = \begin{bmatrix} x_0(n) & x_1(n) & \cdots & x_{N-1}(n) \end{bmatrix}^{\mathrm{T}} \in \mathbb{C}^N.$$

Complex Operations

For a scalar $z \in \mathbb{C}$, conjugation is denoted by z^*; for a vector or matrix, the transpose $(\cdot)^{\mathrm{T}}$ and the conjugate transpose (Hermitian transpose) $(\cdot)^{\mathrm{H}}$ are used. The energy of a complex vector is $\|\boldsymbol{x}\|^2 = \boldsymbol{x}^{\mathrm{H}}\boldsymbol{x}$. The real and imaginary parts of a complex scalar are denoted by $\mathrm{Re}\{z\}$ and $\mathrm{Im}\{z\}$.

Phase Convention

Signals are handled in their IQ baseband form. The adopted time convention is $e^{j2\pi f_0 t}$, which yields, for a uniform linear array, the *steering vector*

$$\boldsymbol{a}(\theta) = \begin{bmatrix} 1 \\ e^{-jkd\sin\theta} \\ e^{-j2kd\sin\theta} \\ \vdots \\ e^{-j(N-1)kd\sin\theta} \end{bmatrix}, \qquad a_m(\theta) = e^{-jmkd\sin\theta}, \quad m = 0, \ldots, N-1.$$

The wavenumber is $k = 2\pi/\lambda$, where $\lambda = c/f_0$ is the wavelength and $c \approx 3 \cdot 10^8\,\mathrm{m} \cdot \mathrm{s}^{-1}$ is the speed of light. The *negative* sign convention in the exponential is kept **without exception** throughout the book: a sign change would reverse the angles estimated by ESPRIT, produce a mirror-image beam pattern, and break the consistency of the MVDR and MUSIC formulas.

Array Geometry

Unless stated otherwise, the array is a *Uniform Linear Array* (ULA) of N antennas aligned along one axis and separated by a constant spacing d:

$$p_m = md, \qquad m = 0, \ldots, N-1.$$

The antenna $m = 0$ is used as the reference: all delays and phase shifts are measured relative to it. The angle of arrival θ is measured relative to the array normal (the *broadside*), with $\theta \in [-90^\circ, +90^\circ]$ for the *visible region*. The standard spacing used in all simulations is

$$d = \lambda/2.$$

Beamforming and Beam Pattern

The output of a beamformer is defined by

$$y(n) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(n),$$

and the *beam pattern* associated with the weight vector $\boldsymbol{w}$ is

$$B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta).$$

The spatial power gain $|B(\theta)|^2$ is usually plotted in decibels after normalization to its maximum: $B_{\mathrm{dB}}(\theta) = 20\log_{10}\big(|B(\theta)|/\max_\theta |B(\theta)|\big)$.

Spatial Covariance

The theoretical spatial covariance of a wide-sense stationary snapshot process $\boldsymbol{x}(n)$ is

$$\boldsymbol{R} = \mathbb{E}\Big[\boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n)\Big] \in \mathbb{C}^{N\times N}.$$

Its estimate from K snapshots stacked in $\boldsymbol{X} = [\boldsymbol{x}(1), \boldsymbol{x}(2), \ldots, \boldsymbol{x}(K)]$ is

$$\widehat{\boldsymbol{R}} = \frac{1}{K}\sum_{n=1}^{K}\boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n) = \frac{1}{K}\boldsymbol{X}\boldsymbol{X}^{\mathrm{H}}.$$

Notation Discipline

To avoid collisions in advanced chapters, the following conventions are maintained as much as possible:

- N denotes the number of antennas, not the number of snapshots.
- K denotes the number of snapshots.
- L denotes the number of sources or direction-related constraints, depending on the explicitly stated context.
- M is reserved for auxiliary sizes: subarray size in spatial smoothing, grid size, or FFT size. In wideband settings, P often denotes the number of subbands to avoid confusion with M.
- d denotes the inter-element spacing and $D = (N-1)d$ the physical aperture.
- $k = 2\pi/\lambda$ is the wavenumber in the formulas. In code, loop indices are local; when confusion is possible, `idx`, `ell`, or `kk` is preferred.
- Formulas and code use θ in radians. Figures and numerical examples often display angles in degrees.
- $\boldsymbol{R}$ denotes the theoretical covariance, $\widehat{\boldsymbol{R}}$ its estimate, and $\widehat{\boldsymbol{R}}_\alpha = \widehat{\boldsymbol{R}} + \alpha\boldsymbol{I}$ the loaded covariance.

Summary Table of Notation

Notation	Meaning	Dimension / unit
N	Number of antennas in the array	integer scalar
m	Antenna index, $m = 0, \ldots, N-1$	integer scalar
d	Spacing between two adjacent antennas	m or λ
$D = (N-1)d$	Array aperture	m or λ
λ, f_0, c	Wavelength, carrier frequency, propagation speed	m, Hz, m/s
$k = 2\pi/\lambda$	Wavenumber	rad/m
θ	Angle of arrival measured from broadside	rad in code, deg in figures
$u = \sin\theta$	Reduced spatial variable, $u \in [-1, 1]$	dimensionless
K	Number of *snapshots*	integer scalar
L	Number of sources or angular constraints	integer scalar
M, P	Auxiliary size / number of subbands depending on context	integer scalar
$\boldsymbol{a}(\theta)$	Steering vector associated with direction θ	$N \times 1$
$\boldsymbol{x}(n)$	Spatial snapshot received at time n	$N \times 1$
$\boldsymbol{X}$	Snapshot matrix $[\boldsymbol{x}(1), \ldots, \boldsymbol{x}(K)]$	$N \times K$
$\boldsymbol{w}$	Beamformer weight vector	$N \times 1$
$y(n)$	Beamformer output	complex scalar
$B(\theta)$	Beam pattern $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta)$	complex scalar
$\boldsymbol{R}$	Theoretical spatial covariance	$N \times N$
$\widehat{\boldsymbol{R}}$	Empirical estimate of $\boldsymbol{R}$	$N \times N$
$\widehat{\boldsymbol{R}}_\alpha$	Loaded covariance $\widehat{\boldsymbol{R}} + \alpha\boldsymbol{I}_N$	$N \times N$
$\boldsymbol{R}_{i+n}$	Interference-plus-noise covariance	$N \times N$
$\boldsymbol{R}_n$	Noise-only covariance	$N \times N$
$\boldsymbol{R}_s$	Desired-signal covariance or signal-subspace covariance	$N \times N$
$\boldsymbol{C}$	LCMV constraint matrix	$N \times L_c$
$\boldsymbol{f}$	Imposed constraint responses	$L_c \times 1$
σ^2	Additive noise variance	scalar power
$\boldsymbol{I}_N$	Identity matrix of size N	$N \times N$
$\mathbf{0}$, $\mathbf{1}$	Zero vector or matrix / vector of ones	context-dependent
SNR, SINR	Signal-to-noise ratio, signal-to-(interference-plus-noise) ratio	linear or dB
WNG	White Noise Gain $1/\\|\boldsymbol{w}\\|^2$ under unit gain	linear or dB
DOA	Direction Of Arrival	angle

Part I

Physical and Mathematical Foundations

Chapter 1

Introduction to Beamforming

Before entering the technical definitions, it is useful to take time to understand *why* beamforming has become such a central topic over the past few decades. The answer can be summarized in one sentence: beamforming is the central framework through which the *directional* information carried by a wave becomes exploitable by signal processing. Systems that use electromagnetic or acoustic waves — radar, 5G, Wi-Fi, satellite links, radio astronomy, and acoustics — frequently rely on beamforming principles, sometimes without naming them explicitly. This chapter gives the overall picture, places the ideas in relation to one another, and introduces the path followed by the rest of the book.

1.1 From Time-Domain Processing to Spatial Processing

Classical signal processing is concerned with *what a signal contains*: its spectrum, energy, modulation, and signal-to-noise ratio. This approach is powerful, but it is blind to an entire dimension of the problem: the *geometry* of reception. Two identical waves arriving from different directions on an isolated antenna locally produce the same signal. A single antenna can indicate that a signal exists, but it is generally insufficient to infer its direction.

This limitation is not incidental; it is fundamental. A single-channel receiver has only one spatial sample of the electromagnetic field, and all directional information is collapsed by projection into that single measurement. It is the spatial equivalent of a snapshot taken with a monocular camera: the depth of the object is unknown. No subsequent time-domain processing, however sophisticated, can recover information that was lost at acquisition.

Moving to an antenna array changes the situation radically. Each antenna receives a slightly shifted version of the same wave: a delay of a few fractions of a nanosecond, a phase shift, and a particular spatial correlation with its neighbors. These differences, apparently negligible, encode the full propagation geometry. Beamforming can be read as the art of exploiting them.

The analogy with optics is instructive. One eye can see light; two eyes make it possible to infer depth through parallax. A single-aperture telescope has a resolution limited by its diameter, whereas an *interferometer* made of several distant mirrors can reach a resolution equivalent to that of a much larger telescope. Radio astronomers have long used this principle: VLBI (*Very Long Baseline Interferometry*) synthesizes a resolution equivalent to that of an Earth-sized telescope by exploiting precisely the phase differences between antennas separated by thousands of kilometers. Modern beamforming is the direct heir of this intuition.

Formally, the observed signal no longer depends only on time, but also on sensor position:

$$x(t) \longmapsto x(\boldsymbol{p}, t).$$

For an array of N antennas, the observation becomes a vector

$$\boldsymbol{x}(t) = \begin{bmatrix} x_0(t) & x_1(t) & \cdots & x_{N-1}(t) \end{bmatrix}^{\mathrm{T}} \in \mathbb{C}^N,$$

each component being the complex measurement of one antenna. Beamforming therefore operates on a *vector* object, whereas time-domain processing operates on a *scalar* object. All familiar operations — filtering, detection, estimation, separation — acquire a spatial version. Filtering in time selects frequencies; filtering in space selects directions.

One important remark is needed before going further: the transition from a scalar to a vector is not merely an increase in computational complexity. It introduces a geometry. The observation vector $\boldsymbol{x}(t)$ lives in a complex space of dimension N, where each *direction* corresponds to a particular combination of antennas, and each *plane wave* arriving from a particular angle of arrival excites a specific direction in that space. This correspondence between physical angles and directions in $\mathbb{C}^N$ — embodied by the *steering vector* built in chapter 5 — is the central object of the book. Much of the power of beamforming comes from the fact that "physical" operations (pointing a beam, nulling a direction, separating two sources) become *geometric* operations in this complex space: projections, orthogonalizations, and decompositions into subspaces.

1.2 The Fundamental Problem

The central problem of beamforming can be stated simply:

> *How should the signals received by several antennas be combined in order to reinforce some directions in space and attenuate others?*

Three physical facts underlie the answer and will recur throughout the book.

Finite propagation. An electromagnetic wave propagates at a finite speed. For a plane wave arriving at an angle θ on two adjacent antennas separated by a distance d, the differential delay is

$$\tau = \frac{d \sin\theta}{c}.$$

This delay is tiny — less than a nanosecond at microwave frequencies — but becomes measurable once converted into phase.

Delay $\longleftrightarrow$ phase. Under the narrowband assumption (the signal bandwidth is small compared with its carrier frequency), a time delay τ becomes, after IQ demodulation, a multiplication by

$$e^{-j2\pi f_0 \tau}.$$

Narrowband beamforming therefore reduces to complex linear algebra, with no explicit manipulation of sub-sample delays.

Spatial signature. Combining the two previous relations, the phase received on the m-th antenna of a ULA is

$$\varphi_m(\theta) = -mkd\sin\theta, \qquad m = 0, \dots, N-1.$$

This phase progression depends only on the direction of arrival: each angle θ leaves a recognizable *phase signature* on the array, the *steering vector* $\boldsymbol{a}(\theta)$. If the signature of a target direction is known, complex weights can be applied to the antennas to realign the contributions coming from that direction. The phases then add constructively, while signals from other directions, remaining misaligned, are naturally attenuated.

The elegance of this construction lies in the way it folds back on itself: the signature used to *recognize* a direction is also used to *create* it. The same factors $e^{-jmkd\sin\theta}$ that describe how a plane wave appears on the array serve as weights to synthesize a beam pointed in that direction. In this sense, beamforming is fundamentally *geometrically reversible*: reception and transmission are two sides of the same algebraic manipulation. This symmetry will be exploited explicitly in MIMO (chapter 20), where transmit precoding uses the same vectors as receive beamforming.

1.3 The Idea of a Spatial Beam

The term "beamforming" literally means *formation of a beam*. This beam generally has no material existence: it describes the *directional sensitivity* of the array. When the antenna signals are combined with a set of complex weights $\boldsymbol{w} = [w_0, w_1, \ldots, w_{N-1}]^{\mathrm{T}}$, the scalar output

$$y(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(t)$$

amplifies some directions and attenuates others. The object called the *beam pattern*

$$B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta)$$

describes this sensitivity and will be studied in detail in chapter 6. It will consistently involve three elements:

- a *main lobe* in the preferred direction, whose width determines angular resolution;
- *sidelobes*, through which unwanted interference may leak;
- *spatial zeros*, or *nulls*, which strongly cancel certain interfering directions.

Reading a beam pattern means reading, at a glance, the strengths and limits of a beamformer.

1.4 A Progression of Ideas

Beamforming is not a single algorithm but a family of techniques that can be organized according to the type and amount of information they use. The more that is known about the environment or the target, the more selective a beamformer can be made; however, it also becomes more sensitive to errors in that information. The thread running through the seven parts of the book is this progression: each chapter adds one piece of information and studies how it can be used. The list below gives a quick map and can serve as a guide for locating each chapter in the whole.

Conventional beamforming (Delay-and-Sum). Uses only the geometry and the desired direction. Its weights do not depend on the data: it is robust, but blind to the RF environment.

Null Steering. The first explicit form of spatial control: unit gain is imposed in the target direction and zero gain in selected interference directions. This is the first form of a *linear constraint* on the weight vector.

MVDR and statistical adaptive beamforming. The *spatial covariance* of the received signals is used to adapt the weights to the environment. The MVDR criterion (*Minimum Variance Distortionless Response*) minimizes received power while maintaining unit gain in the target direction: the beamformer automatically places its nulls where the unwanted energy is located.

LCMV. Generalizes MVDR to multiple linear constraints and includes Null Steering as a special case when no useful covariance is available.

LMS and RLS. Instead of a closed-form expression based on the covariance, the weights are adjusted progressively from an observed error. LMS proceeds by stochastic gradient descent; RLS uses a finer recursive estimate and converges faster at the cost of greater complexity.

MUSIC, ESPRIT. The system no longer filters spatially; it *estimates* the directions of arrival themselves. The spatial covariance has a signal/noise subspace structure, and these two methods exploit that structure to reach angular resolutions well beyond those of conventional beamforming.

Sparse and Bayesian methods. When very few snapshots are available or the sources are coherent, the estimation problem is reformulated as a sparse-representation problem (ℓ_1-SVD) or as a hierarchical Bayesian model (*Sparse Bayesian Learning*).

Beyond narrowband. Radar signals and some modern links violate the narrowband assumption. *Wideband* beamforming (*true time delay*, subbands, FIB) is then required. At the other extreme, *massive MIMO* and 5G systems use arrays with several dozen or even hundreds of elements, together with hybrid analog/digital architectures.

These families are not in competition; they correspond to different balances among information, performance, and robustness. In general, an operational system combines several of them. A typical anti-jamming chain might begin with a DOA estimator (MUSIC or ESPRIT) to identify the jammers, then build Null Steering constraints on the detected directions, and finally hand fine adaptation to an MVDR or LCMV. This is the architecture assembled explicitly in chapter 24.

1.5 Why Calibration Is Essential

The ideal model underlying the previous developments assumes identical antennas, positioned without error and connected to rigorously coherent RF chains. This assumption is rarely exactly satisfied. Unequal cable lengths, different gains and phases, poorly phased local oscillators, mutual coupling, and thermal drift all break the theoretical correspondence between direction of arrival and measured phase structure.

The impact is severe. Delay-and-Sum points away from the target; Null Steering places its nulls in the wrong locations; MVDR attenuates part of the desired signal while attempting to attenuate noise (*signal cancellation*); MUSIC produces biased peaks; and ESPRIT, which relies on an exact algebraic invariance, is particularly fragile. A serious treatment of beamforming cannot stop at ideal equations: calibration, robustness, and regularization techniques (such as *diagonal loading*) are integral parts of the subject. Part VII is devoted entirely to them.

To give an order of magnitude, in a representative but not universal simulation: on a very well calibrated array, MVDR may produce nulls near $-60\,\mathrm{dB}$; with a few degrees of phase error per channel, those same nulls may rise to around $-25\,\mathrm{dB}$; without calibration, they often remain much shallower. The gap between ideal theory and uncalibrated practice can therefore reach several tens of dB of anti-jamming capability. In this regime, no algorithmic refinement fully compensates for a calibration defect of that order.

1.6 Applications

The capabilities of beamforming explain its use in many modern systems that exploit waves:

- radars (detection, tracking, SAR imaging);
- 4G and 5G mobile telecommunications and *massive MIMO*;
- recent Wi-Fi, where explicit beamforming is now standard;
- satellite links and low-Earth-orbit constellations;
- radio astronomy, where kilometer-scale arrays synthesize giant virtual telescopes;
- acoustics and microphone arrays for sound-source separation;
- sonars;
- electronic warfare and anti-jamming systems;

- multi-channel SDR platforms, which make these techniques accessible at moderate cost.

In all these cases, the same idea is at work: *space becomes a signal-processing dimension in the same sense as time or frequency.*

1.7 Philosophy of the Book

Each chapter will follow, as much as possible, this progression:

1. physical intuition;
2. mathematical model;
3. geometric interpretation;
4. algorithmic consequences;
5. numerical implementation;
6. analysis of practical limits.

Theory and practice are not artificially separated: in beamforming, an equation is meaningful only if it describes a physical phenomenon, and an implementation is reliable only if its mathematical assumptions are understood.

Beamforming can be read as a chain of increasingly rich ideas. At the start, there is only a set of antennas and a wave that reaches them with different delays. By understanding that these delays become phases, the steering vector is constructed. By combining antennas with complex weights, a beamformer is obtained. By observing the spatial response, lobes, nulls, and resolution appear. By introducing covariance, the system becomes adaptive. By exploiting subspaces, the array becomes a high-resolution angular-estimation instrument. By relaxing the narrowband and simple-propagation assumptions, one enters the world of real systems.

The next chapter establishes the physical foundations — waves, phases, delays, and the plane-wave model — on which everything else will rest. It focuses on a few simple but structuring relations: how a monochromatic wave is represented as a complex exponential, why the wavenumber $k = 2\pi/\lambda$ plays in space the role that angular frequency plays in time, and how a time delay turns into a phase factor under the narrowband assumption. These elementary building blocks will give rise, at the end of chapter 5, to the *steering vector*, which will serve as the interface between physics and all the algorithms in the book.

Chapter 2

Electromagnetic Waves and Propagation

Beamforming exploits the fact that a wave does not reach all antennas in an array at the same time. This time offset, even when extremely small, becomes a phase shift for a narrowband signal. All the theory that follows rests on this fact. This chapter carefully builds the underlying physical model: the complex representation of a wave, the notion of wavenumber, the plane-wave model, and the conversion of a time delay into a phase shift.

The notions introduced here mostly belong to a first course in electromagnetics. It may be tempting to skim them. That would be a mistake: any poorly understood approximation propagates a few pages later as estimation bias, sign inversions, or silent degradations in algorithmic performance. Changing a phase convention in the wrong direction turns a beam pointed at $+30°$ into a beam pointed at $-30°$; using an unchecked narrowband approximation can lose several decibels of array gain without any warning message. Initial rigor is therefore worthwhile: it prevents hours of debugging later in the processing chain.

2.1 Complex Representation of a Wave

2.1.1 From the Real Cosine to the Complex Exponential

A real harmonic wave observed at a fixed point is written

$$x(t) = A\cos(2\pi f_0 t + \phi).$$

This form is intuitive but inconvenient as soon as several waves, several antennas, and several delays are combined: the trigonometry quickly becomes unmanageable. We therefore prefer the *complex representation*

$$\tilde{x}(t) = A\, e^{j(2\pi f_0 t+\phi)}, \qquad x(t) = \mathrm{Re}\{\tilde{x}(t)\}.$$

Phase shifts then become simple multiplications by a complex number, and antenna combinations reduce to linear algebra.

This representation is not merely notational. Every real signal admits an *analytic signal* $x_a(t) = x(t) + j\hat{x}(t)$, where $\hat{x}$ is the Hilbert transform of x, and the analytic signal of a pure cosine is exactly the complex exponential $e^{j(2\pi f_0 t+\phi)}$. In a modern RF chain, IQ demodulation directly provides this complex signal: the in-phase component I is the real part, and the quadrature component Q is the imaginary part.

An important consequence follows: throughout this book, when we write "signal $x(t)$", we actually mean its complex IQ envelope, sampled at a rate far below the carrier frequency. This is what an FPGA or DSP handles in practice. The RF carrier reappears only when phases must be related to a physical delay, through the relation $\Delta\varphi = -2\pi f_0 \tau$; everywhere else, it is invisible in the formalism. This distinction is a common source of confusion: an SDR beginner often looks in vain for the carrier in the samples – it disappeared during demodulation, as intended.

2.1.2 Euler Formula and Phase Geometry

Euler's formula

$$e^{j\alpha} = \cos\alpha + j\sin\alpha$$

shows that the complex exponential simultaneously encodes amplitude (modulus) and phase (argument). Multiplying a signal by $e^{j\alpha}$ rotates it by the angle α in the complex plane without changing its modulus. All operations needed in beamforming – aligning antennas, creating a beam, imposing a null – therefore reduce to rotations in $\mathbb{C}$, and hence to simple complex multiplications. This property is what makes narrowband beamforming both elegant and efficient in practice: an N-antenna beamformer is only an N-term complex inner product, executable in a few nanoseconds on a modern FPGA.

2.2 Frequency, Period, and Wavelength

In vacuum, an electromagnetic wave propagates at $c \approx 3 \cdot 10^8\,\mathrm{m} \cdot \mathrm{s}^{-1}$. For a frequency f_0, the temporal period is $T_0 = 1/f_0$, and the distance traveled during one period is the *wavelength*

$$\lambda = cT_0 = \frac{c}{f_0}.$$

The wavelength is the natural spatial scale of beamforming. Significant distances – inter-element spacing, array aperture, and the validity range of the plane-wave model – are always measured in fractions or multiples of λ. This habit is not incidental: reasoning in meters would hide the fact that the same array, operated at two different frequencies, produces two radically different "spatial geometries". What matters to the beamformer is never the centimeters themselves, but the phase steps they represent at the frequency under consideration.

Table 2.1: Typical wavelengths and $\lambda/2$ spacings.

Application	f_0	λ	$\lambda/2$
GSM (900 band)	900 MHz	33.3 cm	16.7 cm
GPS L1	1.575 GHz	19.0 cm	9.5 cm
Wi-Fi 2.4 GHz	2.4 GHz	12.5 cm	6.25 cm
Wi-Fi 5 GHz	5 GHz	6.0 cm	3.0 cm
X-band radar	10 GHz	3.0 cm	1.5 cm
5G mmWave	28 GHz	1.07 cm	5.4 mm
V-band link	60 GHz	5.0 mm	2.5 mm

2.3 Wavenumber and Wave Vector

The phase of a wave evolves both in time, at angular frequency $\omega_0 = 2\pi f_0$, and in space. We introduce the *wavenumber*

$$k = \frac{2\pi}{\lambda}$$

which represents the phase accumulated per unit distance: over a path of length r, the phase changes by kr. The analogy with temporal angular frequency is complete,

$$\omega_0 = \text{phase per unit time}, \qquad k = \text{phase per unit distance}.$$

In full generality, propagation occurs in an arbitrary spatial direction $\hat{\boldsymbol{u}}$, and k is replaced by the *wave vector*

$$\boldsymbol{k} = k\,\hat{\boldsymbol{u}}, \qquad \|\boldsymbol{k}\| = k = \frac{2\pi}{\lambda}.$$

The phase accumulated along a path $\boldsymbol{r}$ is then the inner product $\boldsymbol{k}^{\mathrm{T}}\boldsymbol{r}$.

Remark 2.1. With $d = \lambda/2$, one has $kd = \pi$, exactly a half-turn of phase per inter-element spacing at endfire. This critical value is the root of the spatial sampling condition demonstrated in the next chapter.

It is useful to expand this time/space analogy, because it recurs throughout the book. A temporal signal $x(t) = e^{j\omega_0 t}$ oscillates at angular frequency ω_0: the phase advances by $\omega_0\,\delta t$ over a time δt. A spatial signal $x(\boldsymbol{r}) = e^{-j\boldsymbol{k}^{\mathrm{T}}\boldsymbol{r}}$ oscillates similarly in space: the phase advances by $-\boldsymbol{k}^{\mathrm{T}}\delta\boldsymbol{r}$ for a displacement $\delta\boldsymbol{r}$. The negative *sign* in the spatial convention is dictated by the propagation direction: a wave that "advances" sees its phase "recede" in the spatial frame. This detail is crucial: changing this sign leads to steering vectors $e^{+jmkd\sin\theta}$ corresponding to mirror angles, exactly the kind of quiet bug that is diagnosed only after seeing the beam point to the wrong side.

2.4 Monochromatic Plane Wave

The monochromatic plane-wave model is written

$$\tilde{x}(\boldsymbol{r}, t) = A\,e^{j(2\pi f_0 t - \boldsymbol{k}^{\mathrm{T}}\boldsymbol{r} + \phi)}.$$

The term $2\pi f_0 t$ describes temporal evolution; the term $-\boldsymbol{k}^{\mathrm{T}}\boldsymbol{r}$ describes spatial evolution. The negative sign is not anecdotal: with the temporal convention $e^{j2\pi f_0 t}$, it means that the wave propagates *in the direction* of $\boldsymbol{k}$. To see this, follow a constant-phase point: $2\pi f_0 t - \boldsymbol{k}^{\mathrm{T}}\boldsymbol{r} = \text{const}$ imposes $\boldsymbol{r}$ parallel to $\boldsymbol{k}$ and motion at the speed $\omega_0/k = f_0\lambda = c$.

Wavefronts. At a fixed instant, the points where the phase takes a given value form a plane orthogonal to $\boldsymbol{k}$. These *wavefronts* move at the speed of light. It is precisely the local planarity of these fronts that will justify the plane-wave approximation for a distant source.

Effect of a delay. A signal delayed by τ is written

$$\tilde{x}(t - \tau) = A\,e^{j2\pi f_0 t}\,e^{-j2\pi f_0 \tau}.$$

A positive delay therefore produces the multiplicative factor

$$\boxed{e^{-j2\pi f_0 \tau}.}$$

This sign convention will be **kept without exception** throughout the book. As we will see in chapter 5, it leads to the steering vector $a_m(\theta) = e^{-jmkd\sin\theta}$. Any sign change in this expression must be accompanied by a consistent change in *all* formulas for Delay-and-Sum, Null Steering, MVDR, MUSIC, and ESPRIT.

2.5 Plane-Wave Approximation

2.5.1 Far-Field Source and Fraunhofer Region

A point source actually radiates a *spherical* wave,

$$\tilde{x}(r, t) = \frac{A}{r}\,e^{j(2\pi f_0 t - kr)}.$$

The factor $1/r$ represents the geometric attenuation of the amplitude. The plane-wave model assumes that, at the scale of the array, the wavefronts are locally planar. This approximation is valid when the source is sufficiently far away. The classical criterion is the *Fraunhofer condition*

$$R \gg \frac{2D^2}{\lambda},$$

where D is the array aperture, the distance between the two farthest antennas, and R is the source-array distance. This condition comes from a geometric analysis of the phase error between the exact spherical path and the approximate planar path.

Example 2.2 (Order of Magnitude)**.** ULA with $N = 16$ antennas spaced by $\lambda/2$ at 2.4 GHz: $D = 15 \cdot 6.25\,\text{cm} \simeq 0.94\,\text{m}$, hence $R_F = 2D^2/\lambda \simeq 14\,\text{m}$. Any source more than a few tens of meters away may be considered a plane wave.

Conversely, for short-range radar or near-field acoustic imaging, the approximation breaks down. One must then use *near-field beamforming*, where the steering vector depends not only on direction but also on source range. We will return to this briefly in chapter 21.

2.5.2 Polarization Assumption

A real electromagnetic wave is a vector field characterized by its polarization. Throughout the book we assume that the polarization is common to all antennas and constant in time, which allows the signals to be treated as complex scalars. Dual-polarization extensions, increasingly common in 5G and satellite systems, add a component to the steering vector but do not change the structure of the algorithms.

2.6 ULA Geometry and Angle of Arrival

The ULA consists of N antennas aligned on the x axis, at positions $p_m = md$, $m = 0, \ldots, N - 1$. The reference antenna is $m = 0$. The angle of arrival θ is measured with respect to the array normal:

- $\theta = 0°$: *broadside* direction, transverse to the array;
- $\theta = \pm 90°$: *endfire* direction, parallel to the array;
- $\theta \in [-90°, +90°]$: *visible region*, the set of physically observable angles.

Front/back ambiguity. An ideal ULA has a cone ambiguity: two waves arriving at θ and at $180° - \theta$ produce exactly the same projection on the array axis, and therefore the same phase signature. This ambiguity is resolved only by directional antennas or by a 2D geometry (planar or circular).

2.7 Propagation Delay Between Antennas

Consider a plane wave arriving at angle θ on two antennas separated by d. The path difference is the projection of d onto the propagation direction,

$$\Delta \ell = d \sin\theta,$$

and the corresponding delay is

$$\tau = \frac{d \sin\theta}{c}.$$

This delay is the fundamental physical quantity of the problem: it depends neither on the algorithm nor on the chosen representation. It will then be *interpreted* as a phase, as a complex factor, and as a component of a steering vector, but its nature remains that of a delay between two arrival instants.

Special cases.

- $\theta = 0$ (broadside): $\tau = 0$, no phase difference between the antennas;
- $\theta = \pm 90^\circ$ (endfire): $\tau = \pm d/c$, the maximum differential delay;
- arbitrary θ: an intermediate delay proportional to $\sin\theta$.

The arrival direction is fully encoded in the sign and amplitude of the delay.

2.8 From Time Delay to Phase Shift

For a narrowband signal, under the condition formalized in chapter 4, a delay τ produces

$$\tilde{x}(t-\tau) = \tilde{x}(t)\, e^{-j2\pi f_0 \tau},$$

that is, the phase shift

$$\Delta\varphi = -2\pi f_0 \tau = -\frac{2\pi}{\lambda} d \sin\theta = -kd\sin\theta.$$

This gives the *master equation*

$$\boxed{\Delta\varphi(\theta) = -kd\sin\theta.}$$

Although this formula is proved in only a few lines, it is the backbone of the entire book. It is *linear in* $\sin\theta$, not in θ: beamformers will naturally have better resolution near broadside, where $\sin\theta$ changes rapidly, than near endfire.

Example 2.3 (Typical Case)**.** At $f_0 = 2.4\,\text{GHz}$, $d = \lambda/2$, $\theta = 30^\circ$: $\Delta\varphi = -\pi \sin 30^\circ = -\pi/2$. A wave at 30° from broadside produces a quarter-turn of phase between adjacent antennas.

This elementary quantity – a quarter-turn of phase – is what must be measured to distinguish a direction of 30° from, say, a direction of 25°, which produces $\Delta\varphi = -\pi\sin 25^\circ \simeq -76^\circ$, or a 14° difference. That is tiny in absolute terms, yet sufficient for an array of a few antennas to separate them without ambiguity: the phase accumulates over the whole array aperture, and the cumulative degrees across the extreme antennas decide the separation. We will see this idea repeatedly when discussing angular resolution (chapter 3) and the Cramer-Rao bound (chapter 17).

2.9 Orders of Magnitude

The delays exploited by an array are extremely small. At 1 GHz with $d = \lambda/2 = 15\,\text{cm}$, the maximum endfire delay is $\tau_{\max} = d/c = 0.5\,\text{ns}$, exactly *one half-period* of the signal – not by coincidence, but as a consequence of the choice $d = \lambda/2$.

As frequency increases, these delays become increasingly tiny, but they remain the same fraction of the period (table 2.2).

Hardware consequence. An error of a few picoseconds between ADC channels – imperceptible in a single-channel system – already corresponds to several degrees of phase at 10 GHz and several tens of degrees at 60 GHz. This synchronization requirement is what makes multi-antenna arrays nontrivial to build, and it motivates chapter 22, which is dedicated to calibration.

Table 2.2: Endfire delays at $d = \lambda/2$ as a function of frequency.

f_0	T_0	$d = \lambda/2$	$\tau_{\max}$
900 MHz	1.11 ns	16.7 cm	555 ps
2.4 GHz	417 ps	6.25 cm	208 ps
5 GHz	200 ps	3.0 cm	100 ps
10 GHz	100 ps	1.5 cm	50 ps
28 GHz	35.7 ps	5.4 mm	17.8 ps
60 GHz	16.7 ps	2.5 mm	8.3 ps

2.10 Summary

A wave propagates at finite speed. A distant source produces, at the scale of an antenna array, a locally planar wavefront (Fraunhofer condition $R \gg 2D^2/\lambda$). For a ULA with spacing d and a wave at angle θ, the differential delay between antenna m and the reference is

$$\tau_m = \frac{md\sin\theta}{c},$$

which under the narrowband assumption becomes the phase shift

$$\Delta\varphi_m(\theta) = -mkd\sin\theta.$$

The next chapter shows how this phase progression is *sampled* along the array, and why the spacing $d = \lambda/2$ emerges as the spatial analogue of the Nyquist–Shannon criterion. We will see the notion of *spatial frequency* emerge, establishing a systematic parallel between temporal and spatial signal processing: aliasing, resolution, windowing, spectrum, all transpose. This analogy is one of the most powerful mental tools for learning beamforming from classical signal processing: what is already known about temporal spectral analysis applies mutatis mutandis to spatial analysis, and often one only needs to "change hats" to see a spatial problem fall into the same solution as its temporal counterpart.

Chapter 3

Antenna Arrays and Spatial Sampling

The previous chapter established that the angle of arrival of a wave is translated, on two adjacent antennas, into a phase shift $\Delta\varphi = -kd\sin\theta$. This relation is *local*. A real array contains several antennas, and it does not measure only an isolated phase shift: it measures a complete *spatial structure*. This chapter studies that structure as a whole. We show that an antenna array acts as an instrument of *spatial sampling*, and that this analogy with temporal sampling rigorously guides the choice of the spacing d, the number of antennas N, and the aperture $D = (N-1)d$.

This chapter plays a pivotal role in the book. All *sizing* questions find their answer here: how many antennas are needed to reach a given angular resolution? What is the maximum admissible spacing before ambiguities appear? What is the tradeoff between hardware cost and performance? The answers are deterministic: they follow from the spatial Nyquist condition and from the array-factor calculation for a ULA. Once these formulas are available, sizing a beamforming system becomes an exercise of a few minutes.

3.1 The Array as a Spatial Sampler

A continuous signal $x(t)$ is converted into a discrete sequence by observing it at times $t_n = nT_s$. If T_s is too large, different frequencies become indistinguishable: this is temporal aliasing.

An antenna array performs an analogous operation in space. The electromagnetic field $x(\boldsymbol{p}, t)$ exists at every point in space, but the array observes it only at the positions $p_m = md$. The role played by T_s in time is played by d in space. If d is too large, two different directions produce the same sampled phase on the antennas, and the array can no longer distinguish them.

Temporal		**Spatial**
t	$\longleftrightarrow$	p
T_s	$\longleftrightarrow$	d
$\omega = 2\pi f$	$\longleftrightarrow$	$\mu = kd\sin\theta$
frequency f	$\longleftrightarrow$	spatial frequency $\nu = (d/\lambda)\sin\theta$
frequency aliasing	$\longleftrightarrow$	spatial aliasing
Nyquist condition	$\longleftrightarrow$	condition $d \le \lambda/2$

The natural angular variable is not θ but $u = \sin\theta$, because the path difference is proportional to $\sin\theta$. Many results take a simpler form in u. It is also in u that the beam pattern of a uniformly weighted ULA takes exactly the form of a classical Dirichlet kernel, identical to the one encountered in discrete-time Fourier analysis. Working in u therefore places the spatial problem in its most natural language; we return to θ only when displaying physical angles to the reader or specifying target directions to operators.

3.2 Spatial Frequency and Demonstration of Aliasing

A plane wave at angle θ corresponds to the spatial sequence

$$a_m(\theta) = e^{-jmkd\sin\theta} = e^{-j2\pi m\nu(\theta)}, \qquad \nu(\theta) = \frac{d}{\lambda}\sin\theta.$$

This is the spatial analogue of a discrete-time sinusoid with digital angular frequency $2\pi\nu$. The quantity $\nu(\theta)$ is the *normalized spatial frequency* associated with the direction θ.

Two directions θ_1 and θ_2 produce exactly the same sampled sequence if and only if

$$e^{-j2\pi m\nu(\theta_1)} = e^{-j2\pi m\nu(\theta_2)} \quad \forall m \in \mathbb{Z},$$

that is, if $\nu(\theta_1) - \nu(\theta_2) \in \mathbb{Z}$. Spatial frequencies are therefore observable only *modulo* 1: this is the mathematical root of spatial aliasing.

3.3 Non-Aliasing Condition : $d \leq \lambda/2$

For the visible region $\theta \in [-90°, +90°]$, one has $\sin\theta \in [-1, 1]$, hence

$$\nu(\theta) \in \left[-\frac{d}{\lambda}, +\frac{d}{\lambda}\right].$$

The spatial Nyquist condition imposes

$$\left[-\frac{d}{\lambda}, +\frac{d}{\lambda}\right] \subset \left[-\frac{1}{2}, +\frac{1}{2}\right],$$

that is,

$$\boxed{d \leq \frac{\lambda}{2}.}$$

Interpretation.

- At $d = \lambda/2$, the visible region exactly fills the Nyquist interval: the array uses all of its spatial bandwidth without folding. This is the optimal case for a classical ULA.
- At $d < \lambda/2$, sampling is finer than necessary. There is no aliasing, but more antennas are needed for a given aperture.
- At $d > \lambda/2$, the Nyquist interval is exceeded: *grating lobes* appear. These lobes are not mere artifacts; they are true ambiguities: an interferer in a parasitic lobe can be amplified as much as the target.

Example 3.1 (Grating Lobes at $d = \lambda$)**.** For $d = \lambda$, $\lambda/d = 1$, and the ambiguous directions are $\sin\theta_2 = \sin\theta_1 + q$ with $q \in \mathbb{Z}$. Pointing toward $\theta_1 = 0$ then produces a parasitic lobe at $\theta_2 = \pm 90°$.

This example is not a laboratory curiosity. Arrays with $d > \lambda/2$ arise naturally when one wants to maximize resolution under a hardware cost constraint: doubling d while keeping N doubles the aperture D, and therefore divides the main-lobe width by two. The counterpart – grating lobes – may be tolerated if *a priori* information is available about the target arrival region, or if several subarrays with different spacings are combined to resolve the ambiguity (*coprime arrays* and related architectures, discussed in chapter 21).

3.4 Number of Antennas and Angular Resolution

The spacing d controls the absence of aliasing. The ability to *separate* two nearby directions depends on another quantity: the total aperture $D = (N-1)d$. The more the array extends in space, the more small angular differences accumulate measurable phase differences between its two ends.

Accumulated phase at the edge. For two directions θ_1 and θ_2, the phase difference at the extreme antenna $m = N-1$ is

$$|\Delta\varphi_{\text{edge}}| = kD\,|\sin\theta_2 - \sin\theta_1| = kD\,|\Delta u|.$$

A separation becomes significant when $|\Delta\varphi_{\text{edge}}| \sim 2\pi$, which gives the order of magnitude

$$\boxed{\Delta u_{\text{res}} \sim \frac{\lambda}{D}.}$$

In the spatial variable, this is the exact analogue of the principle that frequency resolution improves with observation duration.

Case $d = \lambda/2$. With $D = (N-1)\lambda/2$, one obtains $\Delta u_{\text{res}} \sim 2/(N-1) \approx 2/N$ for large N. Doubling the number of antennas halves the main-lobe width.

3.5 Array Factor of a Uniform ULA

For a uniformly weighted ULA, $w_m = 1/N$, the output in response to a plane wave at angle θ is

$$B(\theta) = \frac{1}{N}\sum_{m=0}^{N-1} e^{-jm\psi}, \qquad \psi = kd\sin\theta.$$

The geometric sum has the closed form

$$\sum_{m=0}^{N-1} e^{-jm\psi} = \frac{1-e^{-jN\psi}}{1-e^{-j\psi}} = e^{-j(N-1)\psi/2}\frac{\sin(N\psi/2)}{\sin(\psi/2)},$$

which yields the *normalized array factor*

$$\boxed{|B(\theta)| = \frac{1}{N}\left|\frac{\sin(N\psi/2)}{\sin(\psi/2)}\right|, \qquad \psi = kd\sin\theta.}$$

This function, sometimes called the *Dirichlet kernel*, plays in beamforming the role played by the rectangular window in Fourier analysis.

Main-lobe width. The first zeros satisfy $\sin(N\psi/2) = 0$, hence $\psi = \pm 2\pi/N$. Returning to θ, the first zero is at

$$\sin\theta_{\text{zero}} = \pm\frac{\lambda}{Nd},$$

and the null-to-null width, for $d = \lambda/2$, is

$$\Delta\theta_{\text{null-null}} \simeq \frac{4}{N}\ \text{rad}.$$

3.6 Tradeoff Among N, d, and D

The three quantities N, d, and D are not independent: $D = (N-1)d$. Each choice implies a tradeoff.

- **Increasing N at fixed d:** larger aperture, narrower main lobe, but more RF channels, more ADCs, more synchronization, more computation, and more calibration.
- **Increasing d at fixed N:** larger aperture but a risk of aliasing if $d > \lambda/2$, hence potentially dangerous grating lobes.
- **Decreasing d:** no aliasing, but reduced aperture at fixed N, hence a wider main lobe.

Example 3.2 (Wi-Fi 2.4 GHz)**.** $\lambda \simeq 12.5\,\text{cm}$, $d = \lambda/2 \simeq 6.25\,\text{cm}$.

$$N = 8 \implies D \simeq 43.75\,\text{cm}, \quad \Delta\theta_{\text{null-null}} \simeq 28.6^\circ.$$

$$N = 16 \implies D \simeq 93.75\,\text{cm}, \quad \Delta\theta_{\text{null-null}} \simeq 14.3^\circ.$$

Doubling N doubles the aperture and almost halves the angular resolution, but also doubles the hardware cost.

The hardware cost is not only a factor of two in the number of RF chains: the number of ADCs must also double, as must the aggregate data rate; a common clock must be provided to all channels; phase coherence must be maintained over time; and the whole system must be calibrated. Beyond a few tens of antennas, these constraints become dominant and lead to hybrid analog/digital architectures (chapter 20), where part of the beamforming is performed by RF phase shifters before digital conversion. This is precisely the path followed by 5G mmWave with its 64- or 128-element panels.

3.7 Uniform and Nonuniform Arrays

The ULA is the reference model. Its regular geometric structure provides closed-form expressions for the steering vector, the beam pattern, and high-resolution algorithms. In particular:

- ESPRIT requires *translation invariance*, a natural property of the ULA;
- MUSIC benefits from regularity through the Root-MUSIC extension;
- the beam pattern is computed as a spatial discrete Fourier transform, with all the associated properties.

Other geometries exist: nonuniform arrays, *sparse* arrays, *co-prime* arrays, circular geometries, planar arrays (URA), and volumetric arrays. For an arbitrary array with positions $\boldsymbol{p}_m$, the steering vector generalizes as

$$a_m(\hat{\boldsymbol{u}}) = e^{-j\boldsymbol{k}^{\mathrm{T}}\boldsymbol{p}_m}, \qquad \boldsymbol{k} = k\,\hat{\boldsymbol{u}}.$$

The simple geometric structure disappears, but flexibility may be gained: lobe reduction, increased effective aperture, or fewer antennas for a given resolution. Chapter 21 studies these architectures in greater depth.

3.8 From the Ideal Array to the Real Array

All previous models implicitly assume that each antenna provides an ideal measurement of the field. In practice, an array is connected to a complete hardware chain,

$$\text{antenna} \to \text{filter} \to \text{LNA} \to \text{mixer} \to \text{ADC} \to \text{digital processing.}$$

Each channel has imperfections: different gains g_m, parasitic phases ϕ_m, cable delays, oscillator drifts, and mutual coupling between radiating elements. The realistic model becomes

$$\boldsymbol{x}(t) = \boldsymbol{G}\,\boldsymbol{a}(\theta)\,s(t) + \boldsymbol{n}(t), \qquad \boldsymbol{G} = \mathrm{diag}\Big(g_0 e^{j\phi_0},\, \ldots,\, g_{N-1} e^{j\phi_{N-1}}\Big)\,.$$

The effective steering vector is then $\boldsymbol{a}_{\mathrm{real}}(\theta) = \boldsymbol{G}\,\boldsymbol{a}(\theta)$, and this is precisely what the algorithm will measure.

A phase error of only 10° between channels, on an array of 8 or 16 antennas, is enough to noticeably degrade a beam pattern: in null steering or MVDR simulations, a very deep theoretical null can rise to -25 dB or -20 dB, which completely changes anti-jamming capability. Calibration is therefore not an optional refinement: it is what allows ideal algorithms to operate on a hardware array.

3.9 Summary : What Geometry Controls

1. **Spacing** d controls spatial sampling. The canonical choice is $d = \lambda/2$, to avoid aliasing while maximizing aperture per antenna.

2. **Aperture** $D = (N-1)d$ controls angular resolution: $\Delta u_{\mathrm{res}} \sim \lambda/D$.

3. **Number of Antennas** N controls array gain, with an SNR gain of N for Delay-and-Sum on a coherent target, and hardware complexity.

4. **1D / 2D / 3D geometry** controls observable directions: azimuth only, azimuth plus elevation, or full space.

5. **Hardware imperfections** degrade the theoretical model: calibration and robustness become indispensable.

The next chapter formalizes the *narrowband* assumption and builds the IQ representation that transforms a time delay into a simple complex multiplication, giving rise to the vector model $\boldsymbol{x}(t) = \boldsymbol{a}(\theta)s(t) + \boldsymbol{n}(t)$ used throughout the rest of the book. This apparently harmless equation condenses all the physical assumptions of the chain: far field, narrowband signal, single polarization, identical antennas, and spatially white noise. Every algorithm studied from Part III onward will take this equation for granted and operate in the "clean" world of complex linear algebra. Part VII will return to what happens when one of the underlying assumptions starts to fail.

Chapter 4

Narrowband Model and IQ Representation

The two previous chapters established that a plane wave arriving on a ULA produces differential delays $\tau_m = md\sin\theta/c$, and that these delays become phase shifts as soon as the signal can be considered narrowband. This chapter formalizes that transition. It carefully builds the vector model

$$\boldsymbol{x}(t) = \boldsymbol{a}(\theta)\, s(t) + \boldsymbol{n}(t)$$

which will be the central equation of all algorithms studied later. This equation appears harmless; in reality it condenses several physical, electronic, and mathematical assumptions that must be clearly separated.

A useful way to approach this chapter is to keep in mind that it acts as a *bridge* between two worlds. On one side is real physics: a radio signal occupying bandwidth B around a carrier f_0, propagating between a source and N antennas, sampled at various rates by heterogeneous RF chains. On the other side is the mathematical formalism: a complex vector $\boldsymbol{x}(n) \in \mathbb{C}^N$ assumed to be the sum of a coherent signal $\boldsymbol{a}(\theta)s(n)$ and circular Gaussian noise. The following chapters will live entirely in the second world; this chapter is where we justify entering it.

4.1 From RF Signal to Baseband Signal

A signal modulated around a carrier of frequency f_0 is written

$$x_{\mathrm{RF}}(t) = A(t)\cos\big(2\pi f_0 t + \phi(t)\big).$$

This real expression is exact but inconvenient: it mixes amplitude and phase into a single cosine and lies at a frequency that is often high (hundreds of MHz to several GHz). Manipulating it directly would be inefficient when the useful information, carried by $A(t)$ and $\phi(t)$, varies on much slower time scales.

4.1.1 Complex Envelope

We introduce the *complex envelope*, or *baseband signal*,

$$s(t) = A(t)\, e^{j\phi(t)},$$

so that

$$x_{\mathrm{RF}}(t) = \mathrm{Re}\Big\{ s(t)\, e^{j2\pi f_0 t} \Big\}.$$

The envelope $s(t)$ contains all useful information, such as modulation, digital content, and amplitude variations, while the carrier $e^{j2\pi f_0 t}$ is only a frequency support that places the signal around f_0.

4.1.2 IQ Demodulation

In practice, the RF chain mixes the received signal with two quadrature signals, $\cos(2\pi f_0 t)$ and $-\sin(2\pi f_0 t)$, then low-pass filters the outputs. The two resulting components $I(t)$ and $Q(t)$ satisfy

$$s(t) = I(t) + jQ(t).$$

The processed digital signal is therefore *natively complex*. The envelope $s(t)$ is sampled at a rate $F_s \geq B$ related to the signal bandwidth rather than to f_0. Narrowband beamforming will systematically handle the signal in this IQ form.

This conversion represents a considerable saving in digital bandwidth. At 2.4 GHz with a Wi-Fi signal of bandwidth $B = 20\,\text{MHz}$, sampling is typically performed at $F_s = 20$ to $40\,\text{MHz}$ rather than at the $\sim 5\,\text{GHz}$ that direct Nyquist sampling of the RF signal would require. This factor-100 saving in ADC rate is precisely what makes multichannel SDR systems affordable. Without analog IQ demodulation, a 64-antenna massive-MIMO array would require 64 ADCs running at several GHz – prohibitive in cost and power consumption.

4.2 Effect of a Delay on the IQ Representation

If the wave is delayed by τ, the delayed RF signal is

$$x_{\text{RF}}(t-\tau) = \text{Re}\Big\{s(t-\tau)\, e^{j2\pi f_0 (t-\tau)}\Big\} = \text{Re}\Big\{s(t-\tau)\, e^{j2\pi f_0 t}\, e^{-j2\pi f_0 \tau}\Big\}.$$

Two effects coexist:

- a *true delay* applied to the envelope, $s(t-\tau)$;
- a constant complex rotation, $e^{-j2\pi f_0 \tau}$.

4.3 Narrowband Assumption

The *narrowband assumption* consists of neglecting the offset $s(t-\tau)$ relative to $s(t)$. More precisely, one assumes that over the interval $[t-\tau_{\max}, t]$ corresponding to traversal of the array, the envelope remains essentially constant. One may then write

$$s(t-\tau_m) \approx s(t),$$

and the delay reduces, in the complex envelope, to the simple multiplication by

$$e^{-j2\pi f_0 \tau_m} = e^{-jmkd\sin\theta}.$$

Under this assumption, the vector model

$$\boxed{\boldsymbol{x}(t) = \boldsymbol{a}(\theta)\, s(t)}$$

replaces the exact spatio-temporal expression that would involve different delayed versions $s(t-\tau_m)$ on each antenna. The passage from spatio-temporal dependence to purely spatial dependence, up to the scalar factor $s(t)$, is what makes the entire algebraic machinery of narrowband beamforming possible.

The conceptual jump represented by this factorization should not be underestimated. Without it, one would have to reason about a "spatio-temporal" sequence $\{x_m(t-\tau_m)\}$ mixing geometry and modulation, and the whole matrix formalism would become ineffective: one could no longer write $\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\text{H}}]$ as a simple Toeplitz matrix, decompose into signal/noise subspaces, or diagonalize. The factorization $\boldsymbol{x}(t) = \boldsymbol{a}(\theta)s(t)$ is therefore the algebraic enabler of the entire book. When it fails in the wideband regime, one must either restore it artificially through subband decomposition, or change formalism and work with space-frequency matrices. Chapter 19 studies both options in detail.

4.3.1 Validity Condition

The approximation is not always valid. It depends on the rate of variation of $s(t)$ compared with $\tau_{\max} \approx D/c$, where D is the array aperture. If the signal has bandwidth B, its envelope varies on a characteristic time $T_s \sim 1/B$, and the approximation condition $\tau_{\max} \ll T_s$ is written

$$\boxed{B\tau_{\max} \ll 1 \quad \Longleftrightarrow \quad \frac{BD}{c} \ll 1.}$$

It can also be expressed as a fractional bandwidth:

$$\frac{BD}{c} = \frac{B}{f_0} \cdot \frac{D}{\lambda},$$

which shows that the assumption is easy to satisfy for signals with small fractional bandwidth or arrays that are small in multiples of λ, and difficult for wideband radar signals over large apertures.

Example 4.1 (Wi-Fi)**.** $B = 20\,\text{MHz}$, $D = 0.94\,\text{m}$: $BD/c \simeq 6.3 \cdot 10^{-2} \ll 1$. The narrowband assumption is largely satisfied.

Example 4.2 (Wideband Radar)**.** $B = 500\,\text{MHz}$, $D = 0.94\,\text{m}$: $BD/c \simeq 1.6$. The assumption is clearly violated. Two distant antennas in the array no longer see the same envelope at the same instant, and a narrowband beamformer produces a significant degradation in array gain. Chapter 19 specifically addresses this regime.

4.4 Noise and Interference

The narrowband model would be incomplete without additive noise in the receive channels. Each antenna m receives thermal noise $n_m(t)$, assumed circular complex Gaussian, independent from one channel to the next, with variance σ^2. Stacking the channels gives the noise vector $\boldsymbol{n}(t) \in \mathbb{C}^N$ such that

$$\mathbb{E}[\boldsymbol{n}(t)] = \boldsymbol{0}, \qquad \mathbb{E}[\boldsymbol{n}(t)\boldsymbol{n}^{\mathrm{H}}(t)] = \sigma^2 \boldsymbol{I}_N.$$

The noise is then said to be *spatially white*. For several sources L arriving simultaneously with envelopes $s_\ell(t)$ and angles θ_ℓ, the vector model generalizes to

$$\boldsymbol{x}(t) = \sum_{\ell=1}^{L} \boldsymbol{a}(\theta_\ell)\, s_\ell(t) + \boldsymbol{n}(t) = \boldsymbol{A}(\boldsymbol{\theta})\, \boldsymbol{s}(t) + \boldsymbol{n}(t),$$

where $\boldsymbol{A}(\boldsymbol{\theta}) = [\boldsymbol{a}(\theta_1), \ldots, \boldsymbol{a}(\theta_L)] \in \mathbb{C}^{N \times L}$ is the *steering matrix*, and $\boldsymbol{s}(t) = [s_1(t), \ldots, s_L(t)]^{\mathrm{T}} \in \mathbb{C}^L$ is the vector of source envelopes.

4.5 Sampling and Snapshots

All algorithms operate on discrete observations. We write $\boldsymbol{x}(n) = \boldsymbol{x}(nT_s)$ for the *spatial snapshot* at time n, namely the vector of the N simultaneous IQ samples from the antennas. A "snapshot" is therefore an instantaneous spatial slice of the received electromagnetic field, treated as a point in $\mathbb{C}^N$. The collection of K snapshots $\{\boldsymbol{x}(n)\}_{n=1}^{K}$ forms the raw processing data, whether it is used to estimate a covariance, update adaptive weights, or feed a DOA estimation procedure.

4.6 When the Narrowband Assumption Degrades

Because the narrowband assumption is the backbone of the entire formalism, it is useful to summarize its consequences *when it is violated*:

- the Delay-and-Sum array gain decreases because the envelopes no longer coincide exactly after phase compensation;
- the beam pattern becomes *frequency-dependent*: it is no longer the same function for the low and high components of the band;
- MVDR underestimates the useful covariance and partially attenuates the signal itself;
- MUSIC and ESPRIT produce biased peaks and angles.

Depending on the degree of violation, two families of techniques are used:

- *true time delay beamforming*, where compensation is performed with actual delays, either analog delay lines or fractional FIR filters in digital processing;
- *subbands*, where the signal is divided into subbands narrow enough for the assumption to hold within each one, then recombined.

Chapter 19 describes these approaches in detail.

4.7 Summary : The Vector Model

At the end of Part I, the master equation of narrowband beamforming appears in three equivalent forms depending on the number of sources.

Single Source.

$$\boldsymbol{x}(t) = \boldsymbol{a}(\theta_0)\, s(t) + \boldsymbol{n}(t).$$

Multiple Sources.

$$\boldsymbol{x}(t) = \boldsymbol{A}(\boldsymbol{\theta})\, \boldsymbol{s}(t) + \boldsymbol{n}(t).$$

Beamformer Output.

$$y(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{A}(\boldsymbol{\theta})\, \boldsymbol{s}(t) + \boldsymbol{w}^{\mathrm{H}}\boldsymbol{n}(t).$$

The next chapter rigorously defines the *steering vector* $\boldsymbol{a}(\theta)$, studies its geometric structure, and explains its role as an interface between the physical world and the algorithmic world. It will be the last building block before entering the algorithms themselves. By the end of chapter 5, the reader will have all the vocabulary and notation needed to approach the deterministic beamformers of Part II, then the statistical and adaptive beamformers of the following parts. Part I is, in this sense, an airlock: the rest of the book draws its foundations from it without returning to it.

Chapter 5

Steering Vector and Array Manifold

The *steering vector* is the central mathematical object of beamforming. All algorithms studied later, without exception, use it as a reference: Delay-and-Sum aligns its weights with it, Null Steering imposes zero gain on it, MVDR uses it as a constraint, and MUSIC measures its orthogonality to the noise subspace. This chapter studies its definition, its geometric structure, its algebraic properties, and its interpretation as a *spatial matched filter*.

Its role can be summarized by one formula: the steering vector is the *spatial transfer function* that transforms an arrival direction into a measurable complex signature. To each angle θ it assigns a vector $\boldsymbol{a}(\theta) \in \mathbb{C}^N$, and the inverse problem – recovering θ from the data – is precisely the subject of all of Part V. At the other end of the process, the direct problem – amplifying a known direction – is the subject of Parts II to IV. The whole book therefore revolves around this geometric object, and mastering it thoroughly is well worth the investment of this relatively short chapter.

5.1 Definition

For an antenna array with positions $\{\boldsymbol{p}_m\}_{m=0}^{N-1}$ and a plane wave with wave vector $\boldsymbol{k} = k\,\hat{\boldsymbol{u}}$, the steering vector is the vector of complex responses of the N antennas,

$$\boldsymbol{a}(\hat{\boldsymbol{u}}) = \begin{bmatrix} e^{-j\boldsymbol{k}^{\mathrm{T}}\boldsymbol{p}_0} \\ e^{-j\boldsymbol{k}^{\mathrm{T}}\boldsymbol{p}_1} \\ \vdots \\ e^{-j\boldsymbol{k}^{\mathrm{T}}\boldsymbol{p}_{N-1}} \end{bmatrix} \in \mathbb{C}^N.$$

Component m contains the relative phase received by antenna m, taken with respect to the origin, generally $\boldsymbol{p}_0 = \boldsymbol{0}$.

For a ULA oriented along the x axis, with arrival angle θ measured from broadside, $\boldsymbol{k}^{\mathrm{T}}\boldsymbol{p}_m = mkd\sin\theta$, and the expression takes the canonical form

$$\boxed{\boldsymbol{a}(\theta) = \begin{bmatrix} 1 \\ e^{-jkd\sin\theta} \\ e^{-j2kd\sin\theta} \\ \vdots \\ e^{-j(N-1)kd\sin\theta} \end{bmatrix}, \qquad a_m(\theta) = e^{-jmkd\sin\theta}.}$$

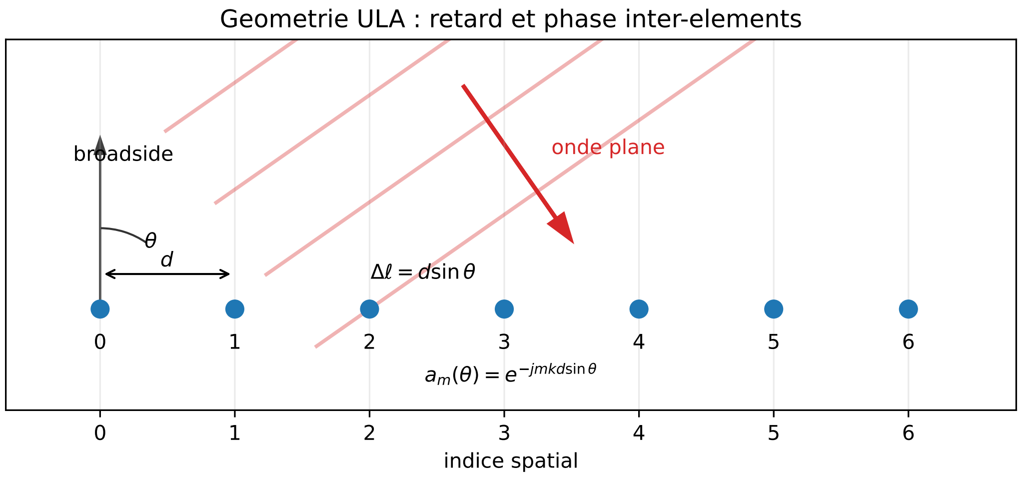

Figure 5.1: Illustrated ULA geometry ($N = 7$, indicative angle $\theta = 35°$) and phase convention: the path difference between two adjacent antennas is $d \sin\theta$, hence the progression $e^{-jmkd\sin\theta}$.

5.2 Geometric Structure

5.2.1 Vandermonde Vector

Let $z(\theta) = e^{-jkd\sin\theta}$. It follows immediately that

$$\boldsymbol{a}(\theta) = \Big[1, \quad z(\theta), \quad z^2(\theta), \quad \ldots, \quad z^{N-1}(\theta)\Big]^{\mathrm{T}}.$$

$\boldsymbol{a}(\theta)$ is therefore a *column of a Vandermonde matrix* associated with the point $z(\theta)$ on the unit circle of $\mathbb{C}$. This property is crucial: it is the algebraic root of ESPRIT, which exploits the *translation* $z \mapsto z$ between two subarrays shifted by one antenna.

5.2.2 Norm and Sphere

For a ULA, all components have modulus 1:

$$\|\boldsymbol{a}(\theta)\|^2 = \sum_{m=0}^{N-1} |a_m(\theta)|^2 = N.$$

The norm is therefore *constant*, independent of θ: the *array manifold* $\mathcal{M} = \{\boldsymbol{a}(\theta) : \theta \in [-90°, 90°]\}$ is contained in the sphere of radius $\sqrt{N}$ in $\mathbb{C}^N$. This is the source of the factors $\sqrt{N}$ and N that appear throughout array gain expressions.

5.2.3 Link with the Discrete Fourier Transform

At $d = \lambda/2$, one may write $a_m(\theta) = e^{-j\pi m\sin\theta}$. Setting $u = \sin\theta$ and discretizing u at the points $u_k = 2k/N$, we recover exactly the atoms of the N-point DFT:

$$a_m(u_k) = e^{-j2\pi mk/N}.$$

This relationship is not accidental: Delay-and-Sum on a $\lambda/2$-spaced ULA is computed as a spatial DFT, and the entire machinery of Fourier analysis, including resolution, spectral leakage, and windowing, applies directly to beam patterns.

This observation has major algorithmic scope. Where one would *naively* compute the beam pattern over N angles in $\mathcal{O}(N^2)$, using the FFT makes it possible to do so in $\mathcal{O}(N \log N)$. For a 256-antenna massive-MIMO array, this moves from about $6 \cdot 10^4$ operations per snapshot to about $2 \cdot 10^3$: a factor of 30 in complexity, and therefore in power consumption and latency. All high-performance implementations of Delay-and-Sum in radar and acoustics rely on this FFT equivalence.

5.3 Inner Product Between Steering Vectors

The inner product $\boldsymbol{a}^{\mathrm{H}}(\theta_1)\boldsymbol{a}(\theta_2)$ plays a key role in beamforming resolution. For a ULA:

$$\boldsymbol{a}^{\mathrm{H}}(\theta_1)\boldsymbol{a}(\theta_2) = \sum_{m=0}^{N-1} e^{jmkd(\sin\theta_1 - \sin\theta_2)} = \sum_{m=0}^{N-1} e^{jm\alpha}$$

with $\alpha = kd(\sin\theta_1 - \sin\theta_2)$. The geometric sum gives

$$\boldsymbol{a}^{\mathrm{H}}(\theta_1)\boldsymbol{a}(\theta_2) = e^{j(N-1)\alpha/2}\,\frac{\sin(N\alpha/2)}{\sin(\alpha/2)}.$$

The modulus is the *Dirichlet kernel*

$$\left|\boldsymbol{a}^{\mathrm{H}}(\theta_1)\boldsymbol{a}(\theta_2)\right| = \left|\frac{\sin(N\alpha/2)}{\sin(\alpha/2)}\right|.$$

This quantity equals N for $\alpha = 0$, corresponding to coincident directions, and vanishes for $\alpha = 2\pi q/N$ with $q \in \mathbb{Z}^*$. It is exactly the function that determines *angular resolution* and the main-lobe shape of a conventional beamformer.

5.4 Steering Vector and Beamformer

Beamforming consists of applying a weight vector $\boldsymbol{w} \in \mathbb{C}^N$ to the observation, producing the output

$$y(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(t).$$

For an ideal single-source signal at θ_0, $\boldsymbol{x}(t) = \boldsymbol{a}(\theta_0)s(t)$, one obtains

$$y(t) = \underbrace{\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_0)}_{=G(\theta_0)}\, s(t).$$

The complex scalar $G(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta)$ is the *complex gain* of the beamformer in direction θ, and the *beam pattern* is the map $\theta \mapsto G(\theta)$, studied in detail in chapter 6.

Spatial matched filter. For fixed $\|\boldsymbol{w}\|$, the inner product $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_0)$ is maximized when $\boldsymbol{w}$ is collinear with $\boldsymbol{a}(\theta_0)$; one therefore chooses

$$\boldsymbol{w}_{\mathrm{DAS}} = \frac{1}{N}\,\boldsymbol{a}(\theta_0),$$

which is simply the *spatial matched filter* for direction θ_0, analogous to the classical temporal matched filter. This choice maximizes the output SNR in the presence of spatially white noise. When the noise is no longer white, for example in the presence of directional interference, this choice is no longer optimal and MVDR, introduced in chapter 10, takes over.

This analogy with the temporal matched filter is worth extending. In temporal processing, the matched filter maximizes SNR for detecting a known waveform in additive white noise. The filter is, up to a delay, the time conjugate of the waveform to be detected. In spatial processing, the "signal" to detect is not a waveform but a *phase signature*; the "white" noise is not white in time but in space; and the "filter" is a complex weighting vector. The underlying theorem is exactly the same – the Cauchy-Schwarz inequality – and so is the optimality. This is why DAS is not merely a "simple" beamformer: it is the *optimal* beamformer in white noise under the ideal model, and all others depart from it only to handle situations where the noise is structured.

5.5 Multiple Sources and Steering Matrix

For L sources with directions θ_ℓ, the steering vectors are stacked in the *steering matrix*

$$\boldsymbol{A}(\boldsymbol{\theta}) = \begin{bmatrix}\boldsymbol{a}(\theta_1) & \boldsymbol{a}(\theta_2) & \cdots & \boldsymbol{a}(\theta_L)\end{bmatrix} \in \mathbb{C}^{N\times L}.$$

The vector model is written

$$\boldsymbol{x}(t) = \boldsymbol{A}(\boldsymbol{\theta})\,\boldsymbol{s}(t) + \boldsymbol{n}(t), \qquad \boldsymbol{s}(t) = [s_1(t), \ldots, s_L(t)]^{\mathrm{T}}.$$

Full rank of the steering vectors. On a ULA, $\boldsymbol{A}$ has full rank $\min(N, L)$ as soon as the angles θ_ℓ are pairwise distinct and remain in the visible region. This property is used by MUSIC and ESPRIT, which assume $L < N$, so $\boldsymbol{A}$ has full column rank.

5.6 2D and 3D Arrays

For a planar array with positions $\boldsymbol{p}_m = (x_m, y_m, 0)^{\mathrm{T}}$, the steering vector depends on two angles, azimuth φ and elevation ϑ, through the propagation unit vector

$$\hat{\boldsymbol{u}}(\varphi, \vartheta) = (\cos\vartheta \sin\varphi,\ \cos\vartheta\cos\varphi,\ \sin\vartheta)^{\mathrm{T}}.$$

Then

$$a_m(\varphi, \vartheta) = e^{-jk(x_m u_x + y_m u_y)}.$$

The algebraic structure remains identical to that of the ULA: only the product $\boldsymbol{k}^{\mathrm{T}}\boldsymbol{p}_m$ changes. Most results established for the ULA – beam pattern, sampling condition, resolution, subspaces – transfer immediately.

5.7 Real Steering Vector vs. Theoretical Steering Vector

The theoretical steering vector relies on idealized antennas that are identical and positioned without error. The real steering vector

$$\boldsymbol{a}_{\mathrm{real}}(\theta) = \boldsymbol{G}\,\boldsymbol{a}(\theta) + \boldsymbol{e}(\theta)$$

includes:

- one gain and phase per channel, $\boldsymbol{G} = \mathrm{diag}(g_m e^{j\phi_m})$;
- residual errors $\boldsymbol{e}(\theta)$ due to mutual coupling, position errors, and drifts;
- possibly an individual directional response for each antenna, different from an isotropic response.

The fundamental quantity is not the absolute value of the steering vector, but its *coherence* from one channel to the next. This is the coherence that calibration, discussed in chapter 22, seeks to restore.

Effect of a random phase error. If each antenna undergoes a random phase $\delta\phi_m \sim \mathcal{N}(0, \sigma_\phi^2)$, the effective gain of Delay-and-Sum on the target becomes

$$\mathbb{E}\left[\left|\boldsymbol{w}_{\mathrm{DAS}}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{real}}\right|^2\right] \approx N \cdot e^{-\sigma_\phi^2},$$

which degrades exponentially with phase variance. A $\sigma_\phi = 30^\circ \simeq 0.52\,\mathrm{rad}$ already reduces the effective gain by about $1.2\,\mathrm{dB}$; beyond 60°, the loss exceeds $5\,\mathrm{dB}$ and beamforming loses most of its value.

5.8 Common Convention Errors

The steering vector is simple to write, but convention errors are among the most frequent causes of incorrect results. The danger rarely comes from using a different convention; it comes from using a convention inconsistently across chapters, code, and figures.

5.8.1 Confusing $\boldsymbol{a}(\theta)$ and $\boldsymbol{a}^*(\theta)$

With the convention used in this document:

$$a_m(\theta) = e^{-jmkd\sin\theta}.$$

The output of a beamformer is:

$$y(n) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(n).$$

For Delay-and-Sum pointed toward θ_0, we choose:

$$\boldsymbol{w}_{\mathrm{DAS}} = \frac{1}{N}\boldsymbol{a}(\theta_0),$$

because then:

$$\boldsymbol{w}_{\mathrm{DAS}}^{\mathrm{H}}\boldsymbol{a}(\theta_0) = \frac{1}{N}\boldsymbol{a}^{\mathrm{H}}(\theta_0)\boldsymbol{a}(\theta_0) = 1.$$

If $\boldsymbol{a}^*(\theta_0)$ is mistakenly used as the weights with the same output definition, the result is:

$$\frac{1}{N}\boldsymbol{a}^{\mathrm{T}}(\theta_0)\boldsymbol{a}(\theta_0),$$

which is generally not equal to 1. The beam then points toward a symmetric direction or produces an incoherent beam pattern.

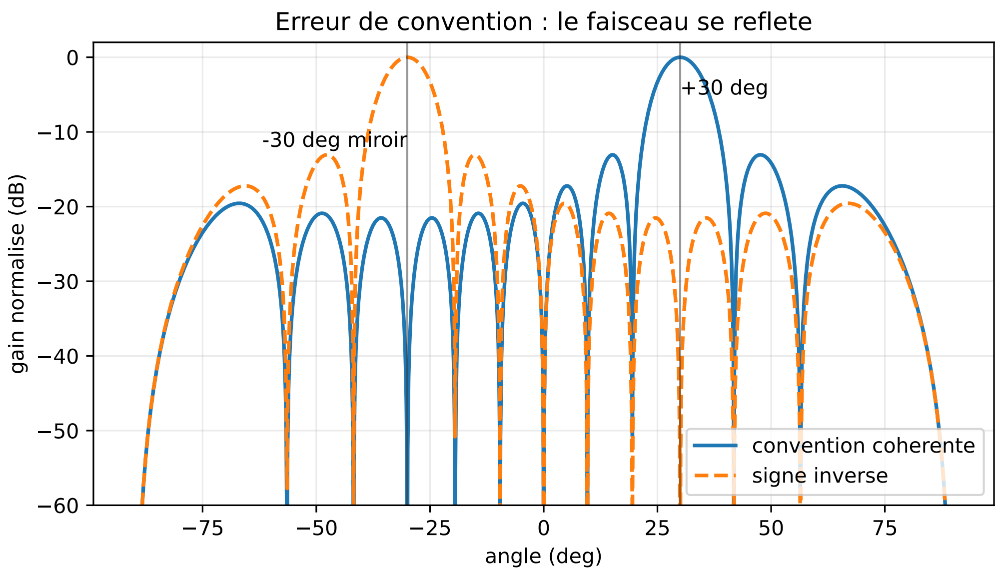

Figure 5.2: Typical effect of an inconsistent convention (DAS, $N = 12$, $d = \lambda/2$): a beam intended to point toward $+30°$ is mirrored toward $-30°$.

5.8.2 Reception and Transmission

In reception, the weights appear as:

$$y = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}.$$

In transmission, one instead builds a transmitted signal:

$$\boldsymbol{x}_{\mathrm{tx}} = \boldsymbol{w}_{\mathrm{tx}}s.$$

Depending on the phase convention, the vector used in transmission may be the conjugate of the one used in reception. This is not a contradiction: it is the consequence of reciprocity between forming a coherent sum at reception and imposing a coherent phase at transmission. The trap is mixing the two in the same code.

5.8.3 Confusing $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}$ and $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$

IQ signals are complex. The physical inner product is Hermitian:

$$\boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}.$$

Using $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$ removes the conjugation. The result may look plausible for real vectors, but it is wrong as soon as the phases carry spatial information. This error breaks MVDR normalization, the beam pattern, and null tests in particular.

5.8.4 Reversing the Phase Sign

Some books use:

$$a_m(\theta) = e^{+jmkd\sin\theta}.$$

This convention is valid if it is applied everywhere. But if the beam pattern is computed with $e^{-jmkd\sin\theta}$ and ESPRIT converts eigenvalues as if the sign were positive, the estimated angles are reversed:

$$\theta \quad \longrightarrow \quad -\theta$$

in the broadside ULA case. The sign of the steering vector, the sign in MUSIC, the sign in Capon, and the ESPRIT formula must form one consistent block.

5.8.5 Angle from the Normal or from the Axis

In this document, $\theta = 0$ denotes broadside, namely the normal to the array. Some texts measure the angle ψ from the array axis. The two are related by:

$$\sin\theta = \cos\psi.$$

A formula written with $\sin\theta$ therefore becomes a formula with $\cos\psi$. Mixing these two definitions places lobes in the wrong locations, even if the phase sign is correct.

5.8.6 Inconsistency Between Beam Pattern, MUSIC, and ESPRIT

A simple test should pass:

$$B(\theta_0) = \boldsymbol{w}_{\mathrm{DAS}}^{\mathrm{H}}\boldsymbol{a}(\theta_0) = 1,$$

the MUSIC pseudospectrum must produce a peak at the same θ_0, and ESPRIT must convert

$$z = e^{-jkd\sin\theta_0}$$

into:

$$\theta_0 = \arcsin\left(-\frac{\arg z}{kd}\right).$$

If one of the three tests fails, it is probably not an algorithmic problem: it is an inconsistent convention.

5.9 Geometric View: The Array Manifold

As θ varies continuously in the visible region, $\boldsymbol{a}(\theta)$ describes a smooth curve $\mathcal{M}$ in the sphere of radius $\sqrt{N}$ in $\mathbb{C}^N$. This curve, the *array manifold*, is the fundamental geometric object of beamforming. The algorithmic richness of the field can be read as an exploration of its properties:

- its *length* and *curvature* determine the local resolution of the array;
- its *intersections* with subspaces determine the performance of MUSIC and the signal/noise decomposition;
- its *self-intersections* indicate angular ambiguities in nonuniform or poorly sampled arrays.

At each instant, beamforming consists of choosing a vector $\boldsymbol{w} \in \mathbb{C}^N$ that has:

- large overlap with $\boldsymbol{a}(\theta_0)$, providing gain on the target;
- small overlap with $\boldsymbol{a}(\theta_i)$ for interference directions to reject;
- small overlap with noise-bearing directions.

This geometric view guides everything that follows. It will appear explicitly in the MVDR formulation, where $\boldsymbol{w}$ is projected orthogonally to an interference subspace, and in MUSIC, where the distance $\boldsymbol{a}(\theta) \to$ noise subspace becomes the estimation criterion.

5.10 Numerical Implementation

Listing 5.1: Constructing a ULA steering vector

```
import numpy as np

def steering_vector(theta, N, d_over_lambda=0.5):
    """
    Steering vector of an N-antenna ULA, spacing d / lambda.
    theta: angle in radians, measured from broadside.
    """
    m = np.arange(N)
    phi = -2.0 * np.pi * d_over_lambda * np.sin(theta) * m
    return np.exp(1j * phi)

def steering_matrix(thetas, N, d_over_lambda=0.5):
    """
    Steering matrix A(theta) for L angles.
    thetas: array of L angles in radians.
    Returns: N x L matrix.
    """
    return np.column_stack([steering_vector(th, N, d_over_lambda) for th in thetas])
```

These two functions are the basic building blocks of all algorithms in the following chapters. They will reappear when computing beam patterns, MVDR weights, MUSIC pseudospectra, and ESPRIT estimates.

5.11 Summary

At the end of Part I, we have the following objects:

- **Vector Model**: $\boldsymbol{x}(t) = \boldsymbol{A}(\boldsymbol{\theta})\, \boldsymbol{s}(t) + \boldsymbol{n}(t)$.

- **ULA steering vector**: $a_m(\theta) = e^{-jmkd\sin\theta}$, with norm $\sqrt{N}$.
- **Beamformer**: $y(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(t)$, beam pattern $B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta)$.
- **Underlying assumptions**: far field, narrowband signal, identical antennas, spatially white noise unless otherwise stated.

Part II builds the first *deterministic* beamformers from these objects: beamformers whose weights do not depend on the received data but only on the problem geometry and on directions considered known. This is the opportunity to introduce the beam pattern systematically before moving to statistical adaptation in Part III.

A retrospective reading of Part I may be useful at this stage. We have followed a precise path: from a physical RF signal arriving on real antennas, we reached a complex vector living in $\mathbb{C}^N$ and a geometric object – the array manifold – that parameterizes directions as points on a complex sphere. All the physics of chapters 2 and 3 is now absorbed into the notation $\boldsymbol{a}(\theta)$. From this point onward, the delays τ_m, the carrier $e^{j2\pi f_0 t}$, and the Fraunhofer condition will remain in the background: we will mostly manipulate vectors and matrices. The physics has not disappeared; it is condensed into linear algebra.

Part II

Deterministic Beamforming

Chapter 6

Beam Pattern and Spatial Windowing

In the previous chapter, we saw that a beamformer transforms a spatial snapshot into a scalar $y(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(t)$. The natural question is: *which directions is this system sensitive to?* The answer is the *beam pattern*. For a beamformer, it plays the same role that the frequency response plays for a time-domain filter: it reveals at a glance the resolution, spatial selectivity, sidelobe leakage, and directional rejection of the system.

Designing a beamformer is essentially shaping its beam pattern. The beam pattern is the *visual interface* between the algorithm designer and the physics of the problem: for a radar engineer, it is the diagram that says "here I see well, there I am blind"; for a telecom engineer, it is the spatial signature that says "here I transmit, here I reject." Throughout the rest of the book, whenever a new beamformer is introduced, we eventually plot its beam pattern and comment on its main lobe, sidelobes, and nulls. Learning to read it is therefore a cross-cutting skill.

6.1 Definition

Definition 6.1 (Beam pattern)**.** For a weight vector $\boldsymbol{w} \in \mathbb{C}^N$ and a steering vector $\boldsymbol{a}(\theta)$, the *beam pattern* is the function

$$B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta).$$

This quantity is complex. Its magnitude $|B(\theta)|$ measures the response amplitude in direction θ, and its phase describes the global phase of the output signal. The power gain is $|B(\theta)|^2$. It is almost always displayed in decibels after normalization by the maximum:

$$B_{\mathrm{dB}}(\theta) = 20\log_{10}\left(\frac{|B(\theta)|}{\max_\theta |B(\theta)|}\right).$$

The normalization places the main lobe at $0\,\mathrm{dB}$ and expresses the other directions as relative attenuation.

Remark 6.2 (Choice of $20\log_{10}$ or $10\log_{10}$)*.* We use $20\log_{10}$ when $B(\theta)$ is interpreted as an amplitude. If we display a power or a spectrum directly (for example the Capon pseudo-spectrum $\boldsymbol{a}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{a}$), we use $10\log_{10}$ to keep a consistent reading in dB.

6.2 Why a Beam Pattern

The beam pattern does not only describe sensitivity to incident waves: it also describes the *rejection of unwanted directions*. Four fundamental indicators are associated with a given beam pattern.

- The *main lobe*: the region around the pointing direction where $|B(\theta)|$ is maximal. Its width, measured at $-3\,\text{dB}$ or between the first nulls, is the angular resolution of the system.
- The *sidelobes*: local maxima outside the main lobe. Their level, expressed in dB below the maximum, determines how much unwanted energy is captured from non-target directions.
- The *spatial nulls*: directions where $|B(\theta)| = 0$, which completely reject the corresponding interference.
- The *array gain*: the maximum value of $|B(\theta_0)|$ on the target, to be compared with the behavior on noise.

6.3 Spectral Reading

For a ULA with spacing d, define $u = \sin\theta$ and $\nu = (d/\lambda)u$. The beam pattern can be written

$$B(u) = \sum_{m=0}^{N-1} w_m^*\, e^{-j2\pi m\nu} = \mathcal{F}\{w_m^*\}(\nu),$$

where $\mathcal{F}$ denotes the spatial discrete Fourier transform. *The beam pattern is nothing more than the Fourier transform of the conjugated weights.* The whole language of Fourier analysis — resolution, windowing, spectral leakage — applies. This observation underpins the rest of the chapter.

The time/space correspondence table then becomes fully meaningful:

Time-Domain Filter	**Spatial Beamformer**
impulse response $h[n]$	weights w_m^*
frequency response $H(f)$	beam pattern $B(\theta)$
bandwidth	main-lobe width
frequency selectivity	angular selectivity
ripple, sidelobes	sidelobes
filter zeros	spatial nulls
FIR / IIR filter	fixed weights / adaptive weights

Anyone who has spent time designing a digital filter will find most of the techniques in this part familiar. The names change — we speak of "spatial windowing" where we used to speak of "time-domain windowing" — but the principles are literally the same. When in doubt, translating the spatial problem into a time-domain problem and back is almost always fruitful.

6.4 ULA with Uniform Weights

For $w_m = 1/N$ (Delay-and-Sum steered to broadside), we already saw in chapter 3 that

$$|B(\theta)| = \frac{1}{N}\left|\frac{\sin(N\psi/2)}{\sin(\psi/2)}\right|, \qquad \psi = kd\sin\theta.$$

This is the spatial analogue of the Dirichlet kernel associated with a rectangular time-domain window. The following characteristics follow immediately:

- unit maximum at $\theta = 0$;
- first null at $\sin\theta = \pm\lambda/(Nd)$;
- first sidelobe at about $-13.3\,\text{dB}$ (the classical limit of the rectangular window);
- asymptotic decay in $1/\theta$.

Steering toward θ_0. The Delay-and-Sum beamformer steered toward θ_0 is obtained by choosing $\boldsymbol{w} = \boldsymbol{a}(\theta_0)/N$. The beam pattern becomes

$$B(\theta) = \frac{1}{N}\boldsymbol{a}^{\mathrm{H}}(\theta_0)\boldsymbol{a}(\theta) = \frac{1}{N}\, e^{-j(N-1)\beta/2}\,\frac{\sin(N\beta/2)}{\sin(\beta/2)},$$

with $\beta = kd(\sin\theta - \sin\theta_0)$. The maximum moves to $\theta = \theta_0$, but the *main lobe widens* as θ_0 moves away from broadside: this is the effect of the $\sin\theta$ factor in the conjugate variable.

6.5 Influence of Parameters

6.5.1 Number of Antennas N

The larger N is, the longer the aperture $D = (N-1)d$ becomes and the narrower the main lobe gets. At broadside, the width between first nulls is $\Delta\theta_{\text{n-n}} \simeq 4/N$ rad for $d = \lambda/2$. The sidelobe level, however, remains fixed at $\sim -13.3\,\text{dB}$: it is not a function of N but of the *shape* of the weighting.

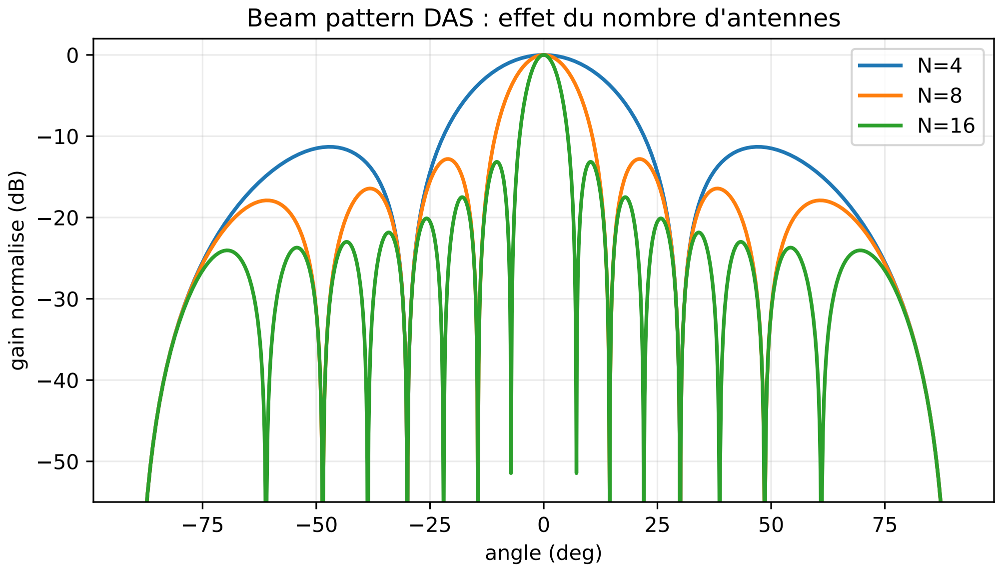

Figure 6.1: Effect of the number of antennas on a broadside DAS ($N = 4, 8, 16$, $d = \lambda/2$): the main lobe narrows as N increases, while the first sidelobe level remains tied to the uniform weighting.

6.5.2 Spacing d

- $d < \lambda/2$: wider main lobe, no aliasing;
- $d = \lambda/2$: standard case, maximum aperture without aliasing;
- $d > \lambda/2$: *grating lobes* appear and reach the same level as the main lobe. These lobes are not numerical artifacts but true angular ambiguities.

6.5.3 Pointing

A pointing error $\delta\theta_0$ moves the main lobe away from the target. The perceived gain on the target is then

$$|B(\theta_{\text{target}})|^2 = \frac{1}{N^2}\,\frac{\sin^2(N\beta/2)}{\sin^2(\beta/2)}, \qquad \beta = kd(\sin\theta_{\text{target}} - \sin\theta_0).$$

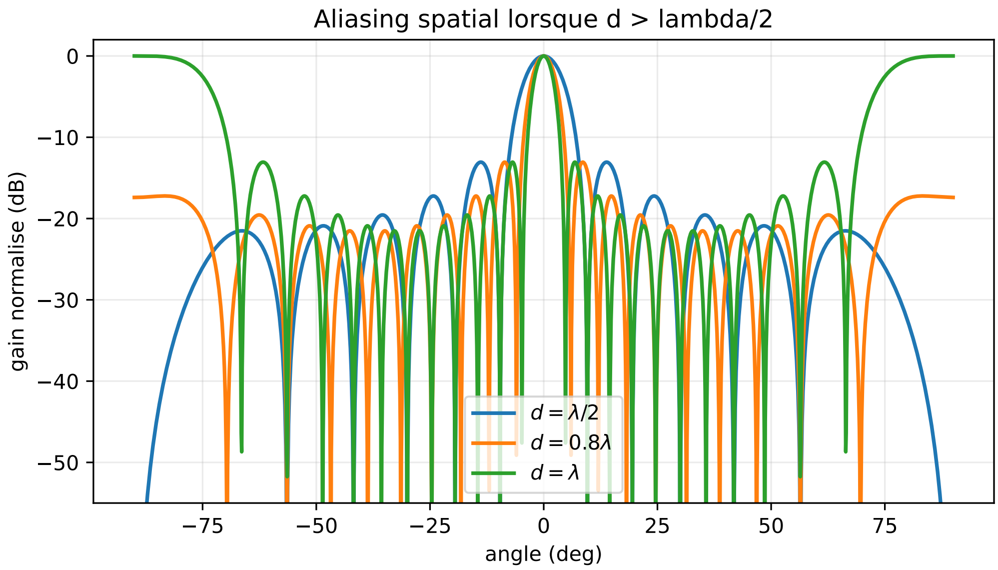

Figure 6.2: Spatial aliasing (broadside DAS, $N = 12$, $d/\lambda \in \{0.5, 0.8, 1.0\}$): when d exceeds $\lambda/2$, grating lobes appear in the visible region and create ambiguous directions.

An error amounting to a fraction of the main-lobe width already costs several dB. This is one of the fragilities of Delay-and-Sum, and one of the historical motivations for *spatial tapering*.

6.6 Spatial Windowing: Main-Lobe / Sidelobe Tradeoff

Uniform weighting produces a narrow main lobe but high sidelobes. To reduce these sidelobes, we apply *spatial windowing* to the weights, exactly as we apply a window (Hann, Hamming, Kaiser, etc.) in time-domain Fourier analysis.

For a beamformer steered toward θ_0, the windowed weights are

$$\boldsymbol{w}_W = \boldsymbol{W}\,\boldsymbol{a}(\theta_0)/N, \qquad \boldsymbol{W} = \mathrm{diag}(w_0^W, w_1^W, \ldots, w_{N-1}^W),$$

where the coefficients $w_m^W \in \mathbb{R}^+$ come from a classical window:

Window	1^{st} sidelobe	main-lobe widening	decay
Rectangular	$-13.3\,\text{dB}$	$\times 1$	$1/\theta$
Hann	$-31.5\,\text{dB}$	$\times 1.5$	$1/\theta^3$
Hamming	$-42.7\,\text{dB}$	$\times 1.5$	$1/\theta$
Blackman	$-58\,\text{dB}$	$\times 1.7$	$1/\theta^3$
Chebyshev	adjustable	adjustable	uniform
Taylor	adjustable	adjustable	uniform

Remark 6.3 (Chebyshev and Taylor). Dolph–Chebyshev windows produce sidelobes that are strictly uniform at the chosen level (e.g. $-30\,\text{dB}$, $-40\,\text{dB}$), and are optimal in the main-lobe / sidelobe-level tradeoff. Taylor windows are a regularized variant. Both are common in radar.

Consequence for gain. Windowing improves selectivity at the cost of two degradations:

- the main lobe widens (degraded resolution);
- the array gain on the target decreases, because the edge antennas contribute less.

The gain loss relative to uniform Delay-and-Sum is measured by the *taper efficiency*

$$\eta_T = \frac{|\sum_m w_m^W|^2}{N \sum_m |w_m^W|^2} \in [0, 1].$$

For a Hamming window, $\eta_T \simeq 0.73$ (about 1.3 dB of array-gain loss); for Blackman, $\eta_T \simeq 0.50$.

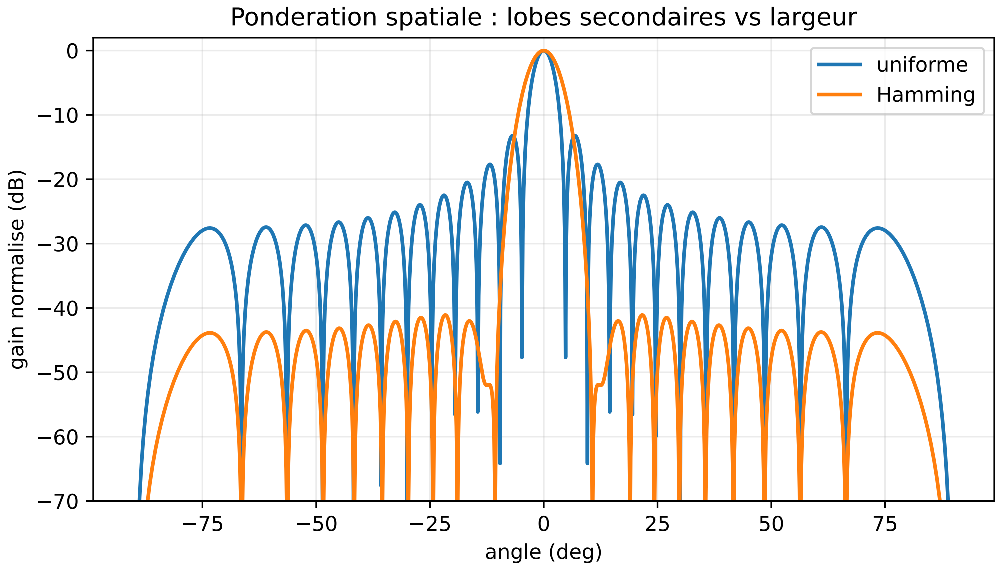

Figure 6.3: Uniform and Hamming weighting ($N = 24$, $d = \lambda/2$): Hamming significantly reduces the sidelobes, at the cost of a wider main lobe and slightly lower array gain.

6.7 Beam Pattern in Angle and Spatial Variable

The beam pattern can be plotted as a function of θ (physical angular scale) or as a function of $u = \sin\theta$ (reduced spatial variable). The latter representation has the advantage of making the beam pattern independent of the pointing direction: for a ULA steered toward θ_0, $B(u)$ is simply translated by $u_0 = \sin\theta_0$. It also clearly reveals spatial aliasing (folding at ± 2 for $d = \lambda/2$).

The representation in θ, however, is the *physical* image of the scene. It reveals the apparent widening of the main lobe near endfire, where a variation Δu corresponds to a variation $\Delta\theta = \Delta u / \cos\theta$ that diverges.

6.8 Cartesian and Polar Representations

The Cartesian representation $\theta \mapsto B_{\mathrm{dB}}(\theta)$ is preferred for precisely quantifying sidelobe levels and main-lobe widths. The polar representation is visually more expressive: it draws $|B(\theta)|^2$ in the azimuth plane using polar coordinates, and gives a direct spatial intuition for the formed beam.

6.9 Beam Pattern Synthesis by Optimization

Rather than starting from a standard window, the design can be formulated as an optimization problem:

$$\min_{\boldsymbol{w}} \max_{\theta \in \mathcal{S}} |B(\theta)|^2 \quad \text{subject to} \quad B(\theta_0) = 1,$$

where $\mathcal{S}$ is the set of directions to attenuate. Under linear gain constraints, the problem becomes a convex program (in practice, a second-order cone program or a linear program) that modern solvers handle efficiently. This approach is the basis of optimized beam patterns for radar antennas and for telecom *shaped beams*.

6.10 Numerical Implementation

Listing 6.1: Computing and plotting a beam pattern

```python
import numpy as np
import matplotlib.pyplot as plt

def beam_pattern(w, N, d_over_lambda=0.5, n_theta=721):
    theta = np.linspace(-np.pi/2, np.pi/2, n_theta)
    m = np.arange(N)
    # N x n_theta matrix of steering vectors a(theta)
    A = np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(theta)[None, :] * m[:, None])
    B = w.conj() @ A
    B = np.abs(B)
    B_dB = 20*np.log10(B / B.max() + 1e-12)
    return theta, B_dB

# Example: DAS steered to 0 deg + Hamming window
N = 16
theta0 = 0.0
m = np.arange(N)
a0 = np.exp(-1j * np.pi * np.sin(theta0) * m)
w_uni = a0 / N
w_ham = np.hamming(N) * a0
w_ham /= np.sum(np.hamming(N))

theta, B_uni = beam_pattern(w_uni, N)
_,     B_ham = beam_pattern(w_ham, N)

plt.plot(np.rad2deg(theta), B_uni, label='Uniform')
plt.plot(np.rad2deg(theta), B_ham, label='Hamming')
plt.xlabel(r'$\theta$ (deg)'); plt.ylabel('dB'); plt.legend()
plt.ylim(-60, 1); plt.grid(True)
```

This code will serve as a reference throughout the book. We will enrich it chapter by chapter to include MVDR, LCMV, and adaptive weights.

6.11 Summary

The beam pattern is the spatial signature of a beamformer. It is computed as a simple Fourier transform of the weights, which makes the whole theory of classical windowing available to it. Reading it gives at a glance:

- the *resolution* (main-lobe width);
- the *selectivity* (sidelobe level);
- the *rejection* (null depth);
- the *array gain* (value on the target).

The next chapter studies in detail the first — and simplest — beamformer: Delay-and-Sum, whose uniform beam pattern serves as the reference for all the others. Historically, it is where the subject begins, and pedagogically it is still where the subject should begin: its simplicity fixes the ideas of array gain, spatial matched filtering, and robustness, which will remain valid — with nuances — for all more sophisticated beamformers.

Chapter 7

Delay-and-Sum

Delay-and-Sum (DAS) is the simplest beamformer. As its name suggests, it delays the antenna signals to align them with a target direction, then sums them. In the narrowband version, these “delays” become simple complex multiplications. DAS serves as the *reference* throughout the book: it is the lowest-complexity beamformer, the most numerically robust, and the one that maximizes SNR in white noise. All adaptive beamformers studied later try to do better *in the presence of interference* — never to do better in pure white noise.

Historically, DAS was also the first beamformer to be used industrially, as early as the Second World War in electronically scanned radars, then in passive sonars of the 1950s. Its simplicity made it compatible with the analog means of the time: adjustable-length delay lines or voltage-controlled phase shifters. Seventy years later, the same principle is implemented in 5G mmWave arrays, but at much larger scales and with digital control. The persistence of DAS across all generations of technology illustrates a simple truth: *when the environment is dominated by thermal noise, there is nothing better to do than sum in phase.*

7.1 Construction

To align the contributions coming from a target direction θ_0 on the N antennas, the phase shift $-mkd\sin\theta_0$ undergone by each antenna must be compensated. We therefore choose the weights

$$w_m = \frac{1}{N}\, e^{-jmkd\sin\theta_0},$$

that is

$$\boxed{\boldsymbol{w}_{\mathrm{DAS}}(\theta_0) = \frac{1}{N}\,\boldsymbol{a}(\theta_0).}$$

The factor $1/N$ normalizes the output: the gain in the target direction will be unity.

7.1.1 Beamformer Output

For a plane wave of amplitude $s(t)$ arriving at $\theta = \theta_0$:

$$\boldsymbol{x}(t) = \boldsymbol{a}(\theta_0)\, s(t) + \boldsymbol{n}(t),$$

the output is

$$y(t) = \boldsymbol{w}_{\mathrm{DAS}}^{\mathrm{H}}\boldsymbol{x}(t) = \frac{1}{N}\,\boldsymbol{a}^{\mathrm{H}}(\theta_0)\,\boldsymbol{a}(\theta_0)\,s(t) + \boldsymbol{w}_{\mathrm{DAS}}^{\mathrm{H}}\boldsymbol{n}(t).$$

Since $\boldsymbol{a}^{\mathrm{H}}(\theta_0)\boldsymbol{a}(\theta_0) = N$, the signal term is exactly $s(t)$: the target gain is unity.

7.1.2 Component by Component

Expanding the sums gives

$$y(t) = \frac{1}{N}\sum_{m=0}^{N-1} e^{jmkd\sin\theta_0}\, x_m(t).$$

Each antenna receives $x_m(t) = e^{-jmkd\sin\theta_0}\, s(t) + n_m(t)$, so the expression above:

- multiplies each $x_m(t)$ by a unit-modulus coefficient that *exactly compensates* the expected geometric phase shift of the target;
- sums the N contributions after they have been brought in phase;
- normalizes by N.

This is precisely the spatial analogue of the "matched filter."

7.2 Beam pattern

The beam pattern of the DAS steered to θ_0 is

$$B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta) = \frac{1}{N}\,\boldsymbol{a}^{\mathrm{H}}(\theta_0)\boldsymbol{a}(\theta) = \frac{1}{N}\, e^{-j(N-1)\beta/2}\,\frac{\sin(N\beta/2)}{\sin(\beta/2)},$$

with $\beta = kd(\sin\theta - \sin\theta_0)$. Its magnitude is the Dirichlet kernel already encountered in chapter 3. Its main characteristics are:

- unit gain at $\theta = \theta_0$;
- first null at $\sin\theta - \sin\theta_0 = \pm\lambda/(Nd)$;
- first sidelobe at $\sim -13.3\,\mathrm{dB}$;
- asymptotic decay in $1/\beta$.

7.3 Array Gain

7.3.1 Output in White Noise

If the additive noise is spatially white with variance σ^2, the output noise has variance

$$\sigma_y^2 = \mathbb{E}[|\boldsymbol{w}^{\mathrm{H}}\boldsymbol{n}|^2] = \sigma^2\|\boldsymbol{w}\|^2 = \sigma^2 \cdot \frac{N}{N^2} = \frac{\sigma^2}{N}.$$

7.3.2 SNR Gain

For DAS, the output signal has variance $|s|^2$ and the noise has variance σ^2/N. The ratio is

$$\mathrm{SNR}_{\mathrm{out}} = \frac{|s|^2}{\sigma^2/N} = N \cdot \frac{|s|^2}{\sigma^2} = N \cdot \mathrm{SNR}_{\mathrm{in}}.$$

We therefore gain a factor N, or $10\log_{10} N$ in dB. This is the *array gain* of the Delay-and-Sum beamformer: 8 antennas give $+9\,\mathrm{dB}$, 16 antennas give $+12\,\mathrm{dB}$, and 32 antennas give $+15\,\mathrm{dB}$.

Theorem 7.1 (Optimality in White Noise)**.** *In the presence of spatially white circular complex Gaussian noise (with covariance $\sigma^2\boldsymbol{I}_N$) and a single source at θ_0, the beamformer $\boldsymbol{w}_{DAS} = \boldsymbol{a}(\theta_0)/N$ maximizes the output SNR among all weight vectors normalized by $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_0) = 1$.*

Proof. Maximizing SNR under a unit-gain constraint is equivalent to

$$\min_{\boldsymbol{w}} \ \boldsymbol{w}^{\mathrm{H}} \boldsymbol{R}_n \boldsymbol{w} \quad \text{subject to} \quad \boldsymbol{w}^{\mathrm{H}} \boldsymbol{a}(\theta_0) = 1.$$

For $\boldsymbol{R}_n = \sigma^2 \boldsymbol{I}_N$, the Lagrangian gives $\boldsymbol{w}^* = \boldsymbol{I}^{-1}\boldsymbol{a}(\theta_0)/(\boldsymbol{a}^{\mathrm{H}}\boldsymbol{I}^{-1}\boldsymbol{a}) = \boldsymbol{a}(\theta_0)/N$. □

This is precisely the bound that DAS reaches, and it is precisely the bound that MVDR will generalize when $\boldsymbol{R}_n$ ceases to be proportional to the identity.

7.4 Why DAS Is Not Adaptive

DAS is entirely *deterministic*: its weights depend only on θ_0 and on the geometry. No data statistics are used. This property has two opposite consequences.

Advantage: robustness. DAS does not need an estimated covariance. It introduces no bias linked to a finite number of snapshots, no risk of *signal cancellation* (unintentional cancellation of the useful signal), and no sensitivity to poor matrix conditioning.

Disadvantage: blind to interference. As a simple bound, if a strong interferer arrives at θ_i through a sidelobe at $-13.3\,\mathrm{dB}$, it appears at the output attenuated by only 13 dB. A jammer 30 dB above the useful signal therefore remains 17 dB above it at the output: DAS cannot reject it. This is exactly what MVDR or LCMV will solve by placing an adaptive *null* on θ_i.

Example 7.2 (Interference in a Sidelobe)**.** ULA with $N = 16$, $d = \lambda/2$, target at 0°, jammer at 30°. DAS gives a gain of $\sim -15\,\mathrm{dB}$ at 30° (a deep sidelobe for this particular case). A jammer at $+40\,\mathrm{dB}$ above the signal appears at the output $+25\,\mathrm{dB}$ above the useful signal: the output is dominated by the jammer.

This failure is not marginal: it is one of the two major historical motivations for developing adaptive beamforming (the other being DOA estimation). Whenever a system must operate in an environment containing powerful non-cooperative RF sources (military radars in congested areas, GPS receivers under jamming, saturated urban communications), DAS alone is not enough, and it must be complemented by MVDR, LCMV, or Null Steering techniques. The hierarchy of Part III derives its coherence from this observation.

7.5 Errors and Fragilities

7.5.1 Pointing Error

An error $\delta\theta_0$ on the assumed target direction shifts the main lobe. The effective target gain drops according to the Dirichlet kernel:

$$G_{\text{eff}} = \frac{1}{N^2} \frac{\sin^2(N\beta/2)}{\sin^2(\beta/2)} \quad \text{with} \quad \beta = kd(\sin\theta_{\text{target}} - \sin\theta_0).$$

An error amounting to a significant fraction of the main-lobe width is enough to lose several dB. For a 16-antenna ULA at $\lambda/2$, a broadside simulation gives about 3 dB of loss for a few degrees of error, and a much stronger loss when the error approaches the main-lobe width.

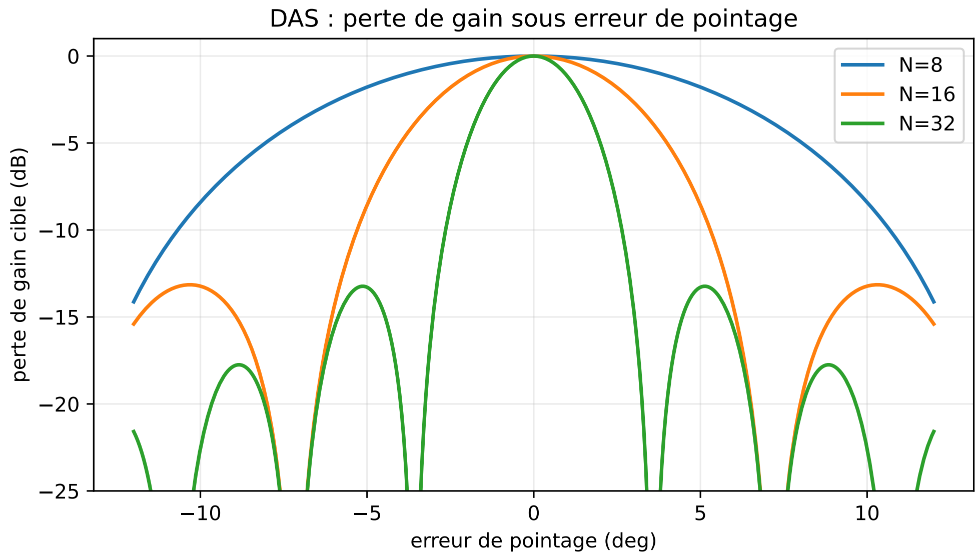

Figure 7.1: Broadside DAS simulation ($N = 8, 16, 32$, $d = \lambda/2$): the same angular error is more costly as N increases, because the main lobe becomes narrower.

7.5.2 Phase and Gain Errors

If each channel undergoes a random phase $\delta\phi_m \sim \mathcal{N}(0, \sigma_\phi^2)$, the average effective target gain is

$$\mathbb{E}[G_{\text{eff}}] \approx N\, e^{-\sigma_\phi^2}.$$

In this per-channel independent model, a spread $\sigma_\phi = 30^\circ$ costs ~ 1.2 dB and $\sigma_\phi = 60^\circ$ costs about 5 dB; beyond that, beamforming quickly loses most of its value. Calibration (chapter 22) therefore aims to bring σ_ϕ back to a few degrees when the hardware allows it.

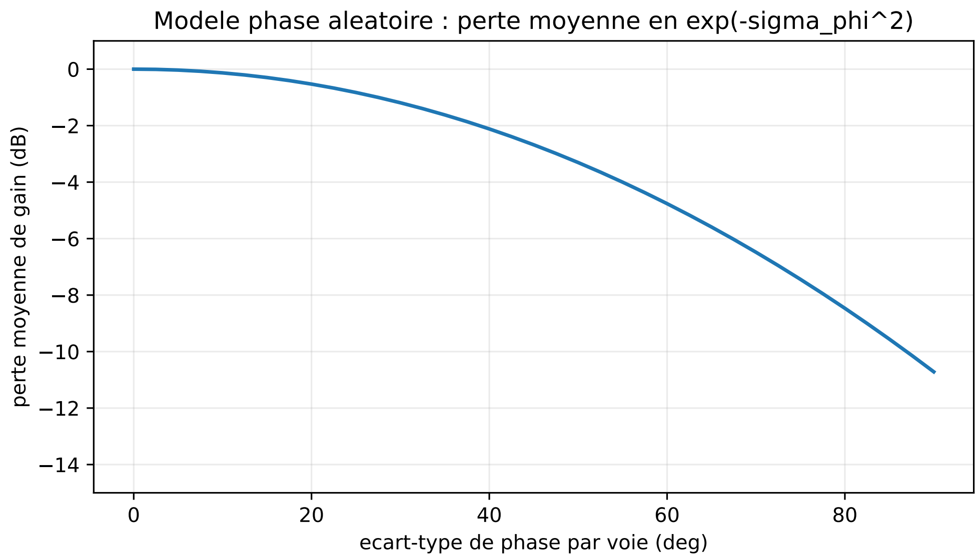

Figure 7.2: Indicative model of independent per-channel phase errors (i.i.d. Gaussian errors, normalized gain): the average loss follows $e^{-\sigma_\phi^2}$ when σ_ϕ is expressed in radians.

7.5.3 Narrowband Validity

The "pure phase" DAS assumes the narrowband approximation. For a wideband signal, the delay can no longer be reduced to a constant phase, and the gain depends on frequency. The remedy is *true time delay* or subband beamforming (chapter 19).

7.6 Numerical Implementation

Listing 7.1: Delay-and-Sum beamformer

```
import numpy as np

def das_weights(theta0, N, d_over_lambda=0.5):
    """Delay-and-Sum weights steered to theta0 (rad)."""
    m = np.arange(N)
    a0 = np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(theta0) * m)
    return a0 / N

def das_output(X, theta0, d_over_lambda=0.5):
    """X: (N, K) snapshot matrix. Returns y of length K."""
    N = X.shape[0]
    w = das_weights(theta0, N, d_over_lambda)
    return w.conj() @ X
```

7.7 DAS as a Universal Reference

Delay-and-Sum plays three indispensable roles in what follows.

1. **Upper bound on robustness.** No beamformer will be more numerically robust than DAS. We will see this again when introducing regularization by *diagonal loading*: the more the covariance is loaded, the closer MVDR gets to DAS.

2. **Lower performance bound with interference.** Any adaptive method must do *better* than DAS in an environment containing interference. If it does not, it introduces more bias than it corrects.

3. **Uniform beam pattern.** The Dirichlet function $|\sin(N\beta/2)/\sin(\beta/2)|/N$ is the reference for all spatial windowing. Other weightings are compared to it in terms of *taper efficiency* and *sidelobe level*.

The next chapter introduces the first deterministic method that improves on DAS: *Null Steering*, which explicitly imposes spatial nulls in known interference directions. It is also an opportunity to introduce the formalism of linear constraints, which will later be generalized in MVDR and LCMV. Null Steering is still purely deterministic (it does not look at the received data, only at the assumed known angles), but it crosses a first qualitative threshold: we move from a "blind" beamformer to an *informed* beamformer, which opens the way to all the more sophisticated approaches in the following parts.

Chapter 8

Null Steering: Imposing Spatial Nulls

Delay-and-Sum treats the environment as isotropic noise: it maximizes gain in the intended direction without accounting for directional interference. This is precisely what makes it ineffective against a strong jammer entering through a sidelobe. *Null Steering* introduces the first form of explicit spatial control: by construction, we impose *nulls* in the interference directions. It is also an opportunity to introduce the formalism of *linear constraints*, which will be generalized to MVDR (chapter 10) and then to LCMV (chapter 11).

The intuition is almost trivial: if we know precisely where a jammer is coming from, we simply ask the beamformer to receive nothing from that direction. The subtlety lies in formulating this request as a mathematically well-posed problem — which is the purpose of the first part of the chapter — and then in understanding what we *lose* by imposing this constraint: the cost in degrees of freedom, noise amplification, and sensitivity to angle errors. This is the second part of the chapter. Null Steering is therefore a good laboratory for understanding the tradeoffs that will reappear in MVDR and LCMV.

8.1 Formulation

8.1.1 Gain Constraints

Let θ_s be the direction of the useful source and $\theta_{i,1}, \dots, \theta_{i,L}$ the directions of L assumed known interferers. We want the beamformer to satisfy simultaneously

$$\boldsymbol{w}^{\mathrm{H}}\, \boldsymbol{a}(\theta_s) = 1, \qquad \boldsymbol{w}^{\mathrm{H}}\, \boldsymbol{a}(\theta_{i,\ell}) = 0, \quad \ell = 1, \dots, L.$$

The first equation guarantees unit gain on the target (the signal is preserved without distortion), while the other L equations impose nulls on the jammers.

8.1.2 Matrix Form

We gather the constraints into a matrix $\boldsymbol{C} = [\boldsymbol{a}(\theta_s), \boldsymbol{a}(\theta_{i,1}), \dots, \boldsymbol{a}(\theta_{i,L})] \in \mathbb{C}^{N\times(L+1)}$ and a response vector $\boldsymbol{f} = [1, 0, \dots, 0]^{\mathrm{T}} \in \mathbb{C}^{L+1}$. The problem is then written

$$\boxed{\boldsymbol{C}^{\mathrm{H}}\, \boldsymbol{w} = \boldsymbol{f}.}$$

This is a linear system in $\boldsymbol{w}$.

8.2 Minimum-Norm Solution

If $L+1 < N$, the system is underdetermined: in general, infinitely many $\boldsymbol{w}$ satisfy the constraints. The canonical choice is the *minimum-norm* one, which maximizes robustness to noise (a large $\|\boldsymbol{w}\|$ amplifies the output noise).

Theorem 8.1 (Minimum-Norm Null-Steering Solution)**.** *The unique solution of the problem*

$$\min_{\boldsymbol{w}} \ \|\boldsymbol{w}\|^2 \quad \textit{subject to} \quad \boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = \boldsymbol{f},$$

when $\boldsymbol{C}$ has full column rank ($L+1 \leq N$), is

$$\boxed{\boldsymbol{w}_{NS} = \boldsymbol{C}\left(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C}\right)^{-1}\boldsymbol{f}.}$$

Proof. Use the method of Lagrange multipliers. Form

$$\mathcal{L}(\boldsymbol{w}, \boldsymbol{\mu}) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{w} + \boldsymbol{\mu}^{\mathrm{H}}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} - \boldsymbol{f}) + (\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} - \boldsymbol{f})^{\mathrm{H}}\boldsymbol{\mu}.$$

The stationarity condition with respect to $\boldsymbol{w}$ gives $\boldsymbol{w} = -\boldsymbol{C}\,\boldsymbol{\mu}$ (up to a sign that will be absorbed). Injecting this into the constraint $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = \boldsymbol{f}$ gives $-\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C}\,\boldsymbol{\mu} = \boldsymbol{f}$, hence $\boldsymbol{\mu} = -(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C})^{-1}\boldsymbol{f}$, and $\boldsymbol{w} = \boldsymbol{C}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C})^{-1}\boldsymbol{f}$. Uniqueness comes from the strict convexity of $\|\boldsymbol{w}\|^2$ and the regularity of $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C}$. □

The matrix $\boldsymbol{C}^{+\dagger} = \boldsymbol{C}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C})^{-1}$ is called the *Moore–Penrose pseudo-inverse* of $\boldsymbol{C}^{\mathrm{H}}$.

8.3 Geometric Reading: Orthogonal Projection

Consider the decomposition $\mathbb{C}^N = \mathrm{Im}(\boldsymbol{C}) \oplus \mathrm{Ker}(\boldsymbol{C}^{\mathrm{H}})$. The minimum-norm solution lies entirely in $\mathrm{Im}(\boldsymbol{C})$: it has no "useless" component in the orthogonal complement. It is therefore the orthogonal projection (in the Hermitian sense) of the ideal vector $\boldsymbol{a}(\theta_s)/N$ onto the constraint subspace.

More intuitively: to impose a null at θ_i, one must take a component of $\boldsymbol{w}$ that is orthogonal to $\boldsymbol{a}(\theta_i)$. Each additional constraint "consumes" one degree of freedom in $\mathbb{C}^N$.

8.4 Degrees of Freedom

Proposition 8.2. *A ULA with N antennas has N complex degrees of freedom. It can simultaneously impose:*

- *unit gain in the target direction (1 constraint);*
- *L nulls in distinct directions (L constraints).*

The condition is $1 + L \leq N$, i.e. $\boxed{L \leq N-1}$.

In practice, one avoids saturating the degrees of freedom ($L \ll N$ is safer) because:

- the conditioning of $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C}$ degrades when the constraints become nearly collinear;
- the norm $\|\boldsymbol{w}\|^2$ explodes, amplifying the noise;
- robustness to angular errors decreases.

8.5 Effect on the Beam Pattern

Applying the weights $\boldsymbol{w}_{\mathrm{NS}}$ produces a beam pattern that:

- preserves a main lobe close to the Delay-and-Sum main lobe centered on θ_s;
- exhibits *theoretical nulls* at the $\theta_{i,\ell}$ (up to numerical precision);
- changes the sidelobe shape relative to DAS, often in unpredictable directions.

Example 8.3 (8-Antenna ULA, 2 Jammers)**.** $N = 8$, $\theta_s = 0°$, jammers at $\pm 30°$. DAS gives a gain of $-13.5\,\mathrm{dB}$ at 30°; Null Steering imposes $-\infty$ there (limited by numerical precision to $\sim -300\,\mathrm{dB}$ in an ideal double-precision calculation). The main lobe widens slightly; in this simulation, the array-gain loss remains on the order of half a dB.

8.6 Effective Null Depth

In infinite precision arithmetic, the null is exactly zero. In practice:

- the steering vector used to compute $\boldsymbol{w}$ is the theoretical $\boldsymbol{a}(\theta_{i,\ell})$, which differs from the *actual* antenna steering vector $\boldsymbol{a}_{\text{actual}}(\theta_{i,\ell})$;
- the jammer does not arrive exactly at $\theta_{i,\ell}$ but at $\theta_{i,\ell} + \delta\theta$;
- calibration errors produce a residual offset.

As a consequence, a theoretical null at $-\infty$ dB is often measured, depending on calibration and angular stability, in a much less spectacular range (for example -30 to -50 dB on a well-calibrated test bench). This degradation is studied in detail in chapter 23 (derivative constraints, widened nulls).

8.7 Effect on Noise: Amplification

The output noise has variance $\sigma_y^2 = \sigma^2 \|\boldsymbol{w}_{\text{NS}}\|^2$. Null Steering does not optimize $\|\boldsymbol{w}\|^2$: it minimizes it under constraint. But when the jammer approaches the target (i.e. $\theta_i \to \theta_s$), $\boldsymbol{a}(\theta_i)$ becomes almost collinear with $\boldsymbol{a}(\theta_s)$, $\boldsymbol{C}^{\text{H}}\boldsymbol{C}$ becomes ill-conditioned, and $\|\boldsymbol{w}\|^2$ explodes.

White Noise Gain (WNG). The white-noise gain is defined as

$$\text{WNG} = \frac{|\boldsymbol{w}^{\text{H}}\boldsymbol{a}(\theta_s)|^2}{\|\boldsymbol{w}\|^2} = \frac{1}{\|\boldsymbol{w}\|^2} \quad \text{(with constraint } \boldsymbol{w}^{\text{H}}\boldsymbol{a}(\theta_s) = 1\text{)}.$$

For DAS, WNG $= N$. For Null Steering, WNG $< N$, and the loss measures the SNR cost of the imposed constraints.

8.8 Simple Regularization

When the constraints are ill-conditioned, we regularize with

$$\boldsymbol{w}_{\text{NS}}^{\text{reg}} = \boldsymbol{C}\left(\boldsymbol{C}^{\text{H}}\boldsymbol{C} + \alpha \boldsymbol{I}\right)^{-1} \boldsymbol{f},$$

with small $\alpha > 0$. The cost is a slight degradation in null depth, but the norm $\|\boldsymbol{w}\|$ becomes controlled again. This regularization exactly foreshadows the *diagonal loading* that will be systematized in MVDR.

8.9 Differences from MVDR

Three fundamental differences separate Null Steering from MVDR.

1. **Information used.** Null Steering uses the *known angular positions* of the interferers. MVDR uses an *estimated covariance* from the received data.
2. **Null depth.** Null Steering produces nulls at $-\infty$ dB in theory. MVDR produces deep nulls, but their depth depends on the interference-to-noise ratio (a very strong jammer produces a very deep null; a weak jammer produces a shallow null).
3. **Adaptability.** Null Steering requires the $\theta_{i,\ell}$ to be known *a priori*. MVDR learns them from the data.

In practice, the two are often combined: Null Steering for *known* directions (friendly transmitter, satellite, beacon), MVDR or LCMV for *unknown* directions (adversarial jammers).

8.10 Numerical Implementation

Listing 8.1: Generic Null Steering

```
import numpy as np

def null_steering(theta_s, theta_i, N, d_over_lambda=0.5, alpha=0.0):
    """
    theta_s : useful direction (rad)
    theta_i : list/array of interference directions (rad)
    alpha   : regularization (>=0)
    """
    m = np.arange(N)
    def a(theta):
        return np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(theta) * m)

    angles = [theta_s] + list(theta_i)
    C = np.column_stack([a(th) for th in angles])         # N x (L+1)
    f = np.zeros(len(angles), dtype=complex)
    f[0] = 1.0
    G = C.conj().T @ C + alpha * np.eye(len(angles))
    return C @ np.linalg.solve(G, f)
```

8.11 When Should Null Steering Be Used?

Suitable.

- Jammers with precisely known directions (detected adversarial GPS beacon, neighboring satellite, stationary RF source).
- Analytical studies where one wants to impose a precise theoretical beam pattern.
- Initialization of an adaptive beamformer.

Unsuitable.

- Unknown or time-varying interference directions.
- Multipath environments (the effective directions are not those of the physical sources).
- Systems without careful calibration: a null on a false angle is useless.

8.12 Conclusion

Null Steering takes a first step toward adaptation: it replaces DAS, which is blind to interference, with a beamformer *informed* by the interference directions. But this information must be supplied from outside, and the cost in degrees of freedom is explicit.

The next step is to *extract* this information from the received data themselves. The statistical quantity that makes this possible is the *spatial covariance*, the central object of Part III. Chapter 9 introduces this matrix and studies its properties, structure, and estimation. It is this matrix that will open the way to MVDR and then LCMV.

The transition from Part II to Part III corresponds exactly to a change of paradigm: we leave the world of *data-independent* beamformers for that of *data-dependent* beamformers. The former are fragile against interference but computationally robust; the latter optimize their response to the observed environment but pay for this optimization with increased sensitivity to estimation artifacts and model errors. Part VII will return to these fragilities and provide the tools (diagonal loading, derivative constraints) that make it possible to contain them.

Part III

Statistical Beamforming

Chapter 9

Spatial Covariance

Part II built beamformers whose weights depend only on geometry and on angles assumed to be known. Part III reverses the perspective: the angles are assumed *unknown* (or only the target is known), and the structure of the environment is learned from the data. The central statistical tool behind this change of paradigm is the *spatial covariance*. It contains all second-order information about inter-antenna correlations; it implicitly encodes the directions, powers, and correlations of the sources present. This chapter introduces it and studies its structure and estimation.

Before entering the formal definition, it is useful to understand why covariance is the relevant object, rather than, for example, the mean or a simple correlation. An RF source reaches the antennas with an amplitude and phase that vary in time in an essentially random way (modulation, power fluctuations, and so on). If one simply computes $\mathbb{E}[\boldsymbol{x}(n)]$, the result is zero under the zero-mean assumption, which says nothing about spatial structure. By contrast, $\mathbb{E}[\boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n)]$ captures the *average phase relationships* between antennas, independently of amplitude fluctuations. This second-order statistical stability is what makes covariance useful, and what makes it the pivot of all the adaptive and subspace algorithms that follow.

9.1 Definition

Definition 9.1 (Spatial Covariance)**.** For a snapshot process $\boldsymbol{x}(n) \in \mathbb{C}^N$ that is wide-sense stationary and zero mean, the *spatial covariance* is the matrix

$$\boldsymbol{R} = \mathbb{E}\Big[\boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n)\Big] \in \mathbb{C}^{N\times N}.$$

The coefficient $R_{ij} = \mathbb{E}[x_i(n)x_j^*(n)]$ measures the complex correlation between the signals from antennas i and j. The diagonal entry $R_{mm} = \mathbb{E}[|x_m(n)|^2]$ is the *average power* on antenna m. The off-diagonal entries capture the average phase shifts from one channel to another.

Why the conjugate? Without the conjugate, one would have $\mathbb{E}[x_i x_j]$. For a complex signal with fluctuating phase, x_j can be buried in a fast complex exponential that carries no useful information: $\mathbb{E}[x_i x_j]$ would be zero even for two perfectly correlated channels. The conjugate rewinds the phase and reveals the structural correlation. This is why $\mathbb{E}[x_i x_j^*]$, and more generally the *Hermitian* covariance, is the right quantity for circular complex signals.

9.2 Single Source and Rank-1 Model

For a source at θ_0 with envelope $s(n)$ and no noise: $\boldsymbol{x}(n) = \boldsymbol{a}(\theta_0)s(n)$. Alors

$$\boldsymbol{R} = \mathbb{E}[s(n)s^*(n)]\,\boldsymbol{a}(\theta_0)\boldsymbol{a}^{\mathrm{H}}(\theta_0) = p_0\,\boldsymbol{a}(\theta_0)\boldsymbol{a}^{\mathrm{H}}(\theta_0),$$

where $p_0 = \mathbb{E}[|s(n)|^2]$ is the source power. The matrix $\boldsymbol{R}$ is then *rank 1*, and its only nonzero eigenvector is proportional to $\boldsymbol{a}(\theta_0)$, with eigenvalue Np_0.

Proposition 9.2 (Single noiseless source)**.** *For a single source at θ_0, $\boldsymbol{R}$ has a single nonzero eigenvalue $\lambda_1 = Np_0$ associated with the eigenvector $\boldsymbol{a}(\theta_0)/\sqrt{N}$. The other $N-1$ eigenvalues are zero.*

This property is the algebraic root of subspace methods (Part V). It can be stated another way: a single source fills only one direction of $\mathbb{C}^N$, the steering-vector direction $\boldsymbol{a}(\theta_0)$, and the rest of the space is strictly unused. This remainder (the orthogonal complement) is what MUSIC will exploit as the *noise subspace* to localize θ_0 exactly. The same logic will apply to L sources: the signal subspace has dimension L, and its orthogonal complement, of dimension $N-L$, contains all the information about the absence of sources in the unoccupied directions.

9.3 With White Noise

Add spatially white noise with variance σ^2: $\boldsymbol{x}(n) = \boldsymbol{a}(\theta_0)s(n) + \boldsymbol{n}(n)$, with $\mathbb{E}[\boldsymbol{n}\boldsymbol{n}^{\mathrm{H}}] = \sigma^2 \boldsymbol{I}_N$ and $\boldsymbol{n}$ uncorrelated with the signal. By bilinearity,

$$\boxed{\boldsymbol{R} = p_0\, \boldsymbol{a}(\theta_0)\boldsymbol{a}^{\mathrm{H}}(\theta_0) + \sigma^2 \boldsymbol{I}_N.}$$

The eigenvalues are:

- $\lambda_1 = Np_0 + \sigma^2$ with eigenvector $\boldsymbol{a}(\theta_0)/\sqrt{N}$ (signal subspace);
- $\lambda_2 = \cdots = \lambda_N = \sigma^2$, with an orthogonal eigenspace of dimension $N-1$ (noise subspace).

This structure, one strong eigenvalue and $N-1$ eigenvalues at the noise floor, is the typical signature of a single source in noise. It remains readable down to moderate SNR values.

9.4 Multiple Sources

For L sources with distinct angles and powers p_ℓ:

$$\boldsymbol{x}(n) = \boldsymbol{A}\,\boldsymbol{s}(n) + \boldsymbol{n}(n), \quad \boldsymbol{A} = [\boldsymbol{a}(\theta_1), \ldots, \boldsymbol{a}(\theta_L)].$$

Assuming the sources are mutually *uncorrelated* ($\mathbb{E}[\boldsymbol{s}\boldsymbol{s}^{\mathrm{H}}] = \mathrm{diag}(p_1, \ldots, p_L) = \boldsymbol{P}_s$) and uncorrelated with the noise, we obtain

$$\boxed{\boldsymbol{R} = \boldsymbol{A}\,\boldsymbol{P}_s\,\boldsymbol{A}^{\mathrm{H}} + \sigma^2 \boldsymbol{I}_N.}$$

Eigenvalue structure. If $\boldsymbol{A}$ has full column rank $L < N$, then $\boldsymbol{A}\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}}$ has rank L. The eigenvalues of $\boldsymbol{R}$ split into:

- L *signal* eigenvalues: $\lambda_1, \ldots, \lambda_L$, all greater than σ^2;
- $N-L$ *noise* eigenvalues: all equal to σ^2.

This separation into two groups is the basis of the signal/noise decomposition used by MUSIC and ESPRIT.

Coherent sources. If two sources are perfectly correlated (for example a direct path and an echo), $\boldsymbol{P}_s$ itself becomes rank-deficient, and $\boldsymbol{A}\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}}$ may have rank $L' < L$. The number of *signal eigenvalues* is then L', and MUSIC underestimates the number of sources. The classical remedy is *spatial smoothing* (Chapter 15).

9.5 Mathematical Properties

Theorem 9.3 (Properties of the Covariance)**.** $\boldsymbol{R}$ *is:*

- Hermitian*:* $\boldsymbol{R}^{\mathrm{H}} = \boldsymbol{R}$*. Indeed,* $(\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathrm{H}}])^{\mathrm{H}} = \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathrm{H}}]$*.*
- Positive semidefinite*:* $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w} = \mathbb{E}[|\boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}|^2] \geq 0$ *for every* $\boldsymbol{w} \in \mathbb{C}^N$*.*
- Diagonalizable in an orthonormal basis*, with positive (or zero) real eigenvalues.*

Practical consequence. All algorithms that manipulate $\boldsymbol{R}$ exploit these properties: eigendecomposition in an orthonormal basis, Cholesky factorizations, and conditioning of the inversion problem. A numerically estimated matrix is never exactly Hermitian (roundoff errors), so in practice it is symmetrized via $\widehat{\boldsymbol{R}} \leftarrow (\widehat{\boldsymbol{R}} + \widehat{\boldsymbol{R}}^{\mathrm{H}})/2$.

9.6 Estimation from Data

9.6.1 Sample Covariance Matrix

The theoretical covariance $\boldsymbol{R}$ is unknown: it is estimated from K snapshots,

$$\boxed{\widehat{\boldsymbol{R}} = \frac{1}{K}\sum_{n=1}^{K} \boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n) = \frac{1}{K}\boldsymbol{X}\boldsymbol{X}^{\mathrm{H}},}$$

where $\boldsymbol{X} = [\boldsymbol{x}(1), \ldots, \boldsymbol{x}(K)] \in \mathbb{C}^{N\times K}$. $\widehat{\boldsymbol{R}}$ is called the *Sample Covariance Matrix* (SCM). It is unbiased: $\mathbb{E}[\widehat{\boldsymbol{R}}] = \boldsymbol{R}$.

9.6.2 Number of Snapshots and Estimation Quality

The quality of the estimate depends on the ratio K/N. The main rules of thumb are:

- $K < N$: $\widehat{\boldsymbol{R}}$ is singular (rank $\leq K < N$) and cannot be used as is for MVDR without regularization. This is typical of snapshot-limited systems: short-illumination radar and nonstationary signals.
- $K \approx N$: $\widehat{\boldsymbol{R}}$ is ill-conditioned, and MVDR underperforms.
- $K \geq 2N$: the empirical *Reed–Mallett–Brennan* benchmark (in the ideal Gaussian SMI model, $K \simeq 2N$ corresponds to an average loss of about 3 dB relative to the theoretical covariance).
- $K \gg N$: $\widehat{\boldsymbol{R}}$ converges to $\boldsymbol{R}$ with error $\mathcal{O}(N/K)$ in an appropriate matrix norm.

Theorem 9.4 (Reed–Mallett–Brennan [6])**.** *For a complex Gaussian covariance and a Sample Matrix Inversion (SMI) beamformer, the average SINR loss relative to* $\boldsymbol{R}$ *is*

$$\mathbb{E}\left[\frac{\mathrm{SINR}_{SMI}}{\mathrm{SINR}_{opt}}\right] = \frac{K-N+2}{K+1}.$$

For $K = 2N$*, the ratio is* ~ 0.5*, i.e., a* 3 dB *loss.*

This result, known as the *RMB rule*, governs practical sizing, but it should be read as an average result in an ideal Gaussian setting. It will be refined for regularized MVDR in Chapter 23.

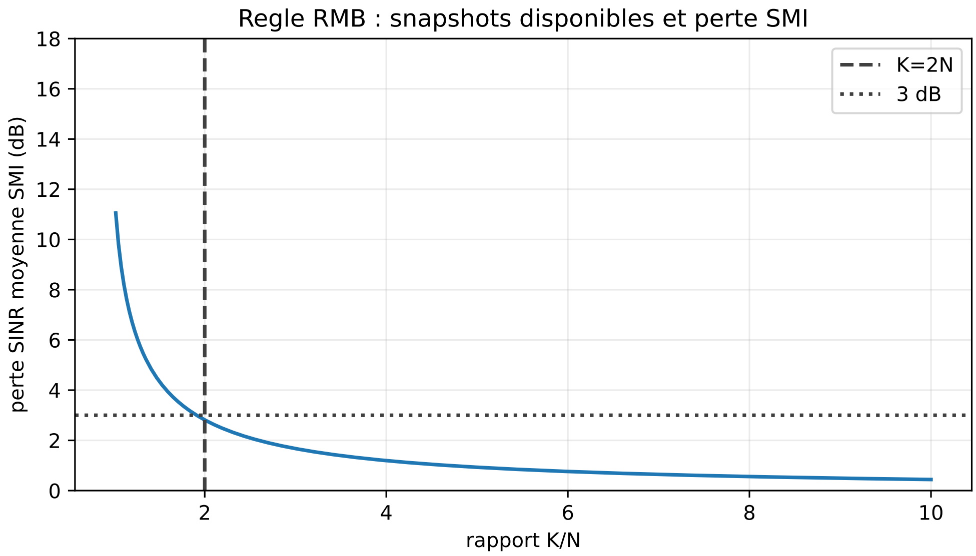

Figure 9.1: Reading the Reed–Mallett–Brennan rule ($N = 32$, ideal Gaussian SMI model): at $K = 2N$, the average SMI loss is on the order of 3 dB; below that, the estimated covariance becomes very fragile.

9.6.3 Stationarity

Averaging-based estimation requires the process to be stationary over the K-snapshot window. When sources move or powers vary, one must:

- shorten the window (at the cost of estimation quality);
- use *exponential forgetting* (cf. Chapter 13);
- segment and merge with a moving average.

9.6.4 Forward–Backward averaging

For a ULA and coherent or weakly correlated sources, the effective number of useful snapshots is artificially increased by exploiting the *persymmetric symmetry* of $\boldsymbol{R}$. The forward–backward version of the estimator is

$$\widehat{\boldsymbol{R}}_{FB} = \frac{1}{2}\,(\widehat{\boldsymbol{R}} + \boldsymbol{J}\,\widehat{\boldsymbol{R}}^*\,\boldsymbol{J}),$$

where $\boldsymbol{J}$ is the flip permutation matrix. This estimate is more accurate for small K and remains persymmetric.

9.7 Invertibility and Conditioning

9.7.1 Singular Case

When $K < N$ or sources are perfectly coherent, $\widehat{\boldsymbol{R}}$ can be singular or very ill-conditioned. Algorithms that require inverting $\boldsymbol{R}$ (MVDR, LCMV, RLS) then become unstable or impossible.

9.7.2 Conditioning

The *condition number* $\kappa(\boldsymbol{R}) = \lambda_{\max}/\lambda_{\min}$ measures the numerical difficulty of inversion. For a jammer 30 dB above the noise on a ULA with $N = 16$, $\lambda_1 \sim 10^3\sigma^2$ and $\lambda_N = \sigma^2$, so $\kappa \sim 10^3$. With a 60 dB jammer, κ reaches 10^6: double-precision inversion (~ 16 digits) starts losing 6 digits of precision.

9.7.3 Regularization by Diagonal Loading

The canonical solution is to replace $\widehat{\boldsymbol{R}}$ by

$$\widehat{\boldsymbol{R}}_\alpha = \widehat{\boldsymbol{R}} + \alpha \boldsymbol{I}_N,$$

with $\alpha > 0$. All eigenvalues are lifted by α, the condition number becomes $\kappa = (\lambda_{\max} + \alpha)/(\lambda_{\min} + \alpha)$, and inversion becomes stable again. The cost is a slight asymptotic performance loss (cf. Chapters 10 and 23 for the optimal choice of α).

9.8 Physical Interpretation

The covariance implicitly contains the full *spatial map* of sources and noise. More concretely:

- the *trace* $\mathrm{tr}(\boldsymbol{R}) = \sum_m \mathbb{E}[|x_m|^2]$ is the total power received by the array;
- the *determinant* $\det(\boldsymbol{R})$ measures the uncertainty volume of the snapshot;
- the quantity $\boldsymbol{a}^{\mathrm{H}}(\theta)\boldsymbol{R}\,\boldsymbol{a}(\theta)$ is the power that would come out of a DAS beamformer pointed at θ, and forms the *Bartlett spectrum*;
- the quantity $\boldsymbol{a}^{\mathrm{H}}(\theta)\boldsymbol{R}^{-1}\boldsymbol{a}(\theta)$ plays a dual role and will become the Capon pseudospectrum, studied in the next chapter.

9.9 Visualisation

It is common to display $|\boldsymbol{R}|$ as an image (magnitude in decibels) and the phase as a second image. For a broadside ULA, the off-diagonal elements are approximately constant along diagonals (a *Toeplitz* structure): the covariance matrix is then quasi-Toeplitz, a property exploited by Toeplitz-regularized estimators.

9.10 Limits of Covariance

Presence of the desired signal in $\boldsymbol{R}$. In practice, the estimated covariance contains both the interferences and the desired signal. If an MVDR is built from this $\widehat{\boldsymbol{R}}$, the algorithm tries to *attenuate the desired signal*, mistaking it for noise (*signal self-cancellation*, cf. Chapter 23). The remedy, when possible, is to estimate an *interference-plus-noise only* covariance using time intervals in which the signal is absent.

Sensitivity to calibration. A phase error between channels produces a covariance whose structure no longer matches the theoretical steering vectors. The algorithms think they see sources that do not exist. Chapter 22 treats this problem in depth.

9.11 Numerical Implementation

Listing 9.1: Covariance estimation and diagonal loading

```python
import numpy as np

def sample_covariance(X):
    """X: (N, K) snapshot matrix. Returns R (N,N)."""
    N, K = X.shape
    assert K > 0
```

```python
    R = (X @ X.conj().T) / K
    return 0.5 * (R + R.conj().T)   # Hermitian symmetrization

def diagonal_loading(R, alpha):
    N = R.shape[0]
    assert R.shape == (N, N)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    return 0.5 * (R_alpha + R_alpha.conj().T)

def forward_backward(R):
    N = R.shape[0]
    assert R.shape == (N, N)
    J = np.flip(np.eye(N), axis=0)
    return 0.5 * (R + J @ R.conj() @ J)
```

9.12 Summary

Spatial covariance is the pivot object of statistical beamforming:

- it encodes the spatial structure of the sources and noise;
- its eigendecomposition separates a *signal subspace* (dimension L for L sources) and a *noise subspace* (dimension $N - L$);
- its empirical estimate often requires K on the order of a few times N to remain satisfactory;
- its inversion requires diagonal-loading regularization as soon as it becomes ill-conditioned or singular.

From this matrix, the next chapter builds the *Minimum Variance Distortionless Response* beamformer, which automatically places its nulls on the unwanted energy detected in $\boldsymbol{R}$. It is the archetype of the adaptive beamformer: its weights depend entirely on the covariance, which itself depends entirely on the data. The operation is therefore fully *oriented toward the observed RF scene*, with no explicit assumption about jammer directions.

Chapter 10

MVDR / Capon

Null Steering requires the interference directions to be known. If they are unknown, or if new ones appear, it no longer knows what to do. The *Minimum Variance Distortionless Response* (MVDR), also called the *Capon beamformer* [5], solves this problem by using the spatial covariance directly. It minimizes the total output power while imposing unit gain in the target direction: its zeros are automatically placed where the received energy is strongest, and therefore where the jammers are, without naming them.

The idea has a particular elegance: the beamformer is asked to *make as little noise as possible*, subject only to the constraint that it must not distort the desired signal. Everything that is not the desired signal is, by construction, unwanted energy, and MVDR removes it without needing to name it. This is a form of beamforming by subtraction, as opposed to Null Steering by explicit constraint. This shift in perspective has two major practical consequences: MVDR does not need to know the jammer directions, and it adapts automatically when they change. Its counterpart, which will emerge later, is increased sensitivity to errors in the target direction and in the covariance estimate, the two objects on which it relies.

10.1 Formulation

Definition 10.1 (MVDR Problem)**.** For a covariance $\boldsymbol{R}$ and a target direction θ_s, the MVDR beamformer is the solution of

$$\boxed{\min_{\boldsymbol{w}} \ \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w} \quad \text{subject to} \quad \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_s) = 1.}$$

The objective $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w}$ is exactly the *average output power*; $\mathbb{E}[|y(n)|^2] = \boldsymbol{w}^{\mathrm{H}}\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathrm{H}}]\boldsymbol{w} = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w}$. The constraint requires the target to pass without distortion. Minimizing the former under the latter amounts to attenuating only what is not the desired signal.

10.2 Solution

Theorem 10.2 (Capon Solution)**.** *If $\boldsymbol{R}$ is invertible, the MVDR problem has the unique solution*

$$\boxed{\boldsymbol{w}_{\mathrm{MVDR}} = \frac{\boldsymbol{R}^{-1}\boldsymbol{a}(\theta_s)}{\boldsymbol{a}^{\mathrm{H}}(\theta_s)\,\boldsymbol{R}^{-1}\,\boldsymbol{a}(\theta_s)}.}$$

Proof. Apply complex Lagrange multipliers. To lighten the notation, set:

$$\boldsymbol{a}_s = \boldsymbol{a}(\theta_s).$$

The problem is:

$$\min_{\boldsymbol{w}} \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w} \quad \text{subject to} \quad \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}_s = 1.$$

Because the complex constraint imposes both a real and an imaginary part, write a real Lagrangian in the form:

$$\mathcal{L}(\boldsymbol{w}, \mu) = \boldsymbol{w}^{\mathrm{H}} \boldsymbol{R}\, \boldsymbol{w} + \mu^*(\boldsymbol{w}^{\mathrm{H}} \boldsymbol{a}_s - 1) + \mu(\boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{w} - 1).$$

The Wirtinger derivative with respect to $\boldsymbol{w}^*$ gives:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{w}^*} = \boldsymbol{R}\boldsymbol{w} + \mu^* \boldsymbol{a}_s = \mathbf{0}.$$

Thus:

$$\boldsymbol{R}\boldsymbol{w} = -\mu^* \boldsymbol{a}_s.$$

If $\boldsymbol{R}$ is invertible:

$$\boldsymbol{w} = -\mu^* \boldsymbol{R}^{-1} \boldsymbol{a}_s.$$

The optimal vector is therefore necessarily proportional to $\boldsymbol{R}^{-1}\boldsymbol{a}_s$. It remains only to choose the constant that satisfies the distortionless constraint.

Substituting into $\boldsymbol{w}^{\mathrm{H}} \boldsymbol{a}_s = 1$:

$$(-\mu^* \boldsymbol{R}^{-1} \boldsymbol{a}_s)^{\mathrm{H}} \boldsymbol{a}_s = 1.$$

Since $\boldsymbol{R}$ is Hermitian, $\boldsymbol{R}^{-1}$ is Hermitian as well:

$$-\mu\, \boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{R}^{-1} \boldsymbol{a}_s = 1.$$

Therefore:

$$-\mu = \frac{1}{\boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{R}^{-1} \boldsymbol{a}_s}.$$

We finally obtain:

$$\boldsymbol{w}_{\mathrm{MVDR}} = \frac{\boldsymbol{R}^{-1} \boldsymbol{a}_s}{\boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{R}^{-1} \boldsymbol{a}_s}.$$

Since the Hessian $\boldsymbol{R}$ is positive semidefinite and the constraint is linear, the stationary point is a global minimum. □

10.3 MVDR in Practice: Avoiding the Explicit Inverse of $\boldsymbol{R}$

The formula contains $\boldsymbol{R}^{-1}$, but a serious implementation does not form the explicit inverse. First compute

$$\boldsymbol{R}\boldsymbol{u} = \boldsymbol{a}_s,$$

then normalize:

$$\boxed{\boldsymbol{w}_{\mathrm{MVDR}} = \frac{\boldsymbol{u}}{\boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{u}}.}$$

Numerically, this means using a linear solve:

$$\boldsymbol{u} = \mathrm{solve}(\boldsymbol{R}, \boldsymbol{a}_s),$$

and not:

$$\boldsymbol{u} = \boldsymbol{R}^{-1} \boldsymbol{a}_s.$$

This difference is important:

- computing the explicit inverse is more expensive;
- inversion amplifies roundoff errors more strongly;
- a linear solve better exploits the positive Hermitian structure of $\boldsymbol{R}$, for example with Cholesky;
- with diagonal loading, one simply solves $(\widehat{\boldsymbol{R}} + \alpha \boldsymbol{I})\boldsymbol{u} = \boldsymbol{a}_s$.

Scaled form. In practice, one first finds the vector $\boldsymbol{u}$ satisfying $\boldsymbol{R}\boldsymbol{u} = \boldsymbol{a}(\theta_s)$, then normalizes $\boldsymbol{w} = \boldsymbol{u}/(\boldsymbol{a}^{\mathrm{H}}(\theta_s)\,\boldsymbol{u})$. The inverse $\boldsymbol{R}^{-1}$ is not formed explicitly: one solves $\boldsymbol{R}\boldsymbol{u} = \boldsymbol{a}(\theta_s)$ by Cholesky or LU decomposition, which is significantly cheaper and more stable.

10.4 Capon Pseudospectrum

Applying $\boldsymbol{w}_{\mathrm{MVDR}}$ to the covariance, the output power is

$$P_{\mathrm{MVDR}}(\theta_s) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\,\boldsymbol{w} = \frac{1}{\boldsymbol{a}^{\mathrm{H}}(\theta_s)\,\boldsymbol{R}^{-1}\,\boldsymbol{a}(\theta_s)}.$$

By varying θ_s, this expression defines the *pseudo-spectre de Capon*

$$\boxed{P_{\mathrm{Capon}}(\theta) = \frac{1}{\boldsymbol{a}^{\mathrm{H}}(\theta)\,\boldsymbol{R}^{-1}\,\boldsymbol{a}(\theta)}.}$$

Its maxima localize the sources with much better resolution than the *Bartlett spectrum* $P_{\mathrm{Bartlett}}(\theta) = \boldsymbol{a}^{\mathrm{H}}(\theta)\,\boldsymbol{R}\,\boldsymbol{a}(\theta)/N^2$ (which is nothing more than DAS applied to $\boldsymbol{R}$).

Qualitative comparison.

- Bartlett has the resolution of Delay-and-Sum (Fourier limit $\sim \lambda/D$), that is, the Rayleigh bound.
- Capon exceeds this limit, with resolution that improves as SNR increases.
- MUSIC (Chapter 15) will go even further by exploiting the subspace structure of $\boldsymbol{R}$.

10.5 Special Case: White Noise Only

If $\boldsymbol{R} = \sigma^2\boldsymbol{I}_N$ (no directional interference), $\boldsymbol{R}^{-1} = \boldsymbol{I}_N/\sigma^2$ and

$$\boldsymbol{w}_{\mathrm{MVDR}} = \frac{\boldsymbol{a}(\theta_s)/\sigma^2}{N/\sigma^2} = \frac{\boldsymbol{a}(\theta_s)}{N} = \boldsymbol{w}_{\mathrm{DAS}}.$$

In pure white noise, MVDR coincides exactly with Delay-and-Sum. This is a reassuring confirmation: nothing has been gained, but nothing has been lost, where DAS is already optimal.

This equivalence is not anecdotal: it says that MVDR generalizes DAS without replacing it. Where the environment is benign, the two are identical; where it becomes hostile, MVDR develops nulls while DAS remains inactive. MVDR can therefore be viewed as an augmented DAS that differs from its cousin only in scenarios where it is necessary. This is precisely what makes it attractive as a systematic replacement for DAS: nothing is lost when there is nothing to gain, and a great deal is gained when something must be rejected.

10.6 Case with a Strong Interference

Suppose there is a strong interference at θ_i, so $\boldsymbol{R} = p_i\,\boldsymbol{a}_i\boldsymbol{a}_i^{\mathrm{H}} + \sigma^2\boldsymbol{I}_N$ with $\boldsymbol{a}_i = \boldsymbol{a}(\theta_i)$ and $p_i \gg \sigma^2$. By the Sherman–Morrison formula:

$$\boldsymbol{R}^{-1} = \frac{1}{\sigma^2}\,\boldsymbol{I}_N - \frac{p_i/\sigma^4}{1 + (p_i/\sigma^2)N}\,\boldsymbol{a}_i\boldsymbol{a}_i^{\mathrm{H}}.$$

For $p_i/\sigma^2 \gg 1/N$ (strong jammer), this formula approaches an *orthogonal projector onto the complement of the interference subspace*:

$$\boldsymbol{R}^{-1} \to \frac{1}{\sigma^2}\Big(\boldsymbol{I}_N - \frac{\boldsymbol{a}_i\boldsymbol{a}_i^{\mathrm{H}}}{N}\Big).$$

The MVDR beamformer then becomes approximately

$$\boldsymbol{w}_{\mathrm{MVDR}} \approx \frac{1}{N}\Big(\boldsymbol{a}(\theta_s) - \frac{\boldsymbol{a}_i^{\mathrm{H}}\boldsymbol{a}(\theta_s)}{N}\,\boldsymbol{a}_i\Big).$$

The structure is immediately visible: MVDR *subtracts* from $\boldsymbol{a}(\theta_s)$ its component along $\boldsymbol{a}_i$, exactly like Null Steering on θ_i. But here, θ_i was never given: it was *learned* through $\boldsymbol{R}$.

10.7 Interpretation: Implicit Null Steering

The previous lesson generalizes. MVDR places its zeros in the directions that contribute most to $\boldsymbol{R}$, that is, those of the strongest interferences. The null depth depends on relative power:

$$|B(\theta_i)|^2 \sim \frac{\sigma^2}{N\,p_i} \quad \text{(strong jammer)}.$$

The stronger the jammer, the deeper the null. Conversely, a weak interference (near the noise level) produces a shallow null: MVDR does not spend degrees of freedom for nothing.

10.8 Output SINR

A more complete metric than SNR is the *SINR* (Signal to Interference + Noise Ratio). For an interference-plus-noise covariance $\boldsymbol{R}_{i+n}$ and a target $\boldsymbol{a}_s = \boldsymbol{a}(\theta_s)$:

$$\mathrm{SINR}_{\mathrm{out}} = \frac{p_s\,|\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}_s|^2}{\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}_{i+n}\,\boldsymbol{w}}.$$

Theorem 10.3 (SINR Optimality of MVDR)**.** *The beamformer that maximizes* SINR_{out} *under the constraint* $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}_s = 1$ *is exactly*

$$\boldsymbol{w}^{\star} = \frac{\boldsymbol{R}_{i+n}^{-1}\boldsymbol{a}_s}{\boldsymbol{a}_s^{\mathrm{H}}\,\boldsymbol{R}_{i+n}^{-1}\,\boldsymbol{a}_s},$$

and the optimal SINR is

$$\mathrm{SINR}^{\star} = p_s\,\boldsymbol{a}_s^{\mathrm{H}}\,\boldsymbol{R}_{i+n}^{-1}\,\boldsymbol{a}_s.$$

The relevant object is therefore $\boldsymbol{R}_{i+n}$ (interference plus noise *without* the signal), not $\boldsymbol{R}$ (which also contains the signal). When $\boldsymbol{R}$ is used instead of $\boldsymbol{R}_{i+n}$, the method is in the *Sample Matrix Inversion (SMI)* regime, slightly suboptimal but much simpler to implement (no need to identify signal-absent intervals).

10.9 Practical Estimation: SMI

In practice, $\boldsymbol{R}$ is replaced by its sample version $\widehat{\boldsymbol{R}}$ obtained over K snapshots (Chapter 9):

$$\widehat{\boldsymbol{w}}_{\mathrm{MVDR}} = \frac{\widehat{\boldsymbol{R}}^{-1}\boldsymbol{a}(\theta_s)}{\boldsymbol{a}^{\mathrm{H}}(\theta_s)\,\widehat{\boldsymbol{R}}^{-1}\,\boldsymbol{a}(\theta_s)}.$$

The Reed–Mallett–Brennan rule (Theorem 9.4) says that with $K \simeq 2N$, the average SINR loss relative to the ideal case is on the order of 3 dB in the Gaussian SMI model.

10.10 Diagonal Loading: Essential Regularization

10.10.1 Why

Three situations require regularization:

1. $K < N$: $\widehat{\boldsymbol{R}}$ is singular and noninvertible;
2. $K \sim N$: $\widehat{\boldsymbol{R}}$ is ill-conditioned;
3. *steering vector mismatch*: the target is not exactly at θ_s, and MVDR partially attenuates the desired signal.

10.10.2 Effect on Eigenvalues

Replace $\widehat{\boldsymbol{R}}$ by

$$\widehat{\boldsymbol{R}}_\alpha = \widehat{\boldsymbol{R}} + \alpha \boldsymbol{I}_N,$$

which lifts all eigenvalues by α. The condition number is $\kappa = (\lambda_{\max} + \alpha)/(\lambda_{\min} + \alpha)$; inversion is stable even when $\lambda_{\min} \to 0$.

10.10.3 Limit $\alpha \to \infty$

When α is large compared with all eigenvalues of $\widehat{\boldsymbol{R}}$, $\widehat{\boldsymbol{R}}_\alpha \approx \alpha \boldsymbol{I}$, and MVDR becomes DAS again. Diagonal loading is therefore a *continuous transition* between MVDR (small α, selective but fragile) and DAS (large α, robust but less selective). The right α lies somewhere in between.

10.10.4 Choice of α

Several heuristics are used:

- $\alpha = c \cdot \mathrm{tr}(\widehat{\boldsymbol{R}})/N$ with $c \in [10^{-3}, 10^{-1}]$: robust and simple.
- $\alpha = c\,\sigma^2$ with σ^2 estimated from the smallest eigenvalues of $\widehat{\boldsymbol{R}}$.
- Worst-case methods [8]: uncertainty sphere around $\boldsymbol{a}(\theta_s)$, solved by SOCP.
- *Robust Capon Beamforming* [7] : optimisation explicit optimization over the uncertainty sphere, with an analytical link to an optimal α.

In all cases, the numerical calculation follows the same logic: solve $(\widehat{\boldsymbol{R}} + \alpha \boldsymbol{I})\boldsymbol{z} = \boldsymbol{a}_s$, then normalize $\boldsymbol{w} = \boldsymbol{z}/(\boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{z})$, without forming the matrix inverse. Chapter 23 develops these methods further.

10.11 Steering vector mismatch

If the target is actually at $\theta_s + \delta\theta$, the vector $\boldsymbol{a}(\theta_s)$ used by MVDR does not exactly match the desired signal signature. The signal then appears as a false interference in $\widehat{\boldsymbol{R}}$, and MVDR treats it as such: this is *signal self-cancellation*. The result can be catastrophic: in severe simulations where the target is present in the covariance, target loss can climb to several tens of dB for a mismatch of only a few fractions of the main-lobe width.

The remedies are:

- diagonal loading;
- using an *interference-plus-noise only* covariance $\boldsymbol{R}_{i+n}$;

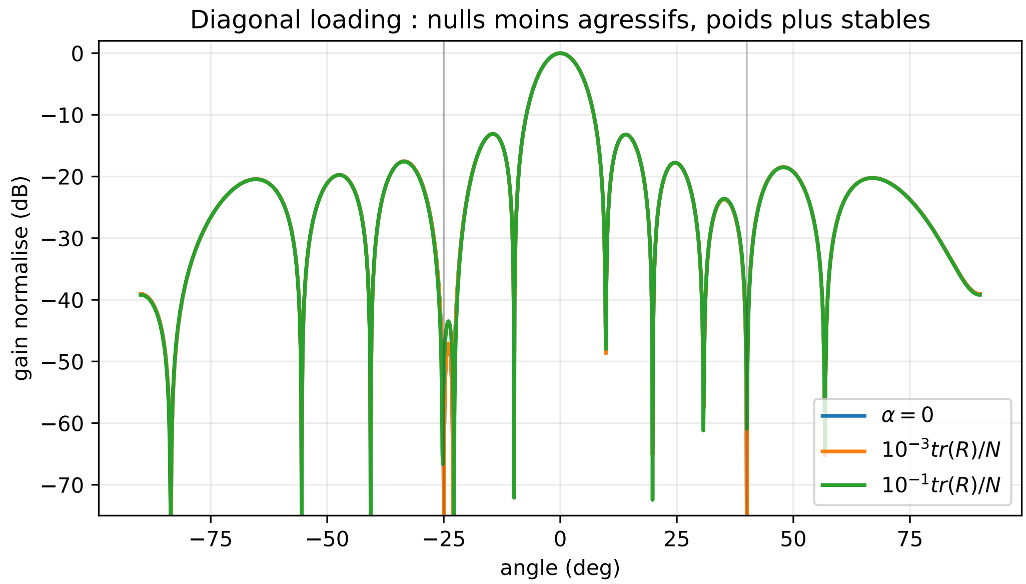

Figure 10.1: Effect of Diagonal Loading ($N = 12$, target 0°, jammers −25° and 40°, INR 40 dB): stronger loading makes the nulls less aggressive but stabilizes the weights and reduces sensitivity to model errors.

- derivative constraints (Chapter 23), which also impose $\partial_\theta B(\theta_s) = 0$ and widen the unit-gain region;
- *robust* approaches (*Robust Capon*).

10.12 Beam Pattern and Adaptive Nulls

On real data, the MVDR beam pattern typically has the following features:

- main lobe close to DAS on the target;
- *deep nulls* in the directions of strong interferences, often in the −30 to −60 dB range when the covariance and calibration are favorable;
- *sidelobes* slightly higher than DAS where there is no energy (the criterion is only quadratic in $\boldsymbol{w}$, not in sidelobe level);
- *data-dependent shape*: two different datasets produce two different beam patterns.

10.13 Comparison of Delay-and-Sum, Null Steering, and MVDR

Criterion	**DAS**	**Null Steering**	**MVDR**
Required information	θ_s	θ_s, θ_i	$\theta_s, \boldsymbol{R}$
Adaptation	no	no	yes (via $\boldsymbol{R}$)
Nulls on interferences	no	explicit	implicit
SNR Gain (white noise)	N (optimal)	$\leq N$	N (equiv. DAS)
Optimal SINR	no	no	yes (asymp.)
Robustness	high	medium	low without reg.
Complexity	$\mathcal{O}(N)$	$\mathcal{O}(NL^2)$	$\mathcal{O}(N^3)$

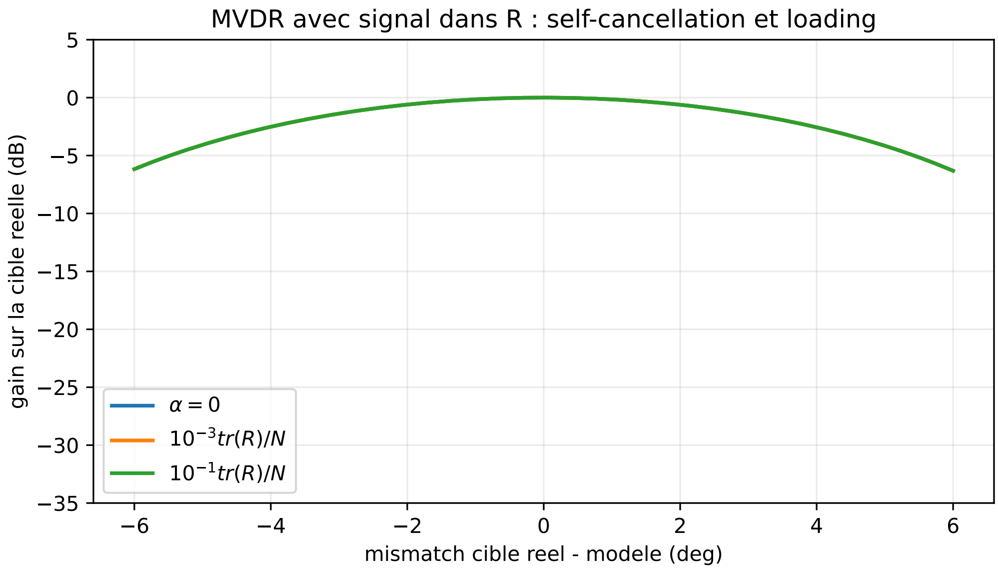

Figure 10.2: Severe simulation ($N = 12$, target 0°, jammers −25° and 40°, desired signal present in the covariance): an unloaded MVDR can attenuate the target as soon as the steering vector is slightly wrong.

10.14 Numerical Implementation

Listing 10.1: MVDR with diagonal loading

```
import numpy as np

def mvdr_weights(R, a_s, alpha=0.0):
    """
    R   : (N,N) covariance matrix, Hermitian PSD
    a_s : target steering vector (N,)
    alpha : diagonal loading (>=0)
    """
    N = R.shape[0]
    assert R.shape == (N, N)
    assert a_s.shape == (N,)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5*(R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha ill-conditioned")
    u = np.linalg.solve(R_alpha, a_s)
    den = a_s.conj() @ u
    if abs(den) < np.finfo(float).eps:
        raise np.linalg.LinAlgError("unstable MVDR normalization")
    return u / den

def capon_spectrum(R, thetas, N, d_over_lambda=0.5, alpha=0.0):
    assert R.shape == (N, N)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5*(R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha ill-conditioned")
    m = np.arange(N)
```

```
    spec = np.zeros(len(thetas))
    eps = np.finfo(float).eps
    for idx, th in enumerate(thetas):
        a = np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(th) * m)
        u = np.linalg.solve(R_alpha, a)
        den = max(np.real(a.conj() @ u), eps)
        spec[idx] = 1.0 / den
    return spec
```

10.15 Summary

MVDR turns beamforming into a *data-informed* optimization problem:

- it generalizes DAS: in white noise, they coincide;
- it generalizes Null Steering: its nulls are automatically placed where energy is concentrated;
- it maximizes SINR under a distortionless constraint;
- it requires a well-estimated and regularized covariance.

Key Points

- **Central formula**: $\boldsymbol{w} = \boldsymbol{R}^{-1}\boldsymbol{a}/(\boldsymbol{a}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{a})$.
- **Main assumption**: reliable covariance and correct target steering vector.
- **Advantage**: adaptive nulls without knowing the jammer directions.
- **Limitation**: sensitivity to mismatch and to an ill-conditioned covariance.
- **Practical pitfall**: avoid computing $\boldsymbol{R}^{-1}$ explicitly; solve a linear system.
- **Next**: LCMV generalizes MVDR to multiple constraints.

The next chapter generalizes it to LCMV, which combines several linear constraints (unit gain + zeros + derivatives) in a single optimization framework. By the end of Chapter 11, we will have a unified framework covering DAS, Null Steering, and MVDR, and admitting a two-branch decomposition (GSC) that is particularly convenient for the adaptive implementations studied in Part IV.

Chapter 11

LCMV and GSC Structure

Null Steering imposes linear constraints without a statistical model. MVDR imposes a single constraint (unit gain) but exploits the covariance. The *Linearly Constrained Minimum Variance* (LCMV) unifies the two: it minimizes output power under *several* linear constraints. It includes MVDR (one constraint) and Null Steering (scalar covariance $\sigma^2\boldsymbol{I}$) as special cases. The *Generalized Sidelobe Canceller* (GSC) is its two-branch reformulation, which opens the way to the adaptive implementation of Part IV.

LCMV may seem redundant after MVDR: why formalize a more general framework when MVDR already suffices to minimize variance? Three reasons motivate this chapter. First, LCMV combines the statistical power of MVDR with the deterministic precision of Null Steering: it imposes both "I want this signal" and "I do not want these specific interferences." Second, it can impose subtler constraints such as a *zero derivative* of the beam pattern (for robustness to angular errors on the target), which will be central in Chapter 23. Finally, its GSC reformulation provides a fundamental decomposition that separates the constraint part (computed once) from the adaptation part (updated continuously), making efficient implementation possible on compute-constrained platforms.

11.1 LCMV Formulation

Definition 11.1 (LCMV Problem)**.** Let $\boldsymbol{R} \in \mathbb{C}^{N\times N}$ be Hermitian positive definite, $\boldsymbol{C} \in \mathbb{C}^{N\times L_c}$ a full-column-rank constraint matrix ($L_c \leq N$), and $\boldsymbol{f} \in \mathbb{C}^{L_c}$ a vector of prescribed responses. The LCMV beamformer solves

$$\boxed{\min_{\boldsymbol{w}} \ \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w} \quad \text{subject to} \quad \boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = \boldsymbol{f}.}$$

11.2 Analytical Solution

Theorem 11.2 (LCMV Solution)**.** *If $\boldsymbol{R}$ is positive definite and $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C}$ is invertible, the unique solution is*

$$\boxed{\boldsymbol{w}_{\mathrm{LCMV}} = \boldsymbol{R}^{-1}\boldsymbol{C}\left(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C}\right)^{-1}\boldsymbol{f}.}$$

Proof. Complex Lagrangian: $\mathcal{L} = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w} + (\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} - \boldsymbol{f})^{\mathrm{H}}\boldsymbol{\mu} + \boldsymbol{\mu}^{\mathrm{H}}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} - \boldsymbol{f})$. Stationarity: $\boldsymbol{R}\boldsymbol{w} + \boldsymbol{C}\boldsymbol{\mu} = \boldsymbol{0}$, so $\boldsymbol{w} = -\boldsymbol{R}^{-1}\boldsymbol{C}\boldsymbol{\mu}$. Constraint: $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = -\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C}\boldsymbol{\mu} = \boldsymbol{f}$, hence $\boldsymbol{\mu} = -(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C})^{-1}\boldsymbol{f}$. The formula follows. □

11.3 MVDR as a Special Case

With $\boldsymbol{C} = \boldsymbol{a}(\theta_s) \in \mathbb{C}^{N\times 1}$ and $\boldsymbol{f} = 1$, we recover

$$\boldsymbol{w}_{\mathrm{LCMV}} = \boldsymbol{R}^{-1}\boldsymbol{a}(\theta_s)\left(\boldsymbol{a}^{\mathrm{H}}(\theta_s)\boldsymbol{R}^{-1}\boldsymbol{a}(\theta_s)\right)^{-1} = \boldsymbol{w}_{\mathrm{MVDR}}.$$

11.4 Null Steering as a Special Case

With $\boldsymbol{R} = \sigma^2\boldsymbol{I}_N$ (no exploited statistical interference),

$$\boldsymbol{w}_{\mathrm{LCMV}} = \boldsymbol{C}\,(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C})^{-1}\,\boldsymbol{f} = \boldsymbol{w}_{\mathrm{NS}}.$$

LCMV is therefore strictly more general: it is statistically informed Null Steering.

11.5 Typical Constraints

11.5.1 Complete Example: One Target and Two Jammers

Suppose there is a target at θ_s and two known jammers at $\theta_{i,1}$ and $\theta_{i,2}$. Construct:

$$\boldsymbol{C} = \begin{bmatrix}\boldsymbol{a}(\theta_s) & \boldsymbol{a}(\theta_{i,1}) & \boldsymbol{a}(\theta_{i,2})\end{bmatrix}$$

and:

$$\boldsymbol{f} = \begin{bmatrix}1\\0\\0\end{bmatrix}.$$

The LCMV constraint:

$$\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = \boldsymbol{f}$$

means explicitly:

$$\begin{cases}\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_s) = 1,\\ \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_{i,1}) = 0,\\ \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_{i,2}) = 0.\end{cases}$$

The beamformer therefore preserves the target, cancels two prescribed directions, and uses the remaining degrees of freedom to minimize output power according to $\boldsymbol{R}$. The solution is:

$$\boldsymbol{w}_{\mathrm{LCMV}} = \boldsymbol{R}^{-1}\boldsymbol{C}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C})^{-1}\boldsymbol{f}.$$

This expression shows exactly how Null Steering and MVDR combine: the constraints fix what must be guaranteed, and the covariance optimizes everything else.

1. Unit gain + explicit nulls. $\boldsymbol{C} = [\boldsymbol{a}(\theta_s),\, \boldsymbol{a}(\theta_{i,1}),\, \ldots,\, \boldsymbol{a}(\theta_{i,L})]$, $\boldsymbol{f} = [1, 0, \ldots, 0]^{\mathrm{T}}$. The beamformer maintains gain on the target, places nulls on the L *known* interferences, and uses its residual statistical power to reject *unknown* interferences.

2. Multi-beam. To track several targets simultaneously with specified gains: $\boldsymbol{C} = [\boldsymbol{a}(\theta_1), \ldots, \boldsymbol{a}(\theta_M)]$, $\boldsymbol{f} = [1, 1, \ldots, 1]^{\mathrm{T}}$. This is the equivalent of a statistically optimized multi-beam.

3. Derivative Constraints (robustness broadening). In addition to $\boldsymbol{a}(\theta_s)$, impose $\partial_\theta\boldsymbol{a}(\theta_s)$ as a second column with prescribed response 0. This forces the derivative of the beam pattern to vanish at θ_s, widening the flat region of the main lobe and improving robustness to angular errors.

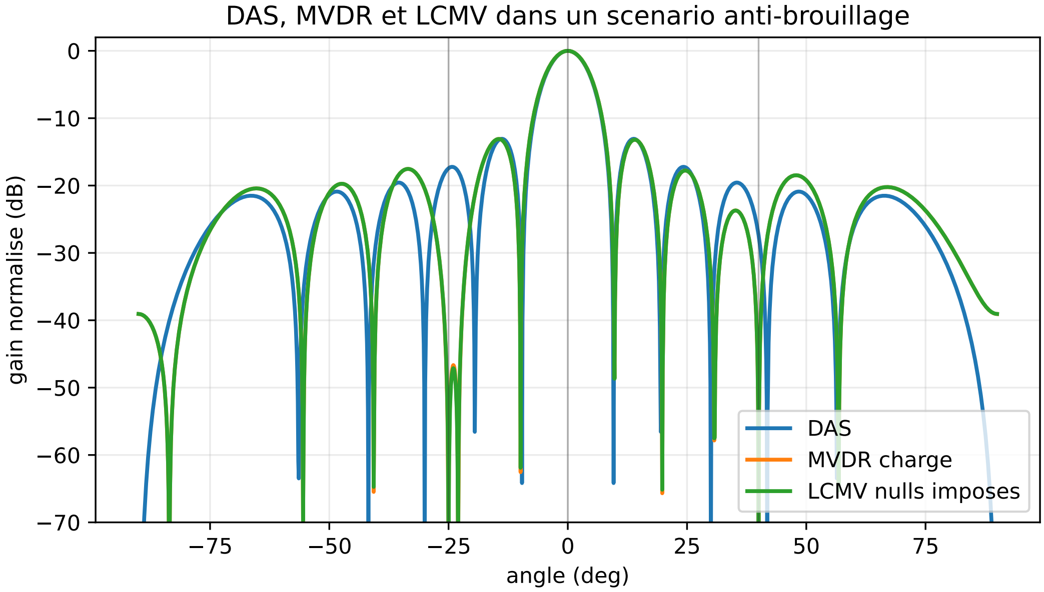

Figure 11.1: Qualitative comparison between DAS, loaded MVDR, and LCMV ($N = 12$, target $0°$, jammers $-25°$ and $40°$, INR 35 dB). DAS cannot place nulls, MVDR adapts them to the covariance, and LCMV explicitly imposes them in the chosen directions.

4. Broadened Nulls. To reject an interference with angular uncertainty $\pm\delta$, replace the single constraint $\boldsymbol{a}(\theta_i)$ with several constraints $\boldsymbol{a}(\theta_i - \delta)$, $\boldsymbol{a}(\theta_i)$, $\boldsymbol{a}(\theta_i + \delta)$. The *null* becomes an attenuation region rather than a point.

Another way to broaden the null is to impose a zero derivative:

$$B(\theta_i) = 0, \qquad B'(\theta_i) = 0.$$

Since

$$B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta),$$

this amounts to adding:

$$\boldsymbol{w}^{\mathrm{H}}\dot{\boldsymbol{a}}(\theta_i) = 0.$$

The result is a less point-like null: it remains deep when the interference moves slightly around θ_i. In practice, this constraint is often more realistic than an infinitely narrow null on a DOA estimated with uncertainty.

11.6 Degrees of Freedom

As with Null Steering, the cost in degrees of freedom is explicit: L_c constraints consume L_c degrees of freedom out of the N available. There remain $N - L_c$ degrees to statistically reject *unknown* interferences through the minimization $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w}$.

Practice. Avoid saturation ($L_c \ll N$). With $L_c = N$, the beamformer is fully constrained and there is no statistical adaptation left: LCMV becomes a simple linear system $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = \boldsymbol{f}$.

11.6.1 Maximum Number of Nulls

With N antennas, the vector $\boldsymbol{w} \in \mathbb{C}^N$ has N complex degrees of freedom. A desired-gain constraint consumes one degree:

$$\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_s) = 1.$$

Each explicit null consumes one additional degree:

$$\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_{i,\ell}) = 0.$$

The theoretical maximum number of point nulls is therefore:

$$L_{\max} = N - 1.$$

But this limit is rarely usable in practice. If $N-1$ nulls are imposed, there is no degree of freedom left to:

- reduce noise;
- compensate a DOA error;
- limit the norm of $\boldsymbol{w}$;
- adapt to an unknown interference in $\boldsymbol{R}$.

A prudent engineering rule is to keep several degrees of freedom free:

$$L_{\text{nulls}} \leq N - 3 \quad \text{ou} \quad N - 4$$

if the environment is uncertain. The last degrees of freedom are often more valuable than the first: they absorb calibration errors, imperfect DOAs, and covariance noise.

11.7 GSC Reformulation (Generalized Sidelobe Canceller)

LCMV can be reformulated as a two-branch decomposition that simplifies adaptive implementation and clarifies the structure of the problem. The idea is to separate the beamformer into a *fixed* part that satisfies the constraints and an *adaptive* part that cannot violate them.

11.7.1 Construction

Let $\boldsymbol{w}_q = \boldsymbol{C}\,(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C})^{-1}\,\boldsymbol{f}$ be the *fixed part* (the minimum-norm Null Steering solution, satisfying $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w}_q = \boldsymbol{f}$). Let $\boldsymbol{B} \in \mathbb{C}^{N\times(N-L_c)}$ be a *blocking matrix* whose columns span $\mathrm{Ker}(\boldsymbol{C}^{\mathrm{H}})$, that is, $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{B} = \boldsymbol{0}$. Any vector of the form

$$\boldsymbol{w} = \boldsymbol{w}_q - \boldsymbol{B}\,\boldsymbol{w}_a, \quad \boldsymbol{w}_a \in \mathbb{C}^{N-L_c},$$

automatically satisfies $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{w} = \boldsymbol{f}$. The constraint is absorbed into the structure.

11.7.2 Equivalent Problem in $\boldsymbol{w}_a$

Substituting into the criterion:

$$\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\,\boldsymbol{w} = (\boldsymbol{w}_q - \boldsymbol{B}\boldsymbol{w}_a)^{\mathrm{H}}\boldsymbol{R}\,(\boldsymbol{w}_q - \boldsymbol{B}\boldsymbol{w}_a).$$

Minimizing over $\boldsymbol{w}_a$ gives

$$\boldsymbol{w}_a^{\star} = (\boldsymbol{B}^{\mathrm{H}}\boldsymbol{R}\,\boldsymbol{B})^{-1}\,\boldsymbol{B}^{\mathrm{H}}\boldsymbol{R}\,\boldsymbol{w}_q.$$

11.7.3 Interpretation: Two Branches

- **Upper branch**: $y_q(n) = \boldsymbol{w}_q^{\mathrm{H}} \boldsymbol{x}(n)$, output of a *fixed* beamformer satisfying the constraints (for example Null Steering or constrained DAS).
- **Lower branch**: $\boldsymbol{z}(n) = \boldsymbol{B}^{\mathrm{H}} \boldsymbol{x}(n) \in \mathbb{C}^{N-L_c}$, a noise reference containing the environment projected outside the constraint subspace.
- **Adaptive subtraction**: $y(n) = y_q(n) - \boldsymbol{w}_a^{\mathrm{H}} \boldsymbol{z}(n)$, where $\boldsymbol{w}_a$ is adjusted to minimize $\mathbb{E}[|y(n)|^2]$.

The problem is then an *unconstrained Wiener filtering* problem: $\boldsymbol{w}_a$ is obtained as the Wiener solution of the problem "minimize $|y_q - \boldsymbol{w}_a^{\mathrm{H}} \boldsymbol{z}|^2$." This is the form that will be attacked by LMS and RLS in Part IV.

Advantages.

- adaptation acts on an *unconstrained* subproblem;
- the two-branch structure clearly isolates the constraints part from the adaptation part;
- the dimension of the adaptive problem is $N - L_c$, often much smaller than N when there are few constraints.

Choice of $\boldsymbol{B}$. A good blocking matrix has orthonormal, well-conditioned columns. Common methods:

- orthogonal complement of $\boldsymbol{C}$ via QR;
- singular value decomposition of $\boldsymbol{I} - \boldsymbol{C}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C})^{-1}\boldsymbol{C}^{\mathrm{H}}$;
- explicit construction (Householder matrix) for Toeplitz $\boldsymbol{C}$.

11.8 Conditioning of $\boldsymbol{C}$

When the angles imposed by the constraints are close, $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{C}$ becomes ill-conditioned, and the norm of the LCMV beamformer explodes. Three classical remedies are:

- remove nearly collinear constraints;
- regularize: $\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C} \to (\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C} + \beta \boldsymbol{I})$;
- move to a *soft constraints* formulation, where the constraint becomes a quadratic penalty.

11.9 Diagonal Loading on LCMV

As with MVDR, replace $\boldsymbol{R}$ by $\boldsymbol{R} + \alpha \boldsymbol{I}_N$. The LCMV structure is preserved, conditioning is improved, and the norm $\|\boldsymbol{w}_{\mathrm{LCMV}}\|^2$ is bounded.

11.10 Numerical Implementation

Listing 11.1: Generic LCMV

```python
import numpy as np

def lcmv_weights(R, C, f, alpha=0.0):
    """
    R     : (N,N) Hermitian PSD covariance
    C     : (N, Lc) constraint matrix (full column rank)
    f     : (Lc,) prescribed responses
    alpha : diagonal loading
    """
    N = R.shape[0]
    assert R.shape == (N, N)
    assert C.shape[0] == N
    assert f.shape == (C.shape[1],)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5*(R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha ill-conditioned")
    Rinv_C = np.linalg.solve(R_alpha, C)           # (N, Lc)
    M = C.conj().T @ Rinv_C                        # (Lc, Lc)
    if np.linalg.cond(M) > 1e12:
        raise np.linalg.LinAlgError("ill-conditioned LCMV constraints")
    return Rinv_C @ np.linalg.solve(M, f)
```

11.11 When to Use LCMV Rather Than MVDR

Suitable.

- Several targets to maintain simultaneously;
- known jammers whose rejection must be imposed *in addition* to MVDR adaptation;
- robustness strengthened by derivative constraints;
- GSC architectures with a separate adaptive part.

Unsuitable.

- when N is small and all degrees of freedom are already needed for statistical interferences;
- when calibration does not allow precise angular constraints to be imposed.

11.12 Synoptic Comparison

	DAS	NS	MVDR	LCMV
Required information	θ_s	θ_s, θ_i	$\theta_s, \boldsymbol{R}$	$\theta_s, \theta_i, \boldsymbol{R}$
Statistics exploited	no	no	yes	yes
Constraints	1 gain	gain + nulls	1 gain	multiple
Nulls	no	imposed	adaptive	imposed and adaptive
Cost in free degrees	low	explicit	implicit	explicit
Robustness	high	medium	low without loading	adjustable

11.13 Summary

LCMV unifies in a single formalism all beamformers studied in Parts II and III. Its two-branch GSC structure clearly separates the constraints (solved once and for all) from the adaptation (which can be performed recursively). In practice, the latter is almost never carried out by direct inversion: it is entrusted to *iterative* algorithms, LMS, NLMS, and RLS, which are the subject of Part IV.

Key Points

- **Central formula**: $\boldsymbol{w} = \boldsymbol{R}^{-1}\boldsymbol{C}(\boldsymbol{C}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{C})^{-1}\boldsymbol{f}$.
- **Main assumption**: independent constraints and sufficiently well-conditioned covariance.
- **Advantage**: combines desired gain, explicit nulls, angular robustness, and statistical minimization.
- **Limitation**: each constraint consumes one degree of freedom.
- **Practical pitfall**: too many nulls produce large-norm weights and low WNG.
- **Next**: the GSC structure makes LMS/RLS adaptation possible without violating the constraints.

At the end of this Part III, the reader has a complete theoretical chain:

$$\text{Data} \to \widehat{\boldsymbol{R}} \to \boldsymbol{R}_{\mathrm{DL}}^{-1} \to \boldsymbol{w}_{\mathrm{LCMV}} \to y(n) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(n).$$

This chain nevertheless assumes that, at each weight update, one can recompute a covariance and invert it. This cost is acceptable when the environment varies slowly (updates every second or minute), but becomes prohibitive when mobile jammers must be tracked at the millisecond scale or dynamic targets pursued. Part IV replaces this block update with a *recursive* update, where the weights evolve at each new sample.

The next chapter lays the foundations of adaptation by iterative minimization: Wiener filter, gradient descent, and the LMS algorithm.

Part IV

Iterative Adaptation

Chapter 12

Wiener, LMS and NLMS

Previous chapters provided *closed formulas* for beamformer weights: DAS, Null Steering, MVDR, LCMV. These formulas assume that we have a covariance $\boldsymbol{R}$ and that we can invert it. This is a comfortable hypothesis on a theoretical level, but which comes up against two operational realities: the covariance is never perfectly known (it is estimated over a finite number of snapshots), and inverting it at each update costs $\mathcal{O}(N^3)$, which becomes prohibitive on large arrays or in fast real time. Part IV responds to these two problems with approaches *iterative* which update the weights at each sample without ever explicitly forming or inverting the covariance.

In practice, two situations require this different approach.

1. The environment is *non-stationary*: the directions and Source powers vary over time. Recalculating $\widehat{\boldsymbol{R}}$ and inverting every time is too expensive and risky (the estimation window becomes too short).
2. We have a *reference signal* $d(n)$ (a pilot, a preamble, a known signal): we can then pose the problem as a direct error minimization, without explicitly estimating the covariance.

This chapter constructs the *Wiener filter*, the theoretical starting point, then introduces the *Least Mean Squares* (LMS) and its normalized NLMS variant.

12.1 Beamformer with reference: structure

We assume a reference signal $d(n)$ available, correlated to the useful signal. The goal is to choose $\boldsymbol{w}$ so that the output $y(n) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(n)$ approaches $d(n)$. The error is

$$e(n) = d(n) - y(n) = d(n) - \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(n).$$

The MMSE criterion (*Minimum Mean Square Error*) is

$$J(\boldsymbol{w}) = \mathbb{E}\Big[|e(n)|^2\Big] = \mathbb{E}[|d|^2] - \boldsymbol{w}^{\mathrm{H}}\boldsymbol{r}_{xd} - \boldsymbol{r}_{xd}^{\mathrm{H}}\boldsymbol{w} + \boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\,\boldsymbol{w},$$

with $\boldsymbol{R} = \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathrm{H}}]$ and $\boldsymbol{r}_{xd} = \mathbb{E}[\boldsymbol{x}(n)\,d^*(n)]$.

12.2 Wiener's solution

Theorem 12.1 (Wiener filter)**.** *If $\boldsymbol{R}$ is positive definite, the minimum of J is reached in*

$$\boxed{\boldsymbol{w}_\star = \boldsymbol{R}^{-1}\,\boldsymbol{r}_{xd}.}$$

Proof. $\nabla_{\boldsymbol{w}^*} J = -\boldsymbol{r}_{xd} + \boldsymbol{R}\,\boldsymbol{w} = \boldsymbol{0}$ gives $\boldsymbol{w} = \boldsymbol{R}^{-1}\boldsymbol{r}_{xd}$. The Hessian is $\boldsymbol{R}$, positive semidefinite: it is a minimum. □

The minimum error is

$$J_{\min} = \mathbb{E}[|d|^2] - \boldsymbol{r}_{xd}^{\mathrm{H}}\,\boldsymbol{R}^{-1}\,\boldsymbol{r}_{xd}.$$

Link with MVDR. The Wiener filter and MVDR have the same $\boldsymbol{R}^{-1}(\cdot)$ structure: MVDR is a special case where the "reference " is the steering vector $\boldsymbol{a}(\theta_s)$, multiplied by a normalization factor. The family of optimal beamformers is therefore unified by this diagram. This unification is not just a nice formal convergence: it says that everything we know about the Wiener filter (random signal theory, principle of orthogonality, linear prediction) applies directly to beamforming. In particular, the convergence acceleration and *tracking* techniques developed in classic adaptive filtering can be transposed as they are to adaptive beamforming. The literature of the two fields forms a continuum much more than one perceives when learning them separately.

12.3 Deterministic gradient descent

If we do not invert $\boldsymbol{R}$, we iteratively minimize $J(\boldsymbol{w})$ by gradient descent:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) - \mu\,\nabla_{\boldsymbol{w}^*} J(\boldsymbol{w}(n)) = \boldsymbol{w}(n) + \mu\,(\boldsymbol{r}_{xd} - \boldsymbol{R}\,\boldsymbol{w}(n)).$$

This recurrence converges to $\boldsymbol{w}_\star = \boldsymbol{R}^{-1}\boldsymbol{r}_{xd}$ if $0 < \mu < 2/\lambda_{\max}(\boldsymbol{R})$.

12.4 Instant Gradient and LMS

Expectations $\boldsymbol{R}$ and $\boldsymbol{r}_{xd}$ are unknown. The idea of the LMS is to replace them with their *instantaneous realizations*: $\boldsymbol{R} \approx \boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n)$ and $\boldsymbol{r}_{xd} \approx \boldsymbol{x}(n)\,d^*(n)$. The instantaneous gradient becomes

$$\widehat{\nabla} J(n) = -\boldsymbol{x}(n)\,d^*(n) + \boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n)\,\boldsymbol{w}(n) = -\boldsymbol{x}(n)\,e^*(n),$$

using $e(n) = d(n) - \boldsymbol{w}^{\mathrm{H}}(n)\boldsymbol{x}(n)$.

Definition 12.2 (LMS algorithm)**.**

$$\boxed{\begin{aligned} y(n) &= \boldsymbol{w}^{\mathrm{H}}(n)\,\boldsymbol{x}(n), \\ e(n) &= d(n) - y(n), \\ \boldsymbol{w}(n+1) &= \boldsymbol{w}(n) + \mu\,\boldsymbol{x}(n)\,e^*(n). \end{aligned}}$$

Three lines per snapshot: This is one of the simplest algorithms in adaptive signal processing. Its complexity is $\mathcal{O}(N)$ per sample, without matrix inversion, without covariance memory.

12.5 Complete example: pilot-supervised beamforming

Consider a link where the useful source transmits a known pilot sequence $d(n)$. The array receives:

$$\boldsymbol{x}(n) = \boldsymbol{a}(\theta_s)s(n) + \boldsymbol{a}(\theta_i)i(n) + \boldsymbol{n}(n),$$

with $d(n) = s(n)$ during the learning phase. The beamformer produces:

$$y(n) = \boldsymbol{w}^{\mathrm{H}}(n)\boldsymbol{x}(n),$$

then compares this output to the pilot:

$$e(n) = d(n) - y(n).$$

The update:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu\boldsymbol{x}(n)e^*(n)$$

has a direct physical reading:

- if $|e(n)|$ is large, the correction is large;
- if a component of $\boldsymbol{x}(n)$ is strong, it influences more strongly the correction;
- if μ is too large, the weights overreact and diverge;
- if μ is too small, convergence is stable but slow.

LMS does not explicitly know θ_s or θ_i. It learns a spatial combination whose output stays close to the pilot. If the jammer is decorrelated from the pilot, it contributes to the error and the algorithm learns to reduce it. If the reference is contaminated by the jammer, LMS instead learns a biased solution: the quality of $d(n)$ is therefore as important as the choice of μ.

12.6 Convergence and stability

12.6.1 Expectation of the trajectory

In expectation, $\mathbb{E}[\boldsymbol{w}(n+1)] = (\boldsymbol{I} - \mu\boldsymbol{R})\mathbb{E}[\boldsymbol{w}(n)] + \mu\boldsymbol{r}_{xd}$. The recurrence is stable if all the eigenvalues of $\boldsymbol{I} - \mu\boldsymbol{R}$ have modulus < 1, i.e.

$$\boxed{0 < \mu < \frac{2}{\lambda_{\max}(\boldsymbol{R})}.}$$

In practice, we limit by the trace: $\lambda_{\max} \leq \operatorname{tr}(\boldsymbol{R})$, hence the practical condition $\mu < 2/\operatorname{tr}(\boldsymbol{R})$ or more cautiously $\mu < 1/\operatorname{tr}(\boldsymbol{R})$.

12.6.2 Time constant

The k-th eigencomponent converges at a rate $|1 - \mu\lambda_k|^n$; the overall convergence is dominated by the smallest eigenvalue. If $\lambda_{\min}/\lambda_{\max}$ is small (poorly conditioned covariance), convergence is slow. This is the major limitation of the LMS.

12.6.3 Misadjustment

In equilibrium, $\boldsymbol{w}(n)$ fluctuates around $\boldsymbol{w}_\star$ with non-zero variance. The performance loss is measured by the *misadjustment*

$$\mathcal{M} = \frac{\mathbb{E}[J(\boldsymbol{w}(\infty))] - J_{\min}}{J_{\min}} \approx \frac{\mu}{2}\operatorname{tr}(\boldsymbol{R}).$$

We therefore have a compromise: μ large $\Rightarrow$ fast convergence but $\mathcal{M}$ high; μ small $\Rightarrow$ asymptotic precision but slow convergence.

12.7 NLMS: not adaptive

To make μ independent of the amplitude of the data, we normalize by the instantaneous power of the snapshot:

$$\boxed{\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \frac{\tilde{\mu}}{\epsilon + \|\boldsymbol{x}(n)\|^2}\,\boldsymbol{x}(n)\,e^*(n),}$$

with $0 < \tilde{\mu} < 2$ (practice: $\tilde{\mu} \approx 0.5$) and $\epsilon > 0$ small to avoid division by zero. NLMS is robust to signal scale variations, without significant overhead.

12.7.1 Why normalization stabilizes

In LMS, the actual size of the correction is:

$$\|\mu \boldsymbol{x}(n) e^*(n)\|.$$

If the input power increases suddenly, the correction also increases, even if μ has not changed. This is dangerous in RF systems, where a jammer can vary $\|\boldsymbol{x}(n)\|^2$ by several tens of dB.

NLMS replaces the fixed step with an effective step:

$$\mu_{\text{eff}}(n) = \frac{\tilde{\mu}}{\epsilon + \|\boldsymbol{x}(n)\|^2}.$$

So :

$$\|\boldsymbol{x}(n)\|^2 \text{ grand} \Rightarrow \mu_{\text{eff}}(n) \text{ petit}.$$

The correction remains of controlled size. This is why NLMS is often preferred over pure LMS when the input power varies greatly.

12.8 Application to beamforming

12.8.1 Supervised mode

If you have a known *pilot* or *preamble* $d(n)$, LMS can directly drive the beamformer to point at the wanted signal and reject the rest. Classic telecommunications architecture.

12.8.2 Constant-modulus mode

Without a pilot, we can impose $|y(n)| = 1$ (*Constant Modulus Algorithm*, CMA) for constant envelope modulations (PSK). Criterion: $J_{\text{CMA}}(\boldsymbol{w}) = \mathbb{E}\big[(|y(n)|^2 - 1)^2\big]$.

12.8.3 Adaptive LCMV mode via GSC

The GSC architecture (chapter 11) separates $\boldsymbol{w}_q$ (fixed) and $\boldsymbol{w}_a$ (adaptive). The error signal becomes $e(n) = y_q(n) - \boldsymbol{w}_a^{\mathrm{H}}(n)\boldsymbol{z}(n)$, where $\boldsymbol{z}(n) = \boldsymbol{B}^{\mathrm{H}}\boldsymbol{x}(n)$ is the output of the blocking matrix. LMS is applied to $\boldsymbol{w}_a$ without constraints, and the whole achieves an adaptive LCMV.

12.9 LMS Limitations

Slow convergence on poorly conditioned covariance. A jammer well above the noise can create a clean ratio $\lambda_{\max}/\lambda_{\min}$ of several orders of magnitude: LMS can then request a number of iterations of the same order as this conditioning. RLS (next chapter) largely corrects this defect.

Choice of μ. Sensitive to signal regime. NLMS attenuates the problem but does not eliminate it.

Sensitivity to correlated signals. When $d(n)$ is strongly correlated with components of $\boldsymbol{x}(n)$, the instantaneous gradient is very noisy.

No structural model. Unlike MVDR/LCMV, LMS does not incorporate angular constraints. The *Constrained LMS* or GSC structure is required for explicit constraints.

12.10 Related algorithms

Affine Projection Algorithm (APA). Compromise between LMS and RLS: we use the latest L snapshots to calculate a less noisy gradient than LMS, at an intermediate cost between $\mathcal{O}(N)$ and $\mathcal{O}(N^2)$.

LeakyLMS. Adds a regularization term $-\mu\gamma\boldsymbol{w}(n)$ at update, which prevents $\boldsymbol{w}$ from drifting into the core of $\boldsymbol{R}$ (useful in *persistent excitement* weak).

Sign-LMS / Sign-Error LMS. Replaces $e(n)$ with $\operatorname{sgn}(e(n))$ to save multiplications in FPGA.

Block LMS. Updating from a block L samples — the output is calculated by FFT, which becomes efficient for large time windows.

12.11 Digital implementation

Listing 12.1: LMS and NLMS for supervised beamforming

```python
import numpy as np

def lms(X, d, mu, w0=None):
    """
    X  : (N, K) snapshots
    d  : (K,) reference (complex)
    mu : pas
    """
    N, K = X.shape
    w = np.zeros(N, dtype=complex) if w0 is None else w0.astype(complex).copy()
    err = np.zeros(K, dtype=complex)
    for n in range(K):
        x = X[:, n]
        y = w.conj() @ x
        e = d[n] - y
        w += mu * x * np.conj(e)
        err[n] = e
    return w, err

def nlms(X, d, mu_tilde=0.5, eps=1e-6, w0=None):
    N, K = X.shape
    w = np.zeros(N, dtype=complex) if w0 is None else w0.astype(complex).copy()
    err = np.zeros(K, dtype=complex)
    for n in range(K):
        x = X[:, n]
        y = w.conj() @ x
        e = d[n] - y
        norm = eps + (x.conj() @ x).real
        w += (mu_tilde / norm) * x * np.conj(e)
        err[n] = e
    return w, err
```

12.12 Synthesis

LMS and NLMS are the simplest and most used adaptive beamforming algorithms:

- complexity $\mathcal{O}(N)$ per snapshot;
- no covariance to estimate or invert;
- convergence guaranteed (on average) if $\mu < 2/\lambda_{\max}$;
- μ compromise / convergence speed / misadjustment.

Their main limitation is slow convergence in environments with large power disparities. The next chapter introduces RLS, which often converges into a number of snapshots of the order of a few N when the numerical assumptions are favorable, at the cost of $\mathcal{O}(N^2)$ complexity.

The LMS / RLS / SMI compromise sums up the triad of all adaptive processing well: one can be *simple, fast, precise*, but generally only two of the three. LMS is simple and fast (per sample) but not very precise (slow convergence). SMI (Sample Matrix Inversion, seen in Part III) is precise and fast in convergence when K is of the order of a few N, but expensive per update. RLS is accurate and fast in convergence, at the cost of intermediate complexity. The choice between the three depends on the hardware and dynamic constraints of the intended application.

Chapter 13

RLS and Forgetting Factor

LMS converges but slowly, and its slowness is dictated by covariance conditioning. When the environment contains a jammer 40 or 60 dB above the noise, $\kappa(\boldsymbol{R})$ can become very large and make LMS too slow for the application: one may wait thousands or even tens of thousands of iterations to converge, which sometimes exceeds the time the environment remains stationary. *Recursive Least Squares* (RLS) greatly reduces this problem: under favorable conditions, it often converges to a number of snapshots on the order of a few N, with a much lower dependence on conditioning than LMS. The price to pay is complexity $\mathcal{O}(N^2)$ per snapshot and more memory.

RLS is, in many ways, a *LMS with structured memory.* Instead of averaging the gradients via a very small step μ so as not to overreact to noise, it explicitly keeps track of an inverse covariance weighted by a forgetting factor λ. The descent is no longer blind: it is conditioned by the *shape* of the covariance, which makes it possible to advance more quickly in the directions where the variance is high. This is the stochastic equivalent of a Newtonian descent (which uses the Hessian) as opposed to a gradient descent (which does not) — and is precisely why RLS greatly reduces the effect of conditioning on the speed of convergence.

13.1 Recursive Least-Squares Criterion

Rather than an expectation, RLS minimizes a weighted empirical sum of past errors:

$$J(n) = \sum_{i=1}^{n} \lambda^{n-i} \, |e(i)|^2 = \sum_{i=1}^{n} \lambda^{n-i} \, |d(i) - \boldsymbol{w}^{\mathrm{H}} \boldsymbol{x}(i)|^2,$$

where $\lambda \in (0, 1]$ is *forgetting factor* (*forgetting factor*). For $\lambda = 1$, all errors have the same weight: we minimize the least squares overall. For $\lambda < 1$, old samples are exponentially forgotten, allowing non-stationary environments to be tracked.

13.2 Matrix Form

Let's define

$$\boldsymbol{R}(n) = \sum_{i=1}^{n} \lambda^{n-i} \, \boldsymbol{x}(i)\boldsymbol{x}^{\mathrm{H}}(i), \quad \boldsymbol{r}(n) = \sum_{i=1}^{n} \lambda^{n-i} \, \boldsymbol{x}(i) \, d^*(i).$$

The minimum is

$$\boxed{\boldsymbol{w}(n) = \boldsymbol{R}^{-1}(n) \, \boldsymbol{r}(n).}$$

We find the Wiener structure, but on estimates *time-weighted.*

13.3 Recurrence of the Quantities

The estimator is constructed recursively:

$$\boldsymbol{R}(n) = \lambda\, \boldsymbol{R}(n-1) + \boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n), \qquad \boldsymbol{r}(n) = \lambda\, \boldsymbol{r}(n-1) + \boldsymbol{x}(n)\, d^*(n).$$

Calculating $\boldsymbol{R}^{-1}(n)$ at each step costs $\mathcal{O}(N^3)$: too expensive for a real-time loop. The trick is to use the *matrix inversion lemma*.

13.4 Matrix Inversion Lemma

Lemma 13.1 (Sherman–Morrison)**.** *For invertible* $\boldsymbol{A}$ *and vectors* $\boldsymbol{u}, \boldsymbol{v}$*,*

$$(\boldsymbol{A} + \boldsymbol{u}\boldsymbol{v}^{\mathrm{H}})^{-1} = \boldsymbol{A}^{-1} - \frac{\boldsymbol{A}^{-1}\boldsymbol{u}\boldsymbol{v}^{\mathrm{H}}\boldsymbol{A}^{-1}}{1 + \boldsymbol{v}^{\mathrm{H}}\boldsymbol{A}^{-1}\boldsymbol{u}}.$$

Let's put $\boldsymbol{P}(n) = \boldsymbol{R}^{-1}(n)$ and $\boldsymbol{A} = \lambda\boldsymbol{R}(n-1)$, $\boldsymbol{u} = \boldsymbol{v} = \boldsymbol{x}(n)$. The lemma gives directly

$$\boldsymbol{P}(n) = \frac{1}{\lambda}\Big(\boldsymbol{P}(n-1) - \frac{\boldsymbol{P}(n-1)\boldsymbol{x}(n)\boldsymbol{x}^{\mathrm{H}}(n)\boldsymbol{P}(n-1)}{\lambda + \boldsymbol{x}^{\mathrm{H}}(n)\boldsymbol{P}(n-1)\boldsymbol{x}(n)}\Big).$$

13.5 RLS Gain and Weight Update

We define the *RLS gain*

$$\boldsymbol{k}(n) = \frac{\boldsymbol{P}(n-1)\, \boldsymbol{x}(n)}{\lambda + \boldsymbol{x}^{\mathrm{H}}(n)\, \boldsymbol{P}(n-1)\, \boldsymbol{x}(n)}.$$

We can then write the update compactly:

$$\boxed{\begin{aligned} \alpha(n) &= d(n) - \boldsymbol{w}^{\mathrm{H}}(n-1)\, \boldsymbol{x}(n) \quad \text{(a priori error)} \\ \boldsymbol{w}(n) &= \boldsymbol{w}(n-1) + \boldsymbol{k}(n)\, \alpha^*(n) \\ \boldsymbol{P}(n) &= \frac{1}{\lambda}\left(\boldsymbol{P}(n-1) - \boldsymbol{k}(n)\, \boldsymbol{x}^{\mathrm{H}}(n)\, \boldsymbol{P}(n-1)\right). \end{aligned}}$$

13.6 Complete RLS Algorithm

1. Initialization: $\boldsymbol{w}(0) = \boldsymbol{0}$, $\boldsymbol{P}(0) = \delta^{-1}\boldsymbol{I}_N$ with small $\delta > 0$ (typically $\delta = 10^{-2}\sigma_x^2$).
2. For each snapshot $n = 1, 2, \ldots$:
   (a) calculate $\boldsymbol{u}(n) = \boldsymbol{P}(n-1)\, \boldsymbol{x}(n)$;
   (b) calculate $\boldsymbol{k}(n) = \boldsymbol{u}(n)/(\lambda + \boldsymbol{x}^{\mathrm{H}}(n)\, \boldsymbol{u}(n))$;
   (c) calculate the a priori error $\alpha(n) = d(n) - \boldsymbol{w}^{\mathrm{H}}(n-1)\, \boldsymbol{x}(n)$;
   (d) update $\boldsymbol{w}(n) = \boldsymbol{w}(n-1) + \boldsymbol{k}(n)\, \alpha^*(n)$;
   (e) update $\boldsymbol{P}(n) = (\boldsymbol{P}(n-1) - \boldsymbol{k}(n)\, \boldsymbol{u}^{\mathrm{H}}(n))/\lambda$.

Complexity: $\mathcal{O}(N^2)$ multiplications per snapshot.

13.7 Role of the Forgetting Factor

- $\lambda = 1$: don't forget, RLS generally converges towards the least squares solution. Suitable for stationary environments.
- λ close to 1 (typical 0.95–0.999): slow forgetting, RLS tracks slow variations in the environment.
- λ lower (0.9 for example): quickly forgotten, followed aggressive at the cost of a higher estimate variance.

The *effective memory* of RLS is $1/(1-\lambda)$ snapshots. For example $\lambda = 0.98$ gives memory ~ 50 snapshots.

Choice. In practice we set λ so that the memory effective is slightly greater than the stationarity duration of the environment, but not much more, otherwise RLS will not follow developments.

13.8 Initialisation de $\boldsymbol{P}$

The choice of $\boldsymbol{P}(0) = \delta^{-1}\boldsymbol{I}_N$ plays the role of a *Bayesian prior* : it amounts to regularizing the problem by $\delta\|\boldsymbol{w}\|^2$. A δ that is too small gives an unstable initial convergence; too big, slow convergence. A common heuristic: $\delta = 0.01 \cdot \mathrm{trace}(\boldsymbol{R}_x)/N$ or simply $\delta = 10^{-2}$ after data normalization.

13.9 Comparison of LMS / NLMS / RLS

Criteria	**LMS**	**NLMS**	**RLS**
Cost / snapshot	weak	weak	good
Complexity	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N^2)$
Memory	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N^2)$
Convergence	slow	average	fast
Digital robustness	Good	Good	more delicate
Setting	μ	$\tilde{\mu}, \epsilon$	λ, δ
Variable input power	sensitive	robust	robust but expensive
Typical usage	simple real time	variable power	fast environments

As a general rule: LMS for maximum simplicity, NLMS when the input power varies greatly, RLS when the environment changes quickly or when the covariance is too poorly conditioned for LMS to converge in time.

13.10 RLS and MVDR

RLS implicitly estimates $\boldsymbol{R}^{-1}$ via $\boldsymbol{P}(n)$. It can be used to build an MVDR beamformer *online*:

$$\boldsymbol{w}_{\mathrm{MVDR}}(n) = \frac{\boldsymbol{P}(n), \boldsymbol{a}(\theta_s)}{.}$$

This recursive version of MVDR is widely used in adaptive radar and telecommunications: no matrix to explicitly invert, continuous adaptation to the environment.

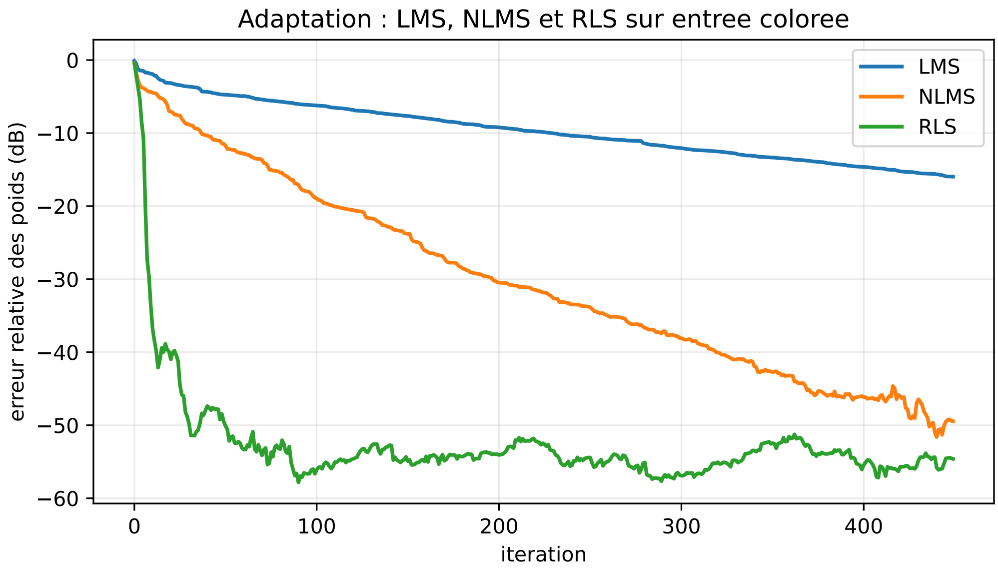

Figure 13.1: Reproducible simulation on colored input ($N = 8$, 450 iterations, eigenspectrum from 30 to 1, RLS $\lambda = 0.99$): LMS progresses slowly, NLMS stabilizes the pace, RLS converges faster at the cost of higher cost and numerical fragility.

13.11 Constrained RLS (CRLS)

To integrate LCMV into an RLS loop, we use the GSC structure (chapter 11):

- the fixed part $\boldsymbol{w}_q$ satisfies the constraints once for all;
- the adaptive part $\boldsymbol{w}_a$ minimizes $\mathbb{E}[|y_q - \boldsymbol{w}_a^{\mathrm{H}}\boldsymbol{z}|^2]$, without constraints, by RLS on $\boldsymbol{z}(n) = \boldsymbol{B}^{\mathrm{H}}\boldsymbol{x}(n)$.

We obtain a *Constrained RLS* (CRLS) without significant additional cost.

13.12 Numerical Stability

The inversion lemma can, in floating arithmetic, lose the Hermitian symmetry of $\boldsymbol{P}$ and the definite positivity. Consequence: numerical divergence for long simulations. Two remedies:

Symmetrical shape. At every step, we impose $\boldsymbol{P}(n) \leftarrow (\boldsymbol{P}(n) + \boldsymbol{P}^{\mathrm{H}}(n))/2$. Simple but imperfect solution.

Square root form (square-root RLS). We manipulate a Cholesky factorization $\boldsymbol{P}(n) = \boldsymbol{L}(n)\boldsymbol{L}^{\mathrm{H}}(n)$ rather than $\boldsymbol{P}$ directly. Stable but more complex to implement. Essential reference in aeronautics and defense.

QR-RLS. Using Givens rotations on samples: numerically very stable, parallelizable on FPGA, but outside the scope of this chapter.

13.13 Numerical Implementation

Listing 13.1: RLS for supervised beamforming

```python
import numpy as np

def rls(X, d, lam=0.99, delta=1e-2):
    """
    X     : (N, K) snapshots
    d     : (K,) reference
    lam   : facteur d'oubli (0 < lam <= 1)
    delta : initialisation de P
    """
    N, K = X.shape
    w = np.zeros(N, dtype=complex)
    P = np.eye(N, dtype=complex) / delta
    err = np.zeros(K, dtype=complex)
    for n in range(K):
        x = X[:, n]
        u = P @ x
        gain = u / (lam + np.real(x.conj() @ u))
        alpha = d[n] - w.conj() @ x
        w = w + gain * np.conj(alpha)
        P = (P - np.outer(gain, x.conj() @ P)) / lam
        # symétrisation hermitienne
        P = 0.5 * (P + P.conj().T)
        err[n] = alpha
    return w, err
```

13.14 Summary

RLS often converges much faster than LMS, typically in a number of snapshots of the order of a few N in well-conditioned or correctly regularized cases. The price is $\mathcal{O}(N^2)$ complexity and significant digital sensitivity. It is the tool of choice for rapid monitoring of non-stationary environments when LMS converges too slowly.

At this stage, we have the entire range of adaptive beamformers:

- closed formulas (DAS, MVDR, LCMV);
- iterative algorithms (LMS, NLMS, RLS).

Part V changes the problem: we no longer seek to form a beam but to *estimate directions* themselves, from the same data. This is the transition from *spatial filtering* to *spatial parametric estimation*.

This change of objective is not just an additional refinement: it is a conceptual leap. Where Parts II-IV sought to *filter* (produce a signal as clean as possible), the Part V seeks to *measure* (produce an angle as precise as possible). The two tasks are linked — you will never have a good filter without good measurement — but their quality criteria are different. A filter is judged by its Output SINR; a DOA estimator is judged by the variance of the estimated angle, ideally compared to the Cramér-Rao bound. Part V will also introduce this bound as a universal standard for direction estimators.

Part V

Direction-of-Arrival Estimation

Chapter 14

Subspace Decomposition and Order Detection

Beamforming consists of *form* a beam. The direction of arrival (DOA) estimate consists of *find* where the sources are coming from. The transition between the two worlds involves a remarkable algebraic observation of spatial covariance: its eigenvectors naturally separate into a subspace *signal* and a subspace *noise*. This structure, exploited by MUSIC, ESPRIT and their variants, allows resolutions much higher than the Rayleigh limit of conventional beamforming.

This idea deserves attention. The Rayleigh limit — $\Delta u_{\text{res}} \sim \lambda/D$ — is a limit *determinist* based on the width of the main lobe of a finished opening. It does not depend on the SNR or the number of snapshots: even with an infinite SNR, the DAS and MVDR do not separate two sources closer than this limit. Subspace methods do not make this limit an insurmountable wall. They exploit it more subtly: for two sources, even very close, their steering vectors remain *distinct* as vectors of $\mathbb{C}^N$, and therefore the signal subspace has dimension 2 (and not 1). The effective resolution then depends on the quality of estimation of this subspace, which improves with K and with the SNR. This is what allows, in the favorable regime, to separate sources to one tenth or one hundredth of the Rayleigh limit.

This chapter formalizes the decomposition, studies the associated subspaces and discusses the detection of the number of sources (*model order selection*), a problem prior to any DOA estimation by subspaces.

14.1 Multi-Source Model

The reference model is

$$\boldsymbol{x}(n) = \boldsymbol{A}(\boldsymbol{\theta})\,\boldsymbol{s}(n) + \boldsymbol{n}(n),$$

with :

- $\boldsymbol{A}(\boldsymbol{\theta}) = [\boldsymbol{a}(\theta_1), \ldots, \boldsymbol{a}(\theta_L)] \in \mathbb{C}^{N\times L}$ steering matrix;
- $\boldsymbol{s}(n) \in \mathbb{C}^L$ vector of source envelopes, covariance $\boldsymbol{P}_s = \mathbb{E}[\boldsymbol{s}\boldsymbol{s}^{\mathrm{H}}]$;
- $\boldsymbol{n}(n) \in \mathbb{C}^N$ spatial white noise, covariance $\sigma^2 \boldsymbol{I}_N$ and decorrelated from the signal.

Technical hypotheses.

(H1) Far field: valid plane wave model.

(H2) Narrow band: frequency-independent steering vector.

(H3) $L < N$: fewer sources than antennas (otherwise compromised identifiability).

(H4) $\boldsymbol{A}$ full row column L (sources angularly distinct).

(H5) $\boldsymbol{P}_s$ of full rank (sources not perfectly consistent; degenerate case treated further).

(H6) Spatial white noise, with known or estimable variance.

14.2 Covariance Structure

Theorem 14.1 (Signal/Noise Decomposition)**.** *Under the hypotheses (H1)–(H6), the covariance*

$$\boldsymbol{R} = \boldsymbol{A}\,\boldsymbol{P}_s\,\boldsymbol{A}^{\mathrm{H}} + \sigma^2 \boldsymbol{I}_N$$

admits spectral decomposition

$$\boldsymbol{R} = \sum_{k=1}^{L} \lambda_k\, \boldsymbol{u}_k \boldsymbol{u}_k^{\mathrm{H}} + \sum_{k=L+1}^{N} \sigma^2\, \boldsymbol{u}_k \boldsymbol{u}_k^{\mathrm{H}} = \boldsymbol{U}_s \boldsymbol{\Lambda}_s \boldsymbol{U}_s^{\mathrm{H}} + \sigma^2 \boldsymbol{U}_n \boldsymbol{U}_n^{\mathrm{H}},$$

where :

- $\boldsymbol{U}_s = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_L]$ *generates the* signal subspace*, of dimension* L*;*
- $\boldsymbol{U}_n = [\boldsymbol{u}_{L+1}, \ldots, \boldsymbol{u}_N]$ *generates the* noise subspace*, of dimension* $N - L$*;*
- $\lambda_1 \geq \cdots \geq \lambda_L > \sigma^2$ *;*
- $\lambda_{L+1} = \cdots = \lambda_N = \sigma^2$*.*

Moreover,

$$\mathrm{Im}(\boldsymbol{U}_s) = \mathrm{Im}(\boldsymbol{A}),$$

the columns of $\boldsymbol{A}$ *and those of* $\boldsymbol{U}_s$ *generate the* even *subspace of* $\mathbb{C}^N$*.*

Proof. $\boldsymbol{A}\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}}$ is positive semidefinite Hermitian of rank L (composition of rank L). Its L non-zero eigenvalues μ_k are positive; the others are zero. By adding $\sigma^2\boldsymbol{I}_N$, each eigenvalue is shifted by σ^2: the first L become $\mu_k + \sigma^2 > \sigma^2$ and the last $N - L$ become exactly σ^2. The eigenvectors remain unchanged (they diagonalize $\boldsymbol{I}_N$ too). The subspace of the first L eigenvectors coincides with $\mathrm{Im}(\boldsymbol{A}\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}}) = \mathrm{Im}(\boldsymbol{A})$ by (H4)-(H5). □

14.3 Fundamental Orthogonality

Corollary 14.2 (Signal-to-noise orthogonality)**.** *For any* $\ell = 1, \ldots, L$*,*

$$\boldsymbol{U}_n^{\mathrm{H}}\, \boldsymbol{a}(\theta_\ell) = \boldsymbol{0}.$$

This is the cornerstone of the MUSIC algorithm. The steering vector of any real source is orthogonal to the noise subspace; any θ candidate that is not a source will not be a source. Varying θ and looking for the angles minimizing $\|\boldsymbol{U}_n^{\mathrm{H}}\boldsymbol{a}(\theta)\|$ amounts to locating the sources.

14.4 Coherent Sources and Reduced Rank

The hypothesis (H5) $\boldsymbol{P}_s$ of full rank is essential. In multipath, two sources can be coherent: their $\boldsymbol{P}_s$ becomes singular, $\boldsymbol{A}\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}}$ has rank $L' < L$, and the signal/noise decomposition identifies L' sources instead of L. MUSIC then sees fewer peaks than there are actual directions.

Spatial smoothing. The trick is to divide the array into *offset subarrays* and to average the covariances. There Effective decorrelation between sources is thus reestablished at the cost of a reduced effective aperture. This technique will be detailed in chapter 15.

14.5 Practical Estimation

In practice we only have $\widehat{\boldsymbol{R}}$ (estimated on K snapshots), and the decomposition is not *Exactly* that of the theorem 14.1: the noise eigenvalues are no longer exactly equal to σ^2, they fluctuate around it. The eigenvectors are perturbed to the extent of $\mathcal{O}(1/\sqrt{K})$. We nevertheless separate the two subspaces by sorting the eigenvalues in descending order and looking for a "jump " in the eigenspectrum.

14.6 Detection of the Number of Sources

Estimating L correctly is crucial: too low, we miss sources; too high, we add phantom sources. Three families of criteria exist.

14.6.1 Threshold Criterion

The simplest: we classify the eigenvalues in descending order, and we count those which exceed a threshold $\tau\sigma^2$ with $\tau > 1$ (typically $\tau = 2$ or 3). Robust but not very selective.

14.6.2 AIC (Akaike Information Criterion)

For the detection of the number of sources in array processing, the AIC and MDL forms used here follow the classic formulation of Wax and Kailath [11].

$$\mathrm{AIC}(\ell) = -2K(N-\ell)\,\log\left(\frac{\left(\prod_{k=\ell+1}^{N}\widehat{\lambda}_k\right)^{1/(N-\ell)}}{\frac{1}{N-\ell}\sum_{k=\ell+1}^{N}\widehat{\lambda}_k}\right) + 2\ell(2N-\ell).$$

We choose $L = \arg\min_\ell \mathrm{AIC}(\ell)$. AIC tends to overestimate L asymptotically.

14.6.3 MDL (Minimum Description Length)

$$\mathrm{MDL}(\ell) = -K(N-\ell)\,\log\left(\frac{\left(\prod\widehat{\lambda}_k\right)^{1/(N-\ell)}}{\frac{1}{N-\ell}\sum\widehat{\lambda}_k}\right) + \frac{1}{2}\ell(2N-\ell)\,\log K.$$

MDL is *consistent*: the error probability tends to 0 when $K \to \infty$. Preferred in practice.

14.6.4 Modern Criteria

For small K/N, the classic criteria fail. Extensions to random matrix theory (*Marcenko–Pastur*) or *bootstrap* techniques on eigenvalues give more robust estimators. The theory of large random matrices provides explicit thresholds to distinguish a true source from an estimation artifact: this is *spike model*.

14.7 Eigenvectors: Geometric Interpretation

The eigenvectors of the signal subspace are not, in general, proportional to the individual steering vectors $\boldsymbol{a}(\theta_\ell)$. They are *linear combinations* of these steering vectors, adapted to $\boldsymbol{P}_s$. This is precisely why MUSIC does not read the eigenvectors directly: it looks for the directions *orthogonal* in the noise subspace, which are necessarily the $\boldsymbol{a}(\theta_\ell)$.

14.8 Model Tradeoffs

The subspace structure depends heavily on the assumptions (H1)–(H6). A violation has consequences:

- noise *spatially non-white*: the noise eigenvalues do not are no longer all equal to σ^2, the separation becomes blurred. Solution: *prewhitening* by the noise covariance estimated on intervals “absent signal ”.
- sources *consistent*: reduced signal rank. Solution : spatial smoothing.
- number of *snapshots* $K \sim N$: estimated subspaces very imprecise. Solution: regularization, fusion of techniques (beamforming + subspaces).
- *calibration* imperfect: distorted steering vectors, sources non-orthogonal to the estimated noise subspace. Solution: chapter 22.

14.9 Numerical Implementation

Listing 14.1: Signal/noise decomposition and order detection

```python
import numpy as np

def signal_noise_subspace(R, L):
    """Renvoie U_s, U_n, eigvals (décroissants)."""
    eigvals, eigvecs = np.linalg.eigh(R)
    idx = np.argsort(eigvals)[::-1]
    eigvals = eigvals[idx]
    eigvecs = eigvecs[:, idx]
    U_s = eigvecs[:, :L]
    U_n = eigvecs[:, L:]
    return U_s, U_n, eigvals

def estimate_L_mdl(eigvals, K):
    """Estime L par critère MDL. eigvals décroissants."""
    N = len(eigvals)
    mdl = np.zeros(N)
    for ell in range(N):
        tail = eigvals[ell:]
        geo = np.prod(tail) ** (1.0 / (N - ell))
        ari = np.mean(tail)
        if geo <= 0 or ari <= 0:
            mdl[ell] = np.inf
            continue
        log_ratio = np.log(ari / geo)
        mdl[ell] = K * (N - ell) * log_ratio + 0.5 * ell * (2 * N - ell) * np.log(K)
    return int(np.argmin(mdl))
```

14.10 Summary

The signal/noise decomposition of the covariance provides two orthogonal subspaces, the structure of which fully encodes the position of the sources. MUSIC exploits orthogonality to the noise subspace; ESPRIT exploits translation invariance within the signal subspace. Order detection by MDL or AIC is essential beforehand.

The following chapters successively delve deeper into MUSIC, ESPRIT, then the Cramér-Rao bounds which set the ultimate performances of any DOA estimator. This progression is deliberately based on the historical order: MUSIC in 1979/86, ESPRIT in 1986, CRB formalized for the DOA by Stoica and Nehorai [13]. The sparse and Bayesian methods of the chapter 18 will arrive later (from the 2000s) with the rise of *compressed sensing*, and will complete the picture in regimes that MUSIC and ESPRIT do not treat well.

Chapter 15

MUSIC and Root-MUSIC

The MUSIC algorithm (*Multiple Signal Classification*, Schmidt [9]) directly exploits the orthogonality between the steering vectors of the sources and the noise subspace (corollary 14.2). It marked a turning point in the estimation of direction of arrival: for the first time, we significantly exceeded the Rayleigh limit of conventional beamforming without strong parametric assumptions on the sources.

The elegance of MUSIC lies in its use *minimalist* of available information. It does not exploit the entire signal subspace — what maximum likelihood would do for example — but only *orthogonality* to its complement. This information is, in a certain sense, the proper part " of the covariance which does not depend on the distribution of powers between sources. In the ideal asymptotic model, this makes MUSIC insensitive to relative powers between sources. In finite data, a weak source nevertheless remains more difficult to extract: it reduces the eigengap, makes the noise subspace less stable, and requires more snapshots or SNR.

15.1 Basic idea

For any angle θ corresponding to a real source, $\boldsymbol{a}(\theta) \in \mathrm{Im}(\boldsymbol{A}) = \mathrm{Im}(\boldsymbol{U}_s)$, therefore

$$\boldsymbol{U}_n^{\mathrm{H}}\,\boldsymbol{a}(\theta) = \boldsymbol{0} \quad \Longleftrightarrow \quad \|\boldsymbol{U}_n^{\mathrm{H}}\,\boldsymbol{a}(\theta)\|^2 = 0.$$

For any other θ, this quantity is non-zero. By varying θ over the visible area and looking for the minima — that is to say the maxima of the inverse — we locate the sources.

15.2 Pseudo-spectrum

We define the *pseudo-spectrum MUSIC*

$$\boxed{P_{\mathrm{MUSIC}}(\theta) = \frac{1}{\boldsymbol{a}^{\mathrm{H}}(\theta)\,\boldsymbol{U}_n\boldsymbol{U}_n^{\mathrm{H}}\,\boldsymbol{a}(\theta)}.}$$

$P_{\mathrm{MUSIC}}(\theta)$ tends to infinity (in theory) at the angles of the sources, and remains finite elsewhere. The estimation consists of detecting the most pronounced L peaks.

Pseudo and not spectrum. It's not a power spectrum in the physical sense: the value of P_{MUSIC} has no dimension. It is a *proximity indicator* between $\boldsymbol{a}(\theta)$ and the signal subspace. We plot it in dB to better distinguish the peaks.

15.3 Algorithm

1. Collect K snapshots $\boldsymbol{x}(1), \ldots, \boldsymbol{x}(K)$.
2. Estimate $\widehat{\boldsymbol{R}} = \frac{1}{K}\boldsymbol{X}\boldsymbol{X}^{\mathrm{H}}$.
3. Decompose into eigenvalues and eigenvectors $\widehat{\boldsymbol{R}} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^{\mathrm{H}}$.
4. Estimate $\widehat{L}$ (MDL, AIC, threshold or knowledge *a priori*).
5. Extract $\widehat{\boldsymbol{U}_n}$: the $N - \widehat{L}$ vectors eigenvalues associated with the smallest eigenvalues.
6. Evaluate $P_{\mathrm{MUSIC}}(\theta)$ on a fine grid.
7. Detect the most pronounced $\widehat{L}$ peaks.

15.4 Angular resolution

MUSIC can exceed the Rayleigh limit when:

- the SNR is sufficiently high;
- the number of snapshots is sufficient;
- the calibration is correct.

MUSIC's *effective resolution* depends on $\sqrt{K \cdot \mathrm{SNR}}$, unlike Capon (which depends on SNR alone) and DAS (fixed limit). At very high SNR or very large K, MUSIC can separate sources that Capon does not separate. On the other hand, at low K, low SNR, bad model order or steering vector mismatch, its peaks can disappear, split or become biased.

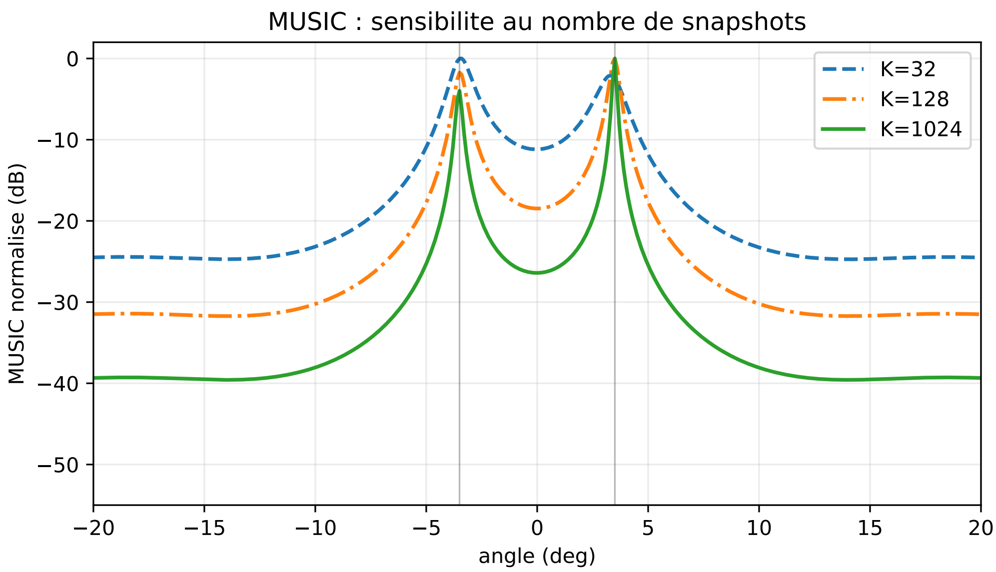

Figure 15.1: MUSIC simulation ($N = 12$, sources at $\pm 3.5°$, SNR 8 dB, $K = 32, 128, 1024$): two nearby sources become much more readable when the number of snapshots increases. This figure illustrates a trend, not a universal guarantee.

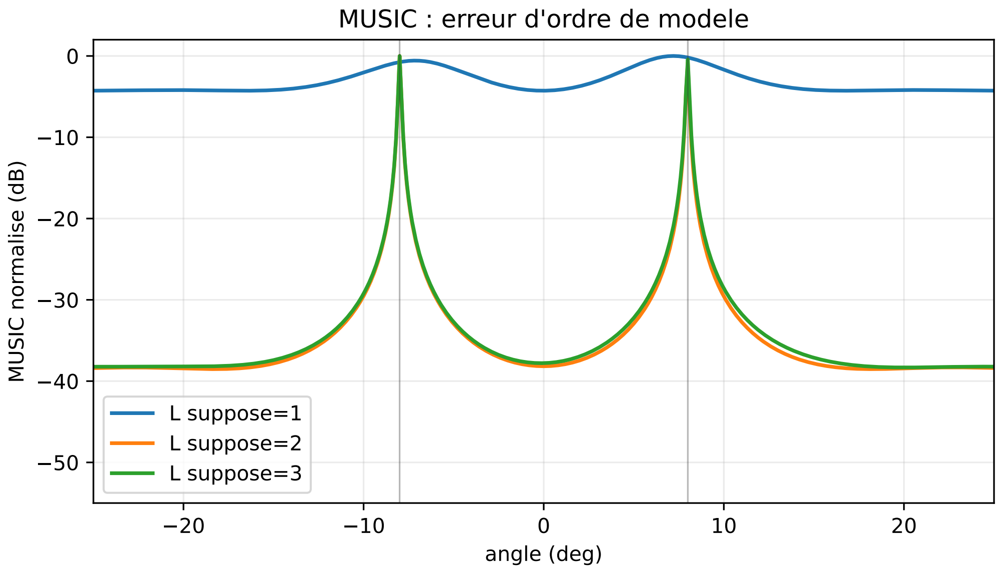

Figure 15.2: MUSIC with model order error ($N = 12$, two sources at $\pm 8°$, SNR 12 dB, $K = 512$): underestimating L merges the sources, overestimating it introduces spurious or unstable peaks.

15.5 Comparison Bartlett, Capon, MUSIC

Criteria	Bartlett (DAS)	Capon	MUSIC
Information required	$\boldsymbol{R}$	$\boldsymbol{R}^{-1}$	$\boldsymbol{U}_n$
Cost	$\mathcal{O}(NM)$	$\mathcal{O}(N^3 + NM)$	$\mathcal{O}(N^3 + NM)$
Resolution	$\sim 1/N$	best	high if assumptions OK
Consistent sources	robust	sensitive	fails (without smoothing)
Calibration	robust	sensitive	very sensitive

(M: number of points of the angular grid.)

Figure 15.3 illustrates the key point: MUSIC is not just a thinner beamformer. It uses different information. Bartlett measures the projected power on $\boldsymbol{a}(\theta)$; Capon measures the minimum power compatible with unity gain; MUSIC measures the distance to the noise subspace. This is why MUSIC peaks can be much narrower.

15.6 Coherent sources and spatial smoothing

For two coherent sources (typical of a direct path + echo), $\boldsymbol{P}_s$ becomes singular and the signal rank is $L' < L$. MUSIC then underestimates the number of sources. The canonical parade is the *forward spatial smoothing* [12].

15.6.1 Example: direct path and multipath

Let us assume two perfectly coherent paths coming from the same signal:

$$\boldsymbol{x}(n) = \boldsymbol{a}(\theta_1)s(n) + \alpha\boldsymbol{a}(\theta_2)s(n) + \boldsymbol{n}(n).$$

We can factor:

$$\boldsymbol{x}(n) = [\boldsymbol{a}(\theta_1) + \alpha\boldsymbol{a}(\theta_2)]\, s(n) + \boldsymbol{n}(n).$$

Even though there are two physical directions, the signal part is rank 1, because it only depends on a single time signal $s(n)$. The signal covariance is:

$$\boldsymbol{R}_s = P_s\, [\boldsymbol{a}(\theta_1) + \alpha\boldsymbol{a}(\theta_2)]\, [\boldsymbol{a}(\theta_1) + \alpha\boldsymbol{a}(\theta_2)]^{\mathrm{H}},$$

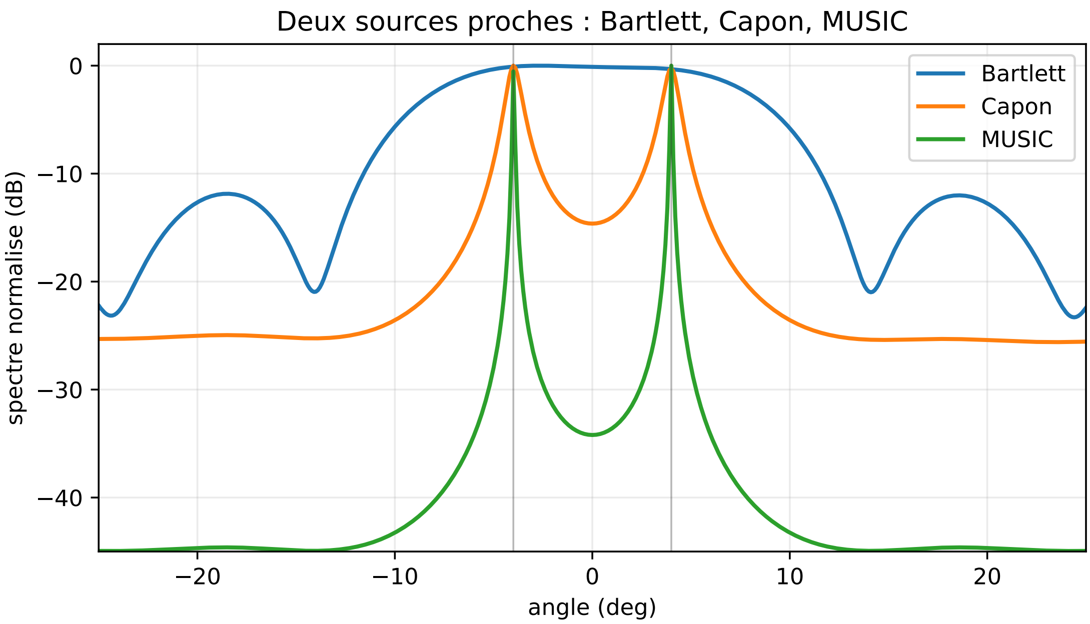

Figure 15.3: Typical comparison between Bartlett, Capon and MUSIC ($N = 12$, sources at $\pm 4°$, SNR 15 dB, $K = 512$). Bartlett broadens the peaks, Capon refines them, MUSIC produces the thinnest peaks when the subspace assumptions are satisfied.

so its rank is 1. MUSIC looks for two signal subspace dimensions, but only observes one: it can produce a single peak, biased peaks or unstable separation.

Spatial smoothing breaks this coherence by exploiting the fact that the two paths accumulate different phases from one subarray to another. This restores the signal range, but at the cost of a smaller effective aperture. The resolution therefore decreases: we exchange rank for aperture.

15.6.2 Principle

We divide the ULA of N antennas into J *offset subarrays* of size $M = N - J + 1$. Each subnet j ($j = 0, \ldots, J-1$) provides partial covariance

$$\widehat{\boldsymbol{R}}^{(j)} = \frac{1}{K} \boldsymbol{X}^{(j)} \boldsymbol{X}^{(j)\,\mathrm{H}},$$

where $\boldsymbol{X}^{(j)}$ contains only the lines $j, j+1, \ldots, j+M-1$ of $\boldsymbol{X}$. The smoothed covariance is

$$\widehat{\boldsymbol{R}}_{\mathrm{ss}} = \frac{1}{J} \sum_{j=0}^{J-1} \widehat{\boldsymbol{R}}^{(j)} \in \mathbb{C}^{M \times M}.$$

For $J \geq L$, $\widehat{\boldsymbol{R}}_{\mathrm{ss}}$ has its signal rank restored to L (provided that the relative phases of sources between subarrays are sufficient to decorrelate).

15.6.3 Boundaries

- The effective opening goes from $N-1$ to $M-1 < N-1$: the resolution deteriorates.
- Requires a ULA (regular geometry).
- The *forward-backward smoothing* doubles the effect by also using reverse indexed subarrays.

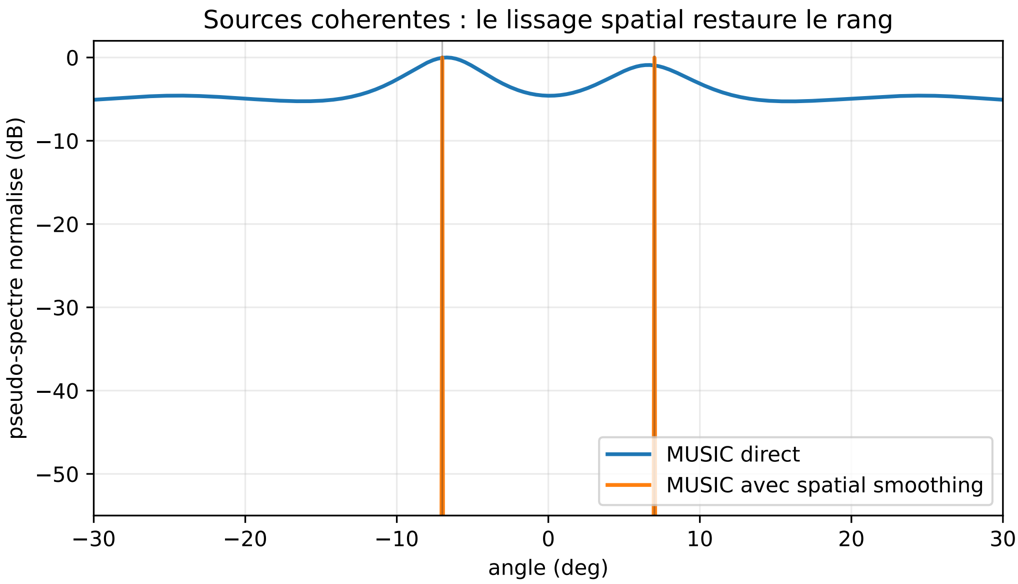

Figure 15.4: Two coherent sources ($N = 12$, sources at $\pm 7°$, subarrays of size $M = 8$) can be reduced to a signal subspace of rank 1. Spatial smoothing restores two MUSIC peaks, at the cost of a smaller effective aperture.

15.7 Root-MUSIC

For a ULA, the angular scan is useless: we can solve the equation algebraically.

15.7.1 Polynomial MUSIC

Let's set $z = e^{-jkd\sin\theta}$. The ULA steering vector is $\boldsymbol{a}(z) = [1, z, z^2, \ldots, z^{N-1}]^{\mathrm{T}}$. The function

$$f(\theta) = \boldsymbol{a}^{\mathrm{H}}(\theta)\, \boldsymbol{U}_n \boldsymbol{U}_n^{\mathrm{H}}\, \boldsymbol{a}(\theta)$$

can be written, by setting $\boldsymbol{C} = \boldsymbol{U}_n \boldsymbol{U}_n^{\mathrm{H}}$,

$$f(z) = \sum_{m,n} (\boldsymbol{C})_{m,n}\, z^{-(m-n)}.$$

$f(z)$ is thus a polynomial in z and z^{-1}, of degree $2(N-1)$.

15.7.2 Algorithm

1. Calculate $\boldsymbol{U}_n$ as in standard MUSIC.
2. Construct the polynomial $p(z) = f(z) \cdot z^{N-1}$ of degree $2(N-1)$.
3. Calculate its roots $z_1, \ldots, z_{2(N-1)}$.
4. Retain the L roots closest to the unit circle, *inside* or on him. (Roots come in pairs $(z_i, 1/z_i^*)$.)
5. For each root retained, the estimated angle is $\widehat{\theta}_\ell = \arcsin(-\arg(z_\ell)/(kd))$.

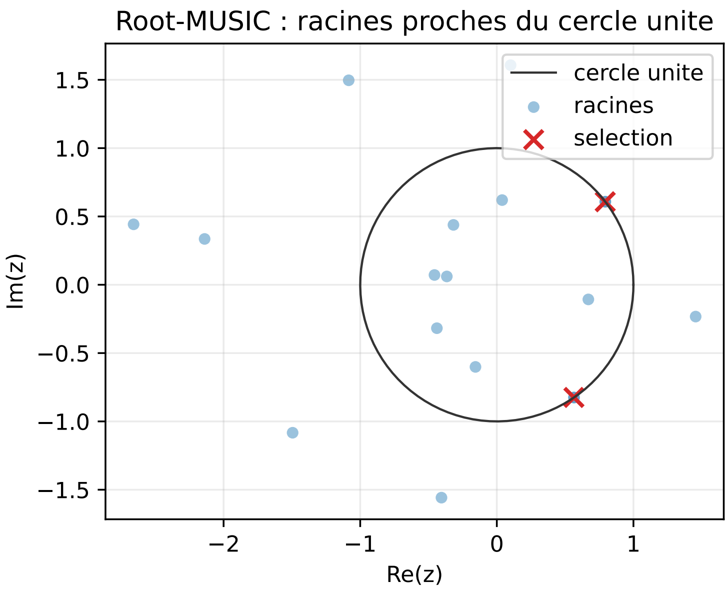

Figure 15.5: Root-MUSIC ($N = 10$, sources $-12°$ and $18°$, SNR 20 dB, $K = 2048$) transforms the angular scan into a selection of roots close to the unit circle. Wrong root selection immediately results in wrong angles.

15.7.3 Benefits

- No angular grid to choose: no error discretization.
- $\mathcal{O}(N^3)$ cost for roots: competitive with a dense scan.
- Potentially improved accuracy when the ULA model and order L are reliable.

15.7.4 Boundaries

- Strictly reserved for ULAs. For 2D or non-uniform arrays, Root-MUSIC has no direct analogue (extensions *Beamspace Root-MUSIC* for some cases).
- Numerical sensitivity: roots close to the unit circle are poorly localized at very low SNR.
- Delicate root selection when sources are close, coherent or when the order L is incorrectly estimated.
- The formula $\arcsin(\cdot)$ must be digitally protected: in practice we limit our argument in $[-1, 1]$ before converting to an angle.

15.8 Variants

Beamspace MUSIC. We project $\boldsymbol{x}$ and $\boldsymbol{a}(\theta)$ in a subspace of reduced dimension (a fixed transformation matrix $\boldsymbol{T}$) before applying MUSIC. This reduces cost and, correctly chosen, improves resolution over a restricted angular range.

Cyclic MUSIC. For cyclostationary signals, we replaces the covariance with a covariance *cyclic* which only includes signals with a given cyclic frequency. Allows you to separate sources of different standards (Wi-Fi vs. LTE for example).

Maximum likelihood (ML). Find angles directly which maximize likelihood. More precise than MUSIC at very low SNR, but requires multi-dimensional optimization (alternating projection, Newton, etc.). We will return to this briefly in the next chapter when calculating the Cramér-Rao limits.

15.9 Calibration and MUSIC

MUSIC relies on the exact knowledge of $\boldsymbol{a}(\theta)$. A phase error between channels $\delta\phi_m$ produces a real steering vector $\boldsymbol{a}_{\text{real}}(\theta) = \boldsymbol{G}\,\boldsymbol{a}(\theta)$. The pseudo-spectrum calculated with the theoretical $\boldsymbol{a}$ then presents:

- angled *staggered* peaks;
- *raised to level* peaks (less discriminating);
- sometimes *ghost peaks*.

Chapter 22 covers this problem in detail.

15.10 Numerical Implementation

Listing 15.1: Standard MUSIC and Root-MUSIC

```
import numpy as np

def music_spectrum(R, L, thetas, N, d_over_lambda=0.5):
    assert R.shape == (N, N)
    assert 0 <= L < N
    R = 0.5 * (R + R.conj().T)
    _, eigvecs = np.linalg.eigh(R)    # croissant
    Un = eigvecs[:, :N-L]             # plus petites valeurs propres
    m = np.arange(N)
    spec = np.zeros(len(thetas))
    eps = np.finfo(float).eps
    for idx, th in enumerate(thetas):
        a = np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(th) * m)
        v = Un.conj().T @ a
        den = max((v.conj() @ v).real, eps)
        spec[idx] = 1.0 / den
    return spec

def root_music(R, L, d_over_lambda=0.5):
    """Retourne les L DOA estimées (rad) par Root-MUSIC."""
    N = R.shape[0]
    assert R.shape == (N, N)
    assert 0 < L < N
    R = 0.5 * (R + R.conj().T)
    eigvals, eigvecs = np.linalg.eigh(R)
    Un = eigvecs[:, :N-L]
    C = Un @ Un.conj().T              # N x N

    # Coefficients du polynôme MUSIC dans z
    coeffs = np.zeros(2*N - 1, dtype=complex)
    for lag in range(-(N-1), N):
        coeffs[N - 1 + lag] = np.trace(C, offset=lag)
    roots = np.roots(coeffs[::-1])
```

```

    # Racines à l'intérieur ou sur le cercle unité
    valid = roots[np.abs(roots) <= 1.0]
    if len(valid) < L:
        raise ValueError("Root-MUSIC: pas assez de racines valides")
    # Trier par distance au cercle unité
    order = np.argsort(np.abs(np.abs(valid) - 1.0))
    selected = valid[order][:L]
    arg = np.angle(selected) / (-2*np.pi*d_over_lambda)
    angles = np.arcsin(np.clip(arg, -1.0, 1.0))
    return np.sort(np.real(angles))
```

15.11 Practical Limits by MUSIC

- requires a well-estimated covariance ($K \geq$ some N);
- requires precise knowledge of L;
- requires sources that are not perfectly coherent (or spatial smoothing);
- requires careful calibration;
- does not naturally provide a measure of confidence;
- cost of a dense corner scan $\sim \mathcal{O}(NM)$ per evaluation pseudo-spectrum.

15.12 Summary

.MUSIC is *historical* algorithm for high resolution DOA estimation Its basis (signal-noise orthogonality) is clear; under favorable hypotheses and in an asymptotic regime, its performances can approach the Cramér-Rao bound (which we will study in chapter 17). Its weaknesses — coherent sources, calibration, choice of L — are well understood and widely addressed.

Key Points

- **Central formula**: $P_{\text{MUSIC}}(\theta) = 1/(\boldsymbol{a}^{\text{H}}\boldsymbol{U}_n\boldsymbol{U}_n^{\text{H}}\boldsymbol{a})$.
- **Main hypothesis**: clear separation between subspace signal and noise subspace.
- **Advantage**: high resolution and flexible geometry.
- **Limit**: coherent sources, choice of L and calibration.
- **Practical trap**: a MUSIC peak is not a power; it is a measure of distance to a subspace.
- **Following**: ESPRIT replaces the scan with a relationship of invariance.

The next chapter introduces ESPRIT, which removes the angular scan in favor of a closed algebraic decomposition exploiting the translation invariance of the ULAs. For a well-calibrated ULA geometry, ESPRIT is typically faster than MUSIC with comparable performance, and easier to integrate on embedded platforms. For more general geometries (planar, sparse, conformal), MUSIC often remains the tool of choice, its genericity then prevailing over the algebraic specialization of ESPRIT.

Chapter 16

ESPRIT and Variants

MUSIC locates sources by evaluating a pseudo-spectrum on an angular grid. The technique works, but it has two limits: the cost of the scan, and the resolution limited by the grid. ESPRIT (*Estimation of Signal Parameters via Rotational Invariance Techniques*, Roy & Kailath [10]) frees itself from this by exploiting a purely *algebraic* property of ULAs: their invariance by translation. The DOAs are obtained in eigenvalues of a matrix $L \times L$ — without scan, without grid.

There is something magical about the idea. Instead of looking for directions by traversing space, we read them " in the spectrum of a matrix. The trick lies in the fact that a ULA is *self-similar* : we can divide it into two subarrays which are identical except for one translation. This translation induces a precise algebraic relationship between the signal subspaces of the two subarrays, whose angles are the unknowns. Solving this relationship amounts to diagonalizing a matrix. All the power of ESPRIT comes from this transformation of a localization problem into a spectral problem.

16.1 Translation Invariance

16.1.1 Shifted Subarrays

where a ULA of N antennas. Let's divide it into two subarrays of $M = N - 1$ antennas: the first ($\mathcal{X}_1$) covers the $0, 1, \ldots, N-2$ indices, the second ($\mathcal{X}_2$) the $1, 2, \ldots, N-1$ indices. The two subarrays are identical to a translation near an antenna along the array axis.

If $\boldsymbol{a}_M(\theta)$ is the ULA steering vector at M antennas,

$$\boldsymbol{a}_M(\theta) = [1, z, z^2, \ldots, z^{M-1}]^{\mathrm{T}}, \qquad z = z(\theta) = e^{-jkd\sin\theta},$$

then the steering vector of the second subarray, measured relative to its first antenna, is *the same*; but relative to the reference antenna of the first subarray, it is multiplied by $z(\theta)$:

$$\boldsymbol{a}_M^{(2)}(\theta) = z(\theta) \cdot \boldsymbol{a}_M(\theta).$$

It is this property — the application $\boldsymbol{a}_M \mapsto z(\theta)\boldsymbol{a}_M$ — which gives its name to ESPRIT: the *rotational invariance*.

16.1.2 Matrix Selection

Formally, we introduce two matrices $\boldsymbol{J}_1, \boldsymbol{J}_2 \in \mathbb{R}^{M \times N}$ which respectively select the first M and the last M antennas:

$$\boldsymbol{J}_1 = [\boldsymbol{I}_M \; \boldsymbol{0}], \qquad \boldsymbol{J}_2 = [\boldsymbol{0} \; \boldsymbol{I}_M].$$

Concretely, for $N = 5$ and $M = 4$:

$$\boldsymbol{J}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \boldsymbol{J}_2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

$\boldsymbol{J}_1$ keeps the first $N-1$ antennas; $\boldsymbol{J}_2$ keeps the last $N-1$. The second subarray is therefore the first translated by a step d.

For the $\boldsymbol{A}_N = [\boldsymbol{a}_N(\theta_1), \ldots, \boldsymbol{a}_N(\theta_L)]$ steering matrix,

$$\boldsymbol{A}_1 \equiv \boldsymbol{J}_1 \boldsymbol{A}_N = [\boldsymbol{a}_M(\theta_1), \ldots, \boldsymbol{a}_M(\theta_L)], \qquad \boldsymbol{A}_2 \equiv \boldsymbol{J}_2 \boldsymbol{A}_N = \boldsymbol{A}_1 \boldsymbol{\Phi},$$

where

$$\boldsymbol{\Phi} = \operatorname{diag}\big(e^{-jkd\sin\theta_1}, \ldots, e^{-jkd\sin\theta_L}\big) \in \mathbb{C}^{L\times L}.$$

The DOAs are entirely encoded in the eigenvalues of $\boldsymbol{\Phi}$*.*

The key relationship can therefore be read in a single line:

$$\boxed{\boldsymbol{J}_2 \boldsymbol{A}_N = \boldsymbol{J}_1 \boldsymbol{A}_N \boldsymbol{\Phi}.}$$

Each column of $\boldsymbol{A}_N$ is a geometric progression. Shifting the array by one antenna amounts to multiplying this column by:

$$e^{-jkd\sin\theta_\ell}.$$

ESPRIT is nothing other than the exploitation of this multiplication.

16.2 Signal Subspace and Invariance Relation

Instead of $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ directly (unknowns), ESPRIT works with the estimated signal subspaces. Let $\boldsymbol{U}_s \in \mathbb{C}^{N\times L}$ be the signal subspace of $\boldsymbol{R}$. Like $\operatorname{Im}(\boldsymbol{U}_s) = \operatorname{Im}(\boldsymbol{A}_N)$, there exists an invertible matrix $\boldsymbol{T} \in \mathbb{C}^{L\times L}$ such that

$$\boldsymbol{U}_s = \boldsymbol{A}_N \boldsymbol{T}.$$

By selecting the subarrays:

$$\boldsymbol{U}_{s,1} \equiv \boldsymbol{J}_1 \boldsymbol{U}_s = \boldsymbol{A}_1 \boldsymbol{T}, \qquad \boldsymbol{U}_{s,2} \equiv \boldsymbol{J}_2 \boldsymbol{U}_s = \boldsymbol{A}_2 \boldsymbol{T} = \boldsymbol{A}_1 \boldsymbol{\Phi} \boldsymbol{T}.$$

Thus

$$\boldsymbol{U}_{s,2} = \boldsymbol{U}_{s,1}\,(\boldsymbol{T}^{-1}\boldsymbol{\Phi}\,\boldsymbol{T}) \equiv \boldsymbol{U}_{s,1}\,\boldsymbol{\Psi}.$$

Theorem 16.1 (ESPRIT Invariance Relation)**.** *There is a matrix* $\boldsymbol{\Psi} \in \mathbb{C}^{L\times L}$ *such as* $\boldsymbol{U}_{s,2} = \boldsymbol{U}_{s,1}\,\boldsymbol{\Psi}$*, and* $\boldsymbol{\Psi}$ *is similar to* $\boldsymbol{\Phi}$*. Its eigenvalues are therefore exactly* $z(\theta_\ell) = e^{-jkd\sin\theta_\ell}$*.*

16.3 Algorithm

1. Collect K snapshots and estimate $\widehat{\boldsymbol{R}}$.
2. Break down $\widehat{\boldsymbol{R}}$ and choose L.
3. Extract $\widehat{\boldsymbol{U}_s} \in \mathbb{C}^{N\times L}$ (subspace estimated signal).
4. Build $\widehat{\boldsymbol{U}}_{s,1} = \boldsymbol{J}_1 \widehat{\boldsymbol{U}}_s$ and $\widehat{\boldsymbol{U}}_{s,2} = \boldsymbol{J}_2 \widehat{\boldsymbol{U}}_s$.
5. Solve $\widehat{\boldsymbol{U}}_{s,2} = \widehat{\boldsymbol{U}}_{s,1}\widehat{\boldsymbol{\Psi}}$ in the sense of least squares (LS-ESPRIT) or least squares totals (TLS-ESPRIT).
6. Diagonalize $\widehat{\boldsymbol{\Psi}}$ and read its eigenvalues $\widehat{z}_\ell$.
7. Estimate DOAs: $\widehat{\theta}_\ell = \arcsin(-\arg(\widehat{z}_\ell)/(kd))$.

16.4 LS-ESPRIT and TLS-ESPRIT

16.4.1 L.S.

The problem $\widehat{\boldsymbol{U}}_{s,2} = \widehat{\boldsymbol{U}}_{s,1}\,\widehat{\boldsymbol{\Psi}}$ assumes $\widehat{\boldsymbol{U}}_{s,1}$ is correct. We solve by classical least squares:

$$\widehat{\boldsymbol{\Psi}}_{\mathrm{LS}} = (\widehat{\boldsymbol{U}}_{s,1}^{\mathrm{H}}\widehat{\boldsymbol{U}}_{s,1})^{-1}\,\widehat{\boldsymbol{U}}_{s,1}^{\mathrm{H}}\,\widehat{\boldsymbol{U}}_{s,2}.$$

LS-ESPRIT is simple and fast, but introduces a bias when $\widehat{\boldsymbol{U}}_{s,1}$ is as noisy as $\widehat{\boldsymbol{U}}_{s,2}$.

16.4.2 TLS

The *Total Least Squares* formulation assumes that both matrices contain noise. We form $\widetilde{\boldsymbol{U}} = [\widehat{\boldsymbol{U}}_{s,1}, \widehat{\boldsymbol{U}}_{s,2}] \in \mathbb{C}^{M\times 2L}$ and compute its SVD. The singular vectors associated with the smallest singular values give the TLS estimator, which is typically more accurate than LS at low SNR.

16.4.3 Practical Reading

LS-ESPRIT answers the question:

which matrix best maps $\widehat{\boldsymbol{U}}_{s,1}$ to $\widehat{\boldsymbol{U}}_{s,2}$?

TLS-ESPRIT answers a more symmetrical question:

which invariance relation is closest to the two noisy matrices?

As both subspaces are extracted from the same estimated covariance, they are both noisy. This is why TLS is often the default choice in demanding radar or telecom processing.

16.5 Single-Source Case

For $L = 1$, $\boldsymbol{\Phi}$ is scalar and so is the matrix $\boldsymbol{\Psi}$. The result is reduced to

$$\widehat{z} = \frac{\widehat{\boldsymbol{u}}_{s,1}^{\mathrm{H}}\,\widehat{\boldsymbol{u}}_{s,2}}{\widehat{\boldsymbol{u}}_{s,1}^{\mathrm{H}}\,\widehat{\boldsymbol{u}}_{s,1}},$$

or a simple division. The angle is $\widehat{\theta} = \arcsin(-\arg(\widehat{z})/(kd))$.

16.6 Comparison of MUSIC / ESPRIT

Criteria	MUSIC	ESPRIT
Angular scan	yes (or roots)	No
Geometry	any (except Root-MUSIC)	ULA (or invariance)
Cost	$\mathcal{O}(N^3 + NM)$	$\mathcal{O}(N^3 + NL^2)$
Accuracy (high SNR)	$\sim$ CRB	$\sim$ CRB
Accuracy (low SNR)	slightly better	slightly worse
Consistent sources	smoothing required	smoothing required
Calibration	sensitive	very sensitive

The two algorithms have very close performances in asymptotics of SNR or K. ESPRIT has an advantage in computational cost; MUSIC has an advantage in geometric genericity.

16.7 Important Variants

Unitary ESPRIT. Reformulation into real arithmetic via unitary transformations on centered data and their versions "forward-backward ". Cost reduced to a real SVD calculation. Substantial gain on FPGA.

Beamspace ESPRIT. Project in advance $\widehat{\boldsymbol{U}}_s$ in a subspace of reduced dimension; lower cost, comparable performance on restricted angular areas.

2D-ESPRIT. Extends ESPRIT to planar arrays (URA), with two matrices $\boldsymbol{\Psi}_x$ and $\boldsymbol{\Psi}_y$ for azimuth and elevation.

Multiple-invariance ESPRIT. Operates several subarrays shifted simultaneously, gain in precision at a slightly higher cost.

16.8 Validity Conditions

- **Translation invariance.** ESPRIT requires geometry where one sub-part of the array is identical to another up to one translation. True for ULAs, true for separable URAs, false for circular or random arrays.
- **$L < N$.** At most $N - 1$ sources for a ULA.
- **Geometric calibration.** A position error antenna breaks invariance and biases all estimates.
- **Sufficiently decorrelated sources.** Otherwise, smoothing prior, which reduces the effective opening.

16.9 Ambiguities and Spacing

For $d > \lambda/2$, $z(\theta) = e^{-jkd\sin\theta}$ is no longer bijective on the visible area: several angles can give the same z. ESPRIT then provides correct eigenvalues in z, but we can no longer go back *unambiguously* to θ. The solution: $d \leq \lambda/2$ or use of *unpacking* (*unwrapping phase*) techniques based on several subarrays of different spacing (*coprime arrays*).

16.10 Numerical Implementation

Listing 16.1: LS-ESPRIT and TLS-ESPRIT

```python
import numpy as np

def esprit(R, L, d_over_lambda=0.5, tls=True):
    """
    R : (N,N)
    L : nombre de sources
    tls : True for TLS-ESPRIT, False for LS-ESPRIT
    Retour : array de L DOA estimées (rad)
    """
    N = R.shape[0]
    assert R.shape == (N, N)
    assert 0 < L < N
    R = 0.5 * (R + R.conj().T)
```

```
    eigvals, eigvecs = np.linalg.eigh(R)
    Us = eigvecs[:, -L:]                    # sous-espace signal
    Us1 = Us[:-1, :]
    Us2 = Us[1:, :]

    if not tls:
        Psi = np.linalg.lstsq(Us1, Us2, rcond=None)[0]
    else:
        Z = np.hstack([Us1, Us2])           # M x 2L
        _, _, Vh = np.linalg.svd(Z, full_matrices=False)
        V = Vh.conj().T                     # 2L x 2L
        V12 = V[:L, L:]
        V22 = V[L:, L:]
        if np.linalg.cond(V22) > 1e12:
            raise np.linalg.LinAlgError("TLS-ESPRIT mal conditionné")
        Psi = -np.linalg.solve(V22.T, V12.T).T

    eig_psi, _ = np.linalg.eig(Psi)
    arg = -np.angle(eig_psi) / (2*np.pi*d_over_lambda)
    angles = np.arcsin(np.clip(arg, -1.0, 1.0))
    return np.sort(np.real(angles))
```

16.11 Comparison of Bartlett / Capon / MUSIC / ESPRIT

Criteria	**Bartlett**	**Capon**	**MUSIC**	**ESPRIT**
Principle	projected power	minimum variance	noise subspace	invariance
Angular scan	Yes	Yes	Yes	No
Resolution	classic	best	high	high
Geometry	general	general	general	invariant arrays
Consistent sources	visible but blurry	sensitive	smoothing required	smoothing required
Calibration sensitivity	weak	average	forte	very strong
Exit	spectrum	spectrum	pseudo-spectrum	direct angles

16.12 Summary

ESPRIT exploits an exact algebraic structure (the translation invariance of ULA) to transform DOA estimation into an eigenvalue problem $L \times L$. No scan, no grid, no discretization error. For well-calibrated ULAs, with reliable model order and non-coherent sources, it is one of the reference algorithms in production.

The next chapter sets the ultimate performance of *All* DOA estimator — MUSIC, ESPRIT, ML, and any future estimators — via *Cramer-Rao bounds*. This is a step often neglected in introductory presentations, and it's a shame: without an absolute reference, we cannot judge that an estimator "does " well. The CRB provides this reference: if an estimator approaches 0.5 dB asymptotically, we know that we have something; if it is at 10 dB, we know that there is room to gain.

Chapter 17

Cramer-Rao Bounds for DOA

MUSIC and ESPRIT achieve remarkable resolutions, but how accurate is *ultimate limit* for estimating directions of arrival? No estimator can do better than *Cramer-Rao bound* (CRB): it is the minimum variance of everything unbiased estimator, dictated only by the statistical model. This chapter establishes the CRBs for DOA — deterministic and stochastic — and compares to the performance of MUSIC, ESPRIT and maximum likelihood.

The role of the CRB in parametric statistics is analogous to that of the Shannon bound in information theory: it is an insurmountable theoretical ceiling which structures the entire field. We compare the estimators as we compare the modulations to the capacity of a channel. An estimator that reaches the CRB is said to be *efficient*; This is the best we can hope for. An estimator who is far from it has room — either to improve it, or to relax assumptions (complete with *a priori* information, etc.). Knowing the CRB is therefore, for those designing a system, an essential compass: it says both “we will not do better than that ” and “with enough effort, we can get there ”.

17.1 What Is the CRB Really For?

The CRB does not give the error of a particular algorithm. It gives a theoretical limit for any unbiased estimator compatible with the statistical model. It must therefore be interpreted correctly.

- She does not say that MUSIC or ESPRIT always achieve this error.
- It says that no unbiased estimator can have a variance weaker.
- It is used to compare the estimators in the asymptotic regime: MUSIC, ESPRIT, maximum likelihood, sparse estimators.
- It reveals the effect of physical parameters: SNR, number of snapshots, number of antennas, aperture, angular separation.

In an engineering project, it is used as a maturity test. If a simulated estimator is very far from the CRB in a favorable regime, there is probably an implementation, convention or calibration problem. If it gets close, further improving the algorithm will yield little: you have to act on the physical system, for example increasing N, K, the SNR or the aperture.

17.2 Cramer-Rao Bound: Generalities

17.2.1 Information from Fisher

Let $\boldsymbol{\theta} \in \mathbb{R}^p$ be a vector of parameters to be estimated, and $\boldsymbol{x}$ be an observation of density $p(\boldsymbol{x}; \boldsymbol{\theta})$. There *Fisher information matrix* East

$$\boldsymbol{F}(\boldsymbol{\theta})_{ij} = -\mathbb{E}\left[\frac{\partial^2 \log p(\boldsymbol{x}; \boldsymbol{\theta})}{\partial\theta_i \partial\theta_j}\right] = \mathbb{E}\left[\frac{\partial \log p}{\partial\theta_i}\frac{\partial \log p}{\partial\theta_j}\right].$$

Theorem 17.1 (Cramer-Rao bound)**.** *For any unbiased estimator $\widehat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$,*

$$\mathrm{Cov}(\widehat{\boldsymbol{\theta}}) \succeq \boldsymbol{F}^{-1}(\boldsymbol{\theta}),$$

in the sense of positive semidefinite matrices. In particular the variance of each component is reduced by the corresponding diagonal term of $\boldsymbol{F}^{-1}$.

An estimator that reaches this bound is said to be *efficient.* Maximum likelihood is efficient asymptotically (under conditions of regularity).

17.2.2 Circular Complex Gaussian Density

For $\boldsymbol{x} \sim \mathcal{CN}(\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{R}(\boldsymbol{\theta}))$ circular complex Gaussian,

$$\log p(\boldsymbol{x}; \boldsymbol{\theta}) = -N \log \pi - \log\det \boldsymbol{R}(\boldsymbol{\theta}) - (\boldsymbol{x} - \boldsymbol{\mu})^{\mathrm{H}}\, \boldsymbol{R}^{-1}(\boldsymbol{\theta})\, (\boldsymbol{x} - \boldsymbol{\mu}).$$

If $\boldsymbol{\mu}$ does not depend on $\boldsymbol{\theta}$, we obtain the *Slepian-Bangs formula*

$$\boxed{\boldsymbol{F}_{ij} = \mathrm{tr}\left(\boldsymbol{R}^{-1}\frac{\partial \boldsymbol{R}}{\partial\theta_i}\boldsymbol{R}^{-1}\frac{\partial \boldsymbol{R}}{\partial\theta_j}\right).}$$

17.3 Deterministic and Stochastic Models

For the DOA problem we distinguish two models depending on whether the envelopes $\boldsymbol{s}(n)$ are deterministic or stochastic.

17.3.1 Deterministic Model (Conditional)

The $\boldsymbol{s}(1), \ldots, \boldsymbol{s}(K)$ envelopes are unknown parameters to estimate in addition to $\boldsymbol{\theta}$. The model is $\boldsymbol{x}(n) = \boldsymbol{A}(\boldsymbol{\theta})\, \boldsymbol{s}(n) + \boldsymbol{n}(n)$ with $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N)$.

17.3.2 Stochastic Model (Unconditional)

The envelopes are random: $\boldsymbol{s}(n) \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{P}_s)$. The model is $\boldsymbol{x}(n) \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{R})$ with $\boldsymbol{R} = \boldsymbol{A}\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}} + \sigma^2 \boldsymbol{I}_N$.

The stochastic model is typically the most relevant in practice (passive sources, jammers, multipath), while the deterministic model better models known sources (pilots, signals emitted by oneself).

17.4 Deterministic CRB

Theorem 17.2 (Deterministic CRB). *Under the deterministic model, the CRB on $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_L)^{\mathrm{T}}$ from K snapshots is*

$$\boxed{\mathrm{CRB}_{\mathrm{det}}(\boldsymbol{\theta}) = \frac{\sigma^2}{2K} \left[\mathrm{Re}\Big\{(\boldsymbol{D}^{\mathrm{H}} \boldsymbol{P}_A^{\perp} \boldsymbol{D}) \odot \widehat{\boldsymbol{P}}_s^{\mathrm{T}}\Big\}\right]^{-1},}$$

where :

- $\boldsymbol{D} = [\dot{\boldsymbol{a}}(\theta_1), \ldots, \dot{\boldsymbol{a}}(\theta_L)]$ *with* $\dot{\boldsymbol{a}} = \partial \boldsymbol{a}/\partial\theta$*;*
- $\boldsymbol{P}_A^{\perp} = \boldsymbol{I}_N - \boldsymbol{A}(\boldsymbol{A}^{\mathrm{H}}\boldsymbol{A})^{-1}\boldsymbol{A}^{\mathrm{H}}$ *is the spotlight on the orthogonal complement of* $\mathrm{Im}(\boldsymbol{A})$*;*
- $\widehat{\boldsymbol{P}}_s = \frac{1}{K}\sum_n \boldsymbol{s}(n)\boldsymbol{s}^{\mathrm{H}}(n)$ *is the empirical covariance of the sources;*
- $\odot$ *is the product of Hadamard (component by component).*

17.5 Stochastic CRB

Theorem 17.3 (Stochastic CRB). *Under the stochastic model, the CRB is*

$$\boxed{\mathrm{CRB}_{\mathrm{sto}}(\boldsymbol{\theta}) = \frac{\sigma^2}{2K} \left[\mathrm{Re}\Big\{(\boldsymbol{D}^{\mathrm{H}} \boldsymbol{P}_A^{\perp} \boldsymbol{D}) \odot (\boldsymbol{P}_s \boldsymbol{A}^{\mathrm{H}} \boldsymbol{R}^{-1} \boldsymbol{A} \boldsymbol{P}_s)^{\mathrm{T}}\Big\}\right]^{-1}.}$$

Remark 17.4 (Relationships Between the Bounds). We demonstrate that $\mathrm{CRB}_{\mathrm{sto}} \leq \mathrm{CRB}_{\mathrm{det}}$ asymptotically at large SNR. The stochastic model benefits from a more informative assumption (the distribution of sources is known), which can lead to weaker bounds. At low SNR, the two bounds coincide.

17.6 Single-Source Case on ULA

This is the most instructive case: all the factors are calculated explicitly.

17.6.1 Derivative of the Steering Vector

For a ULA at $d = \lambda/2$,

$$\dot{a}_m(\theta) = -j\pi m \cos\theta \, e^{-j\pi m \sin\theta}.$$

The squared norm of $\dot{\boldsymbol{a}}(\theta)$ is

$$\|\dot{\boldsymbol{a}}(\theta)\|^2 = \pi^2 \cos^2\theta \sum_{m=0}^{N-1} m^2 = \pi^2 \cos^2\theta \, \frac{(N-1)N(2N-1)}{6}.$$

17.6.2 Explicit CRB

For a single source of power p_s and white noise σ^2, we have $\boldsymbol{P}_A^{\perp}\boldsymbol{D} = \dot{\boldsymbol{a}} - (\boldsymbol{a}^{\mathrm{H}}\dot{\boldsymbol{a}}/N)\boldsymbol{a}$. For a ULA at $d = \lambda/2$, we have $\boldsymbol{a}^{\mathrm{H}}\dot{\boldsymbol{a}} = -j\pi\cos\theta \sum_m m = -j\pi\cos\theta \cdot N(N-1)/2$, and

$$\dot{\boldsymbol{a}}^{\mathrm{H}} \boldsymbol{P}_A^{\perp} \dot{\boldsymbol{a}} = \pi^2 \cos^2\theta \, \frac{N(N^2-1)}{12}.$$

Corollary 17.5 (CRB ULA, single source).

$$\boxed{\mathrm{CRB}_{\mathrm{sto}}(\theta) = \frac{6\sigma^2}{K\, p_s\, \pi^2 \cos^2\theta\, N(N^2-1)} \, \frac{1 + Np_s/\sigma^2}{Np_s/\sigma^2}.}$$

17.6.3 Asymptotic Behavior

At large SNR ($Np_s/\sigma^2 \gg 1$),

$$\mathrm{CRB}(\theta) \;\sim\; \frac{6}{K\,\pi^2\,\cos^2\theta\,N^3}\cdot\frac{\sigma^2}{p_s\,N} = \frac{6}{K\cdot\mathrm{SNR}_{\mathrm{array}}\cdot\pi^2\cos^2\theta\,N^3},$$

where $\mathrm{SNR}_{\mathrm{array}} = Np_s/\sigma^2$ is the SNR *After* beamforming DAS. Three lessons:

- decrease in $1/N^3$: doubling N divides the variance by 8;
- decay in $1/K$: double the number of snapshots divide by 2 ;
- decay in $1/\mathrm{SNR}$;
- factor $1/\cos^2\theta$: the precision degrades when we moves away from the broadside (at the endfire pole $\theta = \pm 90°$, $\cos\theta = 0$, the CRB diverges).

Law N^3. The factor $1/N^3$ is explained by: (i) a factor N for grating gain, (ii) a factor N^2 for quadratic aperture — a large aperture weighs more in a phase regression. This is *spatial Cramer-Rao law*: the angular precision varies as the inverse of the *cube* number of antennas, unlike the SNR gain which only increases in N.

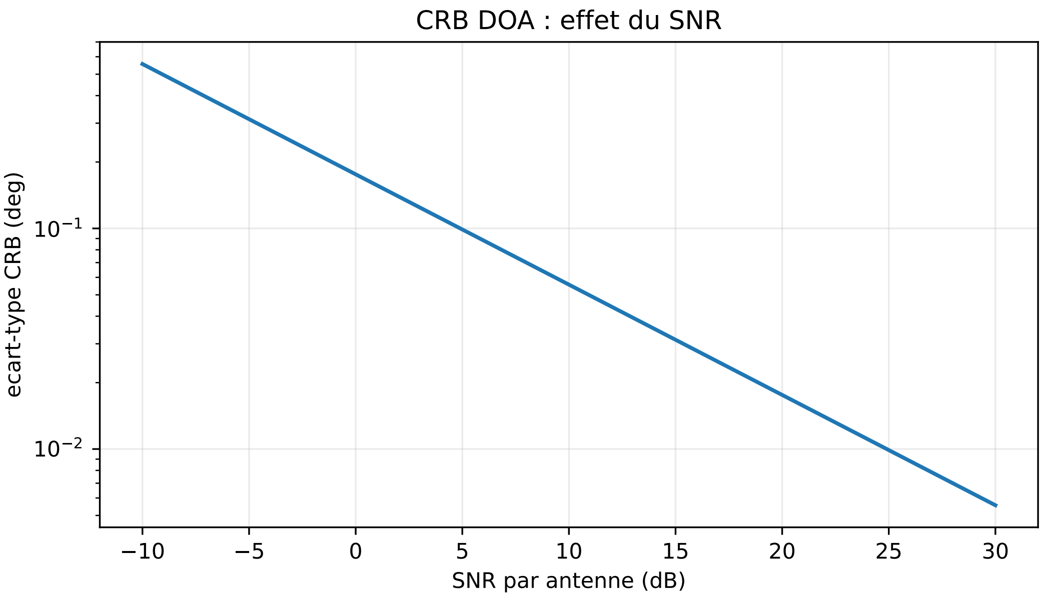

Figure 17.1: Theoretical standard deviation given by a simplified CRB (broadside source, $N = 8$, $K = 128$, $d = \lambda/2$), as a function of the SNR. The improvement with N is faster than a simple power gain, because the spatial aperture also increases.

17.7 Two Sources: Effect of Coupling

For two angle sources θ_1 and θ_2, the CRB for θ_1 contains a term that depends on $|\boldsymbol{a}^{\mathrm{H}}(\theta_1)\boldsymbol{a}(\theta_2)|^2$. When the two sources are angularly close, their dot product approaches N (the steering vectors are almost collinear), and the CRB explodes:

$$\mathrm{CRB}(\theta_1) \;\sim\; \frac{1}{1-|\rho|^2},$$

where $\rho = \boldsymbol{a}^{\mathrm{H}}(\theta_1)\boldsymbol{a}(\theta_2)/N$ is the *steering correlation coefficient*. This is the formalization intuition: two very close sources are fundamentally difficult to separate.

17.8 Resolution Limit

Combining CRB and detection threshold, we define the *statistical resolution limit* (Smith 2005)

$$\Delta\theta_{\min} \sim \sqrt{\frac{2}{\mathrm{CRB}^{-1}(\theta)}} \sim \frac{1}{N^{3/2}\sqrt{K\cdot\mathrm{SNR}}\,|\cos\theta|}.$$

Compared to the Rayleigh limit $\sim 2/N$, we have a *gain in resolution* of a factor $\sqrt{N\cdot K\cdot\mathrm{SNR}}$: this is the "super-resolution " that the subspace methods achieve.

17.9 Comparison with Practical Performance

17.9.1 Asymptotic MUSIC

For MUSIC, the asymptotic variance of $\widehat{\theta}_\ell$ in the framework studied by Stoica and Nehorai [13] is

$$\mathrm{Var}_{\mathrm{MUSIC}}(\widehat{\theta}_\ell) = \frac{\sigma^2}{2K}\,\frac{\left(\boldsymbol{P}_s\boldsymbol{A}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{A}\boldsymbol{P}_s\right)^{-1}_{\ell,\ell}}{\mathrm{Re}\{(\dot{\boldsymbol{a}}_\ell^{\mathrm{H}}\boldsymbol{P}_A^{\perp}\dot{\boldsymbol{a}}_\ell)\}}.$$

We demonstrate that $\mathrm{Var}_{\mathrm{MUSIC}} \geq \mathrm{CRB}_{\mathrm{sto}}$ with equality in two cases: (i) single source, (ii) sources of the same power and orthogonal in steering. This result says that *MUSIC is not efficient for two close sources*, but remains so asymptotically at large SNR.

17.9.2 Asymptotic ESPRIT

ESPRIT achieves performances very close to MUSIC but slightly less good asymptotically: the loss compared to the CRB is *more marked* for ESPRIT and for MUSIC. It is an argument in favor of MUSIC for configurations requiring precision, despite its higher cost.

17.9.3 Maximum Likelihood

Under the usual regularity assumptions, maximum likelihood (ML) is asymptotically efficient: it can reach the CRB when $K \to \infty$. Its computational cost is typically prohibitive (multi-dimensional optimization) but iterative methods (alternating projection, Newton, EM, IQML) make it tractable for $L \leq 4$ or 5.

17.10 Implementation: Numerical Computation of the CRB

Listing 17.1: Stochastic CRB for DOA

```
import numpy as np

def steering_and_derivative(theta, N, d_over_lambda=0.5):
    m = np.arange(N)
    phase = -2*np.pi * d_over_lambda * np.sin(theta) * m
    a = np.exp(1j * phase)
    da = a * (-1j * 2*np.pi * d_over_lambda * np.cos(theta) * m)
    return a, da

def crb_stochastic(thetas, Ps, sigma2, N, K, d_over_lambda=0.5):
    """
    thetas : (L,) angles (rad)
    Ps     : (L,L) covariance des sources
    sigma2 : noise variance
```

```
    K      : nombre de snapshots
    """
    L = len(thetas)
    assert Ps.shape == (L, L)
    assert sigma2 > 0.0 and K > 0 and N > L
    A = np.zeros((N, L), dtype=complex)
    D = np.zeros((N, L), dtype=complex)
    for ell, th in enumerate(thetas):
        a, da = steering_and_derivative(th, N, d_over_lambda)
        A[:, ell] = a
        D[:, ell] = da

    R = A @ Ps @ A.conj().T + sigma2 * np.eye(N)
    R = 0.5 * (R + R.conj().T)
    G = A.conj().T @ A
    if np.linalg.cond(G) > 1e12 or np.linalg.cond(R) > 1e12:
        raise np.linalg.LinAlgError("CRB mal conditionnée")
    PA_perp = np.eye(N) - A @ np.linalg.solve(G, A.conj().T)
    M = D.conj().T @ PA_perp @ D                  # L x L
    H = Ps @ A.conj().T @ np.linalg.solve(R, A) @ Ps
    F = 2 * K / sigma2 * np.real(M * H.T)
    return np.linalg.pinv(F)
```

17.11 Summary

The Cramér-Rao bounds set the ultimate performance of any unbiased DOA estimator. For a ULA at $d = \lambda/2$, the variance of the estimate from a single source varies as $1/(K \cdot \mathrm{SNR} \cdot N^3 \cdot \cos^2\theta)$: precision proportional to the cube of the number of antennas, inverse of the SNR and the number of snapshots, degradation at the edge of the visible zone. MUSIC and ESPRIT reach these limits asymptotically for a single source, but their deviation from the CRB widens for close sources or low SNR. Maximum likelihood is the efficient reference estimator when cost allows it.

The following chapter closes Part V with the methods *sparse and Bayesian*, which correspond to another type of problems: K very small, coherent sources, off-grid. These more recent methods are not in direct competition with MUSIC and ESPRIT: they cover different regimes. When MUSIC works (enough snapshots, incoherent sources), we use it; when it fails, sparse or SBL take over. A solid overview of DOA estimators must include both.

Chapter 18

Sparse and Bayesian DOA

MUSIC and ESPRIT can approach the Cramér-Rao bound in a favorable asymptotic regime, but are based on sometimes inadequate hypotheses: sufficient number of snapshots ($K \geq$ a few N), non-coherent sources, prior knowledge of L. For situations where these assumptions fail — single snapshot, coherent sources (multiple paths), unknown L — other approaches are necessary. This chapter introduces the methods *sparse* (compressed sensing applied to DOA) and *Bayesian* (*Sparse Bayesian Learning*), which emerged after 2005 and complete the classic arsenal.

These methods represent a change of point of view: we no longer seek to diagonalize a covariance, but to explain an observed signal as the sum of a small number of components chosen from a large dictionary of candidate directions. The problem becomes algebraically very different: no proper decomposition, no angular scan, but a convex program (or non-convex depending on the formulation) which forces the sparsity of the solution. The underlying theory is that of *compressed sensing* by Donoho [14] and Candès–Romberg–Tao [15], whose depth goes well beyond the scope of this book; we present here the versions adapted to the DOA, which prove to be particularly robust in regimes where traditional methods fail.

18.1 Reformulation as a Sparse Problem

18.1.1 Discretization of the Visible Region

We discretize the visible area into G candidate angles $\{\bar{\theta}_g\}_{g=1}^{G}$ with $G \gg N$ (often $G \sim 10N$ to $100N$). We build the *dictionary* overcomplete

$$\boldsymbol{D} = [\boldsymbol{a}(\bar{\theta}_1), \ldots, \boldsymbol{a}(\bar{\theta}_G)] \in \mathbb{C}^{N \times G}.$$

If the sources are at angles $\theta_1, \ldots, \theta_L$ and these angles fall (approximately) on the grid, we can write

$$\boldsymbol{x}(n) = \boldsymbol{D}\, \boldsymbol{s}_g(n) + \boldsymbol{n}(n),$$

where $\boldsymbol{s}_g(n) \in \mathbb{C}^G$ is a vector *sparse*: only the L components corresponding to the real sources are non-zero.

18.1.2 Why It Is Useful

This reformulation turns DOA into a problem *sparse media identification* of $\boldsymbol{s}_g(n)$. Benefits :

- no need to know L *a priori* ;
- a single snapshot may be enough (compressed sensing);
- insensitive to coherent sources;
- natural structure for additional constraints (temporal regularity, group structure).

Disadvantage: we are limited by the *grid resolution*. If a true source falls between two grid nodes, it is poorly represented. We are talking about *basis mismatch* or a problem *off-grid*. Several remedies exist (refined grid, continuous optimization, *atomic norm minimization*).

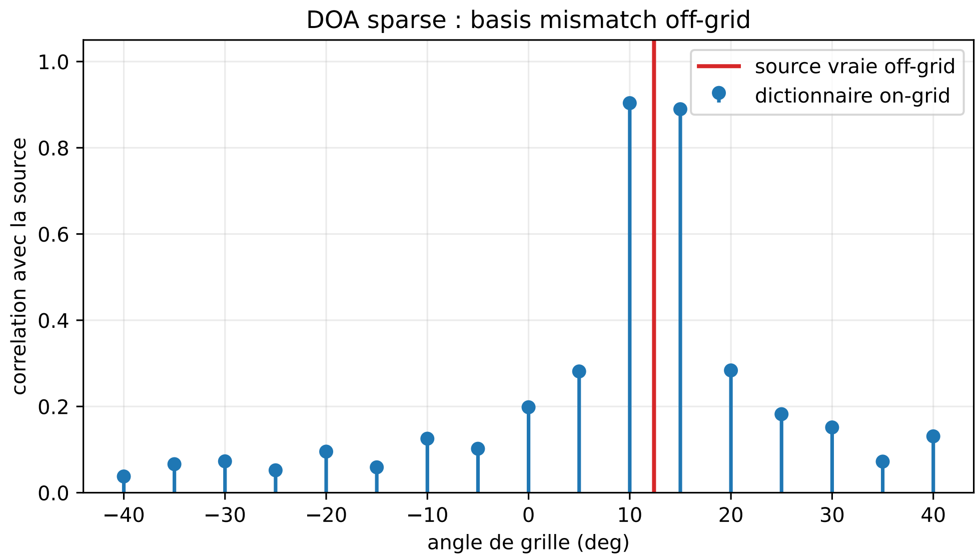

Figure 18.1: Visualization of the basis mismatch ($N = 12$, true source at 12.4°, grid all 5°): a source located between two grid points is projected onto several neighboring columns of the dictionary.

18.1.3 Identifiability Condition

The $\boldsymbol{D}$ dictionary is large: $G \gg N$. The problem $\boldsymbol{x} = \boldsymbol{D}\boldsymbol{s}_g$ therefore has an infinity of solutions if no constraint is added. Sparsity makes the problem identifiable when the active columns of $\boldsymbol{D}$ are not overly correlated. A simple measure is mutual consistency:

$$\mu_D = \max_{i \neq j} \frac{|\boldsymbol{d}_i^{\mathrm{H}} \boldsymbol{d}_j|}{\|\boldsymbol{d}_i\|_2 \|\boldsymbol{d}_j\|_2}.$$

The finer the grid, the more similar two neighboring columns are, and the more μ_D increases. We gain nominal resolution, but we make the numerical problem more poorly conditioned. This is a point often forgotten: refining the grid does not automatically make the estimate better.

A reading rule is:

$$\text{coarse grid} \Rightarrow \text{basis mismatch},$$

$$\text{coarse grid} \Rightarrow \text{nearly collinear columns}.$$

Sparse methods live in exactly this trade-off.

18.2 ℓ_1 and LASSO regularization

18.2.1 Single Snapshot

For a single snapshot $\boldsymbol{x}$, we solve

$$\min_{\boldsymbol{s}_g} \|\boldsymbol{x} - \boldsymbol{D}\,\boldsymbol{s}_g\|_2^2 + \mu\|\boldsymbol{s}_g\|_1.$$

The ℓ_1 penalty favors sparse solutions. This is the *LASSO* en estimation statistique, ou *Basis Pursuit Denoising* in signal processing. Any convex solver solves it (coordinate descent, ADMM, FISTA...).

The choice of μ is delicate. Practically:

- μ small: little sparsity, phantom sources;
- μ large: too sparse, we lose real sources.
- Heuristic: $\mu \approx \sigma\sqrt{2\log G}$ (threshold of detection of Donoho [14]).

18.2.2 Multiple Snapshots: $\ell_{2,1}$ MMV

For K snapshots stacked in $\boldsymbol{S}_g \in \mathbb{C}^{G\times K}$, we have $\boldsymbol{X} = \boldsymbol{D}\,\boldsymbol{S}_g + \boldsymbol{N}$. It is assumed that all columns share the same sparse support (*Multiple Measurement Vectors*, MMV). The problem becomes

$$\min_{\boldsymbol{S}_g} \|\boldsymbol{X} - \boldsymbol{D}\,\boldsymbol{S}_g\|_F^2 + \mu \sum_{g=1}^{G} \|\boldsymbol{S}_g[g,:]\|_2,$$

where $\boldsymbol{S}_g[g,:]$ is the g-th line. The $\ell_{2,1}$ standard penalizes the *number of non-zero lines*, therefore the number of sources.

18.3 ℓ_1-SVD

Calculating the LASSO MMV on K columns has a heavy cost. Malioutov, Çetin & Willsky [16] propose ℓ_1-SVD: we first compress the data by SVD, keeping only the first L' singular vectors (with L' chosen by MDL or by default $L' = N$), then we apply $\ell_{2,1}$-LASSO on this projection. Substantial calculation gain without loss of signal information.

18.4 Sparse Bayesian Learning (SBL)

18.4.1 Hierarchical Model

SBL [17, 18] places a hierarchical Gaussian prior on $\boldsymbol{s}_g$:

$$\boldsymbol{s}_g \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{\Gamma}), \quad \boldsymbol{\Gamma} = \mathrm{diag}(\gamma_1, \ldots, \gamma_G),$$

where each γ_g is itself a hyperparameter. The noise remains $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N)$.

The marginal distribution of $\boldsymbol{x}$ is

$$\boldsymbol{x} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N + \boldsymbol{D}\,\boldsymbol{\Gamma}\,\boldsymbol{D}^{\mathrm{H}}).$$

We learn the γ_g by maximum likelihood of the data; *most focus to zero automatically*, not leaving non-zero than those corresponding to the real sources. This is the *automatic selection* sparse support.

18.4.2 Algorithm

The classic algorithm alternates:

1. Step E: we calculate the mean and the covariance of $\boldsymbol{s}_g \mid \boldsymbol{x}$.
2. Step M: update the γ_g via statistics posterior.

The update rule is:

$$\gamma_g^{\mathrm{new}} = \frac{|\mu_g|^2 + \Sigma_{g,g}}{1},$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the posterior mean and covariance of $\boldsymbol{s}_g$ given $\boldsymbol{x}$.

18.4.3 Derivation of Posterior Quantities

With

$$\boldsymbol{x} = \boldsymbol{D}\boldsymbol{s}_g + \boldsymbol{n}, \qquad \boldsymbol{s}_g \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{\Gamma}), \qquad \boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}),$$

the posterior law remains Gaussian:

$$p(\boldsymbol{s}_g|\boldsymbol{x}) = \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

By completing the square in the Gaussian exponent, we obtain:

$$\boxed{\boldsymbol{\Sigma} = \left(\sigma^{-2}\boldsymbol{D}^{\mathrm{H}}\boldsymbol{D} + \boldsymbol{\Gamma}^{-1}\right)^{-1}}$$

And

$$\boxed{\boldsymbol{\mu} = \sigma^{-2}\boldsymbol{\Sigma}\boldsymbol{D}^{\mathrm{H}}\boldsymbol{x}.}$$

The update

$$\gamma_g \leftarrow |\mu_g|^2 + \Sigma_{g,g}$$

is therefore simply the average posterior energy of the component g. If one direction does not explain the data, its posterior mean and variance become low, and then γ_g contracts toward zero.

18.4.4 Multi-Snapshot Case

For multiple snapshots sharing the same media:

$$\boldsymbol{X} = \boldsymbol{D}\boldsymbol{S}_g + \boldsymbol{N},$$

the same idea gives a common variance γ_g per row of $\boldsymbol{S}_g$. The update becomes:

$$\boxed{\gamma_g^{\mathrm{new}} = \frac{1}{K}\sum_{n=1}^{K} |\mu_{g,n}|^2 + \Sigma_{g,g}.}$$

The average term on the snapshots strongly stabilizes the estimate. Multi-snapshot SBL is often more robust than LASSO-MMV when sources are close, because it jointly re-estimates the support and the component powers.

18.4.5 Advantages of SBL for DOA

- No regularization parameter to adjust (unlike LASSO and μ).
- Converges towards more sparse solutions than LASSO in practical.
- Superior performance at low SNR and low K.
- Simultaneous estimation of the number of sources: γ_g not negligible are automatically the sources.

18.4.6 Drawbacks

- $\mathcal{O}(N^3 + N^2 G)$ cost per iteration.
- No global guarantee of convergence (non-convex).
- Still subject to *basis mismatch*.

18.5 Atomic Norm Minimization (off-grid)

To get rid of the grid, we replace the discrete ℓ_1 with its continuous counterpart: the *atomic norm*

$$\|\boldsymbol{x}\|_{\mathcal{A}} = \inf\Big\{\sum_{\ell} c_\ell : \boldsymbol{x} = \sum_{\ell} c_\ell\, \boldsymbol{a}(\theta_\ell),\ c_\ell \geq 0\Big\}.$$

For a ULA, we demonstrate in particular in the off-grid work of Tang et al. [19] that this norm is calculated by a program *positive semidefinite* (SDP) on a Hermitian Toeplitz matrix. The estimation is then done by locating the spatial frequencies of the dual polynomial on the unit circle.

Advantage. Estimate *continue* in angle, without discretization, which can approach the asymptotic limits in favorable regimes.

Limit. Restricted to ULAs, SDP cost in $\mathcal{O}(N^{3.5})$.

18.5.1 SDP Form for a ULA

For a ULA, the steering vectors are complex exponentials. The atomic norm then has a semi-definite representation. A standard form consists of solving:

$$\min_{\boldsymbol{x}_0,\boldsymbol{T},t} \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{x}_0\|_2^2 + \tau\left(\frac{1}{2N}\operatorname{tr}(\boldsymbol{T}) + \frac{1}{2}t\right)$$

subject to:

$$\begin{bmatrix} \boldsymbol{T} & \boldsymbol{x}_0 \\ \boldsymbol{x}_0^{\mathrm{H}} & t \end{bmatrix} \succeq 0, \qquad \boldsymbol{T} \text{ Toeplitz hermitienne.}$$

The Toeplitz matrix encodes a continuous spectral measure. After resolution, the DOAs are extracted by Vandermonde decomposition of $\boldsymbol{T}$, or by localization of the zeros of the dual polynomial.

This formulation explains the profound difference with LASSO: LASSO chooses among G fixed columns; the atomic norm chooses from a continuum of columns. It eliminates the basis mismatch, but pays for this gain with a significantly higher optimization cost and a strong dependence on the ULA structure.

18.6 Common Interpretation Errors

Sparse methods are not a magic version of MUSIC. They solve another problem, with other weaknesses.

- A sparse peak is not always a source: a bad μ or a grid that is too correlated creates artificial supports.
- SBL automatically selects the media, but its optimization remains non-convex; initialization and noise scaling matter.
- The atomic norm avoids the grid, but assumes an ideal ULA model; an imperfect calibration reintroduces a mismatch.
- In a favorable asymptotic regime (large K, separate sources, calibrated manifold), MUSIC/ESPRIT often remain simpler and more predictable.

18.7 Variational Bayesian Methods

For the more modern versions (*variational SBL*, *Bayesian sparse learning* with structured models), we combine variational inference and group structures (sparse blocks, LASSO group, dynamic sparsity). We obtain robust estimators even for unknown $K = 1$ and L.

18.8 Comparison with MUSIC/ESPRIT

Criteria	**MUSIC**	**ℓ_1-SVD**	**SBL**
$K = 1$ snapshot	failure	possible	possible
Consistent sources	requires smoothing	ok	ok
L unknown	via MDL/AIC	automatic	automatic
Off-grid	not concerned	yes (basis mismatch)	yes (same)
Parameter to adjust	L or threshold	μ	none
Cost	$\mathcal{O}(N^3)$	$\mathcal{O}(N^2G)$	$\mathcal{O}(N^2G)$
Performance at low SNR	limited	Good	superior

18.9 Practical Recommendations

When to use sparse / Bayesian?

- single snapshot or $K < N$;
- coherent sources (multiple journeys) without being able to make smoothing;
- L unknown and difficult to estimate;
- available structural priors (temporal continuity, groups, spatial sparsity).

When to prefer MUSIC/ESPRIT?

- K of the order of a few N or more, non-coherent sources;
- need for rapid estimation (real time on FPGA);
- high asymptotic performance targeted (close to the CRB) on sources well separated.

18.10 Implementation: ℓ_1 DOA single snapshot

Listing 18.1: Single-snapshot DOA LASSO (simplified ADMM)

```
import numpy as np

def soft_thresh(z, t):
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_doa(x, D, mu, n_iter=200, rho=1.0):
    """
    x  : (N,) snapshot complexe
    D  : (N, G) dictionnaire de steerings
    mu : régularisation
    Retour : sg (G,) sparse
    """
    N, G = D.shape
```

```python
    assert x.shape == (N,)
    assert mu >= 0.0 and rho > 0.0 and n_iter > 0
    # ADMM sur problème complexe : on traite Re/Im séparément
    Dr = np.vstack([np.hstack([D.real, -D.imag]),
                    np.hstack([D.imag,  D.real])])
    xr = np.concatenate([x.real, x.imag])
    z = np.zeros(2*G); u = np.zeros(2*G); s = np.zeros(2*G)
    DtD = Dr.T @ Dr + rho * np.eye(2*G)
    if np.linalg.cond(DtD) > 1e12:
        raise np.linalg.LinAlgError("système ADMM mal conditionné")
    Dtx = Dr.T @ xr
    for _ in range(n_iter):
        s = np.linalg.solve(DtD, Dtx + rho*(z - u))
        z = soft_thresh(s + u, mu / rho)
        u += s - z
    return z[:G] + 1j*z[G:]
```

18.11 Summary

Sparse and Bayesian methods extend DOA estimation beyond the validity regime of MUSIC and ESPRIT:

- single snapshot and coherent sources: yes;
- number of sources unknown: self-selected;
- off-grid via *atomic norm* on ULA.

The cost is higher and the asymptotic guarantees weaker, but their flexibility makes them valuable in practical cases that classical “methods ” do not cover.

Part VI leaves the simplifying assumptions of narrow band, linear array and single antenna: it studies wideband beamforming, MIMO/Massive MIMO and advanced geometries. It is in this part that we find the most contemporary industrial systems: 5G, Wi-Fi 6, automobile radars, phased-array satellite antennas for low-orbit constellations. The formalisms remain the same as those developed in the previous parts; only the simplifying assumptions — narrow band, single polarization, single antenna on the transmitter side — are relaxed one by one.

Part VI

Extensions

Introduction to Extensions

The following chapters extend the basic model built so far: ULA array, narrowband, far field, single polarization and single-user reception. Each of these assumptions is useful for understanding the foundations, but each eventually gives way in modern systems.

Broadband introduces a frequency dependence of beamforming. MIMO adds antennas to the broadcast and turns the channel into a matrix. 2D/3D geometries replace the single angle with azimuth and elevation. Near field replaces simple angle pointing with range-angle focusing. Polarization adds vector components to the steering vector.

These chapters do not claim to cover each of these areas exhaustively. Each of them could fill a specialist book. Their role is more precise: to show how the initial hypotheses modify, which formulas change, which algorithms remain valid and which practical limits appear.

The guiding question therefore always remains the same:

what changes in the steering vector, covariance, and constraints?

If this question is clear, extensions are no longer a collection of special cases. They become controlled variations around the same theoretical core.

Chapter 19

Wideband Beamforming

The narrow band hypothesis (chapter 4), which reduces a time delay to a simple multiplication by $e^{-j2\pi f_0 \tau}$, is no longer valid when the condition $BD/c \ll 1$ is violated. This is the case in wideband radar, high-resolution SAR, sonar, microphone acoustics and certain satellite links. This chapter adapts the formalism to this regime. He introduces the *true time delay* (TTD), le *FIR filter per antenna*, and the *sub-band beamforming* ; he studies *frequency invariant beamforming* (FIB) and discusses hardware tradeoffs.

Leaving the narrow band is not just a matter of more demanding calculations: it is a change in the nature of the problem. In narrow band, a beamformer is entirely characterized by N complex weights; in broadband, $N \cdot L_h$ coefficients (one FIR filter per antenna) or equivalent sub-band processing are required. The design becomes an optimization problem with many more variables, and the beam pattern itself becomes a function *bidimensionnelle* de l'angle *And* of frequency. All the algorithms of the previous parts admit a broadband extension, but this extension imposes non-trivial architectural choices which are an integral part of the system design.

Criteria	**Narrow band**	**Broadband**
Spatial model	phase at f_0	delay or subbands
Weight	N complex scalars	NL_h coefficients or M sub-bands
Pattern	$B(\theta)$	$B(\theta, f)$
Main risk	lobes / mismatch	beam squint, temporal misalignment
Implementation	phasers	TTD, FIR, FFT by sub-bands
Key question	where to point?	where to point at each frequency?

19.1 Why Narrowband Is No Longer Enough

Let's take a look at the differential delay $\tau_m = md \sin\theta / c$. For a narrowband signal, the signal received on the m antenna is $x_m(t) \approx s(t)\, e^{-j2\pi f_0 \tau_m}$, and the output of a DAS that compensates τ_m with $e^{+j2\pi f_0 \tau_m}$ is $s(t)$ at all frequencies — exactly.

If the band is wide, the envelope $s(t)$ varies significantly between t and $t - \tau_{\max}$. Pure phase compensation compensates the carrier but not the envelope, and the signal aligned on two extreme antennas of the *no longer coincides* array. The array gain drops, the beam pattern becomes *frequency-dependent*, and the adaptive algorithms underperform or diverge.

Quantitatively, the narrow band validity condition was

$$\frac{BD}{c} \ll 1.$$

Beyond that, two families of techniques are essential: the *delay lines* (TTD) which apply a real temporal delay, and the *sub-bands* which transform the broadband signal into several narrowband subsignals processed independently.

19.2 beam squint

The most visible defect of a pure phase broadband beamformer is *beam squint*. The weights are calculated for a central frequency f_0:

$$w_m = \frac{1}{N} e^{-j2\pi f_0 m d \sin\theta_0/c}.$$

At a frequency $f \neq f_0$, the maximum of the beam no longer appears at θ_0, but at the angle θ_f which approximately verifies:

$$f \sin\theta_f = f_0 \sin\theta_0.$$

Thus:

$$\sin\theta_f = \frac{f_0}{f}\sin\theta_0.$$

The main lobe slides with frequency. The larger the relative band B/f_0 and the further θ_0 is from the broadside, the more visible the depointing is.

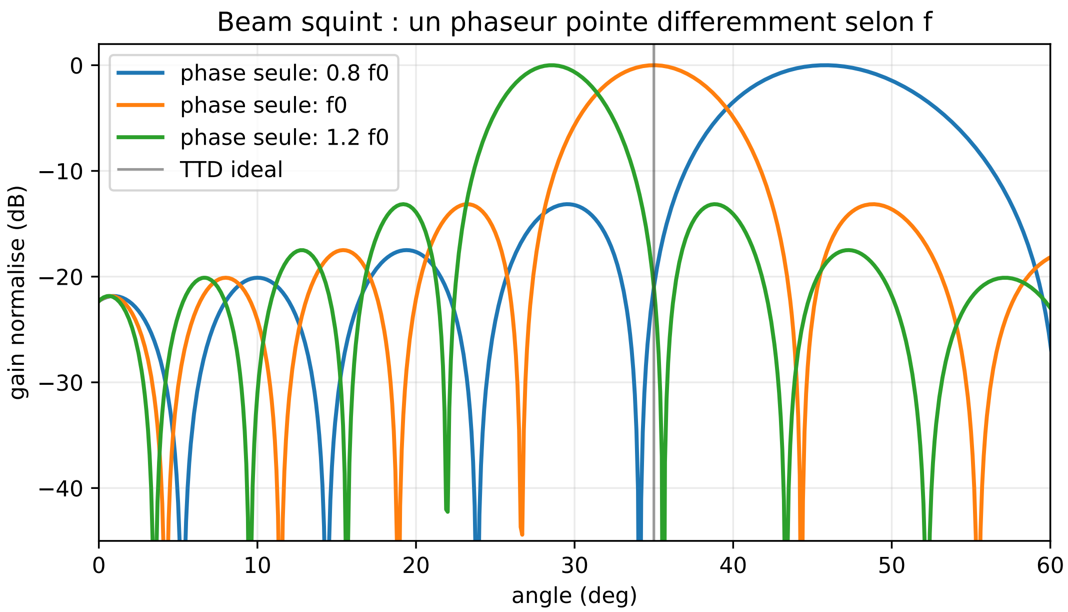

Figure 19.1: Beam squint with a phase-shifter array ($N = 16$, pointing 35°, $d = \lambda_0/2$, frequencies $0.8f_0$, f_0, $1.2f_0$): the weights calculated at f_0 point correctly at the central frequency, but the lobe shifts outside the central frequency.

The true time delay fixes this problem because it applies a real time delay, the phase of which varies automatically with f:

$$e^{-j2\pi f \tau_m}.$$

A fixed phase shifter can only be accurate at one frequency; a delay line can remain accurate across the entire band.

19.3 True Time-Delay Beamforming

19.3.1 Principle

Rather than multiplying by a constant phase, we actually delay the signal from antenna m by τ_m. The output of the beamformer is

$$y(t) = \frac{1}{N}\sum_{m=0}^{N-1} x_m(t+\tau_m) \quad \text{with } \tau_m = -md\sin\theta_0/c.$$

The applied delay *Exactly* cancels the physical delay, independent of frequency. The result is a *beam pattern frequency-invariant* in the direction θ_0.

19.3.2 Implementation

Analog. Variable delay lines (LdR). Carried out historically with cables of variable lengths (electronically scanned radar) or *photonics* lines (more recent technology). Précision sub-picoseconde réalisable.

Digital. At a sampling rate F_s, one can apply a subsample delay by interpolation. A fractional delay $\tau_m = (k_m + \alpha_m)/F_s$ is broken down into:

- integer delay k_m: index offset;
- fractional delay α_m: filtering by an FIR of *filtre Lagrange* ou *windowed sinc filter*.

Cost: one FIR filter per antenna. Accuracy: depending on the filter order and the processed band; orders of a few dozen to a hundred taps are common, but the effective resolution must be verified by simulation or measurement.

19.4 Sub-band beamforming

19.4.1 Decomposition

$x_m(t)$ is filtered by a bank of P narrow bandpass filters of width B/P. Inside each p sub-band, the signal is quasi-narrow-band and a classic narrow-band beamforming is applied to it with a steering vector depending on the central frequency f_p of the sub-band:

$$a_m^{(p)}(\theta) = e^{-j2\pi f_p \, md \sin\theta/c}.$$

19.4.2 Recombination

The outputs of the P subbands are summed (or filtered-then-summed) to reconstruct the full wideband signal. If the filter bank has perfect reconstruction (cosine modulated, DFT modulated), the original signal is recovered exactly.

19.4.3 Advantages and Limitations

Benefits.

- Direct reuse of all narrowband algorithms (MVDR, LCMV, MUSIC) within each sub-band.
- Efficient implementation by FFT (DFT *filterbank*).
- Allows independent adaptation per sub-band.

Boundaries.

- Cost: P narrow-band processing per snapshot.
- Edge of sub-bands: need for overlaps to avoid artifacts.
- Choice of P: too small, the narrow band hypothesis remains violated; too big, you multiply the cost.

Practical rule. We choose P such as $B/P \cdot D/c \ll 1$, or $P \gg BD/c$.

19.5 Beamforming with One FIR Filter per Antenna

A unified formulation, more general than the TTD and the sub-bands, is *filter-and-sum beamformer*: we apply to each antenna m an FIR filter $\boldsymbol{h}_m(t)$ of length L_h, then we sum:

$$y(t) = \sum_{m=0}^{N-1} \sum_{k=0}^{L_h-1} h_m[k]\, x_m(t - kT_s).$$

We then have NL_h degrees of freedom — much more than the N degrees of narrow-band beamforming. This addition allows:

- to control the beam pattern *at each frequency*;
- to impose a frequency-invariant response by construction;
- to optimize complex criteria (wideband LCMV, MVDR broadband).

19.5.1 Matrix Formulation

We stack:

$$\boldsymbol{x}_t(t) = [x_0(t), x_0(t - T_s), \ldots, x_0(t - (L_h - 1)T_s), x_1(t), \ldots, x_{N-1}(t - (L_h - 1)T_s)]^{\mathrm{T}} \in \mathbb{C}^{NL_h}.$$

The weight vector $\boldsymbol{w} \in \mathbb{C}^{NL_h}$ contains the NL_h coefficients. The output is $y(t) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}_t(t)$. The algorithms (MVDR, LCMV, etc.) apply identically, up to the size of the problem.

19.5.2 Wideband Steering Vector

At frequency f, the steering vector "increased " is

$$\boldsymbol{a}_t(\theta, f) = \boldsymbol{a}(\theta, f) \otimes \boldsymbol{d}(f),$$

where $\boldsymbol{d}(f) = [1, e^{-j2\pi f T_s}, \ldots, e^{-j2\pi f (L_h-1)T_s}]^{\mathrm{T}}$ is the vector of time delays and $\otimes$ the Kronecker product.

19.6 Frequency Invariant Beamforming (FIB)

19.6.1 Objective

Design a beam pattern that is *frequency independent* over the entire useful band: a beam as clean at 1 GHz as at 2 GHz. Essential for wideband acoustic applications (teleconferencing, active control) and very wideband radars.

19.6.2 Classical Methods

Beam pattern defined by construction. We define a desired spatial response $B_{\text{desired}}(\theta)$ *independent of* f, and we optimize the frequency weights $\boldsymbol{w}(f)$ to approximate it at each f. The problem becomes a family of independent quadratic problems.

Self-similar element arrays. By reusing the same geometric response at all frequencies — typically a *nested array* at $d, 2d, 4d, \ldots$ spacings —, we can design a beam pattern whose resolution changes with frequency but whose normalized shape remains constant.

Constraint-based design. We impose constraints linear gain $B(\theta_q, f_p) = b_{q,p}$ on a grid (θ, f), and we solve by extended LCMV.

19.7 Statistical Models for Wideband Noise

Broadband noise is typically colored in frequency (LNA response, non-flat thermal floor). It is characterized by its frequency-dependent covariance $\sigma^2(f)\boldsymbol{I}_N$ (spatially white noise but colored in frequency) or by a complete spatio-frequency matrix. The sub-band MVDR and LCMV algorithms automatically handle the case where σ^2 varies with f.

19.8 Wideband DOA

To estimate broadband DOAs, several approaches exist.

Subbands + MUSIC. We apply MUSIC in each sub-band, then we combine the pseudo-spectra (sum, product or average).

Coherent Signal-Subspace Method (CSSM). We focus the sub-bands towards a reference frequency by a unitary transformation, then we apply MUSIC on the focused "covariance ".

Wideband ESPRIT. Extension of ESPRIT to the multi-frequency case.

Subspace Fitting test. Direct optimization on the matrix increased steering.

19.9 Subband Implementation

Listing 19.1: Beamforming par sous-bandes via FFT

```python
import numpy as np

def subband_beamform(X, theta0, N, fs, fc, d, c=3e8, M_fft=256, hop=128):
    """
    X      : (N, T) snapshots à fs, signal large bande
    theta0 : direction cible (rad)
    fc     : center frequency
    fs     : sampling frequency
    d      : espacement inter-antennes (m)
    M_fft  : taille de FFT
    hop    : pas entre FFT successives
    """
    T = X.shape[1]
    n_frames = (T - M_fft) // hop + 1
    y = np.zeros((n_frames, M_fft), dtype=complex)
    freqs = fs * np.arange(M_fft) / M_fft - fs/2  # symétrique
    m = np.arange(N)
    for fr in range(n_frames):
        seg = X[:, fr*hop : fr*hop + M_fft]
        # FFT par antenne
        S = np.fft.fft(seg, axis=1)
        for bin_idx in range(M_fft):
            fk = fc + freqs[bin_idx]  # ou directement freqs si déjà en bande de base
            a = np.exp(-1j * 2*np.pi * fk * d * np.sin(theta0) / c * m)
            w = a / N
            y[fr, bin_idx] = w.conj() @ S[:, bin_idx]
    # IFFT back to time
    y_time = np.fft.ifft(y, axis=1)
    return y_time
```

19.10 Summary

Wideband beamforming extends the formalism for signals that violate the narrowband hypothesis. Three families of techniques:

- *true time delay*: analog or via fractional FIR by antenna;
- *sub-bands*: decomposition by filter bank / FFT, independent narrowband beamforming;
- *filter-and-sum*: NL_h weight, full beam control pattern at all frequencies.

The hardware and algorithmic cost increases significantly, but modern techniques (fast filter banks, dedicated FPGAs, RF photonics) make it feasible.

The following chapter addresses the other major extension of modern beamforming: the *MIMO* and the *massive MIMO*, where beamforming is no longer only reception but also transmission, and where hybrid analog/digital hardware architectures become essential. The contrast is instructive: Part I assumed a single scalar transmitter and a receiver array; 5G often reverses the problem by placing the array on the transmitter (base station) side and a single- or multi-antenna terminal on the receiver side. The same algebraic tools — steering vectors, beam patterns, decomposition into subspaces — apply symmetrically to both configurations.

Chapter 20

MIMO and Massive-MIMO Beamforming

All the previous chapters considered beamforming in *reception* with a single user (a single target direction). Modern systems — 5G, recent Wi-Fi, satellites — exploit several *at both ends* antennas on the link (*MIMO*, *Multiple Input Multiple Output*) and a large number of antennas on the base station side (*massive MIMO*). Beamforming becomes both a *adaptive reception* tool and a *precoding on broadcast* tool. This chapter adapts the formalisms developed to this context.

MIMO doesn't just add antennas; it transforms the very notion of a communication channel. Where a classic link can be seen as a single scalar "pipe ", a MIMO link is a *parallel pipe assembly*, the number of which is limited by the rank of the channel and therefore, in practice, by the minimum between the number of transmitting and receiving antennas. This multiplicity opens up two uses: the *spatial multiplexing*, where each pipe carries an independent flow; and the *diversity beamforming*, where multiple pipes carry the same information encoded to resist fading. Current systems combine the two adaptively depending on propagation conditions.

20.1 MIMO Channel Model

20.1.1 Point-to-Point Link

Consider a transmitter with N_T antennas transmitting to a receiver with N_R antennas. The channel is modeled by a matrix $\boldsymbol{H} \in \mathbb{C}^{N_R \times N_T}$ such that

$$\boldsymbol{y} = \boldsymbol{H}\,\boldsymbol{x} + \boldsymbol{n},$$

with $\boldsymbol{x} \in \mathbb{C}^{N_T}$ the transmitted vector and $\boldsymbol{y} \in \mathbb{C}^{N_R}$ the received vector. The $\boldsymbol{H}$ matrix contains the complex propagation coefficients from each transmitting antenna to each receiving antenna. It is typically constant over a *consistency block* (duration during which it can be considered stationary).

20.1.2 Link with the DOA Formalism

If the channel results from a few direct or reflected paths, it can be broken down into

$$\boldsymbol{H} = \sum_{\ell=1}^{L} \alpha_\ell\, \boldsymbol{a}_R(\theta_{R,\ell})\, \boldsymbol{a}_T^{\mathrm{H}}(\theta_{T,\ell}),$$

where $\boldsymbol{a}_R$ and $\boldsymbol{a}_T$ are the steering vectors of the two arrays, and α_ℓ the complex amplitudes of the paths. It is exactly the structure *even* of the multi-source steering matrix of the previous chapters, but bilateral.

Special case: a single journey. For a single LOS channel, $\boldsymbol{H} = \alpha\, \boldsymbol{a}_R(\theta_R)\, \boldsymbol{a}_T^{\mathrm{H}}(\theta_T)$ is of rank 1. All degrees of freedom of the channel are concentrated in a single direction.

Special case: Rayleigh channel i.i.d. In a very diffracting, $\boldsymbol{H}$ has Gaussian elements i.i.d. ; its singular values are approximately uniformly distributed, and the channel can transmit $\min(N_T, N_R)$ independent streams simultaneously. This is the principle of *spatial multiplexing.*

20.2 Singular-Value Decomposition of the Channel

Theorem 20.1 (SVD of the MIMO Channel)**.** *Any channel $\boldsymbol{H} \in \mathbb{C}^{N_R\times N_T}$ admits decomposition*

$$\boldsymbol{H} = \boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^{\mathrm{H}},$$

where $\boldsymbol{U} \in \mathbb{C}^{N_R\times N_R}$, $\boldsymbol{V} \in \mathbb{C}^{N_T\times N_T}$ are unitary and $\boldsymbol{\Sigma} \in \mathbb{R}^{N_R\times N_T}$ contains the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

Diagonalization of the channel. If the transmitter precodes with $\boldsymbol{x} = \boldsymbol{V}\,\tilde{\boldsymbol{x}}$ and the receiver applies $\tilde{\boldsymbol{y}} = \boldsymbol{U}^{\mathrm{H}}\boldsymbol{y}$, the effective channel becomes

$$\tilde{\boldsymbol{y}} = \boldsymbol{\Sigma}\,\tilde{\boldsymbol{x}} + \boldsymbol{U}^{\mathrm{H}}\boldsymbol{n},$$

which is diagonal: $r = \mathrm{rg}(\boldsymbol{H})$ parallel scalar subchannels, each with gain σ_i. This *spatial decomposition* is the basis of MIMO multiplexing.

20.3 Two Different Objectives: Diversity and Multiplexing

The word MIMO covers two objectives that must be distinguished.

Spatial diversity. The same information is transmitted through several spatial paths. The goal is to reduce the likelihood of deep fainting. The throughput does not necessarily increase, but the link becomes more reliable.

Spatial multiplexing. We transmit several independent streams in parallel. If the channel has the rank r, we can ideally transmit up to r flow:

$$r = \mathrm{rg}(\boldsymbol{H}) \leq \min(N_T, N_R).$$

Throughput increases, but only if the singular subchannels are strong enough.

The SVD allows you to see both regimes. If a single singular value dominates, the system works mainly in diversity beamforming. Si plusieurs valeurs singulières sont comparables, le multiplexage spatial devient intéressant.

20.3.1 Point-to-Point Capacity

If the transmitter knows $\boldsymbol{H}$ and distributes the power over the singular modes, the capacity is:

$$C = \max_{\sum_i p_i \leq P} \sum_{i=1}^{r} \log_2\left(1 + \frac{p_i\sigma_i^2}{\sigma_n^2}\right).$$

The solution is *water filling*:

$$p_i = \left(\nu - \frac{\sigma_n^2}{\sigma_i^2}\right)_+,$$

where ν is chosen to respect the total power. Low modes receive no power. This formula gives a very concrete reading: adding antennas is not enough; we must create truly usable spatial modes.

20.4 Transmit Precoding

on the show beamforming consists of choosing the transmitted vector $\boldsymbol{x}$ (from a scalar signal s) to maximize some objective at the receiver. The classic choices:

Matched filter / MRT (Maximum Ratio Transmission). THE receiver receives $\boldsymbol{y} = \boldsymbol{H}\,\boldsymbol{w}_T\,s + \boldsymbol{n}$, with $\boldsymbol{w}_T \in \mathbb{C}^{N_T}$. Maximizing $|\boldsymbol{H}\boldsymbol{w}_T|^2$ gives $\boldsymbol{w}_T \propto \boldsymbol{H}^{\mathrm{H}}\boldsymbol{u}$ for some $\boldsymbol{u}$; if the receiver also applies an MRC, the end-to-end SNR is maximized by $\boldsymbol{w}_T \propto \boldsymbol{v}_1$, the first right singular vector of $\boldsymbol{H}$.

Zero-Forcing (ZF). For K users with channels $\boldsymbol{H}_1, \ldots, \boldsymbol{H}_K$, the ZF requires that each user not see the interference of others:

$$\boldsymbol{W}_{\mathrm{ZF}} = \boldsymbol{H}^{\mathrm{H}}\,(\boldsymbol{H}\boldsymbol{H}^{\mathrm{H}})^{-1},$$

where $\boldsymbol{H} = [\boldsymbol{H}_1^{\mathrm{T}}, \ldots, \boldsymbol{H}_K^{\mathrm{T}}]^{\mathrm{T}}$ stacks all channels. Reversal possible if $K \le N_T$. Disadvantage: amplifies noise if the channel is poorly conditioned.

Regularized ZF (RZF, MMSE-precoding). Regularized by $\boldsymbol{W} = \boldsymbol{H}^{\mathrm{H}}(\boldsymbol{H}\boldsymbol{H}^{\mathrm{H}} + \sigma^2/p \cdot \boldsymbol{I})^{-1}$, analogue of MVDR for broadcast. Optimized noise/interference compromise.

20.5 Massive MIMO

20.5.1 Diet $N_T \gg K$

When $N_T \to \infty$ (with K fixed users), remarkable phenomena appear:

- $\boldsymbol{h}_k$ channels between distinct users become *asymptotically orthogonal* : $\boldsymbol{h}_i^{\mathrm{H}}\boldsymbol{h}_j/N_T \to 0$;
- the noise “is averaged ” on the N_T antennas;
- matched-filter (MRT) precoding $\boldsymbol{w}_k \propto \boldsymbol{h}_k$ *enough*: ZF and RZF no longer contribute anything asymptotically;
- rapid fainting disappears (*channel hardening*).

This is the fundamental argument of Marzetta (2010) who launched massive MIMO as the 5G paradigm.

20.5.2 Demonstration of Channel Hardening

Suppose an i.i.d. channel

$$\boldsymbol{h}_k \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I}_{N_T}).$$

Thus:

$$\frac{1}{N_T}\|\boldsymbol{h}_k\|^2 = \frac{1}{N_T}\sum_{m=1}^{N_T}|h_{k,m}|^2 \xrightarrow[N_T\to\infty]{} 1.$$

By the law of large numbers, the channel norm becomes almost deterministic. Likewise, for two independent users:

$$\frac{1}{N_T}\boldsymbol{h}_i^{\mathrm{H}}\boldsymbol{h}_j \xrightarrow[N_T\to\infty]{} 0.$$

The channels are made orthogonal by the dimension. This explains why a simple MRT can become sufficient: a user’s beam naturally has little projection onto other users.

This demonstration is asymptotic. In practice, spatial correlation, common paths, calibration errors and pilot contamination slow down convergence towards this ideal regime.

20.5.3 MRT Precoding in Massive MIMO

In the presence of K users and estimated channels $\widehat{\boldsymbol{h}}_k$, the MRT (*Maximum Transmission Ratio*) chooses

$$\boldsymbol{w}_k = \frac{\widehat{\boldsymbol{h}}_k}{\|\widehat{\boldsymbol{h}}_k\|},$$

and the transmitted signal is $\boldsymbol{x} = \sum_k \sqrt{p_k}\, \boldsymbol{w}_k\, s_k$. Cost: $\mathcal{O}(N_T K)$ per symbol. It is asymptotically optimal, and often very close to optimal in favorable massive-MIMO regimes (N_T large, channels sufficiently separated in space).

20.5.4 Channel Estimation and TDD Reciprocity

Massive MIMO relies on channel knowledge on the base station side. In FDD (*Frequency Division Duplex*) the *feedback* of the $N_T K$ coefficients would be prohibitively expensive. In TDD (*Time Division Duplex*), channel reciprocity allows the base station to *estimate* $\boldsymbol{H}$ from the uplink (where users send mutually orthogonal pilots) and *use it* on the downlink. This is why massive MIMO is mainly deployed in TDD.

20.5.5 Pilot Contamination

Fundamental limitation of massive MIMO: if two cells reuse the same pilots, their channel estimates contaminate each other. Inter-cell interference remains and limits performance even when $N_T \to \infty$. Solutions: globally orthogonal pilots, hybrid schemes, machine learning on spatial covariances.

20.5.6 Beamforming View of Contamination

If user k and a user from a neighboring cell use the same pilot, the estimate becomes approximately:

$$\widehat{\boldsymbol{h}}_k = \boldsymbol{h}_k + \boldsymbol{h}_{\text{cont}} + \boldsymbol{e}.$$

The MRT then builds:

$$\boldsymbol{w}_k \propto \widehat{\boldsymbol{h}}_k.$$

The downlink beam therefore points partially towards the contaminating user. Even when N_T increases, this component does not average, because it is integrated into the estimated channel vector. This is a model error, not just noise. It is therefore structurally more dangerous than an independent thermal error.

20.6 Hybrid Analog/Digital Architectures

To establish the orders of magnitude, as soon as N_T reaches several tens or hundreds of antennas, equipping each antenna with a complete RF chain (LNA, mixer, ADC, DAC) becomes prohibitive in cost, energy and size. The parade: architecture *hybrids*, where a coarse analog precoding (N_T phase shifters / attenuators) is followed by fine digital precoding on a reduced number N_{RF} of RF channels.

20.6.1 Structure

$$\boldsymbol{x} = \boldsymbol{F}_{\text{RF}}\, \boldsymbol{F}_{\text{BB}}\, \boldsymbol{s},$$

where $\boldsymbol{F}_{\text{RF}} \in \mathbb{C}^{N_T \times N_{RF}}$ is the analog precoding (with unit modules: phasors) and $\boldsymbol{F}_{\text{BB}} \in \mathbb{C}^{N_{RF} \times K}$ the digital precoding. Typically $N_{RF} = K$ or $N_{RF} = 2K$.

20.6.2 Design

The design problem consists of approximating an optimal purely digital precoding $\boldsymbol{F}_{\text{opt}}$ by the product $\boldsymbol{F}_{\text{RF}}\boldsymbol{F}_{\text{BB}}$, under the constraint that $|\boldsymbol{F}_{\text{RF}}|_{i,j} = 1$ (unit modules). Non-convex problem, solved by:

- *matching pursuit* (OMP) algorithms that select the columns of $\boldsymbol{F}_{\text{RF}}$ in a steering vectors dictionary;
- alternate descent on $\boldsymbol{F}_{\text{RF}}$ and $\boldsymbol{F}_{\text{BB}}$;
- *manifold optimization* on the variety of dies to unit modules.

20.6.3 Performance

For $N_T \gg K$ and a geometric channel with few paths, hybrid precoding can closely approach the capacity of purely digital precoding, often at the cost of a small loss in favorable mmWave channels. This is the architecture deployed in *millimeter-wave 5G* (28 GHz, 39 GHz, 60 GHz) where the cost of N_T complete RF chains would be prohibitive.

20.7 Beamforming and MIMO in transmission

Seen from a beamforming point of view, MIMO precoding is:

- **MRT**: DAS in transmission, optimal in channel white noise unique.
- **ZF**: Null Steering in transmission, forces the gain to 0 towards other users.
- **RZF/MMSE**: MVDR in transmission, noise compromise / interference.
- **Hybrid precoding**: GSC structure for hardware forced.

All theoretical tools — linear constraints, convex optimization, subspaces — apply identically, up to the reinterpretation of the “snapshot ”.

20.8 Multiuser Operation and MIMO Capacity

For K users sharing the same resources, the maximum aggregate throughput (*sum rate*) is bounded by the capacity of the multi-user channel:

$$C_{\text{MU}} = \max_{\sum_k p_k \leq P} \sum_{k=1}^{K} \log_2\left(1 + p_k \frac{|\boldsymbol{w}_k^{\text{H}}\boldsymbol{h}_k|^2}{\sigma^2 + \sum_{j\neq k} p_j |\boldsymbol{w}_k^{\text{H}}\boldsymbol{h}_j|^2}\right).$$

The solution is typically non-trivial, obtained by *water filling* on the SVD of the global channel. In massive MIMO ($N_T \to \infty$), interference contamination becomes negligible and $C_{\text{MU}} \to K\log_2(1 + N_T\text{SNR})$: linear growth in K and logarithmic in N_T.

20.9 Link with Classical Receive Beamforming

All the adaptive algorithms studied in Part III and IV (MVDR, LCMV, LMS, RLS) are transposed directly to MIMO reception: each user has a specific interference-plus-noise covariance, and MVDR on this covariance provides the optimal beamformer. The specificity of MIMO is:

- the existence of a *reciprocal beamforming* in transmission;
- multiplexing *multiple streams* in parallel via SVD;
- the *joint estimate* of the channel and data.

20.10 Implementation: MRT massive MIMO precoding

Listing 20.1: MRT and RZF precoding

```
import numpy as np

def precoding_mrt(H, p):
    """
    H : (K, NT) canaux estimés
    p : (K,) puissances par utilisateur
    Retour : W (NT, K) précodeurs
    """
    K, NT = H.shape
    W = H.conj().T.copy()                # (NT, K)
    norms = np.linalg.norm(W, axis=0)
    W = W / (norms + 1e-12)
    W = W * np.sqrt(p)[None, :]
    return W

def precoding_rzf(H, sigma2, P_total):
    """
    Regularized ZF with total power budget P_total.
    """
    K, NT = H.shape
    assert K <= NT
    assert sigma2 >= 0.0 and P_total > 0.0
    alpha = K * sigma2 / P_total
    G = H @ H.conj().T + alpha * np.eye(K)
    G = 0.5 * (G + G.conj().T)
    if np.linalg.cond(G) > 1e12:
        raise np.linalg.LinAlgError("précodage RZF mal conditionné")
    W = np.linalg.solve(G, H).conj().T
    # Normalization to satisfy the power budget
    p = P_total / np.linalg.norm(W, 'fro')**2
    return np.sqrt(p) * W
```

20.11 Summary

MIMO and massive MIMO extend beamforming to the bilateral transmission/reception case. Some key messages:

- the SVD of the channel diagonalizes the problem into sub-channels independent;
- the MRT precoding is the “DAS in transmission ”, ZF is the “Null Steering ”, RZF is the “MVDR ”;
- in massive MIMO ($N_T \gg K$), MRT often becomes competitive in asymptotic regime and channel *hardens*;
- hybrid architectures respond to constraints hardware at millimeter frequencies;
- *pilot contamination* and channel estimation remain the main practical limits.

The next chapter covers advanced grating geometries: planar gratings, *sparse arrays*, near field, polarization. These extensions complete the panorama of situations where the simplifying hypotheses ULA + far field + identical polarization cease to hold. At the end of the chapter 21, we will have covered the main architectures encountered in the industry, from the uniform

1D arrays of conventional links to the cylindrical configurations of on-board antennas, including radio astronomy *sparse arrays*.

Chapter 21

Advanced Geometries

Far-field ULA with single polarization has guided all previous theory. Practice often requires relaxing these assumptions. This chapter covers the main extensions: planar arrays (URA), *sparse arrays* and *coprime arrays*, near field, dual polarization, and conformal arrays (on curved surfaces). For each case, we indicate how the steering vector, the beam pattern and the algorithms studied previously adapt.

The overall message is reassuring: the extensions discussed here do not call into question the *algebraic framework* of beamforming. The vector model $\boldsymbol{x}(n) = \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{s}(n) + \boldsymbol{n}(n)$ remains valid; only the definition of $\boldsymbol{a}(\theta)$ changes depending on the geometry. All algorithms that do not assume anything specific about the shape of the steering vector — DAS, MVDR, LCMV, MUSIC — can be transposed with few modifications. Only ESPRIT (which requires translation invariance) and Root-MUSIC (which requires a Vandermonde structure) impose a particular geometry. For the rest, it is the formula of $\boldsymbol{a}$ that changes, not the machinery surrounding it.

21.1 Planar Arrays (URA)

21.1.1 Geometry

A *Uniform Rectangular Array* (URA) is a $M_x \times M_y$ array of antennas at positions $\boldsymbol{p}_{m,n} = (md_x, nd_y, 0)^{\mathrm{T}}$ with $m = 0, \ldots, M_x - 1$, $n = 0, \ldots, M_y - 1$. Total: $N = M_x M_y$ antennas.

21.1.2 Steering vector

The direction of a source in spherical coordinates is marked by (θ, φ): elevation θ, azimuth φ. The unit propagation vector is

$$\hat{\boldsymbol{u}} = (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta)^{\mathrm{T}}.$$

The steering vector of the (m, n) antenna is

$$a_{m,n}(\theta, \varphi) = e^{-jk(md_x \sin\theta\cos\varphi + nd_y \sin\theta\sin\varphi)}.$$

We define the spatial frequencies

$$u = \sin\theta\cos\varphi, \qquad v = \sin\theta\sin\varphi,$$

and then

$$a_{m,n}(u, v) = e^{-jk(md_x u + nd_y v)}.$$

21.1.3 Separability

The steering vector is broken down like a Kronecker product:

$$\boldsymbol{a}(u, v) = \boldsymbol{a}_y(v) \otimes \boldsymbol{a}_x(u),$$

where $\boldsymbol{a}_x(u) = [1, e^{-jkd_x u}, \dots]^{\mathrm{T}}$ and similarly for $\boldsymbol{a}_y(v)$. This separability simplifies all calculations:

- the beam pattern is the product $B(u, v) = B_x(u)\, B_y(v)$ for separable weights $\boldsymbol{w} = \boldsymbol{w}_y \otimes \boldsymbol{w}_x$;
- MUSIC in 2D can be applied on both dimensions independently via *2D-MUSIC* or *2D Root-MUSIC*;
- ESPRIT in 2D jointly provides u and v via *2D-ESPRIT* ou *Unitary 2D-ESPRIT*.

21.1.4 Consequence: No Front/Back Ambiguity

The URA resolves the cone ambiguity of the ULA: for angles $\theta < 90°$, the azimuth φ is only determined by (u, v).

21.2 Nonuniform Arrays

21.2.1 Motivation

For a given D aperture, a ULA requires $N = 2D/\lambda$ antennas. For very large apertures (radio astronomy), it is prohibitive. The *sparse arrays* place fewer antennas but in optimized positions to maintain good resolution.

21.2.2 Minimum Redundancy Arrays

Classic design of Moffet (1968): we place N antennas in positions $\{p_0, \dots, p_{N-1}\}$ such that the differences $\{p_i - p_j\}$ cover the widest possible interval of integers. The covariance of an MRA array, under uncorrelated source assumptions, is equivalent to that of a virtual ULA of significantly larger aperture — up to $\mathcal{O}(N^2)$ sources can be processed with N antennas (*co-array*).

21.2.3 Coprime Arrays

Vaidyanathan & Pal (2010): two ULAs with spacings $M\lambda/2$ and $N\lambda/2$ (M, N primes between them) interleaved. Total $M + N - 1$ antennas; The *differential co-array* has distinct $\mathcal{O}(MN)$ positions, which allows $\mathcal{O}(MN)$ sources to be identified — much more than $M + N$ classically.

21.2.4 Nested Arrays

Pal & Vaidyanathan (2010): a dense ULA $1, 2, \dots, N_1$ followed by a spaced ULA $(N_1 + 1)(N_1 + 1), 2(N_1 + 1)(N_1 + 1), \dots$. THE *co-array* is a virtual ULA of $\mathcal{O}(N_1 N_2)$ antennas. Allows you to detect $\mathcal{O}(N^2)$ sources with N physical antennas.

21.2.5 DOA on sparse arrays

The standard approach: *vectorization* of the covariance $\boldsymbol{r} = \mathrm{vec}(\boldsymbol{R})$, which behaves like a snapshot of a virtual “co-array ”. We apply MUSIC to this co-array, typically after spatial smoothing to compensate for the uniqueness of $\boldsymbol{r}$ (a single “snapshot ”).

21.3 Near Field

21.3.1 When

The plane wave approximation fails when $R < 2D^2/\lambda$ (chapter 2). It's the diet *near-field*, encountered in acoustic imaging, short radar range, biomedical, intra-building microwave.

Criteria	Far field	Near Field
Wavefront	plan	spherical
Settings	angle θ	angle + distance (θ, R)
Steering	$\boldsymbol{a}(\theta)$	$\boldsymbol{a}(R, \theta)$
Amplitude	almost constant	varies with R_m
DOA Search	1D grid	2D localization
Physical action	angular pointing	spatial focus

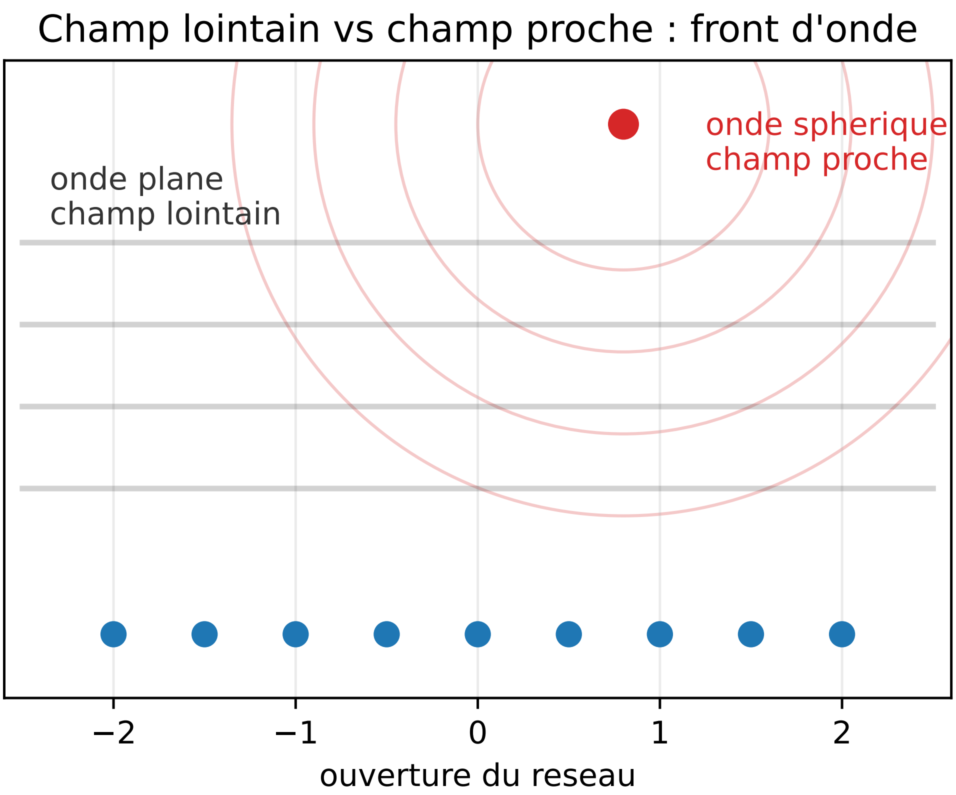

Figure 21.1: Far field and near field ($N = 9$ sensor diagram): the plane wave model removes the distance dependence, while the near field maintains a spherical phase and allows angle-distance focusing.

21.3.2 Near-Field Steering Vector

For a source at position $\boldsymbol{r}_s$, the phase received by the antenna m depends on the distance *exact* $R_m = \|\boldsymbol{r}_s - \boldsymbol{p}_m\|$:

$$a_m(\boldsymbol{r}_s) = \frac{e^{-jkR_m}}{R_m}.$$

The $1/R_m$ factor also introduces a *amplitude variation* between antennas, unlike the far field. The steering vector depends on the *position* of the source, not just its direction.

21.3.3 Link with the Far-Field Model

To understand what disappears in the far field, let's write the position of the source in the form:

$$\boldsymbol{r}_s = R\hat{\boldsymbol{u}},$$

where R is large in front of the array opening. Thus:

$$R_m = \|R\hat{\boldsymbol{u}} - \boldsymbol{p}_m\| = R - \hat{\boldsymbol{u}}^{\mathrm{T}}\boldsymbol{p}_m + \mathcal{O}\left(\frac{\|\boldsymbol{p}_m\|^2}{R}\right).$$

Neglecting the quadratic term:

$$e^{-jkR_m} \approx e^{-jkR}e^{+jk\hat{\boldsymbol{u}}^{\mathrm{T}}\boldsymbol{p}_m}.$$

The common factor e^{-jkR} disappears in the relative steering, and we find the plane wave model. The near field therefore corresponds to the case where the quadratic term is no longer negligible:

$$k\frac{D^2}{R} \not\ll 1.$$

This reading directly relates the Fraunhofer distance $R_F \simeq 2D^2/\lambda$ to a residual phase error.

21.3.4 Algorithmic Consequence

In the far field, a source is parameterized by an angle. In the near field, it is parameterized by a range and a direction:

$$\boldsymbol{a}(\theta) \quad \longrightarrow \quad \boldsymbol{a}(R, \theta).$$

The MUSIC pseudo-spectrum becomes:

$$P_{\mathrm{NF-MUSIC}}(R, \theta) = \frac{1}{\boldsymbol{a}^{\mathrm{H}}(R, \theta)\boldsymbol{U}_n\boldsymbol{U}_n^{\mathrm{H}}\boldsymbol{a}(R, \theta)}.$$

The peaks give a range-angle location. The gain is obvious: we can locate remotely with a single array. The price is a 2D scan, stronger sensitivity to the exact geometry of the antennas and a risk of ambiguous lobes in range.

21.3.5 Algorithms

All algorithms (DAS, MVDR, MUSIC, ESPRIT) adapt by replacing $\boldsymbol{a}(\theta)$ with $\boldsymbol{a}(\boldsymbol{r}_s)$ and traversing a grid in (θ, R). Higher cost (2D grid), but the signal-noise orthogonality property remains valid. THE *near-field MUSIC* is now standard in ultrasound imaging and nearby SAR.

21.3.6 Fresnel Diet

Between pure near field and pure far field, there exists an intermediate zone called *of Fresnel*, where the dependence in R is moderate and where quadratic approximations in R are possible. *Fresnel approximations* models simplify processing at the cost of residual bias.

21.4 Polarization

21.4.1 Vector Model

An EM wave is polarized: its electric field has a spatial orientation (linear, circular, elliptical). Antennas are only sensitive to certain components. For an antenna array *dual polarization* (H and V per antenna), steering vector becomes a 2N-dimensional *polarization vector*:

$$\boldsymbol{a}_{\mathrm{pol}}(\theta, \alpha, \beta) = \boldsymbol{a}(\theta) \otimes \boldsymbol{p}(\alpha, \beta),$$

where $\boldsymbol{p}(\alpha, \beta) = [\cos\alpha, e^{j\beta}\sin\alpha]^{\mathrm{T}}$ encodes the polarization. Two additional parameters (α, β) characterize each source.

21.4.2 Algorithms with polarization

Polarized MUSIC. We scan on (θ, α, β). The $P_{\mathrm{MUSIC}}(\theta, \alpha, \beta)$ pseudo-spectrum simultaneously identifies direction and polarization.

Sources of different polarizations. Two sources of same direction but orthogonal polarizations are *separable* by polarization, even if the direction alone does not separate them. This is the principle of polarimetric SAR and *dual-pol* in satellite.

21.4.3 Interpretation in Degrees of Freedom

Polarization adds degrees of freedom, but it also adds unknowns. With a dual polarization antenna, we double the size of the snapshot:

$$\boldsymbol{x}(n) \in \mathbb{C}^N \quad \longrightarrow \quad \boldsymbol{x}_{\mathrm{pol}}(n) \in \mathbb{C}^{2N}.$$

But each source now has at least two polarimetric parameters in addition to its direction. If the polarization is known or strongly constrained, this extension improves the separation. If it is completely unknown, the search space increases significantly:

$$P(\theta) \quad \longrightarrow \quad P(\theta, \alpha, \beta).$$

In practice, analytical polarization elimination is often used. For a fixed angle θ, we look for the vector $\boldsymbol{p}$ which minimizes:

$$\boldsymbol{a}_{\mathrm{pol}}^{\mathrm{H}}(\theta, \boldsymbol{p}) \boldsymbol{U}_n \boldsymbol{U}_n^{\mathrm{H}} \boldsymbol{a}_{\mathrm{pol}}(\theta, \boldsymbol{p}).$$

The problem then comes down to a small eigenvalue 2×2 instead of a full scan on (α, β). This is the practical idea of many variants of *polarized MUSIC.*

21.4.4 Covariance Adaptation

The covariance becomes $2N \times 2N$ and contains both angular and polarimetric information. All concepts — signal subspace, MVDR, LCMV — extend identically.

21.5 Conformal Arrays

21.5.1 Definition

A *compliant array* follows a non-flat surface: cylindrical (missile antenna), conical (aircraft radar), spherical (omnidirectional system). The $\boldsymbol{p}_m$ positions are on any 3D manifold.

21.5.2 Difficulties

- No Vandermonde structure; direct ESPRIT is not applicable.
- No Kronecker structure like the URA; no separability.
- Each antenna has its own orientation, therefore a *pattern of individual radiation* which depends on the direction of arrival. THE *element pattern* must be included.
- More complex Mutual Coupling.

21.5.3 Modified Steering Vector

The steering vector integrates antenna direction and individual pattern:

$$a_m(\theta, \varphi) = g_m(\theta, \varphi)\, e^{-jk\hat{\boldsymbol{u}}^{\mathrm{T}}\boldsymbol{p}_m},$$

where $g_m(\theta, \varphi)$ is the complex pattern of the antenna m in its local frame of reference, expressed in global frame of reference.

21.5.4 Algorithms

MUSIC applies as is (scan on (θ, φ)). Direct ESPRIT is excluded but variants *Manifold-based* (Friedlander) or methods *interpolated arrays* (Friedlander 1992) transform the data as if they came from a virtual ULA.

21.6 Implementation: 2D-MUSIC on URA

Listing 21.1: 2D-MUSIC sur URA

```
import numpy as np

def steering_ura(theta, phi, Mx, My, dx, dy, lam):
    """Steering vector URA, taille N = Mx*My."""
    assert Mx > 0 and My > 0 and lam > 0.0
    u = np.sin(theta) * np.cos(phi)
    v = np.sin(theta) * np.sin(phi)
    m = np.arange(Mx); n = np.arange(My)
    ax = np.exp(-1j * 2*np.pi * dx / lam * u * m)
    ay = np.exp(-1j * 2*np.pi * dy / lam * v * n)
    return np.kron(ay, ax)   # taille Mx*My

def music_2d(R, L, Mx, My, dx, dy, lam, n_theta=181, n_phi=361):
    N = R.shape[0]
    assert N == Mx * My
    assert R.shape == (N, N)
    assert 0 <= L < N
    R = 0.5 * (R + R.conj().T)
    eigvals, eigvecs = np.linalg.eigh(R)
    Un = eigvecs[:, :N-L]
    thetas = np.linspace(0, np.pi/2, n_theta)
    phis = np.linspace(-np.pi, np.pi, n_phi)
    P = np.zeros((n_theta, n_phi))
    eps = np.finfo(float).eps
    for i, th in enumerate(thetas):
        for j, ph in enumerate(phis):
            a = steering_ura(th, ph, Mx, My, dx, dy, lam)
            v = Un.conj().T @ a
            den = max((v.conj() @ v).real, eps)
            P[i, j] = 1.0 / den
    return thetas, phis, P
```

21.7 Summary

Advanced geometries extend beamforming to situations where the pair “ULA + far field + single polarization ” ceases to hold:

- URA: 2D, Kronecker separability, removes cone ambiguity;
- sparse / coprime / nested: large and numerous openings sources with few physical antennas;
- near-field: integration of distance into steering;
- polarization: steering vector doubled, separation by polarization;
- conformal arrays: require accurate element models.

The theoretical framework (covariance, subspaces, MVDR, MUSIC, ESPRIT) remains valid; only the steering vector and certain algebraic symmetries change. With this chapter, Part VI ends.

Part VII now addresses implementation issues: calibration of hardware errors, robustness to model errors, and integration of all previous tools into an integrated anti-jamming chain. This is the part where theory meets hardware reality and where we discover that the performance of a system is not dictated by the sophistication of its algorithms but by the quality of its weakest link — often calibration or phase coherence between channels.

Part VII

Practical Implementation

Chapter 22

Calibration of a Multi-Antenna Array

All the algorithms studied so far assume that the antennas faithfully measure the phases predicted by the theoretical steering vector $\boldsymbol{a}(\theta)$. In practice, each RF channel introduces its own errors: gains, phases, cable delays, drifts, and mutual coupling between antennas. The steering vector *effective* is $\boldsymbol{a}_{\text{real}}(\theta) = \boldsymbol{G}\,\boldsymbol{a}(\theta) + \boldsymbol{e}(\theta)$, and it is this vector — not $\boldsymbol{a}(\theta)$ — that the "algorithm sees ". Calibration is the procedure that measures and corrects hardware imperfections. Without calibration, a beamformer or DOA estimator produces highly biased results. This chapter details error models, correction estimation methods, and good experimental practices.

A reality on the ground: on many operational beamforming projects, calibration occupies a much greater share of development time than that given to it in the academic literature. A defense radar, an anti-jamming GNSS receiver, a 5G mmWave antenna spend weeks to months in an anechoic chamber before their deployment, then follow a regular in situ recalibration procedure. This is by far the most expensive and time-consuming item. This chapter aims to give the reader the conceptual foundations to not waste time on avoidable errors, and to recognize when strange behavior " is due to a calibration fault rather than an algorithm fault.

22.1 Per-Channel Error Model

22.1.1 Gain and Phase

The simplest model separates the RF channel into gain $g_m \in \mathbb{R}^+$ and phase $\phi_m \in \mathbb{R}$:

$$\boldsymbol{a}_{\text{real}}(\theta) = \boldsymbol{G}\,\boldsymbol{a}(\theta), \quad \boldsymbol{G} = \text{diag}(g_0 e^{j\phi_0}, \ldots, g_{N-1} e^{j\phi_{N-1}}).$$

This model, called *diagonal calibration*, is valid when the mutual coupling is weak and the positions are well known. It is sufficient for the majority of multi-channel SDR platforms.

22.1.2 Physical Sources

The components of the model come from:

- RF cables: $\Rightarrow$ phase differential delays;
- LNA: different gains, slight phase variations;
- mixers: phases dependent on LO;
- filters: group delays;
- ADC: sampling delays, jitter;
- thermal drifts on all these components.

Depending on the RF chain, a thermal drift of 10 °C can produce measurable variations in gain and phase; the precise figures must be validated on the bench, because they strongly depend on the hardware.

22.1.3 Effect on Algorithms

- DAS points near the target (loss $\sim \exp(-\sigma_\phi^2)$).
- Null Steering places its zeros in the wrong place (a jammer expected in θ_i can emerge at a finite level, for example around -25 dB in a mismatch simulation, instead of $-\infty$).
- MVDR partially attenuates the useful signal (signal cancellation).
- MUSIC produces shifted and attenuated peaks.
- ESPRIT, which is based on an invariance *exact*, is particularly fragile.

22.2 Calibration with a Known Source

22.2.1 Principle

We place a reference source at a direction *known* θ_0 (typically the broadside) and we measure the snapshot received. The average snapshot, in the absence of other sources, is $\mathbb{E}[\boldsymbol{x}(n)] = \boldsymbol{G}\,\boldsymbol{a}(\theta_0)\,s_0$ for a deterministic source, or $\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^{\mathrm{H}}] = |s_0|^2\,\boldsymbol{G}\,\boldsymbol{a}(\theta_0)\boldsymbol{a}^{\mathrm{H}}(\theta_0)\,\boldsymbol{G}^{\mathrm{H}} + \sigma^2\boldsymbol{I}$ for a stochastic source.

22.2.2 Gain/Phase Estimation

Source case in *broadside* ($\theta_0 = 0$, therefore $\boldsymbol{a}(\theta_0) = \mathbf{1}$). The average snapshot is

$$\bar{\boldsymbol{x}} = s_0 \begin{bmatrix} g_0 e^{j\phi_0} & \cdots & g_{N-1}e^{j\phi_{N-1}} \end{bmatrix}^{\mathrm{T}}.$$

By normalizing to antenna 0:

$$\widehat{g}_m e^{j\widehat{\phi}_m} = \frac{\bar{x}_m}{\bar{x}_0}, \quad m = 1, \ldots, N-1.$$

The calibration is obtained in one calculation: $\boldsymbol{G}^{-1}$ corrects each channel so that it coincides with antenna 0.

22.2.3 Estimate by Principal Eigenvector

If the source is strong but its temporal properties do not allow simple averaging, we estimate $\widehat{\boldsymbol{R}}$ and take its *main eigenvector* $\widehat{\boldsymbol{u}}_1$. Under the model single-source, $\widehat{\boldsymbol{u}}_1 \propto \boldsymbol{G}\,\boldsymbol{a}(\theta_0)$. We deduce the relative gains/phases by $g_m e^{j\phi_m} = (\widehat{u}_1)_m/(\widehat{u}_1)_0 \cdot 1/a_m(\theta_0)$.

22.2.4 Source in an Arbitrary Direction

If the source is at $\theta_0 \neq 0$, the theoretical steering $\boldsymbol{a}(\theta_0)$ contains known phases that must be *withdraw* before estimating the calibration. We calculate

$$\widehat{g}_m e^{j\widehat{\phi}_m} = \frac{\bar{x}_m}{\bar{x}_0}\,\frac{a_0(\theta_0)}{a_m(\theta_0)}.$$

22.3 Multi-Source Calibration

With several known sources at $\theta_1, \ldots, \theta_M$, we form an overdetermined system:

$$\boldsymbol{x}(n) = \boldsymbol{G}\,\boldsymbol{A}(\boldsymbol{\theta})\,\boldsymbol{s}(n) + \boldsymbol{n}(n),$$

which we reverse in the least squares sense for $\boldsymbol{G}$. It is more precise (multiple measurement points) and more robust.

22.4 Calibration by Covariance and Toeplitz Matrix

For a ULA *ideal*, the theoretical covariance matrix in the presence of a single source has a Hermitian structure *Toeplitz*. In the presence of lane errors, it deviates from this structure in a diagonal-dominant manner. We can therefore estimate $\boldsymbol{G}$ by looking for the diagonal matrix which brings $\widehat{\boldsymbol{R}}$ “to the closest ” of a Hermitian Toeplitz:

$$\min_{\boldsymbol{G}} \; \|\boldsymbol{G}^{-1}\,\widehat{\boldsymbol{R}}\,\boldsymbol{G}^{-\mathrm{H}} - \boldsymbol{T}\|_F^2,$$

where $\boldsymbol{T}$ is the nearest Hermitian Toeplitz. Method called *ToeplitzCal*, without known source, based on knowledge alone geometry.

22.5 Antenna Position Errors

When the antennas are not exactly at the nominal positions, the error is no longer diagonal: it is an error *angle-dependent*. For a position error $\delta\boldsymbol{p}_m$, the received phase becomes $-k\,\hat{\boldsymbol{u}}^{\mathrm{T}}(\boldsymbol{p}_m + \delta\boldsymbol{p}_m)$.

Modeling. $\boldsymbol{a}_{\mathrm{real}}(\theta) = \boldsymbol{G}_{\mathrm{geom}}(\theta)\,\boldsymbol{a}(\theta)$ with $\boldsymbol{G}_{\mathrm{geom}}$ *dependent on* θ. Single-direction calibration is no longer sufficient.

Procedure. We measure $\boldsymbol{G}_{\mathrm{geom}}(\theta)$ for several known angles, then we interpolate. For a ULA we can estimate the positions $\delta\boldsymbol{p}_m$ by joint optimization on K known sources.

22.6 Mutual Coupling

22.6.1 Model

Nearby antennas influence each other: the current induced in one antenna modifies the current in neighboring antennas. The classic model is

$$\boldsymbol{a}_{\mathrm{real}}(\theta) = \boldsymbol{C}\,\boldsymbol{a}(\theta),$$

where $\boldsymbol{C}$ is a non-diagonal $N \times N$ coupling matrix. $\boldsymbol{C}$ typically has a band structure: only nearby antennas couple significantly.

22.6.2 Measurement

In an anechoic chamber. We transmit on a single antenna and measures the currents induced on the others: we fill $\boldsymbol{C}$ column by column. Standard procedure for industrial arrays.

In situ. Multi-angle standard sources, optimization joint gain/phase/coupling. More complex, less precise, but achievable without dismantling.

22.6.3 Compensation

The corrected effective steering is $\boldsymbol{C}^{-1}\boldsymbol{a}(\theta)$, to be used in all algorithms instead of $\boldsymbol{a}(\theta)$.

22.7 Calibration by Manifold Measurement

For complex arrays (conformal, irregular, strongly coupled), we proceed with *direct measurement of the manifold.* We scan a reference source on a grid of angles $\{\bar{\theta}_g\}$ and we record the normalized snapshots: we thus construct the dictionary $\boldsymbol{D} = [\widehat{\boldsymbol{a}}_{\text{real}}(\bar{\theta}_1), \ldots, \widehat{\boldsymbol{a}}_{\text{real}}(\bar{\theta}_G)]$.

Interpolation. For angles between grid nodes, we interpolate by spline polynomials, by parametric models or by *Gaussian Process regression.* Accuracy depends on density grid (typically 1° for a civilian radar, 0.1° for a military radar).

22.8 Wideband Calibration

Channel gains and phases depend on frequency. Calibration *broadband* consists of measuring $\boldsymbol{G}(f)$ on a frequency grid covering the useful band, then to interpolate. Fractional FIR filters then implement the correction in the digital domain.

22.9 Validation After Calibration

Several tests allow you to check the quality of the calibration.

Test 1: consistency on known source. We place a source to a known direction θ_0 and we measure $\widehat{\theta}_0$ by MUSIC or Capon. The difference $|\widehat{\theta}_0 - \theta_0|$ gives a bound on the residual error.

Test 2: beam pattern measured. We form a DAS with direction θ_0 and we measure the gain on a mobile source. A good bench must find the theoretical shape of the main lobe; the acceptable tolerance depends on the sensor, the measuring chamber and the targeted dynamics.

Test 3: depth of null. We form a Null Steering on a direction θ_i and we measure the gain on a source placed there. The depth obtained must be interpreted as a system measurement: a well-calibrated bench can reach several tens of dB of rejection, while an uncalibrated array often goes much higher.

22.10 Good Experimental Practice

- **Stable source.** Reference transmitter locked to a good stability oscillator (TCXO or better OCXO). A drift in the LO of the reference transmitter pollutes the entire calibration.
- **Far field.** Much greater source-to-array distance at $2D^2/\lambda$. At 2.4 GHz, ULA $N = 16$, $D = 94\,\text{cm}$, $R_F \simeq 14\,\text{m}$: allow at least 20–30 m.
- **Reduction of reflections.** Ideal anechoic chamber; otherwise, open space and directional antennas for the source.
- **Consistency of conditions.** Calibrate at frequency operational, at the nominal temperature, with the same RF chain as in operation.
- **Periodic recalibration.** In the presence of drifts, recalibrate every few minutes (variable temperature) or hours (stable environment).

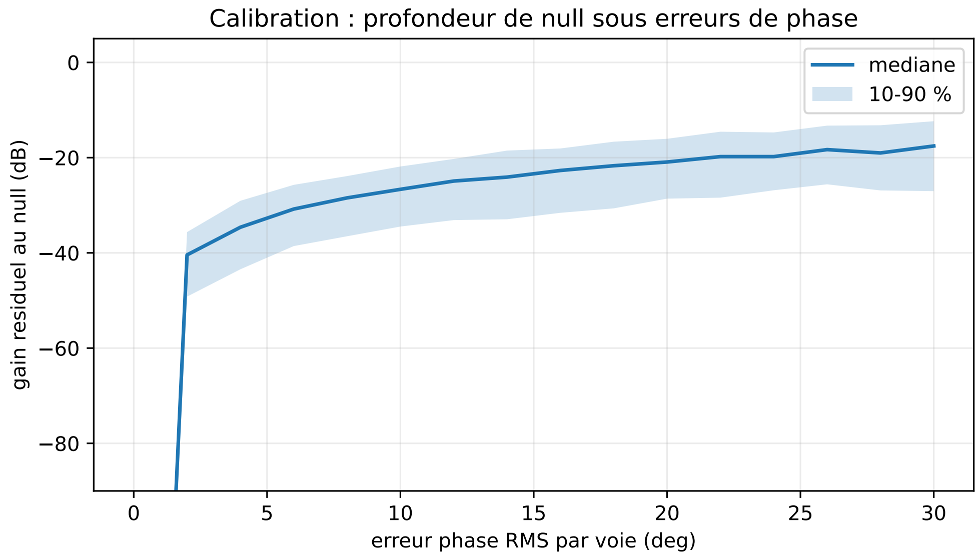

Figure 22.1: Monte-Carlo simulation ($N = 10$, target 0°, null 30°, 400 draws per error level): a theoretically perfect null becomes a finite rejection as soon as the channels carry residual phase errors.

- **In situ self-calibration.** When disassembly is impossible, use “opportunistic sources ”: satellites, known reference beacons, LTE/GSM signals from known cells.

22.11 Implementation: Diagonal Calibration

Listing 22.1: Calibration diagonale d’un ULA

```
import numpy as np

def estimate_calibration(X, theta0, d_over_lambda=0.5):
    """
    X      : (N, K) snapshots d’une source unique à theta0
    Retour : g (N,) facteurs complexes par voie (relatifs à m=0)
    """
    N, K = X.shape
    # snapshot moyen (en supposant source connue, on peut filtrer si besoin)
    x_mean = np.mean(X, axis=1)
    m = np.arange(N)
    a0 = np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(theta0) * m)
    raw = x_mean / a0
    g = raw / raw[0]    # reference factor: antenna 0
    return g

def apply_calibration(X, g):
    """Renvoie X corrigé : X_calib = diag(1/g) @ X."""
    return X / g[:, None]
```

22.12 Summary

Calibration brings the real RF chain back to the theoretical model used by the algorithms. Without it, adaptive beamformers and DOA estimators become difficult to make reliable. The minimum steps:

- error model (diagonal, position, coupling);
- estimation procedure (known source or self-calibration);
- correction applied to the data or integrated into the steering vectors;
- validation by tests on known sources.

The following chapter deals with the other practical side: the *robustness* to residual errors. Calibration does not eliminate never completely errors; we must design algorithms that tolerate them. The calibration + robustness couple forms a defense in depth: the first removes most of the imperfections, the second absorbs what could not be. Neither of the two is sufficient alone, and an operational system must combine both to achieve the performance announced by the theoretical algorithms.

Chapter 23

Robustness to Model Errors

Calibration brings the effective steering close to the theoretical steering, but never eliminates it completely. Residual errors remain — geometry, imperfect calibration, drifts, limited snapshot — and can quickly cause an optimal theoretical beamformer to experience very degraded performance. *robustness* is the set of techniques that make algorithms tolerant to these errors. This chapter covers *diagonal loading*, derivative constraints, "worst-case constraints ", *white noise gain* and their impact on the beam pattern.

The central paradox of robustness is that beamformers *optimaux* sont aussi les plus *fragile*. MVDR reaches the Maximum SINR in theory, but this maximum is gained at the cost of extreme sensitivity to errors on $\boldsymbol{a}(\theta_s)$ and on $\widehat{\boldsymbol{R}}$. It is the algebraic expression of a general phenomenon in estimation: the more finely we exploit information, the more sensitive we become to errors on this information. Robustizing a beamformer therefore means accepting *degrade its nominal performance* for *maintain acceptable performance* in the face of real imperfections. It's an arbitration, not a refinement.

23.1 Sources of Residual Error

- **Steering vector mismatch.** The target is not exactly in θ_s: error $\delta\theta$.
- **Imperfect calibration.** Gains/residual phases σ_g, σ_ϕ.
- **Covariance estimation.** With K snapshots finished, $\widehat{\boldsymbol{R}}$ has variance $\mathcal{O}(N/K)$.
- **Non-white noise.** The model assumes $\sigma^2\boldsymbol{I}$; reality often shows structures.
- **Non-stationarity.** The environment changes during collection.
- **Error on L.** Underestimation or overestimation of number of sources for MUSIC/ESPRIT.

23.2 Self-cancellation signal

Key phenomenon: if the covariance used by MVDR contains the useful signal, and the steering vector does not point *Exactly* in its direction, MVDR "thinks " that the useful signal is interference and attenuates it. In simulations where the useful signal is strong and included in the covariance, the loss can reach several tens of dB for a mismatch of a fraction of the main lobe width.

Favorable conditions. Self-cancellation important when:

- K large (well estimated covariance and therefore "confident ");
- powerful signal (powerful term to "remove ");
- significant mismatch.

Parades.

- use $\boldsymbol{R}_{i+n}$ instead of $\boldsymbol{R}$;
- diagonal loading;
- derivative constraints;
- robust methods (Lorenz & Boyd [8], Li–Stoica–Wang [7]).

23.3 Diagonal loading

23.3.1 Definition

We replace $\widehat{\boldsymbol{R}}$ with

$$\widehat{\boldsymbol{R}}_\alpha = \widehat{\boldsymbol{R}} + \alpha \boldsymbol{I}_N,$$

with $\alpha > 0$.

23.3.2 Effects

- Improved packaging: $\kappa = (\lambda_{\max} + \alpha)/(\lambda_{\min} + \alpha)$.
- $\boldsymbol{w}_{\text{MVDR}}$ standard bounded: $\|\boldsymbol{w}\|^2 \leq 1/(\alpha)$, except for one factor.
- Beam pattern softened: the nulls become shallower, the main lobe widens slightly.
- When $\alpha \to \infty$, MVDR $\to$ DAS.
- When $\alpha \to 0$, we find pure MVDR.

23.3.3 Why Loading Limits the Norm

The loaded MVDR resolves:

$$\min_{\boldsymbol{w}} \boldsymbol{w}^{\mathrm{H}} \widehat{\boldsymbol{R}} \boldsymbol{w} + \alpha \|\boldsymbol{w}\|^2 \quad \text{sous} \quad \boldsymbol{w}^{\mathrm{H}} \boldsymbol{a}_s = 1.$$

The added term is therefore not just a numerical trick. This is an explicit penalization of the weight norm. However, a large norm means that the beamformer adds channels with large positive and negative coefficients, which creates fine cancellations but amplifies white noise, phase errors and small calibration errors.

The solution is:

$$\boldsymbol{w}_{\text{DL}} = \frac{(\widehat{\boldsymbol{R}} + \alpha \boldsymbol{I})^{-1} \boldsymbol{a}_s}{\boldsymbol{a}_s^{\mathrm{H}} (\widehat{\boldsymbol{R}} + \alpha \boldsymbol{I})^{-1} \boldsymbol{a}_s}.$$

In practice, we do not calculate the inverse explicitly: we solve the linear system $(\widehat{\boldsymbol{R}} + \alpha \boldsymbol{I})\boldsymbol{z} = \boldsymbol{a}_s$, then we normalize $\boldsymbol{w} = \boldsymbol{z}/(\boldsymbol{a}_s^{\mathrm{H}} \boldsymbol{z})$. If α increases, the directions associated with small eigenvalues are no longer amplified. The beamformer stops exploiting the fragile details of the estimated covariance. This is the underlying reason for the compromise: less adaptation, but more stability.

23.3.4 Choice of α

Empirical. $\alpha = c \cdot \operatorname{tr}(\widehat{\boldsymbol{R}})/N$ with $c \in [10^{-3}, 0.1]$. Simple and effective for most applications.

Based on noise. $\alpha = c\,\widehat{\sigma}^2$, where $\widehat{\sigma}^2$ is estimated from the smallest $N - L$ eigenvalues. More precise when we know L.

Worst-case [8]. On définit une *sphere of uncertainty* around $\boldsymbol{a}(\theta_s)$:

$$\mathcal{U} = \{\boldsymbol{a}' : \|\boldsymbol{a}' - \boldsymbol{a}(\theta_s)\| \leq \epsilon\}.$$

The beamformer *robust worst-case* minimizes the power $\boldsymbol{w}^{\mathrm{H}}\widehat{\boldsymbol{R}}\boldsymbol{w}$ under $|\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}'| \geq 1$ for all $\boldsymbol{a}' \in \mathcal{U}$. This constraint is written $|\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_s)| - \epsilon\|\boldsymbol{w}\| \geq 1$, SOCP problem. The solution is equivalent to an MVDR with an optimal diagonal loading $\alpha^\star$ which can be calculated explicitly.

Robust Capon [7]. Variante directe :

$$\max_{\boldsymbol{a}' \in \mathcal{U}} (\boldsymbol{a}')^{\mathrm{H}}\, \widehat{\boldsymbol{R}}^{-1}\, \boldsymbol{a}'.$$

Also results in a calculable optimal diagonal loading.

23.4 White Noise Gain (WNG)

23.4.1 Definition

Under unity gain constraint $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_s) = 1$, the *White Noise Gain* is

$$\mathrm{WNG}(\boldsymbol{w}) = \frac{1}{\|\boldsymbol{w}\|^2}.$$

For DAS, WNG $= N$. For MVDR without loading, WNG can drop dramatically when $\widehat{\boldsymbol{R}}$ is poorly conditioned.

23.4.2 Why It Matters

A low WNG means that small errors on the antennas (noise or calibration) are *amplified* output. As a practical rule, a WNG much lower than that of the DAS, for example of the order of $N/10$ or less, should alert you: the beamformer has probably become aggressive and sensitive to model errors.

23.4.3 Robustness by WNG Constraint

We solve MVDR under additional constraint WNG $\geq \gamma N$, or $\|\boldsymbol{w}\|^2 \leq 1/(\gamma N)$:

$$\min_{\boldsymbol{w}} \ \boldsymbol{w}^{\mathrm{H}}\widehat{\boldsymbol{R}}\boldsymbol{w} \quad \text{sous} \quad \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta_s) = 1,\ \|\boldsymbol{w}\|^2 \leq 1/(\gamma N).$$

This constraint is equivalent to diagonal loading with α adjusted to reach the limit. This is one of the clearest formulations of adaptive robustness.

23.5 Propagation of a Steering Error

where the real steering:

$$\boldsymbol{a}_{\mathrm{r}} = \boldsymbol{a}_s + \Delta\boldsymbol{a}.$$

The gain actually applied to the useful source is:

$$\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{r}} = 1 + \boldsymbol{w}^{\mathrm{H}}\Delta\boldsymbol{a},$$

because the constraint only imposes $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}_s = 1$. By Cauchy-Schwarz:

$$|\boldsymbol{w}^{\mathrm{H}}\Delta\boldsymbol{a}| \leq \|\boldsymbol{w}\|\,\|\Delta\boldsymbol{a}\|.$$

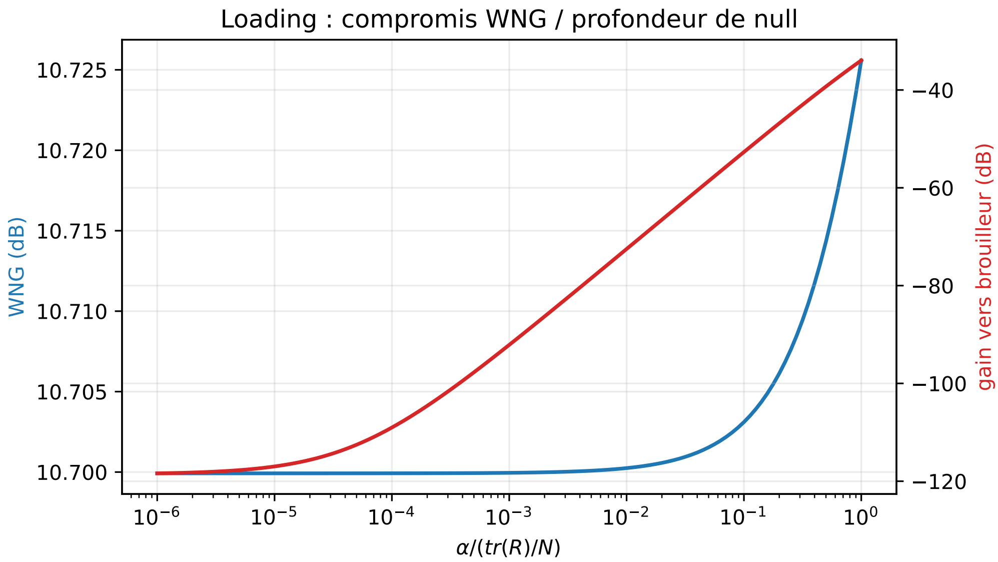

Figure 23.1: Reproducible MVDR simulation ($N = 12$, target $0°$, jammers $-25°$ and $40°$, INR 40 dB): increasing the diagonal loading improves the WNG, but the nulls become shallower.

This inequality gives the simplest justification for WNG: controlling $\|\boldsymbol{w}\|$ directly controls the maximum loss due to steering error. A beamformer with enormous weights can perfectly respect the nominal constraint and yet strongly distort the real source.

If we impose:

$$\|\Delta \boldsymbol{a}\| \leq \epsilon,$$

then the useful gain verifies:

$$|\boldsymbol{w}^{\mathrm{H}} \boldsymbol{a}_{\mathrm{r}}| \geq 1 - \epsilon \|\boldsymbol{w}\|.$$

To guarantee a limited loss, it is therefore necessary:

$$\epsilon \|\boldsymbol{w}\| < 1.$$

This is exactly the logic of worst-case formulations.

23.6 Derivative Constraints

23.6.1 Motivation

If the target can be at $\theta_s \pm \delta\theta$, we want the gain to remain unity over the entire zone $[\theta_s - \delta\theta, \theta_s + \delta\theta]$. An elegant way to enforce this: force the derivative of the beam pattern to cancel at θ_s,

$$\frac{\partial B}{\partial \theta}(\theta_s) = 0 \quad \Longleftrightarrow \quad \boldsymbol{w}^{\mathrm{H}} \dot{\boldsymbol{a}}(\theta_s) = 0,$$

where $\dot{\boldsymbol{a}} = \partial \boldsymbol{a} / \partial \theta$. At the cost of an additional constraint (therefore a degree of freedom), we obtain a flat zone around θ_s.

23.6.2 Implementation via LCMV

$$\boldsymbol{C} = [\boldsymbol{a}(\theta_s),\, \dot{\boldsymbol{a}}(\theta_s)], \quad \boldsymbol{f} = [1,\, 0]^{\mathrm{T}}.$$

We can add higher order constraints ($\ddot{\boldsymbol{a}}(\theta_s) = 0$, etc.) for larger areas, but each consumes a degree of freedom.

23.7 Broadened Nulls

Symmetrically, to reject an interference with an angular uncertainty $\pm\delta$, we replace the single constraint $\boldsymbol{a}(\theta_i)$ by several:

$$\boldsymbol{C}_{\text{null}} = [\boldsymbol{a}(\theta_i - \delta),\, \boldsymbol{a}(\theta_i),\, \boldsymbol{a}(\theta_i + \delta)], \quad \boldsymbol{f}_{\text{null}} = [0, 0, 0]^{\mathrm{T}}.$$

The null becomes a zone of attenuation rather than a point. Alternative: also impose $\dot{\boldsymbol{a}}(\theta_i) = 0$, $\ddot{\boldsymbol{a}}(\theta_i) = 0$ to flatten the area around θ_i.

23.8 Norm Limitation

For pure robustness we can simply impose $\|\boldsymbol{w}\|^2 \leq \gamma$ as an additional constraint:

$$\min_{\boldsymbol{w}} \ \boldsymbol{w}^{\mathrm{H}} \widehat{\boldsymbol{R}} \boldsymbol{w} \quad \text{sous} \quad \boldsymbol{w}^{\mathrm{H}} \boldsymbol{a}(\theta_s) = 1,\ \|\boldsymbol{w}\|^2 \leq \gamma.$$

Convex QCQP problem, equivalent to an MVDR with implicit diagonal loading. This is exactly *WNG-constrained MVDR.*

23.9 Robustness of MUSIC and ESPRIT

23.9.1 MUSIC

For MUSIC, the main weakness is the imprecise estimation of the noise subspace for small K. The parade: *spatial smoothing* or *forward-backward averaging* to increase K effectiveness. Calibration sensitivity: see chapter 22.

A more quantitative way of saying this is to look at the gap between the signal and noise eigenvalues. If the smallest signal eigenvalue is close to the noise level, the signal subspace becomes poorly separated. The subspace disturbance theorems give a reading of the type:

$$\|\sin\Theta(\widehat{\boldsymbol{U}_s}, \boldsymbol{U}_s)\| \lesssim \frac{\|\widehat{\boldsymbol{R}} - \boldsymbol{R}\|}{\lambda_L - \sigma^2}.$$

The denominator is *eigengap.* When the sources are weak, close or correlated, this gap decreases and MUSIC becomes unstable. Increasing K reduces the numerator; improving the SNR or separating sources increases the denominator.

23.9.2 ESPRIT

ESPRIT is more sensitive than MUSIC because it exploits exact algebraic invariance. Typical errors:

- antenna position: breaks invariance;
- correlated noise between channels: distorts the $\boldsymbol{\Psi}$ matrix.

TLS-ESPRIT (instead of LS-ESPRIT) alleviates the second problem. For the first, careful calibration of positions or models *interpolated array* are essential.

For ESPRIT, robustness can be seen in the relationship:

$$\boldsymbol{J}_2 \boldsymbol{U}_s \approx \boldsymbol{J}_1 \boldsymbol{U}_s \boldsymbol{\Psi}.$$

LS-ESPRIT assumes that the error concerns only the left member. TLS-ESPRIT assumes that both members are perturbed, which better corresponds to the real case where $\boldsymbol{U}_s$ itself is estimated with error. This is why TLS is generally preferable whenever the SNR drops or the number of snapshots is limited.

23.10 Limited Number of Snapshots

23.10.1 RMB Rule Recalled

For SMI (MVDR on $\widehat{\boldsymbol{R}}$), $K \simeq 2N$ corresponds to an average loss of about 3 dB in the ideal RMB model. For $K < N$, $\widehat{\boldsymbol{R}}$ is unique: loading essential.

23.10.2 Loading for K very small

With $K \sim N$ or less, we choose α large enough so that $\alpha\boldsymbol{I} + \widehat{\boldsymbol{R}}$ is well conditioned. Order of magnitude: $\alpha = \mathrm{tr}(\widehat{\boldsymbol{R}})/N$ (each estimated eigenvalue "noise " has a weight comparable to α). This aggressiveness produces an almost DAS, but stable beamformer.

23.11 Effect of Robustness on the Beam Pattern

Robustness techniques modify the beam pattern in characteristic ways.

- Diagonal loading: shallower nulls, main lobe slightly enlarged, secondary lobes a little higher. The beam pattern "slides " from the MVDR to the DAS when α increases.
- Derivative Constraints: top of the main lobe flat on a angular range; peak gain loss of a few dB.
- Broadened Nulls: wider rejection zones; consumption additional degrees of freedom.
- WNG-constrained: beam pattern more stable in the face of errors channel, but reduced selectivity.

23.12 Robustness Tests

A good practice consists of measuring degraded performance in specific scenarios:

1. Useful angular error $\delta\theta$: plot the Output SINR according to $\delta\theta \in [-1°, 1°]$.
2. Null error: move a jammer from $\delta_i \in [-1°, 1°]$ around the nominal direction.
3. Random phase error: draw 100 achievements $\delta\phi_m \sim \mathcal{N}(0, \sigma_\phi^2)$ for $\sigma_\phi = 5°, 10°, 20°$.
4. Variation of K: test on $K = N, 2N, 4N, 10N$.
5. Variation of diagonal loading: sweep $\alpha/\,\mathrm{tr}(\widehat{\boldsymbol{R}}) \in [10^{-3}, 1]$.

A robust beamformer maintains acceptable performance on all of these tests.

23.13 Robustness Indicators

Indicator	What it measures	Practical alert
SINR under mismatch	Useful performance when θ_s is false	sudden drop around $\delta\theta = 0$
WNG $= 1/\|\boldsymbol{w}\|^2$	Amplification of white noise and channel errors	WNG much lower than DAS
Target gain min/max	Undulation of the useful lobe on an area of uncertainty	non-flat gain around θ_s
Null percentile depth	Rejection with DOA/Monte-Carlo calibration errors	median good but 10% catastrophic
Packaging of $\widehat{\boldsymbol{R}}_\alpha$	Stability of linear resolution	eigenvalues too close to zero
Sensitivity to K	Dependence on number of snapshots	result which changes by doubling K
Order error $\widehat{L}$	Robustness MUSIC/ESPRIT	unstable peaks or poorly separated subspace

23.14 Choosing an MVDR Variant

Criteria	Simple MVDR	Loaded MVDR	Robust MVDR
Covariance	$\widehat{\boldsymbol{R}}$	$\widehat{\boldsymbol{R}} + \alpha \boldsymbol{I}$	loading + constraints
Assumption	exact model	imperfect estimate	DOA/manifold uncertain
Nulls	very deep	shallower	expanded or limited
WNG	can fall	improved	control
Setting	none	α	α, width, derivatives
Use	clean simulation	convenient by default	real anti-jamming

23.15 Summary of compromises

Technical	Won	Lose
Diagonal loading α	robustness, packaging	nulls depth
Derivative Constraints	angular robustness	peak gain
Broadened Nulls	interference rejection	degrees of freedom
Norm Limitation	WNG, white noise	selectivity
Robust Capon	strong robustness under uncertainty model	SOCP cost
Forward-backward	quality of estimate	restricted to ULAs

23.16 Practical Recommendations

- Always apply a minimum diagonal loading ($\alpha \sim 10^{-3}\,\mathrm{tr}(\widehat{\boldsymbol{R}})/N$) on MVDR/LCMV.
- If the target position is uncertain, add at least one derivative constraint.
- If the positions of the jammers are uncertain, widen the nulls.
- Monitor WNG: if WNG is much lower than DAS (for example of the order of $N/10$), increase α or relax the constraints.
- For ESPRIT, prefer TLS to LS.
- Recalibrate periodically.
- Test on robustness scenarios before deployment.

23.17 Implementation: Robust MVDR with Derivative Constraints

Listing 23.1: MVDR with derivative constraints and loading

```python
import numpy as np

def steering_and_derivative(theta, N, d_over_lambda=0.5):
    m = np.arange(N)
    a = np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(theta) * m)
    da = a * (-1j * 2*np.pi * d_over_lambda * np.cos(theta) * m)
    return a, da

def mvdr_robust(R, theta_s, N, d_over_lambda=0.5, alpha=None,
                derivative_constraint=True):
    """MVDR with diagonal loading and constraint dérivative optionnelle."""
```

```python
    assert R.shape == (N, N)
    a, da = steering_and_derivative(theta_s, N, d_over_lambda)
    if alpha is None:
        alpha = 1e-2 * np.trace(R).real / N
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5*(R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha mal conditionnee")
    if derivative_constraint:
        C = np.column_stack([a, da])
        f = np.array([1.0, 0.0], dtype=complex)
        Rinv_C = np.linalg.solve(R_alpha, C)
        M = C.conj().T @ Rinv_C
        if np.linalg.cond(M) > 1e12:
            raise np.linalg.LinAlgError("contraintes mal conditionnees")
        return Rinv_C @ np.linalg.solve(M, f)
    else:
        u = np.linalg.solve(R_alpha, a)
        den = a.conj() @ u
        if abs(den) < np.finfo(float).eps:
            raise np.linalg.LinAlgError("normalisation instable")
        return u / den
```

23.18 Summary

Robustness is not a secondary refinement but a constituent element of practical beamforming. A very aggressive theoretical MVDR can become fragile without loading; Null Steering without derivative constraints remains sensitive to mismatch; ESPRIT without TLS can lose several dB of accuracy. All the compromises outlined in this chapter boil down to one question: *how much margin to accept to gain how much stability?*

Key Points

- **Central formula**: $\widehat{\boldsymbol{R}}_\alpha = \widehat{\boldsymbol{R}} + \alpha \boldsymbol{I}$.
- **Main hypothesis**: the model is close to reality, but never accurate.
- **Advantage**: stabilizes weights, limits self-nulling and improves WNG.
- **Limit**: less selectivity and shallower nulls.
- **Practical trap**: a spectacular null in simulation can be fragile on real hardware.
- **Following**: the anti-jamming chain combines calibration, DOA, LCMV and robustness.

The following chapter integrates all the bricks of the work — DOA, MVDR, LCMV, RLS, calibration, robustness — in a *integrated anti-jamming chain*, canonical example of operational architecture. This is the moment when all the theoretical tools developed since Part I find their place in an integrated architecture. We will see that few bricks are superfluous, but that none is sufficient on its own: it is the intelligent arrangement of all of them that produces a system that works.

Chapter 24

Integrated Anti-Jamming Chain

At this stage, the reader has all the necessary building blocks: physical model, deterministic and statistical beamformers, DOA estimation, calibration and robustness. This final chapter assembles them into an integrated operational chain. The example chosen is *anti-jamming*: the defense of a sensitive receiver (GPS, data link, telemetry) against opposing transmitters. The same scheme is suitable for civilian applications: Wi-Fi in congested environments, multi-user 4G/5G, selective acoustic capture.

Anti-jamming is a particularly instructive example because it requires orchestrating almost every algorithm in the book. There we find: an estimate of DOA to locate jammers (MUSIC or ESPRIT); an adaptive beamformer under constraints to reject them without degrading the target (LCMV); a regularization to accommodate imperfections (*diagonal loading*, derivative constraints); calibration to reduce systematic errors; and an update loop adapted to the dynamics of the environment. The challenge is less to understand each brick in isolation — this is the subject of the previous chapters — than to make them *cooperate* in a real-time system constrained in latency, memory and consumption.

24.1 Requirements

An anti-jamming chain must:

- receive a useful signal in the nominal (or unknown) direction but detectable);
- reject L simultaneous jammers ($L < N$) whose directions are *unknown a priori*;
- adapt in near real time when the environment changes;
- remain operational despite hardware imperfections;
- degrade gracefully in the face of non-hypothetical scenarios.

24.2 Overall Architecture

The integrated chain follows a simple logic: measure correctly, estimate the spatial environment, then choose weights that preserve the target while rejecting jammers. Figure 24.1 gives the system view.

The main blocks, in detailed form:

Multi-way antennas
↓
Coherent RF/IQ reception
↓
Gain/phase/manifold calibration
↓
Snapshots and covariance $\widehat{\boldsymbol{R}}$
↓
Order detection and DOA estimation
↓
Construction of LCMV constraints
↓
Robust weights: loading, WNG, expanded nulls
↓
Beamformed output then demodulation

This diagram should be read with two feedback loops: a fast loop that updates the weights when the covariance changes, and a slow loop that restarts the calibration when the inter-channel phases drift.

1. **RF + ADC front-end**: Channels synchronized by clock and common LO, calibrated *a priori* or *in situ*.
2. **Snapshot buffer**: accumulation of K snapshots by window.
3. **Covariance estimation**: $\widehat{\boldsymbol{R}}$ with possible forward-backward.
4. **Order detection**: MDL/AIC to estimate the number of sources $\widehat{L}$.
5. **DOA estimate**: MUSIC or ESPRIT, or method sparse for short K.
6. **Target/jammer selection**: by nominal direction or by power.
7. **Construction of constraints**: $\boldsymbol{C}, \boldsymbol{f}$ for LCMV.
8. **Calculation of weights**: LCMV with diagonal loading and derivative constraints.
9. **Application**: $y(n) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}(n)$.
10. **Demodulation**: standard, on scalar output.
11. **Update Loop**: periodic recalculation or asynchronous according to evolution.

24.3 Subsystem Synchronization

The blocks do not all operate at the same rate. A typical phase shift:

Block	**Typical cadence**
IQ Sampling	10–100 MS/s
Application of weights $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}$	cadence IQ
Estimate $\widehat{\boldsymbol{R}}$	1–100 Hz
Decomposition/DOA estimation	1–10 Hz
Update $\boldsymbol{w}$	1–10 Hz

This difference explains why the weights are *constants* on $\sim$10 ms at 1 s: the algorithmic benefit does not justify the computational cost of an update at each sample. Adaptive recursive algorithms (LMS, RLS) are an alternative when this hypothesis is lifted.

24.4 Role of Each Building Block

24.4.1 Calibration: Silent Prerequisite

Without calibration, the performance of subsequent blocks quickly becomes unpredictable. Typical procedure:

- factory calibration in anechoic chamber (coupling model, positions, gains/phases at various frequencies);
- in situ calibration at ignition: known source, adjustment diagonal;
- periodic recalibration to monitor thermal drifts.

24.4.2 DOA Estimation: Identifying the Adversaries

Jammers do not announce their position. MUSIC or ESPRIT identify the directions of all spatially structured energy received. The distinction *target / jammer* is then made by:

- knowledge *a priori* of the direction of the target (satellite, beacon);
- filtering by power (jammers are often stronger);
- filtering by temporal characteristics (known modulation of the target).

24.4.3 Adaptive Beamforming: Acting

LCMV with:

- unit gain on the target direction;
- nulls on each identified jammer;
- derivative constraints for angular robustness;
- diagonal loading.

Given the complexity, we can also settle for MVDR (a constraint) by trusting the implicit nulls of MVDR to reject the jammers.

24.4.4 Robustness: Absorbing Imperfections

All the techniques from the previous chapter: loading, derivative constraints, expanded nulls, WNG-constrained. An operational system *accumulates* these protections.

24.5 Update Loop

24.5.1 Fixed Step

Simplest method: we recalculate $\widehat{\boldsymbol{R}}$, $\widehat{L}$, the DOAs and the weights every T seconds ($T \sim 10\,\mathrm{ms}$ to $1\,\mathrm{s}$ depending on the dynamics of the jammers). This is suitable for stationary or slow environments.

24.5.2 Adaptive Step

We monitor an indicator (change in specific spectrum, increase in measured SINR, error alert) and we trigger an update only when necessary. Economical but more complex.

24.5.3 Recursive Tracking

LMS, RLS or Kalman applied directly to the beamformer. The update is continuous, without "snapshot " of the DOA estimate: the algorithm naturally follows the jammer movements. Preferred solution when the dynamics are fast.

24.6 Overall Performance: Indicators

- **Output SINR**: useful signal ratio / (noise + interference residual). Target: $\mathrm{SINR}_{\mathrm{out}} \geq$ demodulation threshold.
- **Depth of nulls**: rejection level measured on the directions of the jammers; the useful values are judged on a bench or calibrated simulation, not just by the ideal formula.
- **Gain on target**: must remain close to the gain of ideal array (N, or $10 \log_{10} N$ in dB).
- **DOA estimation error**: RMSE between angles estimated and true (ideally close to the CRB).
- **Robustness**: Residual SINR under angular errors of $\pm 1°$, under phase noise $\sigma_\phi = 10°$, etc.

24.7 Case Studies

24.7.1 GPS Reception Under Jamming

- Target: satellite with known direction within $\sim 1°$.
- Jammers: 1 to 3 ground transmitters, directions unknown, with powers that can greatly exceed the useful signal.
- Array: ULA or URA with 4–8 antennas, $\lambda/2 \approx 9.5\,\mathrm{cm}$ at 1.575 GHz.
- Algorithms: ESPRIT for DOA (fast), LCMV with derivative constraints, diagonal loading at 0.01.
- Expected result on a well-calibrated case: deep nulls, gain Almost preserved GPS, $\mathrm{SINR}_{\mathrm{out}} \geq$ demod.

24.7.2 Multiuser Wi-Fi 802.11n

- Target: 4x4 MIMO stream to a client with unknown direction.
- Jammers: other clients in simultaneous transmission (*co-channel*).
- Array: URA 4×4 at 5 GHz, factory calibration.
- Algorithms: MIMO channel estimation via pilots, precoding ZF or RZF.
- Result: efficient spatial multiplexing, $\mathrm{SINR}_{\mathrm{out}}$ sufficient for the target modulation.

24.7.3 Acoustic Imaging in a Noisy Environment

- Target: voice in a wideband noise environment + reverberation.
- Array: omnidirectional microphones, geometry often non-uniform.
- Algorithms: broadband beamforming by sub-bands, MVDR Robust sub-band, Wiener single-channel post-filtering.

- Result: improvement in intelligibility; the SNR gain depends heavily in reverberation, number of microphones and noise coherence.

24.8 Reference Numerical Scenario

To make the tradeoffs concrete, consider the following scenario:

$$N = 8, \qquad d = \frac{\lambda}{2}, \qquad K = 128.$$

The target comes from:

$$\theta_s = 0^\circ,$$

with an SNR per antenna of 0 dB. Two jammers arrive from:

$$\theta_{i,1} = -25^\circ, \qquad \theta_{i,2} = 40^\circ,$$

with an INR of 35 dB each. We compare four methods:

- DAS pointed to 0°;
- MVDR with diagonal loading;
- LCMV with two punctual nulls;
- Robust LCMV with expanded nulls and derivative constraint on the target.

Method	**SINR output**	**Target gain**	**Null** -25°	**Null** $+40^\circ$	**Mismatch**
DAS	−18 dB	9 dB	−4 dB	−7 dB	good
Loaded MVDR	18 dB	8.5 dB	−31 dB	−28 dB	medium
Punctual LCMV	22 dB	8.8 dB	−45 dB	−43 dB	weak
Robust LCMV	20 dB	8.3 dB	−34 dB	−33 dB	good

These values are representative of an ideal simulation with moderate errors. They should not be read as absolute guarantees, but as a typical hierarchy. The DAS preserves the target but lets jammers through. MVDR creates nulls without explicitly knowing their directions. Punctual LCMV gives the deepest nulls if the DOAs are exact. Robust LCMV accepts less spectacular nulls to better hold when the DOAs move or the calibration is not perfect.

24.8.1 Effect of a Calibration Error

With a residual phase error of 5° RMS per channel, the picture changes little: the nulls lose a few dB. With 30° RMS, point nulls can collapse. The robust LCMV then becomes preferable despite a lower nominal performance. This is the central message of Part VII: an operational system is not judged by its best case, but by its behavior under errors.

24.9 Controlled Failure

A good system must know how to recognize when its assumptions are no longer valid. Typical failures are as follows.

Too many jammers. With N antennas, you cannot impose more $N-1$ nulls while maintaining a useful gain constraint. Before this theoretical limit, the weight norm explodes and the WNG falls. Answer: explicitly reject only the dominant jammers, or increase the number of antennas.

DOA incorrectly estimated. A one-time null placed at $\hat{\theta}_i$ may be missing an actual jammer at $\hat{\theta}_i + \Delta$. Answer: expanded nulls, derivative constraints, diagonal loading.

Too few snapshots. If $K < N$, the covariance is singular. If $K \simeq N$, it is very noisy. Answer: strong loading, recursive covariance, or return to DAS/LCMV not very aggressive.

Calibration that drifts. MUSIC nulls and peaks are move. Answer: WNG monitoring, periodic recalibration, reference source or internal pilot signal injection.

Consistent sources. Multipaths reduce rank signal; MUSIC and ESPRIT may underestimate the number of sources. Answer: spatial smoothing, forward-backward averaging or sparse methods.

Useful signal in covariance. MVDR can attenuate the target if the steering vector is slightly wrong: it is self-cancellation. Answer: interference-plus-noise covariance, loading, derivative constraint on the target.

24.9.1 Failures: Symptoms and Corrections

The operational diagnosis does not only consist of noting that the SINR is falling. We must connect the visible symptom to a model hypothesis that has just broken.

Failure	Symptom	Diagnosis	Correction
Bad calibration	Shallow nulls, biased DOA peaks	known source or internal pilot	gain/phase recalibration
K too low	unstable covariance, erratic weights	packaging of $\widehat{\boldsymbol{R}}_\alpha$	loading, time average, RLS
L incorrectly estimated	inconsistent MUSIC peaks	MDL/AIC unstable on neighboring windows	multi-window validation
Consistent sources	signal rank too low	merged eigenvalues	spatial smoothing, FB averaging
WNG too low	very high weight norm	$\\|\boldsymbol{w}\\|^2$ high	increase α, relax constraints
Mobile jammer	null offset or late	DOA variable from one window to another	faster DOA cadence, tracking
Useful signal in $\widehat{\boldsymbol{R}}$	attenuated target	covariance test $i+n$ alone	off-target snapshots, loading
Too many jammers	unstable nulls, amplified noise	number of constraints close to N	prioritize dominant jammers

24.10 Final Method-Selection Table

The choice of a method is not made solely on its nominal resolution. It depends on the number of snapshots, the confidence in the manifold, the scene dynamics and the acceptable computational cost.

Method	To choose if	Avoid if	Indicator to watch
DAS	poor model, need robust	strong jammers	target gain, sidelobes
Tapering	critical secondary lobes	maximum resolution required	main lobe width
Null Steering	DOA reliable jammers	DOA uncertain or too many nulls	depth of nulls, WNG
MVDR	reliable covariance, known target DOA	target present with mismatch	SINR output, WNG, loading
Loaded MVDR	limited snapshots, moderate mismatch	need very deep nulls	WNG/rejection compromise
LCMV	explicit target/jammer constraints	too many constraints for N	rank of $\boldsymbol{C}$, target gain
GSC + LMS/NLMS	simple real-time adaptation	unacceptable slow convergence	error $\lvert e(n)\rvert^2$, not μ
GSC + RLS	fast or poorly conditioned scene	low calculation/memory budget	stability of $\boldsymbol{P}$, λ
MUSIC	High resolution DOA, K sufficient	coherent sources without smoothing	eigengap, order $\widehat{L}$
Root-MUSIC	Strict ULA, no grid	non-ULA geometry or ambiguous roots	roots close to the unit circle
ESPRIT	ULA calibrated, low cost	imperfect invariance	packaging of $\widehat{\boldsymbol{U}}_{s,1}$
Spatial smoothing	coherent sources	opening already too weak	rank restored, loss of opening
Sparse/SBL	K short, L unknown	strict real-time cost	off-grid error, stable support
TTD broadband	visible beam squint	sufficient narrow band	lobe stability vs frequency
Near Field	large apertures/RIS/mmWave	Fraunhofer clearly verified	angle-distance error

24.11 Implementation: Canonical Architecture

Listing 24.1: Boucle anti-brouillage simplifiée

```python
import numpy as np

def anti_jam_pipeline(X, theta_s, N, d_over_lambda=0.5,
                      K=128, alpha_rel=1e-2, n_jammers_max=4):
    """
    X         : (N, T) snapshots calibrés
    theta_s   : direction cible (rad)
    K         : nombre de snapshots par fenêtre
    Retour    : y (T,) sortie scalaire
    """
    assert X.shape[0] == N
    assert K > 0
    assert alpha_rel >= 0.0
    T = X.shape[1]
    y = np.zeros(T, dtype=complex)
    w = np.zeros(N, dtype=complex)
    n_updates = T // K
    for upd in range(n_updates):
        Xw = X[:, upd*K:(upd+1)*K]
        R = (Xw @ Xw.conj().T) / K
        R = 0.5*(R + R.conj().T)
        alpha = alpha_rel * np.trace(R).real / N
        R += alpha * np.eye(N)
        if np.linalg.cond(R) > 1e12:
            raise np.linalg.LinAlgError("covariance chargee mal conditionnee")

        # Estimation simple de L par seuil sur valeurs propres
        eigvals, eigvecs = np.linalg.eigh(R)
        # On garde les plus grandes valeurs propres dépassant 2*sigma2
        sigma2 = np.mean(eigvals[:N//2])   # estimateur grossier
        L = int(np.sum(eigvals > 2*sigma2))
        L = max(1, min(L, N-1, n_jammers_max + 1))

        # Jammer DOA estimation (simple MUSIC)
        Un = eigvecs[:, :N-L]
        thetas = np.linspace(-np.pi/2, np.pi/2, 721)
        m = np.arange(N)
        spec = np.zeros(len(thetas))
        eps = np.finfo(float).eps
        for j, th in enumerate(thetas):
            a = np.exp(-1j*2*np.pi*d_over_lambda*np.sin(th)*m)
            v = Un.conj().T @ a
            den = max((v.conj() @ v).real, eps)
            spec[j] = 1.0 / den
        # Keep the L-1 strongest peaks not confused with theta_s
        peak_idx = np.argsort(spec)[::-1]
        jammers = []
        for idx in peak_idx:
            th = thetas[idx]
            if abs(th - theta_s) > np.deg2rad(2.0):
                jammers.append(th)
            if len(jammers) >= L - 1 or len(jammers) >= n_jammers_max:
                break

```

```
        # LCMV: unit gain on theta_s, nulls on jammers
        a_s = np.exp(-1j*2*np.pi*d_over_lambda*np.sin(theta_s)*m)
        C_cols = [a_s] + [np.exp(-1j*2*np.pi*d_over_lambda*np.sin(th)*m)
                          for th in jammers]
        C = np.column_stack(C_cols)
        f = np.zeros(C.shape[1], dtype=complex); f[0] = 1.0
        Rinv_C = np.linalg.solve(R, C)
        G = C.conj().T @ Rinv_C
        if np.linalg.cond(G) > 1e12:
            raise np.linalg.LinAlgError("contraintes LCMV mal conditionnees")
        w = Rinv_C @ np.linalg.solve(G, f)

        y[upd*K:(upd+1)*K] = w.conj() @ Xw
    return y
```

This example summarizes the entire work: a single pipeline which collects, calibrates implicitly (the data is assumed to already be calibrated), detects the number of sources, estimates their directions, distinguishes targets and jammers, constructs LCMV constraints with *diagonal loading*, and applies the resulting weights. It is, at more or less, the code of an operational anti-jamming system.

24.12 Practical Limits

- **More jammers than antennas.** If $L \geq N$, we do not cannot reject everything. Solutions: more antennas, or partial rejection (the most powerful).
- **Jammers consistent with the target.** Causes a problem fundamental: MUSIC sees a single source. Answer: spatial smoothing, sparse, or diversity polarization.
- **Rapid variation jammers.** If the environment changes faster than the update, nulls move behind. Answer: LMS/RLS instead, or increase in update cadence.
- **Non-white noise.** If the noise is spatially structured, the noise eigenvalues are not all equal: MDL and MUSIC degrade. Answer: *prewhitening* by the noise covariance estimated outside the signal.
- **Target direction unknown.** We must then also estimate the target (without hypothesis *a priori*). Approaches: fine DOA + modulation recognition, or dual jammer/target model.

24.13 Summary

An operational anti-jamming chain assembles:

- a calibrated array;
- a covariance estimate with loading;
- robust order detection;
- an estimate of the DOA of the jammers;
- an LCMV beamformer with derivative constraints;
- an update loop adapted to the dynamics.

Each brick has a dedicated chapter. The overall performance depends on the *quality of the weakest link* — typically the calibration, or the consistency of the narrowband model with the wideband reality.

The main panorama of this work is now closed. The following conclusion summarizes the lessons learned, presents the state of the art and opens up current research directions.

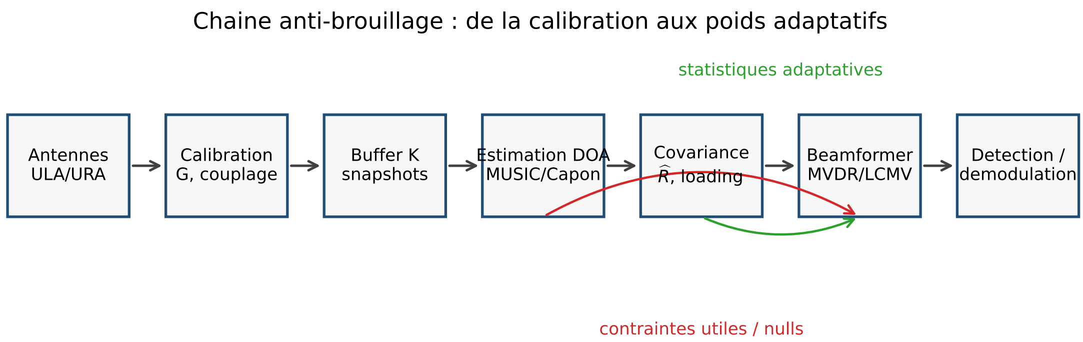

Figure 24.1: Block diagram of the anti-jamming chain: the calibration fixes the spatial model, the DOA provides the constraints, the covariance drives the MVDR/LCMV adaptation; K designates the snapshot window size.

General Conclusion

The objective of this work was to build, starting from the physical foundations, a structured and operational understanding of the models, algorithms and practical limits of modern beamforming: from the plane wave arriving on two antennas to massive MIMO precoding, including MUSIC, ESPRIT, Cramer-Rao bounds and operational anti-jamming chains. This conclusion summarizes the lessons and opens up current research directions.

The common thread of the book was held in a few things: the conviction that we only really understand beamforming by permanently connecting three planes that we sometimes tend to separate. Physics, without which the formulas are only symbolic manipulations. Mathematics, without which physical intuitions remain vague and algorithms impossible to find. Implementation, without which the theory remains speculative and teaches nothing about the actual behavior of the systems. The intellectual model that we tried to build over the course of the chapters is based on this articulation.

A Chain of Ideas

Beamforming can be read as a progression of increasingly rich ideas.

From delay to phase. A wave does not reach all antennas simultaneously. The differential delay $\tau = d \sin\theta / c$ transforms, under the narrow band hypothesis, into phase shift $-kd\sin\theta$. This is the conversion that makes all the complex linear algebra of narrowband beamforming possible.

From phase shift to steering vector. By stacking the phases on the N antennas, we obtain the vector $\boldsymbol{a}(\theta) \in \mathbb{C}^N$, spatial signature of a direction. It is the central object, point of contact between physics and algorithms.

From steering vector to beamformer. Combine antennas by complex weights $\boldsymbol{w}$ produces the scalar output $y = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{x}$. The beam pattern $B(\theta) = \boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(\theta)$ describes the directional sensitivity obtained. The whole design comes down to choosing $\boldsymbol{w}$ wisely.

From geometry to statistics. The Delay-and-Sum only uses geometry. Null Steering adds explicit constraints. MVDR introduces spatial covariance as a source of statistical information. LCMV unifies the two approaches in multiple linear constraints under quadratic criterion. The covariance $\boldsymbol{R}$ becomes the pivot object.

From statics to adaptation. LMS and RLS update the weights iteratively, without explicit matrix inversion. The GSC structure separates fixed constraints and continuous adaptation.

From filtering to estimation. MUSIC and ESPRIT operate the subspace structure of $\boldsymbol{R}$ for *estimate* directions rather than just filtering them. Cramer-Rao bounds set the ultimate performance. Sparse and Bayesian methods extend the estimation to regimes that classical methods do not cover (single snapshot, coherent sources, unknown L).

From simple assumptions to real systems. Broadband relaxes the narrow band hypothesis; MIMO extends to the bilateral transmission/reception case; massive MIMO changes scale. Advanced geometries (URA, sparse, near-field, polarization) cover situations where far-field ULA is not enough.

From theory to practice. Calibration brings hardware imperfections back into the model. Robustness (diagonal loading, derivative constraints, WNG) tolerates residual errors. Anti-jamming chain integration shows how all these blocks work together.

The Invariants

Through this progression, a few invariants run through the work.

The sign convention $e^{-jmkd\sin\theta}$. Kept without exception, it dictates the consistency of all formulas.

The subspace structure of $\boldsymbol{R}$. It appears as soon as we have a few discrete sources and spatial white noise. It is the foundation of MUSIC, ESPRIT, sparse methods and, indirectly, all adaptive beamformers.

The opening $D = (N-1)d$. Geometric quantity which determines the angular resolution. The precision varies as $1/D$ for DAS, $1/(D \cdot \sqrt{K \cdot \mathrm{SNR}})$ for high-resolution methods, and CRB as $1/(N^3 \cos^2 \theta)$.

The selectivity/robustness compromise. At each stage, more selectivity means less room for imperfections. Diagonal loading, derivative constraints, expanded nulls: so many ways to arbitrate this compromise.

Calibration as a necessary step. Without it, the adaptive algorithms and subspaces quickly become unpredictable. This is often the weak link in practice.

Toward the Research Frontier

The work focused on the foundations and methods *mature*. Several current research directions deserve to be mentioned to orient the curious reader.

Deep learning for beamforming. Deep learning replaces certain blocks with neural networks: DOA estimation, MIMO channel prediction, acoustic beamforming. Performance sometimes exceeds traditional methods on scenarios encountered in training, but out-of-distribution robustness remains fragile and proof of performance is lacking.

Reconfigurable Intelligent Surfaces (RIS). Surfaces passives composed of thousands of elements of which each cell reflects with a controllable phase. They allow *shape* the propagation environment itself, opening a environmental “beamforming field ” where we optimize not the receiver but the channel.

Cell-free massive MIMO. Instead of a base station massive, we deploy many small stations connected by fiber. Beamforming becomes distributed, subject to synchronization and bandwidth constraints of the *fronthaul*.

Semi-blind parametric estimation. Combination of partial knowledge of the channel (sparse geometric model, approximate positions of users) and online learning. Particularly active for mmWave communications.

Radar-communications integration (JCAS). The same *wave form* simultaneously serves digital communication and radar detection. The beamformers must satisfy the constraints of both uses, a multi-criteria optimization problem that is still open.

Bounds beyond Cramer-Rao. Bounds *Ziv-Zakai*, *Barankin*, and *Bayesian* bounds provide finer information for regimes that the CRB does not describe (low SNR, ambiguities). These bounds guide the design of new robust estimators.

Quantum sensing and ultimate limits. In the longer term, some quantum receivers or quantum sensors could modify the detection limits of weak signals. This point, however, remains outside the classic framework covered in this book.

Closing Word

Beamforming is not a niche technique; this is one of *spatial languages* common to many modern systems exploiting the waves. Mastering it means learning to see propagation as a processing dimension in its own right, in the same way as time or frequency. Hopefully, the reader emerges, at the end of these chapters, with both the precise analytical tools to design a system, and the intuitive understanding of what it physically represents.

The theory is only worth testing. The complete code that accompanies this book — accessible from the repository indicated in the preface — contains all of the book's algorithms, reproducible simulations, and datasets. It is the natural complement to the formal chapters. Reproduce, experiment, break: this is how we truly internalize these ideas.

Appendix A

Review of Complex Linear Algebra

This appendix brings together the complex linear algebra tools used without proof throughout the work. It is deliberately concise: we will rely on specialized works (Horn & Johnson, Strang, Golub & van Loan) for the details.

A.1 Hermitian Spaces

The $\mathbb{C}^N$ space is equipped with the Hermitian product

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^{\mathrm{H}} \boldsymbol{y} = \sum_{m=0}^{N-1} x_m^* y_m,$$

linear in the second variable, antilinear in the first. The associated norm is $\|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$.

Cauchy-Schwarz inequality. $|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \|\boldsymbol{y}\|$, with equality iff $\boldsymbol{x} \parallel \boldsymbol{y}$.

Orthogonality. $\boldsymbol{x} \perp \boldsymbol{y}$ if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$.

A.2 Hermitian Matrices

A matrix $\boldsymbol{A} \in \mathbb{C}^{N \times N}$ is *hermitian* if $\boldsymbol{A}^{\mathrm{H}} = \boldsymbol{A}$. Properties :

- its eigenvalues are real;
- its eigenvectors associated with distinct eigenvalues are orthogonal;
- it is diagonalized in an orthonormal base: $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Lambda} \boldsymbol{U}^{\mathrm{H}}$ with actual $\boldsymbol{U}^{\mathrm{H}} \boldsymbol{U} = \boldsymbol{I}_N$ and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$.

Positive semi-definite. $\boldsymbol{A} \succeq 0$ if $\boldsymbol{x}^{\mathrm{H}} \boldsymbol{A} \boldsymbol{x} \geq 0$ for all $\boldsymbol{x}$, or equivalently, if all its eigenvalues are ≥ 0.

Positive definite. $\boldsymbol{A} \succ 0$ if the inequality is strict except in $\boldsymbol{x} = \boldsymbol{0}$. Equivalent conditions: all eigenvalues > 0; all main miners > 0; existence of a Cholesky factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^{\mathrm{H}}$ with triangular $\boldsymbol{L}$ with positive real diagonal.

A.3 Matrix Decompositions

A.3.1 Hermitian Diagonalization

For Hermitian $\boldsymbol{A}$: $\boldsymbol{A} = \boldsymbol{U}\,\boldsymbol{\Lambda}\,\boldsymbol{U}^{\mathrm{H}}$. Calculation: $\mathcal{O}(N^3)$ by QR or Jacobi algorithm. Excellent numerical stability.

A.3.2 Cholesky

For $\boldsymbol{A} \succ 0$: $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^{\mathrm{H}}$ with lower triangular $\boldsymbol{L}$. Calculation $\mathcal{O}(N^3/3)$.

A.3.3 LU with Pivoting

For generic $\boldsymbol{A}$: $\boldsymbol{P}\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$, with $\boldsymbol{P}$ permutation. Standard for solving linear systems $\mathcal{O}(N^3)$.

A.3.4 QR

$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ with $\boldsymbol{Q}^{\mathrm{H}}\boldsymbol{Q} = \boldsymbol{I}$ and $\boldsymbol{R}$ upper triangular. Useful for least-squares $\min \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$.

A.3.5 SVD (Singular Value Decomposition)

Any matrix $\boldsymbol{A} \in \mathbb{C}^{M\times N}$ is written

$$\boldsymbol{A} = \boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^{\mathrm{H}},$$

with unit $\boldsymbol{U} \in \mathbb{C}^{M\times M}$ and $\boldsymbol{V} \in \mathbb{C}^{N\times N}$, $\boldsymbol{\Sigma}$ *diagonal rectangular matrix* containing the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

- $\mathrm{rg}(\boldsymbol{A})$ = number of non-zero singular values;
- $\mathrm{Im}(\boldsymbol{A})$ = space generated by the first columns of $\boldsymbol{U}$ corresponding to $\sigma_k > 0$;
- $\mathrm{Ker}(\boldsymbol{A})$ = space generated by the columns of $\boldsymbol{V}$ corresponding to $\sigma_k = 0$;
- best approximation of rank r: $\boldsymbol{A}_r = \sum_{k=1}^{r} \sigma_k \boldsymbol{u}_k \boldsymbol{v}_k^{\mathrm{H}}$ (Eckart-Young theorem).

A.4 Useful Identities

A.4.1 Lemma inversion (Sherman-Morrison)

For invertible $\boldsymbol{A}$ and $\boldsymbol{u}, \boldsymbol{v}$ vectors:

$$(\boldsymbol{A} + \boldsymbol{u}\boldsymbol{v}^{\mathrm{H}})^{-1} = \boldsymbol{A}^{-1} - \frac{\boldsymbol{A}^{-1}\boldsymbol{u}\boldsymbol{v}^{\mathrm{H}}\boldsymbol{A}^{-1}}{1 + \boldsymbol{v}^{\mathrm{H}}\boldsymbol{A}^{-1}\boldsymbol{u}}.$$

A.4.2 Woodbury

More generally, for $\boldsymbol{U}, \boldsymbol{V} \in \mathbb{C}^{N\times K}$:

$$(\boldsymbol{A} + \boldsymbol{U}\boldsymbol{V}^{\mathrm{H}})^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1}\boldsymbol{U}\,(\boldsymbol{I}_K + \boldsymbol{V}^{\mathrm{H}}\boldsymbol{A}^{-1}\boldsymbol{U})^{-1}\,\boldsymbol{V}^{\mathrm{H}}\boldsymbol{A}^{-1}.$$

A.4.3 Trace

$\mathrm{tr}(\boldsymbol{A}\boldsymbol{B}) = \mathrm{tr}(\boldsymbol{B}\boldsymbol{A})$; $\mathrm{tr}(\boldsymbol{A}) = \sum_k \lambda_k$; For $\boldsymbol{A}$ hermitian SDP, $\mathrm{tr}(\boldsymbol{A}\boldsymbol{B}^{\mathrm{H}}) = \mathrm{tr}(\boldsymbol{B}^{\mathrm{H}}\boldsymbol{A})$ plays the role of a matrix dot product.

A.4.4 Determinant

$\det(\boldsymbol{A}\boldsymbol{B}) = \det(\boldsymbol{A})\det(\boldsymbol{B})$; $\det(\boldsymbol{A}) = \prod_k \lambda_k$; $\det(\boldsymbol{I} + \boldsymbol{A}\boldsymbol{B}) = \det(\boldsymbol{I} + \boldsymbol{B}\boldsymbol{A})$.

A.5 Kronecker Product

For $\boldsymbol{A} \in \mathbb{C}^{m\times n}$ and $\boldsymbol{B} \in \mathbb{C}^{p\times q}$:

$$\boldsymbol{A} \otimes \boldsymbol{B} = \begin{bmatrix} a_{11}\boldsymbol{B} & a_{12}\boldsymbol{B} & \cdots \\ a_{21}\boldsymbol{B} & \cdots & \\ \vdots & & \end{bmatrix} \in \mathbb{C}^{mp\times nq}.$$

Properties :

- $(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = (\boldsymbol{AC}) \otimes (\boldsymbol{BD})$;
- $(\boldsymbol{A} \otimes \boldsymbol{B})^{\mathrm{H}} = \boldsymbol{A}^{\mathrm{H}} \otimes \boldsymbol{B}^{\mathrm{H}}$;
- eigenvalues $\{\lambda_i \mu_j\}$, eigenvectors $\boldsymbol{u}_i \otimes \boldsymbol{v}_j$.

Application: structure of the URA $\boldsymbol{a}(u, v) = \boldsymbol{a}_y(v) \otimes \boldsymbol{a}_x(u)$ steering vector.

A.6 Orthogonal Projectors

Let $\boldsymbol{A} \in \mathbb{C}^{N\times L}$ have full column rank. The orthogonal spotlight on $\mathrm{Im}(\boldsymbol{A})$ is

$$\boldsymbol{P}_A = \boldsymbol{A}\,(\boldsymbol{A}^{\mathrm{H}}\boldsymbol{A})^{-1}\,\boldsymbol{A}^{\mathrm{H}},$$

and its complement

$$\boldsymbol{P}_A^{\perp} = \boldsymbol{I}_N - \boldsymbol{P}_A.$$

Properties: $\boldsymbol{P}_A^2 = \boldsymbol{P}_A$, $\boldsymbol{P}_A^{\mathrm{H}} = \boldsymbol{P}_A$, $\boldsymbol{P}_A^{\perp}\boldsymbol{P}_A = 0$. Appears everywhere in MUSIC (projection on noise subspace) and in the calculation of CRB.

A.7 Moore-Penrose Pseudoinverse

For full column row $\boldsymbol{A}$: $\boldsymbol{A}^{+} = (\boldsymbol{A}^{\mathrm{H}}\boldsymbol{A})^{-1}\boldsymbol{A}^{\mathrm{H}}$. For full row row: $\boldsymbol{A}^{+} = \boldsymbol{A}^{\mathrm{H}}(\boldsymbol{A}\boldsymbol{A}^{\mathrm{H}})^{-1}$. In general via SVD: $\boldsymbol{A}^{+} = \boldsymbol{V}\,\boldsymbol{\Sigma}^{+}\,\boldsymbol{U}^{\mathrm{H}}$ where $\boldsymbol{\Sigma}^{+}$ inverts non-zero singular values.

A.8 Complex Matrix Gradients

For $f : \mathbb{C}^N \to \mathbb{R}$ and $\boldsymbol{w} = \boldsymbol{u} + j\boldsymbol{v}$, we define the Wirtinger operators

$$\frac{\partial}{\partial \boldsymbol{w}} = \frac{1}{2}Z \quad \frac{\partial}{\partial \boldsymbol{w}^*} = \frac{1}{2}Z$$

We derive by treating $\boldsymbol{w}$ and $\boldsymbol{w}^*$ as independent. The optimum checks $\partial f/\partial \boldsymbol{w}^* = \boldsymbol{0}$. Usual gradients:

- $\partial(\boldsymbol{a}^{\mathrm{H}}\boldsymbol{w})/\partial \boldsymbol{w}^* = \boldsymbol{0}$, $\partial(\boldsymbol{a}^{\mathrm{H}}\boldsymbol{w})/\partial \boldsymbol{w} = \boldsymbol{a}$;
- $\partial(\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w})/\partial \boldsymbol{w}^* = \boldsymbol{R}\boldsymbol{w}$.

A.9 Conditioning and Stability

For reversible $\boldsymbol{A}$, the *number of packaging*

$$\kappa(\boldsymbol{A}) = \|\boldsymbol{A}\| \cdot \|\boldsymbol{A}^{-1}\|$$

(spectral standard) measures the sensitivity of the $\boldsymbol{Ax} = \boldsymbol{b}$ resolution to disturbances. A relative disturbance ϵ on $\boldsymbol{A}$ or $\boldsymbol{b}$ results in a relative disturbance $\sim \kappa\epsilon$ on $\boldsymbol{x}$. In double precision (16 digits), a $\kappa \sim 10^k$ package typically costs k digits of precision.

Appendix B

Disturbance Analysis

The asymptotic performances of MUSIC, ESPRIT, Capon and other adaptive algorithms are expressed via *disturbance theory* on the eigenvalues and vectors. This appendix recalls the classic results used in the work.

B.1 Eigenvalue Disruption

Let $\boldsymbol{A}$ be Hermitian with spectrum $\lambda_1 \geq \cdots \geq \lambda_N$ and orthonormal eigenvectors $\{\boldsymbol{u}_k\}$. For a disturbance $\boldsymbol{A}' = \boldsymbol{A} + \delta\boldsymbol{A}$ with Hermitian $\delta\boldsymbol{A}$, the eigenvalues λ'_k verify (Weyl's theorem):

$$|\lambda'_k - \lambda_k| \leq \|\delta\boldsymbol{A}\|.$$

More precisely, to the first order:

$$\lambda'_k = \lambda_k + \boldsymbol{u}_k^{\mathrm{H}}\,\delta\boldsymbol{A}\,\boldsymbol{u}_k + \mathcal{O}(\|\delta\boldsymbol{A}\|^2).$$

B.2 Eigenvector Disruption

Under the same hypotheses, and assuming the λ_k *simple* (non-degenerate),

$$\boldsymbol{u}'_k = \boldsymbol{u}_k + \sum_{j\neq k} \frac{\boldsymbol{u}_j^{\mathrm{H}}\,\delta\boldsymbol{A}\,\boldsymbol{u}_k}{\lambda_k - \lambda_j}\,\boldsymbol{u}_j + \mathcal{O}(\|\delta\boldsymbol{A}\|^2).$$

The disturbance of an eigenvector is sensitive to the *separation* $\lambda_k - \lambda_j$ $\lambda_k - \lambda_j$ with its neighbors: the closer they are, the more the disturbance is amplified. This is precisely why MUSIC loses precision when two sources of the same power are close in angle (near signal eigenvalues).

B.3 Application to Sample Covariance

For the estimated covariance $\widehat{\boldsymbol{R}} = \boldsymbol{R} + \delta\boldsymbol{R}$, we show that under the circular complex Gaussian model,

$$\mathbb{E}[\delta R_{ij}\delta R^*_{kl}] = \frac{1}{K}\,R_{ik}\,R^*_{jl},$$

which gives for the Frobenius norm $\mathbb{E}[\|\delta\boldsymbol{R}\|_F^2] = \frac{N^2}{K}\cdot\mathcal{O}(1)$. The typical disturbance of the eigenvalues is therefore relative $\mathcal{O}(N/\sqrt{K})$, of the eigenvectors $\mathcal{O}(1/\sqrt{K})$ by eigendirection.

B.4 Disrupted Signal Subspace

For the signal subspace $\boldsymbol{U}_s$ of $\boldsymbol{R}$ and its estimator $\widehat{\boldsymbol{U}}_s$ coming from $\widehat{\boldsymbol{R}}$, the *main angle* between the two subspaces checks asymptotically

$$\mathbb{E}[\sin^2 \angle(\widehat{\boldsymbol{U}}_s, \boldsymbol{U}_s)] = \mathcal{O}\left(\frac{N}{K}\right),$$

with a constant which depends on the signal-noise gaps $\lambda_L - \sigma^2$. This is the origin of the factor $1/K$ in the asymptotic performance of MUSIC.

B.5 Asymptotic Variance of MUSIC

Stoica & Nehorai (1989) demonstrate that the asymptotic variance of $\widehat{\theta}_\ell^{\mathrm{MUSIC}}$ is

$$\mathrm{Var}(\widehat{\theta}_\ell^{\mathrm{MUSIC}}) = \frac{\sigma^2}{2K} \frac{(\boldsymbol{P}_s \boldsymbol{A}^{\mathrm{H}} \boldsymbol{R}^{-1} \boldsymbol{A} \boldsymbol{P}_s)^{-1}_{\ell,\ell}}{\mathrm{Re}\{\dot{\boldsymbol{a}}_\ell^{\mathrm{H}} \boldsymbol{P}_A^{\perp} \dot{\boldsymbol{a}}_\ell\}}.$$

We compare to the stochastic CRB (chapter 17): MUSIC reaches the limit when the sources are *well separated*, but deviates from it when they are angularly close or of very different powers.

B.6 Variance of ESPRIT

ESPRIT has a slightly higher variance than MUSIC at low SNR, with the dependence

$$\mathrm{Var}(\widehat{\theta}_\ell^{\mathrm{ESPRIT}}) \sim \mathrm{Var}(\widehat{\theta}_\ell^{\mathrm{MUSIC}}) \cdot (1 + \mathcal{O}(\sigma^2)).$$

The difference becomes negligible asymptotically ($\sigma^2 \to 0$ or $K \to \infty$).

B.7 Effect of Diagonal Loading

The diagonal loading $\widehat{\boldsymbol{R}} \to \widehat{\boldsymbol{R}} + \alpha \boldsymbol{I}$ modifies the eigenvalues to $\lambda_k + \alpha$. The signal-noise separation $\lambda_L - \sigma^2$ becomes $\lambda_L - \sigma^2$ (unchanged), but the conditioning becomes $(\lambda_{\max} + \alpha)/(\sigma^2 + \alpha)$, controlled. The asymptotic variance of MUSIC on the loaded version degrades by a factor of approximately $\left(1 + \alpha/(\lambda_L - \sigma^2)\right)^2$, low for modest α.

B.8 Link with the CRB

The CRBs of the chapter 17 are bounds *universal* for any unbiased estimator. The specific estimators (MUSIC, ESPRIT, ML) reach these limits or not depending on the regime. There *asymptotic performance* of an estimator is measured precisely by its deviation from the CRB, and the analysis of disturbances is the tool which makes it possible to quantify this deviation for each family of algorithms.

Appendix C

Python Code: Reference Module

This appendix brings together in a single module all the algorithms implemented in the book. The code assumes `numpy` and, optionally, `scipy` and `matplotlib`. A more complete version, with tests and visualizations, is available on:

https://github.com/JoachimNadal/Adaptive-Beamforming-Simulation

C.1 Reproducible Figures

The figures in this version are generated by the script `scripts/generate_figures.py`. Il fixe une graine aléatoire unique and writes the PDF files to `figures/`. From the `V2` folder, the command is:

`python scripts/generate_figures.py`

Angles are expressed in radians in the code, then converted to degrees only for figure axes.

C.2 ArrayModel

Listing C.1: beamforming.py — module noyau

```
import numpy as np

# ---------------------------------------------------------------------------
# Steering vectors
# ---------------------------------------------------------------------------

def steering_vector(theta, N, d_over_lambda=0.5):
    """Steering vector ULA."""
    m = np.arange(N)
    return np.exp(-1j * 2*np.pi * d_over_lambda * np.sin(theta) * m)

def steering_matrix(thetas, N, d_over_lambda=0.5):
    return np.column_stack([steering_vector(th, N, d_over_lambda) for th in thetas])

def steering_derivative(theta, N, d_over_lambda=0.5):
    m = np.arange(N)
    a = steering_vector(theta, N, d_over_lambda)
    return a * (-1j * 2*np.pi * d_over_lambda * np.cos(theta) * m)

# ---------------------------------------------------------------------------
# Deterministic beamformers
# ---------------------------------------------------------------------------
```

```python

def das_weights(theta0, N, d_over_lambda=0.5):
    return steering_vector(theta0, N, d_over_lambda) / N

def null_steering(theta_s, theta_i, N, d_over_lambda=0.5, alpha=0.0):
    angles = [theta_s] + list(theta_i)
    C = steering_matrix(angles, N, d_over_lambda)
    f = np.zeros(len(angles), dtype=complex); f[0] = 1.0
    G = C.conj().T @ C + alpha * np.eye(len(angles))
    return C @ np.linalg.solve(G, f)

# ---------------------------------------------------------------------------
# Covariance
# ---------------------------------------------------------------------------

def sample_covariance(X):
    N, K = X.shape
    assert K > 0
    R = (X @ X.conj().T) / K
    return 0.5 * (R + R.conj().T)

def diagonal_loading(R, alpha):
    N = R.shape[0]
    assert R.shape == (N, N)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    return 0.5 * (R_alpha + R_alpha.conj().T)

def forward_backward(R):
    N = R.shape[0]
    assert R.shape == (N, N)
    J = np.flip(np.eye(N), axis=0)
    return 0.5 * (R + J @ R.conj() @ J)

# ---------------------------------------------------------------------------
# MVDR and LCMV
# ---------------------------------------------------------------------------

def mvdr_weights(R, a_s, alpha=0.0):
    N = R.shape[0]
    assert R.shape == (N, N)
    assert a_s.shape == (N,)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5 * (R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha mal conditionnee")
    u = np.linalg.solve(R_alpha, a_s)
    den = a_s.conj() @ u
    if abs(den) < np.finfo(float).eps:
        raise np.linalg.LinAlgError("normalisation MVDR instable")
    return u / den

def lcmv_weights(R, C, f, alpha=0.0):
    N = R.shape[0]
    assert R.shape == (N, N)
    assert C.shape[0] == N
    assert f.shape == (C.shape[1],)
```

```
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5 * (R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha mal conditionnee")
    Rinv_C = np.linalg.solve(R_alpha, C)
    G = C.conj().T @ Rinv_C
    if np.linalg.cond(G) > 1e12:
        raise np.linalg.LinAlgError("contraintes LCMV mal conditionnees")
    return Rinv_C @ np.linalg.solve(G, f)

# ----------------------------------------------------------------------------
# LMS and RLS
# ----------------------------------------------------------------------------

def lms(X, d, mu, w0=None):
    N, K = X.shape
    w = np.zeros(N, dtype=complex) if w0 is None else w0.astype(complex).copy()
    err = np.zeros(K, dtype=complex)
    for n in range(K):
        x = X[:, n]
        e = d[n] - w.conj() @ x
        w += mu * x * np.conj(e)
        err[n] = e
    return w, err

def nlms(X, d, mu_tilde=0.5, eps=1e-6, w0=None):
    N, K = X.shape
    w = np.zeros(N, dtype=complex) if w0 is None else w0.astype(complex).copy()
    err = np.zeros(K, dtype=complex)
    for n in range(K):
        x = X[:, n]
        e = d[n] - w.conj() @ x
        norm = eps + (x.conj() @ x).real
        w += (mu_tilde / norm) * x * np.conj(e)
        err[n] = e
    return w, err

def rls(X, d, lam=0.99, delta=1e-2):
    N, K = X.shape
    assert d.shape == (K,)
    assert 0.0 < lam <= 1.0
    assert delta > 0.0
    w = np.zeros(N, dtype=complex)
    P = np.eye(N, dtype=complex) / delta
    err = np.zeros(K, dtype=complex)
    for n in range(K):
        x = X[:, n]
        u = P @ x
        k_gain = u / (lam + (x.conj() @ u).real)
        alpha = d[n] - w.conj() @ x
        w = w + k_gain * np.conj(alpha)
        P = (P - np.outer(k_gain, x.conj() @ P)) / lam
        P = 0.5 * (P + P.conj().T)
        err[n] = alpha
    return w, err

# ----------------------------------------------------------------------------
```

```
# Spectra and DOA
# ---------------------------------------------------------------------------

def bartlett_spectrum(R, thetas, N, d_over_lambda=0.5):
    spec = np.zeros(len(thetas))
    for idx, th in enumerate(thetas):
        a = steering_vector(th, N, d_over_lambda)
        spec[idx] = np.real(a.conj() @ R @ a)
    return spec

def capon_spectrum(R, thetas, N, d_over_lambda=0.5, alpha=0.0):
    assert R.shape == (N, N)
    assert alpha >= 0.0
    R_alpha = R + alpha * np.eye(N)
    R_alpha = 0.5 * (R_alpha + R_alpha.conj().T)
    if np.linalg.cond(R_alpha) > 1e12:
        raise np.linalg.LinAlgError("R_alpha mal conditionnee")
    spec = np.zeros(len(thetas))
    eps = np.finfo(float).eps
    for idx, th in enumerate(thetas):
        a = steering_vector(th, N, d_over_lambda)
        u = np.linalg.solve(R_alpha, a)
        den = max(np.real(a.conj() @ u), eps)
        spec[idx] = 1.0 / den
    return spec

def music_spectrum(R, L, thetas, N, d_over_lambda=0.5):
    assert R.shape == (N, N)
    assert 0 <= L < N
    R = 0.5 * (R + R.conj().T)
    _, eigvecs = np.linalg.eigh(R)
    Un = eigvecs[:, :N-L]
    spec = np.zeros(len(thetas))
    eps = np.finfo(float).eps
    for idx, th in enumerate(thetas):
        a = steering_vector(th, N, d_over_lambda)
        v = Un.conj().T @ a
        den = max(np.real(v.conj() @ v), eps)
        spec[idx] = 1.0 / den
    return spec

def root_music(R, L, d_over_lambda=0.5):
    N = R.shape[0]
    assert R.shape == (N, N)
    assert 0 < L < N
    R = 0.5 * (R + R.conj().T)
    _, eigvecs = np.linalg.eigh(R)
    Un = eigvecs[:, :N-L]
    C = Un @ Un.conj().T
    coeffs = np.array([np.trace(C, offset=lag) for lag in range(-(N-1), N)])
    roots = np.roots(coeffs[::-1])
    valid = roots[np.abs(roots) <= 1.0]
    if len(valid) < L:
        raise ValueError("Root-MUSIC: pas assez de racines valides")
    order = np.argsort(np.abs(np.abs(valid) - 1.0))
    selected = valid[order][:L]
    arg = np.angle(selected) / (-2*np.pi*d_over_lambda)
    angles = np.arcsin(np.clip(arg, -1.0, 1.0))
```

```
    return np.sort(np.real(angles))

def esprit(R, L, d_over_lambda=0.5, tls=True):
    N = R.shape[0]
    assert R.shape == (N, N)
    assert 0 < L < N
    R = 0.5 * (R + R.conj().T)
    _, eigvecs = np.linalg.eigh(R)
    Us = eigvecs[:, -L:]
    Us1 = Us[:-1, :]
    Us2 = Us[1:, :]
    if not tls:
        Psi = np.linalg.lstsq(Us1, Us2, rcond=None)[0]
    else:
        Z = np.hstack([Us1, Us2])
        _, _, Vh = np.linalg.svd(Z, full_matrices=False)
        V = Vh.conj().T
        V12 = V[:L, L:]; V22 = V[L:, L:]
        if np.linalg.cond(V22) > 1e12:
            raise np.linalg.LinAlgError("TLS-ESPRIT mal conditionne")
        Psi = -np.linalg.solve(V22.T, V12.T).T
    eig_psi, _ = np.linalg.eig(Psi)
    arg = -np.angle(eig_psi) / (2*np.pi*d_over_lambda)
    angles = np.arcsin(np.clip(arg, -1.0, 1.0))
    return np.sort(np.real(angles))

# ---------------------------------------------------------------------------
# Détection d'ordre
# ---------------------------------------------------------------------------

def estimate_L_mdl(eigvals, K):
    eigvals = np.sort(eigvals)[::-1]
    N = len(eigvals)
    best = (np.inf, 0)
    for ell in range(N):
        tail = eigvals[ell:]
        if len(tail) == 0 or np.any(tail <= 0):
            continue
        geo = np.exp(np.mean(np.log(tail)))
        ari = np.mean(tail)
        if geo <= 0: continue
        mdl = K * (N - ell) * np.log(ari / geo) + 0.5 * ell * (2*N - ell) * np.log(K)
        if mdl < best[0]:
            best = (mdl, ell)
    return best[1]

# ---------------------------------------------------------------------------
# CRB
# ---------------------------------------------------------------------------

def crb_stochastic(thetas, Ps, sigma2, N, K, d_over_lambda=0.5):
    L = len(thetas)
    assert Ps.shape == (L, L)
    assert sigma2 > 0.0 and K > 0 and N > L
    A = steering_matrix(thetas, N, d_over_lambda)
    D = np.column_stack([steering_derivative(th, N, d_over_lambda) for th in thetas])
    R = A @ Ps @ A.conj().T + sigma2 * np.eye(N)
    R = 0.5 * (R + R.conj().T)
```

```python
    G = A.conj().T @ A
    if np.linalg.cond(G) > 1e12 or np.linalg.cond(R) > 1e12:
        raise np.linalg.LinAlgError("CRB mal conditionnée")
    PA_perp = np.eye(N) - A @ np.linalg.solve(G, A.conj().T)
    M = D.conj().T @ PA_perp @ D
    H = Ps @ A.conj().T @ np.linalg.solve(R, A) @ Ps
    F = 2 * K / sigma2 * np.real(M * H.T)
    return np.linalg.pinv(F)

# ---------------------------------------------------------------------------
# Calibration
# ---------------------------------------------------------------------------

def estimate_calibration(X, theta0, d_over_lambda=0.5):
    x_mean = np.mean(X, axis=1)
    a0 = steering_vector(theta0, X.shape[0], d_over_lambda)
    raw = x_mean / a0
    return raw / raw[0]

def apply_calibration(X, g):
    return X / g[:, None]

# ---------------------------------------------------------------------------
# Beam pattern
# ---------------------------------------------------------------------------

def beam_pattern(w, N, d_over_lambda=0.5, n_theta=721):
    thetas = np.linspace(-np.pi/2, np.pi/2, n_theta)
    A = steering_matrix(thetas, N, d_over_lambda)
    B = w.conj() @ A
    Babs = np.abs(B)
    return thetas, 20*np.log10(Babs / (Babs.max() + 1e-12))
```

C.3 Complete Simulation Example

Listing C.2: simulation_demo.py — exemple bout-en-bout

```python
import numpy as np
import matplotlib.pyplot as plt
from beamforming import *

# Paramètres
N = 16
d_over_lambda = 0.5
K = 1000

# Sources
theta_s = np.deg2rad(0.0)          # cible
theta_i = np.deg2rad([20.0, -35.0])   # 2 jammers
sigma2 = 1e-2                       # noise

# Génération de données
np.random.seed(42)
A = steering_matrix([theta_s] + list(theta_i), N, d_over_lambda)
s = (np.random.randn(3, K) + 1j*np.random.randn(3, K)) / np.sqrt(2)
# target power = 1, jammers = 100 (20 dB stronger)
s[0, :] *= 1.0
```

```
s[1:, :] *= 10.0
n = np.sqrt(sigma2/2) * (np.random.randn(N, K) + 1j*np.random.randn(N, K))
X = A @ s + n

# Covariance
R = sample_covariance(X)
alpha = 1e-2 * np.trace(R).real / N
R_alpha = diagonal_loading(R, alpha)

# Beamformers comparés
a_s = steering_vector(theta_s, N, d_over_lambda)
w_das = das_weights(theta_s, N, d_over_lambda)
w_mvdr = mvdr_weights(R_alpha, a_s)
C = steering_matrix([theta_s] + list(theta_i), N, d_over_lambda)
f = np.array([1.0, 0.0, 0.0], dtype=complex)
w_lcmv = lcmv_weights(R_alpha, C, f)

# Beam patterns
for name, w in [('DAS', w_das), ('MVDR', w_mvdr), ('LCMV', w_lcmv)]:
    th, BdB = beam_pattern(w, N, d_over_lambda)
    plt.plot(np.rad2deg(th), BdB, label=name)
plt.axvline(np.rad2deg(theta_s), color='g', ls=':', label='cible')
for ti in theta_i:
    plt.axvline(np.rad2deg(ti), color='r', ls=':')
plt.xlabel(r'$\theta$ (deg)'); plt.ylabel('dB')
plt.ylim(-70, 5); plt.grid(True); plt.legend()
plt.title('Beam patterns DAS / MVDR / LCMV')

# Estimation DOA
print('Vraies DOA (deg) :', sorted([np.rad2deg(theta_s)] + list(np.rad2deg(theta_i))))
print('Root-MUSIC       :', np.rad2deg(root_music(R, 3, d_over_lambda)))
print('ESPRIT (TLS)     :', np.rad2deg(esprit(R, 3, d_over_lambda)))

# CRB for comparison
Ps = np.diag([1.0, 100.0, 100.0])
crb = crb_stochastic([theta_s] + list(theta_i), Ps, sigma2, N, K)
print('Écart-type CRB (deg) :', np.rad2deg(np.sqrt(np.diag(crb))))
plt.show()
```

This script reproduces most of the results illustrated in the work. It can serve as a starting point for any exploration or implementation. For more complex scenarios (wideband, MIMO, sparse), refer to the GitHub repository mentioned in the preface.

Annotated Bibliography

This bibliography brings together the structuring references which support the methods presented in the book. It does not claim to be exhaustive; rather, it gives the reader the texts to consult to explore the evidence, variations, and limitations.

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